

Particles, Persistence and Derivation of the Mass Spectrum

Structural definition of a particle, classification of states, and construction of the mass operator

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Abstract

A positive mass operator is constructed on the fibres of persistent particles, and its scalar specialisation is determined for the central electron. The result connects particle identity, state classification, and mass evaluation: the magnitude arises from a structure preserving the route, orientation, and memory of its generation. In the electronic sector, two bidirectional readers agree on an explicit trajectory and determine the same return correction; the remaining collections retain the operator, spectral, or scalar scope proved for their own fibres.

The central trajectory visits exactly three nonadic diagonals and twenty-seven cells. Two histories with the same absolute-value histogram retain distinct reversible memories. The finite transform of the return word isolates three nonzero modes and proves spectrally that $(\Omega, P_6) = (80, 6)$. On the multisections, the relative mass law yields a necessary and sufficient operator criterion: a bijective TPK transport preserves mass exactly when it commutes with the mass operator, equivalently when it preserves its complete logarithmic coordinate.

The regional census runs through 104,976 seed pairs, produces 468 emissions, and retains 243 distinct words. Pairwise trit grouping places those words in the 729-position body $S_0 = E_9^3$. The exact threshold difference then determines the shell $243 + 27 + 1 = 271$ and the completion $729 + 243 + 27 + 1 = 1000$, with a normalisation compatible under the projections between dimensions. Regional selection acts afterwards: neither the bijection nor the cubic completion defines a species or a metrological value.

The proposed framework is Triadic Modular Holography, henceforth HMT, and its realisation through Dimensional Mechanics. Its kernel comprises three objects: *All Possible Paths*, henceforth APP, is an arithmetic path structure on $(\mathbb{Z}/9\mathbb{Z})^2$: its additive Cayley graph, with cardinal generators, fixes adjacency, and its cellular realisation is toroidal. Sum and product operations and evaluations are distinguished within that structure; TRIT is the unit of oriented signature $\{-1, 0, +1\}$, distinguishing local regimes; the *Topological Phase Kernel*, henceforth TPK, composes the dynamics of selection, transport, and memory. Its enriched evolution preserves sheets, remainders, quotients, boundary, and carry, and produces the joint discrete structure of the continuum. The composition of 108 updates determines the dodecaphase record, and the orbital transform reversibly reconstructs K , antecedents of the coordinates used by the mass reader.

A particle is defined through the persistence of compatible extensions; the route records its observable history. The atlas distinguishes 81 cells, 56 route configurations, 13 multisections, and 324 oriented conditions. Their quotients, flavour and colour observables, the unique central section, and the operations of conjugation, helicity, and holonomy are constructed, including the Möbius band with its fibre and loop. The integral signature and channel moments determine the positive character; action, the clock, and the unit chart supply the absolute section. Mass ratios preserve the relative character and differences between return exponents.

The marked longitudinal incidence first fixes $N = 5760$, $r_N = 5759/23040$, and $L_\alpha = \alpha_{\text{HMT}}^{16} r_N L_\star$. That length induces the clock, $R_{108} = L_\alpha$, the energy Q_L , the clock mass, and the cotransformation of the electronic factor. On every massive fibre, the identities $H\Lambda = \hbar c_{\text{int}} I$ and $\mathcal{R}_g \Lambda = L_\alpha^2 I$ uniquely determine $G = c_{\text{int}}^3 L_\alpha^2 / \hbar$; its circular expression is a subsequent corollary. The deduction neither uses G as an antecedent of the mass operator nor alters the spectrum tables.

Composition takes place on records before their signatures are evaluated. The development encompasses zero-mass sectors, oriented propagation, mixing, resonances, and Kepler–Sommerfeld orbital realisations. Graded completion produces occupation rules and Pauli exclusion on complete modes, with their spin, sheet, flavour, and colour labels. Metrological comparison brings together each evaluation with its route, operator, and chart, distinguishing central values, intervals, bounds, and widths. Structure, equation, and value are thereby related while preserving the specific content of each result and its physical identification.

Keywords: derivation of the mass spectrum; particle structure; Triadic Modular Holography; Dimensional Mechanics; persistence; holonomy; conjugation; metrological comparison.

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1 Structural identity and the magnitude of mass

Determining a particle's mass requires specifying which object persists and which operation converts its structure into a magnitude. This work constructs that connection: compatible extensions and their records produce fibres of states, on which a positive mass operator is defined. Its electronic specialisation brings together two coincident readers in one scalar evaluation. The thesis is that identity, the mass equation, and the evaluated value belong to a common mathematical chain; physical comparison subsequently examines the realisation of that chain in a measurement chart. The equation's arguments and coefficients thus retain a structural provenance preceding comparison.

Trajectory dynamics incorporates history into that magnitude: it determines the exact cellular support of the central trajectory, reconstructs two memories that a counting statistic cannot distinguish, diagonalises the return reader by finite characters, and characterises mass conservation by the commutator between the mass operator and a TPK transport. These four operations act on the same state and do not replace the atlas, the electron equation, or their prior definitions and proofs.

A formal mathematical kernel called Triadic Modular Holography, henceforth HMT, is introduced through three objects. *All Possible Paths*, henceforth APP, is the arithmetic path structure on $(\mathbb{Z}/9\mathbb{Z})^2$. Adjacency is given by the additive Cayley graph with cardinal generators $\pm(1, 0)$ and $\pm(0, 1)$; the periodic cellular realisation of that support is toroidal. Its operations of additive accumulation and multiplicative composition are distinguished from the evaluations determining their coordinates. TRIT is the unit of oriented signature $\{-1, 0, +1\}$: it determines the local regime and orientation of an operation. The *Topological Phase Kernel*, henceforth TPK, constitutes the kernel's dynamics; it composes selection, transport, memory updating, and return on the enriched state. Dimensional Mechanics develops realisations of that structure on magnitude lines and in physical state spaces. The exposition constructs these objects from their rules before using the internally obtained coordinates in the mass operators.

The development obtains definite results at each of these levels. The composition of 108 TPK updates produces a terminal state that preserves the incidence ledger, its two transverse channels, and the dodecaphase signature; the orbital transform reversibly reconstructs the record K . This genealogy supports the construction of the 81/56/13/324 atlas, the positive mass operator of each fibre, the scalar exception of the central electron, and the composition rules preceding evaluation. The subsequent comparison keeps central values, intervals, bounds, pole masses, running masses, and widths separate. The contribution therefore consists in an effective chain linking the persistent object, its route memory, the operator acting on its fibre, and the precise type of the evaluated magnitude.

In that same causal order, the complete census $104,976 \rightarrow 468 \rightarrow 243 \subset 729$ precedes every regional selection. The bijection $\beta : \mathbb{F}_3^6 \rightarrow E_9^3$ identifies the ambient space of 729 positions, whereas the threshold difference produces the shell $271 = 243 + 27 + 1$ and the exact completion $1000 = 729 + 243 + 27 + 1$. The uniform measure of the completion is compatible with the dimensional projections. These operations construct the ambient space and its boundary; subsequent selection retains its own domain.

1.1 The structure preceding the numerical value

Mass is not the only information carried by a particle. The support of persistence, its orientations, conjugation, representations, and composition rules also enter. In the HMT approach, these data are not appended at the end to an isolated numerical value: they belong to the construction on which the mass functional acts. Two histories can share a scalar reading while preserving different information in their sheets or memory. Equality of values must therefore be distinguished from equality of structures.

This distinction is particularly clear in composition. A table can add coordinates that have already been evaluated; a dynamical construction must first determine whether two histories can compose their phase, boundary, orientations, and predecessors. Records are composed at that level. The signature is extracted afterwards, and the character evaluates that signature. The distinction between these operations provides the mathematical setting for studying composite states, bindings, and resonances.

The question of why a magnitude has the value it does is thus linked to a more precise question: which elements of the construction determine its value, and what information remains available after its evaluation? The article develops this question from the internal rules of the corpus. The subsequent physical comparison examines their realisations without intervening in the selection of the coefficients subjected to comparison.

1.2 Hierarchy of the exposition

The formal kernel is reproduced in common with the preceding articles in the series. This repetition serves self-containment: each text must be readable without access to the others. The specific extensions of the present work follow the kernel and are identified by the objects they construct.

The generated coordinates, the fine-structure constant, and the reduced Planck constant enter when the mass equations require them. The angular pair precedes the channels and their moments; the records precede the signature; the fibres precede the operators acting on them. The central electron is preserved as a structural construction in its own right, even when its magnitude is expressed on a shared chronological scale.

Within this hierarchy, selection, transport, and updating act on the same enriched state. The composition of 108 steps produces the terminal state; its projection onto the dodecaphase ledger, evaluation of the 90/120 channels, the signed extraction U_{12}^{sgn} , the orbital transform, and integral reconstruction establish the chain

$$X_{\text{seed}} \mapsto X_{\text{term}} \mapsto C_{\text{term}} \mapsto U_{\text{term}} \mapsto K.$$

The record, its 12 components, its quotients, and its memory arise from that evolution. The channel, signature, and period charts preserve within one domain the sheet, orientation, carry, and boundary required by mass evaluation. Composition (8.17) brings together the producer and its reconstruction, with the adjacent proofs of updating and inversion; identity (8.13) specifies the reversible transform and its integral sublattice.

The definition of a particle prepares the classification. Conjugation and holonomy, the atlas, the internal observables, and mass evaluation are then developed. The organisation by collections uses that architecture, so that each sector is presented with its support, operation, and reading. Kepler–Sommerfeld is situated within the orbital development of action, where the maps and conditions justifying the relation can be specified; it is not introduced as a historical resemblance devoid of operational content.

1.3 Comparison as the result of an explicit chain

Comparison requires more than two columns of numbers. One must know what has been evaluated, in which unit, and under which definition of the observable. A pole mass, a running mass at a stated scale, and an experimental bound are not interchangeable. A width expresses dynamic information different from the position of a mass. The size of the inventory is useful when it preserves these distinctions, since it makes it possible to examine complete collections rather than only favourable examples.

For this reason, the presentation brings together each output, its provenance, and the conditions of its realisation. An internal identity, the evaluation of a specified signature, and the identification of that signature with a species are related assertions with proofs of their own. Presenting them consecutively shows what the model determines at each level and the specific content of the comparison.

The argumentative unity of the article is, ultimately, the continuity between persistence, classification, operation, and measurement. Mass is an output of this continuity; the structure producing it remains available for understanding its relations with the other readings of the particle.

Section 13 incorporates structural length before the clock: marked incidence produces $N = 5760$, $r_N = 5759/23040$, and $L_\alpha = \alpha_{\text{HMT}}^{16} r_N L_\star$ in (13.2); only then are t_0 , $R_{108} = L_\alpha$, Q_L , m_{clk} , and $b_e(Q_L)$ fixed. The cotransformation law (13.6) shows why the electronic factor cannot be frozen when the longitudinal or action section changes. The operator relations (R01) and (R02) eliminate action length and determine the unique gravitational coefficient (13.7); the circular formula (13.8) follows afterwards. Mass, its route character, and its tables remain causally prior to that radial identification.

2 Formal kernel of Triadic Modular Holography

Triadic Modular Holography, abbreviated HMT, begins with a finite structure of positions and operations. The order of construction matters: positions receive arithmetic evaluations, transitions acquire a ternary signature, and the dynamics preserves the information needed to prolong them. Numerical realizations are formed only afterwards. A constant recognized at the end of this process does not replace the rule that produced its region or the history that determines its evaluation.

The name *All Possible Paths*, APP, expresses the role of paths on the support. It is adopted as the name of the construction, with a conceptual reference to the path formulation of quantum mechanics; it does not imply that its arithmetic law is Feynman's path integral. The *Topological Phase Kernel*, TPK,¹ denotes the composition of selection, transport, memory update, and prolongation. The acronyms are retained only after the objects they represent have been defined.

2.1 APP: positional support and arithmetic evaluations

Consider the nine interior marks on a line graduated from zero to ten,

$$D_9 = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}.$$

Their horizontal and vertical arrangement produces the eighty-one positions of D_9^2 . The interior marks must be distinguished from the ten unit intervals of the reference line. This arrangement fixes an alphabet and two directions; it does not yet use a real expansion to determine a trajectory.

Cyclic identification of the positions yields the support

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2.$$

The chart of positive representatives retains the symbol nine for the zero class. For every positive integer, define

$$\rho_9(n) = 1 + ((n - 1) \bmod 9), \quad q_9(n) = \frac{n - \rho_9(n)}{9}, \quad n = \rho_9(n) + 9q_9(n). \quad (2.1)$$

The positive digital root is therefore a choice of representative of the residue class. By itself, it does not preserve the integer before reduction; the coordinate q_9 supplies precisely the missing information.

At each position (i, j) , accumulating i , repeated j times, produces ij . The two integer evaluations and their projections are

$$S(i, j) = i + j, \quad P(i, j) = ij, \quad \Sigma(i, j) = \rho_9(i + j), \quad \Pi(i, j) = \rho_9(ij).$$

The symmetry $(i, j) \mapsto (j, i)$ interchanges the directions of accumulation and preserves both evaluations. This is a property of a single support, not a subsequent identification between two independent matrices.

Definition 2.1 (Lifted arithmetic evaluation). The APP evaluation that preserves both sheets is

$$\mathcal{A}(i, j) = ((R_{ij}^+, q_{ij}^+), (R_{ij}^\times, q_{ij}^\times)) = ((\rho_9(i + j), q_9(i + j)), (\rho_9(ij), q_9(ij))). \quad (2.2)$$

Here the term *sheet* denotes an evaluation on the same support: additive or multiplicative. The two quotient coordinates belong to the arithmetic state and are not discarded when the residues are reduced.

Proposition 2.2 (Reconstruction of the evaluations). *The two pairs in (2.2) determine $i + j$ and ij exactly. The residues alone do not determine these integer evaluations or their quotients.*

Proof. The first statement is (2.1) applied to each sheet. For the second, $10 = 1 + 9$ and $19 = 1 + 2 \cdot 9$ have the same residue and different quotients. On the two-dimensional support, positions $(5, 5)$ and $(8, 2)$ give the same sum and the same product residue, but products $25 = 7 + 18$ and $16 = 7 + 9$. The residue projection identifies data that the lifted evaluation distinguishes. $\square$

¹The authors also dedicate the initials TPK to Tan Pengkiang. This dedication is independent of the mathematical definition.

	1	2	3	4	5	6	7	8	9	
1	1	2	3	4	5	6	7	8	9	One support: 81 positions.
2	2	4	6	8	1	3	5	7	9	Two evaluations: $S(i, j) = i + j, P(i, j) = ij.$
3	3	6	9	3	6	9	3	6	9	Each row is generated by accumulation: $P(i, j + 1) - P(i, j) = i.$
4	4	8	3	7	2	6	1	5	9	
5	5	1	6	2	7	3	8	4	9	
6	6	3	9	6	3	9	6	3	9	The chart represents $0 \pmod{9}$ by 9.
7	7	5	3	1	8	6	4	2	9	The highlighted block is $B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix},$
8	8	7	6	5	4	3	2	1	9	with eigenvalues 9 and 5.
9	9	9	9	9	9	9	9	9	9	

Figure 1: APP multiplication table in positive representatives. Non-unit rows and columns are distinguished by gray shading; the frame identifies the adjacent diagonal block with equal diagonals. The displayed table is an observable of the arithmetic torus; its representation edges do not constitute an intrinsic topological boundary of the torus.

2.1.1 Cayley graph, toroidal topology, and winding number

The cardinal translations are given, in row and column coordinates, by

$$\nu_N = (-1, 0), \quad \nu_E = (0, 1), \quad \nu_S = (1, 0), \quad \nu_O = (0, -1), \quad T_d(x) = x + \nu_d.$$

The first coordinate increases southwards and the second eastwards. The graph

$$\mathcal{G}_9 = \text{Cay}(X_9, \{\nu_N, \nu_E, \nu_S, \nu_O\})$$

is the two-dimensional cyclic lattice: it has eighty-one vertices and four outgoing oriented edges per vertex. It is a Cayley graph for the *additive* structure of the support. Multiplication of residues has zero divisors and does not turn all APP rows into a multiplicative group.

A cardinal word $w = d_1 \cdots d_r$ and an origin x_0 determine

$$x_k = x_0 + \sum_{j=1}^k \nu_{d_j} \pmod{9}.$$

There are $81 \cdot 4^r$ oriented paths of length r , including returns and backtracking. When the path closes, its integer displacement before reduction defines

$$\text{wind}(w) = \frac{1}{9}(\#S(w) - \#N(w), \#E(w) - \#O(w)) \in \mathbb{Z}^2. \quad (2.3)$$

Path reversal changes the sign of this vector. Returning to the same residue position therefore does not require the winding to vanish. This is the first example of a distinction that will reappear in phase return: one coordinate may close while another records a nontrivial evolution.

2.1.2 Transport quotient and cocycle identity

Let $s_9 : \mathbb{Z}/9\mathbb{Z} \rightarrow D_9$ be the section of positive representatives. The transport quotient associated with additive composition, also called the *carry* in positional arithmetic, is

$$c_9(a, b) = \frac{s_9(a) + s_9(b) - s_9(a + b)}{9}.$$

The compositional character of this memory is expressed by an exact identity.

Lemma 2.3 (Cocycle identity). *For $a, b, c \in \mathbb{Z}/9\mathbb{Z}$,*

$$c_9(a, b) + c_9(a + b, c) = c_9(b, c) + c_9(a, b + c).$$

Proof. Both sums equal

$$\frac{s_9(a) + s_9(b) + s_9(c) - s_9(a + b + c)}{9}.$$

Associativity of the sum of lifted representatives requires precisely this additional term when composing in the quotient. $\square$

With positive representatives, this cocycle is not normalized at the identity: $c_9(0, a) = 1$. If s_0 takes representatives in $\{0, \dots, 8\}$, its normalized carry c_0 satisfies

$$c_9(a, b) = c_0(a, b) + \mathbf{1}_{a=0} + \mathbf{1}_{b=0} - \mathbf{1}_{a+b=0}.$$

The difference is the change of section; it does not alter the cocycle identity. The subsequent decomposition of the calendar into phase and window will use representatives from zero to eight.

The totals of both evaluations distinguish the visible contribution from that retained in the quotient:

$$\begin{aligned} \sum S &= 810, & \sum P &= 2025, & \sum \Sigma &= 405, & \sum \Pi &= 459, \\ \sum q^+ &= 45, & \sum q^\times &= 174. \end{aligned}$$

It follows that

$$2025 - 810 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129. \quad (2.4)$$

The sum-product difference is not exhausted by the visible defect 54. The identity also preserves the quotient defect 129. Removing it would change the arithmetic being transported, even if the residue table remained identical.

2.1.3 Ternary reduction and multiplicative radical

The ring $R = \mathbb{Z}/9\mathbb{Z}$ has unit group $\{1, 2, 4, 5, 7, 8\}$ and maximal ideal

$$\mathfrak{m} = (3) = \{0, 3, 6\}, \quad \mathfrak{m}^2 = 0.$$

Every additive row is a permutation of the nine representatives. A multiplicative row is such a permutation exactly when its index is a unit. Rows three and six contain three repeated values; row nine contains only the representative of the zero class. Thus multiplicative folding already distinguishes the units from the nilpotent sector before any continuous geometry is introduced.

The map $x \mapsto 3x$ has kernel and image both equal to $\mathfrak{m}$. It induces an isomorphism of vector spaces over $\mathbb{F}_3$,

$$R/\mathfrak{m} \longrightarrow \mathfrak{m}, \quad [x] \longmapsto 3x,$$

but not a ring isomorphism. In particular, the visible sequence 3, 6, 9 admits two different readings: its direct reduction modulo three is 0, 0, 0, whereas extracting the factor three and then reducing its internal coordinate gives 1, 2, 0. This second operation preserves a coordinate of the ideal; it is not the same projection as directly reducing the original number.

2.1.4 Latin realization of the additive evaluation

The difference between the two evaluations can also be expressed through operators. Let $(e_r)_{r \in \mathbb{Z}/9\mathbb{Z}}$ be the orthonormal basis of $\mathbb{C}^9$. The assignment $v_{ab} = e_{a+b}$ produces an orthonormal basis in every row and column. The projectors $P_{ab} = v_{ab}v_{ab}^*$ satisfy

$$\sum_b P_{ab} = I_9 = \sum_a P_{ab}, \quad P_{ab}^2 = P_{ab}.$$

Indeed, each translation $b \mapsto a + b$ permutes the nine residues. This yields a realization of a Latin square by rank-one projectors, in the terminology of quantum Latin squares [1, 3]. This construction starts from the additive sheet already defined and makes explicit the map to its operator representation.

The multiplicative evaluation preserves different content: for $a = 3$, the vectors e_{3b} repeat the residues 0, 3, 6, and

$$\sum_{b \in \mathbb{Z}/9\mathbb{Z}} |e_{3b}\rangle\langle e_{3b}| = 3(|e_0\rangle\langle e_0| + |e_3\rangle\langle e_3| + |e_6\rangle\langle e_6|).$$

The multiplicity is recorded in the operator and characterizes the multiplicative radical. Thus the toroidal support, the Latin property, and the operator resolution of the identity are properties requiring different definitions. The quantum magic squares studied by De las Cuevas, Drescher, and Netzer require positive entries and row and column sums equal to the identity; their theory also distinguishes dilations with commuting entries[2]. This reference provides a precise language for comparing realizations of APP.

2.2 TRIT: orientation and quadratic regimes

Definition 2.4 (Ternary signature and local lift). The TRIT signature uses the balanced identification

$$\mathbb{F}_3 \longrightarrow \{-1, 0, +1\}, \quad 0 \mapsto 0, \quad 1 \mapsto +1, \quad 2 \mapsto -1.$$

For $n \in \mathbb{Z}$, its lift preserves $n = r + 3c$, where $r \in \{-1, 0, +1\}$ and $c \in \mathbb{Z}$. In a transition of bounded local amplitude, the three states $(0, -1)$, $(0, 0)$, $(0, +1)$ may exhibit the same zero residue and different carries.

The signature does not replace the trajectory. A zero residue does not mean an absence of orientation, phase, or quotient information. Nor does it introduce a physical quantity by itself. Its function is to distinguish algebraically the transport classes that will be realized next.

2.2.1 Three quadratic regimes and a common exponential

Once the ternary signature has been obtained, its quadratic realization is defined by

$$\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \tau), \quad \tau \in \{-1, 0, +1\}. \quad (2.5)$$

If J_τ is the class of X , then $J_\tau^2 = -\tau I$. The structure distinguishes the elliptic type $\tau = +1$, the parabolic type $\tau = 0$, and the hyperbolic type $\tau = -1$. The real realization does not generate the signature: it expresses the types already produced by oriented reduction.

Proposition 2.5 (Exponentiation of the three types). *In each algebra (2.5),*

$$\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau, \quad (2.6)$$

where

$$C_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n}}{(2n)!}, \quad S_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n+1}}{(2n+1)!}.$$

The three realizations are

τ	J_τ^2	$\exp(tJ_\tau)$
+1	$-I$	$\cos t I + \sin t J_\tau$
0	0	$I + t J_\tau$
-1	I	$\cosh t I + \sinh t J_\tau$

Proof. Separating the even and odd terms of the exponential series and substituting $J_\tau^{2n} = (-\tau)^n I$ and $J_\tau^{2n+1} = (-\tau)^n J_\tau$ gives the formulas. For $\tau = 0$, all terms of degree at least two vanish. In the other two cases, the trigonometric or hyperbolic series are recovered. $\square$

The exponential is common to all three regimes: it is not assigned exclusively to the parabolic branch. The invariants

$$C_\tau(t)^2 + \tau S_\tau(t)^2 = 1$$

follow from $\exp(tJ_\tau)\exp(-tJ_\tau) = I$. The same algebraic principle preserves unity through realizations that differ in their quadratic type.

In the hyperbolic regime, $P_\pm = (I \pm J_{-1})/2$ are complementary projectors and

$$\exp(tJ_{-1}) = e^t P_+ + e^{-t} P_-.$$

Growth and contraction appear jointly in two eigendirections. In the elliptic regime, the parameter describes a rotation; in the parabolic regime, a nilpotent translation. The temporal interpretation adopted by HMT associates $+1$ with return with memory, 0 with the threshold, and -1 with bidirectional opening. This interpretation requires an ordering of the parametrization; it does not add a fourth algebraic type.

2.2.2 Balanced division and composition of lifts

Ternary reduction expresses an integer evaluation of APP through a visible coordinate and a prolongation coordinate. Its function is not merely to abbreviate the notation for an integer: it provides a composition rule determining how the quotient changes when evaluations are added or multiplied. The balanced choice also makes the compatibility of this rule with sign reversal explicit.

Definition 2.6 (Balanced ternary coordinates). For $n \in \mathbb{Z}$, let

$$q_3(n) = \left\lfloor \frac{n+1}{3} \right\rfloor, \quad r_3(n) = n - 3q_3(n).$$

The map

$$\mathcal{B} : \mathbb{Z} \longrightarrow \{-1, 0, +1\} \times \mathbb{Z}, \quad n \longmapsto (r_3(n), q_3(n))$$

is called the balanced ternary decomposition. Its inverse is $\mathcal{L}(r, c) = r + 3c$.

Indeed, $r_3(n) \in \{-1, 0, +1\}$. If two pairs represent the same integer, $r + 3c = s + 3d$, then $r - s$ is a multiple of 3 and belongs to the interval $[-2, 2]$; necessarily $r = s$ and $c = d$. The representation is therefore unique. Division can be repeated on $q_3(n)$, so that each level preserves the data needed to reconstruct the preceding one. The quotient is not a perturbation of the residue: it is its complementary coordinate.

The signature obtained from an evaluation F , with $F = S$ or $F = P$ in APP, is $r_3 \circ F$. When reduction acts on the phase cursor, the same balanced section produces the classes $\{1, 4, 7\}$, $\{2, 5, 8\}$, and $\{3, 6, 9\}$, with values $+1, -1, 0$. The TPK selector defined below uses these classes to determine the operating mode. Reading an evaluation and selecting a mode thus share the same ternary reduction, but have different domains and functions.

Proposition 2.7 (Arithmetic of balanced pairs). Let $n = r + 3c$ and $m = s + 3d$, with $r, s \in \{-1, 0, +1\}$. Define

$$\chi(r, s) = \frac{r + s - r_3(r + s)}{3}.$$

Integer operations are transported to the pairs by

$$(r, c) \boxplus (s, d) = (r_3(r + s), c + d + \chi(r, s)), \quad (2.7)$$

$$(r, c) \boxtimes (s, d) = (rs, rd + sc + 3cd). \quad (2.8)$$

The map $\mathcal{L}$ preserves both operations.

Proof. The equality $r + s = r_3(r + s) + 3\chi(r, s)$ gives

$$n + m = r_3(r + s) + 3(c + d + \chi(r, s)).$$

On the other hand,

$$nm = rs + 3(rd + sc + 3cd).$$

Since the product of two balanced representatives again belongs to $\{-1, 0, +1\}$, it requires no additional residue reduction. Uniqueness of the decomposition proves both formulas. $\square$

For example, $2 = (-1) + 3$ and $4 = 1 + 3$. Their sum and product are reconstructed as

$$(-1, 1) \boxplus (1, 1) = (0, 2), \quad (-1, 1) \boxtimes (1, 1) = (-1, 3).$$

The resulting evaluations are 6 and 8. Adding the quotients suffices in this additive example; the product involves the cross terms rd , sc , and $3cd$. The discrepancy between the two operations also resides in the prolongation coordinate, not only in the residue.

The term χ is a cocycle of the balanced section. If $r \oplus s = r_3(r + s)$, it satisfies

$$\chi(r, s) + \chi(r \oplus s, t) = \chi(s, t) + \chi(r, s \oplus t).$$

Both sides represent the quotient of the same sum of three integers. This identity ensures that regrouping the operations does not change the reconstructed evaluation. At the same time, projection onto the first component suppresses this accounting: $1 + 1 = -1 + 3$, whereas the residue reading retains only -1 . The information removed is determined, in this case, by $\chi(1, 1) = 1$.

The relation with modulus 9 preserves a second discrete scale. The map

$$\beta : \{-1, 0, +1\} \times \mathbb{F}_3 \longrightarrow \mathbb{Z}/9\mathbb{Z}, \quad (r, \bar{c}) \longmapsto r + 3c \pmod{9}$$

is well defined and bijective. Changing the representative of $\bar{c}$ adds a multiple of 9; conversely, equality of images fixes r first, by reduction modulo 3, and then $\bar{c}$. Addition transported by this bijection contains the term $\chi(r, s)$ from (2.7), now reduced modulo 3. Thus the two ternary levels do not compose by independent coordinatewise addition: normalization of the first level modifies the second.

In particular, the classes 3, 6, and 0 of $\mathbb{Z}/9\mathbb{Z}$ have first component 0, but second components 1, 2, and 0, respectively. The distinction already observed between direct reduction and extraction of the ideal coordinate appears as a property of β . Preserving this coordinate prevents the zero residue sector from being artificially collapsed into a single position. Modulus 1000, used later to organize decimal blocks, serves a different positional-representation function; its quotient is preserved through the corresponding division. A change of base preserves the integer when it retains residue and quotient, whereas coincidence of residues in different bases does not by itself identify their transport states.

2.2.3 Inversion, residue class, and non-visible coordinates

Additive inversion lifts unambiguously:

$$\mathcal{B}(-n) = (-r_3(n), -q_3(n)).$$

In particular, 3, 0, and -3 have coordinates $(0, 1)$, $(0, 0)$, and $(0, -1)$. They share a zero signature but represent different evaluations. In a transition restricted to a quotient of maximum amplitude 1, these are the three possibilities of the visible fiber over zero; in a general integer lift, that fiber contains all pairs $(0, c)$, with $c \in \mathbb{Z}$.

The quotient coordinate recovers the evaluated integer, but does not by itself recover the trajectory that produced it. Two APP paths may share evaluation, residue, and quotient while differing in orientation, sequence of modes, or winding. Thus decomposition $\mathcal{B}$ is integrated into the transport state rather than replacing it. The arithmetic memory of a division and the memory of a trajectory are related coordinates, not synonyms.

For a signature sequence of length n , oriented reversal is

$$C_{\text{rev}}(\tau_1, \dots, \tau_n) = (-\tau_n, \dots, -\tau_1).$$

Applying it twice restores the sequence. The operation reverses the order of positions and the sign of each signature; it preserves the length and the zero positions in reversed order. In the adopted parametrization, it interchanges return with memory, associated with $+1$, and bidirectional opening, associated with -1 , while retaining the threshold 0. Transport of the remaining trajectory data is specified by the corresponding TPK involution.

This signature reversal must be distinguished from conjugation within a quadratic algebra. The former may change the type τ ; the latter, constructed below, acts within a type already fixed. This distinction prevents attributing an algebraic equivalence between different regimes to a sign reversal.

2.2.4 Quadratic product, conjugation, and algebraic norm

An element of $\mathbb{A}_\tau$ has a unique expression $aI + bJ_\tau$. The relation $J_\tau^2 = -\tau I$ determines its product:

$$(aI + bJ_\tau)(cI + dJ_\tau) = (ac - \tau bd)I + (ad + bc)J_\tau. \quad (2.9)$$

This is a real realization of the type already selected. The coordinates a, b are not APP residues, and the product (2.9) does not replace (2.8); it expresses another operation in the declared quadratic codomain.

Definition 2.8 (Conjugation and algebraic norm). Conjugation of type τ and its algebraic norm are

$$\overline{aI + bJ_\tau} = aI - bJ_\tau, \quad N_\tau(aI + bJ_\tau) = a^2 + \tau b^2.$$

Proposition 2.9 (Multiplicativity and invertibility). For $z, w \in \mathbb{A}_\tau$,

$$z\bar{z} = N_\tau(z)I, \quad N_\tau(zw) = N_\tau(z)N_\tau(w).$$

Moreover, z is invertible if and only if $N_\tau(z) \neq 0$, in which case

$$z^{-1} = \frac{\bar{z}}{N_\tau(z)}.$$

Proof. The product of $aI + bJ_\tau$ and $aI - bJ_\tau$ is $(a^2 + \tau b^2)I$. Conjugation is multiplicative and the algebra is commutative, so that $(zw)\bar{z}\bar{w} = (z\bar{z})(w\bar{w})$. If the norm does not vanish, the stated formula supplies the inverse. Conversely, if $zw = I$, multiplicativity implies $N_\tau(z)N_\tau(w) = 1$. $\square$

The quadratic form N_τ is positive definite only in the elliptic type. For $\tau = 0$, it equals a^2 , and the element J_0 is nonzero although its square and norm are zero. For $\tau = -1$, it equals $a^2 - b^2$; the nonzero elements $I + J_{-1}$ and $I - J_{-1}$ have zero norm and their product vanishes. Here the algebraic norm is a multiplicative quadratic form: it does not presuppose a positive vector-space norm.

The three types admit faithful realizations by

$$aI + bJ_\tau \mapsto \begin{pmatrix} a & -\tau b \\ b & a \end{pmatrix}.$$

The determinant of this matrix is $N_\tau(aI + bJ_\tau)$. The complex structure, the dual-number structure, and the split real structure are thus represented by a single matrix expression. Each preserves its own law and zero divisors. The form of signature (1, 1) in the last case admits a Lorentzian interpretation in dimension 1 + 1; identification of a physical metric additionally requires a specific space, field, and transport rule.

2.2.5 Multiplicative conservation and composition of evolutions

The exponential already obtained belongs to the set of norm-one elements:

$$N_\tau(\exp(tJ_\tau)) = 1.$$

This conservation persists under composition of parameters. Indeed, all factors are functions of the same generator, and

$$\exp(tJ_\tau)\exp(sJ_\tau) = \exp((t + s)J_\tau).$$

Comparing the two components gives the addition laws

$$C_\tau(t + s) = C_\tau(t)C_\tau(s) - \tau S_\tau(t)S_\tau(s), \quad (2.10)$$

$$S_\tau(t + s) = S_\tau(t)C_\tau(s) + C_\tau(t)S_\tau(s). \quad (2.11)$$

Conjugation changes t to $-t$ and coincides with inversion within this subgroup. It does not change τ : it reverses an evolution of the chosen regime.

In the elliptic type, the preserved form is $a^2 + b^2$, and the exponential trajectory traverses the unit-norm circle. In the hyperbolic type, the eigencoordinates are $u = a + b$, $v = a - b$, with $uv = N_{-1}(z)$. The exponential gives

$$(u, v) = (e^t, e^{-t}), \quad uv = 1.$$

Growth of one coordinate is accompanied by contraction of the other. Conservation expresses a relation between the two coordinates; it does not assert that each remains constant.

In the parabolic type,

$$(I + tJ_0)(I + sJ_0) = I + (t + s)J_0, \quad (I + tJ_0)^{-1} = I - tJ_0.$$

The evolution does not stop because $J_0^2 = 0$. Nilpotency removes terms of order at least two, but preserves the linear term that transports the parameter. The visible symbol 0, the zero element of the algebra, and the nonzero generator J_0 are distinct. This distinction is necessary to describe the threshold without identifying it with an absence of state or dynamics.

Proposition 2.10 (Composition of affine coordinates). *In the chart where the scalar component does not vanish, represent a direction by $I + uJ_\tau$. If $1 - \tau uv \neq 0$, the direction of its product with $I + vJ_\tau$ has coordinate*

$$u \star_\tau v = \frac{u + v}{1 - \tau uv}.$$

Proof. We have

$$(I + uJ_\tau)(I + vJ_\tau) = (1 - \tau uv)I + (u + v)J_\tau.$$

Dividing by the scalar component gives the stated expression. $\square$

The affine chart preserves the ratio of components, not the scalar factor divided out. When $1 - \tau uv = 0$, the product must be examined before changing charts: it may have a well-defined direction with zero scalar component, or be the zero element in the presence of zero divisors. The composition law still belongs to the full algebra; a coordinate singularity does not automatically imply a singularity of the object. This formula will reappear in the elliptic regional reading. Its use requires retaining the denominator and, where applicable, the normalization factor.

2.2.6 Transport of type and subsequent realizations

The algebras $\mathbb{A}_{+1}$, $\mathbb{A}_0$, and $\mathbb{A}_{-1}$ are not isomorphic as unital real algebras. The first is a field; the second contains a nonzero nilpotent; the third has nontrivial idempotents and is isomorphic to $\mathbb{R} \times \mathbb{R}$. Thus a change of signature organizing a route cannot be interpreted as an undifferentiated isomorphism between the three realizations. The TPK preserves the regime index, the domains, and the effective operator acting at each transition.

In the electron chapter, the pair of projections of the APP sheets produces a normalized commutator whose square is $-I$. The same block contains an involution of square I and two nilpotent directions. The joint realization will be proved for that concrete pair of projections; the preceding classification explains why the three quadratic relations can appear within one matrix algebra without being identified with one another.

In the exceptional prolongation, the Paley matrix reduced modulo 3 satisfies $A_W^2 = -I_6$. This relation endows the ternary space with a structure of dimension 3 over $\mathbb{F}_9$. Its connection to a real operator root is expressed through the integral presentation

$$\mathbb{Z}[u]/(u^2 + 1).$$

Base changes to $\mathbb{F}_3$ and $\mathbb{R}$ produce representations of the same relation. Each retains its characteristic, dimension, and incidence information. Developing the respective operators establishes the connection, not equality of their cardinalities.

Norm-one conservation consequently describes the quadratic realizations of evolutions. Conservation of quotients, orientation, and trajectory belongs to the transported arithmetic state. Both properties participate in the construction, with different functions: one determines the identities of the representations; the other makes it possible to compose, reconstruct, and prolong the operations that give rise to them.

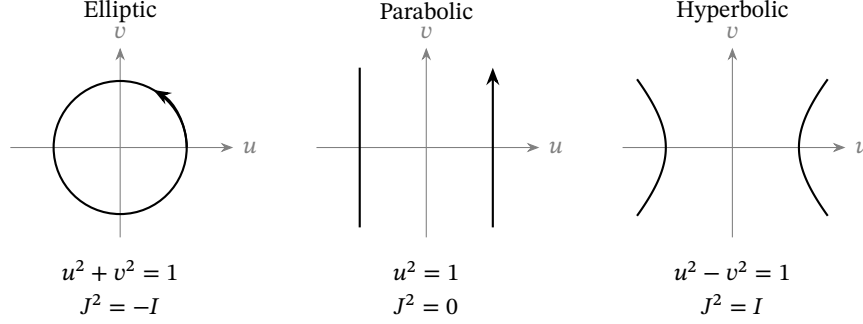


Figure 2: Quadratic realizations of TRIT. For $J_\tau = \begin{pmatrix} 0 & -\tau \\ 1 & 0 \end{pmatrix}$, the exponential preserves $u^2 + \tau v^2$. Parabolic degeneration preserves u and transports v linearly. The three diagrams represent orbits of quadratic forms; their geometric type corresponds to the algebraic relation of the generator.

2.3 TPK: Topological Phase Kernel

The dynamics uses the phase selector

$$M_{\text{ph}}(r) = \begin{cases} +1, & r \in \{1, 4, 7\}, \\ -1, & r \in \{2, 5, 8\}, \\ 0, & r \in \{3, 6, 9\}. \end{cases}$$

Its value determines which evaluation operates. Cardinal transport moves the corresponding cursor; the record preserves the previous value, the new quotient, the orientation, and the transition. Selection and transport are therefore different functions: a ternary sign is not a displacement on the torus.

We shall denote by $\hat{X}$ the space of arithmetic states with trajectory memory. An element contains the positions and directions of both cursors, the active mode, the TRIT signature, the phase, the residues and quotients of both evaluations, the carries, the constructed prefix, and the boundary data. When a refinement cylinder is introduced, it forms part of the state. This notation avoids replacing the object by a reduced version that retains only the visible symbol.

Definition 2.11 (TPK update). Let Sel_t be mode selection through M_{ph} , Tra_t the lifted cardinal transport, and Mem_t the record update. The transition is

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (2.12)$$

Each application preserves the fields it does not modify and updates the remaining fields by Euclidean division and trajectory concatenation.

The notation (2.12) expresses the order of composition; it does not replace the effective rules. The following section specifies its finite two-cursor realization completely. Prolongation will then be obtained not by erasing its memory and repeating the first emission, but by transporting the prefixes and their compatible boundaries. Thus APP supplies the evaluations, TRIT distinguishes their regimes, and TPK composes their evolution.

2.3.1 Dynamic state, observables, and operational dependencies

The Topological Phase Kernel organizes the evolution of the two APP sheets with the orientation determined by TRIT. Its definition comprises the state on which it acts, the transition rules, and the relations that make prolongations compatible. Equation (2.12) acquires operational content when the coordinates received and modified by each factor are specified. Position belongs to the toroidal support; phase belongs to the calendar; orientation determines transport; quotients record the arithmetic evaluation before reduction. These coordinates are related through the transition and retain different mathematical functions.

The observable projection of a state is

$$q = (x_+, d_+, x_-, d_-, \theta) \in \mathcal{Q}_{\text{obs}} := (X_9 \times \mathcal{D})^2 \times \Theta_{108}, \quad \mathcal{D} = \{N, E, S, O\}, \quad \Theta_{108} = \mathbb{Z}/108\mathbb{Z}. \quad (2.13)$$

The subscripts distinguish the additive and multiplicative cursors. The fibers of complete states over q preserve the ordered trajectory, the integer evaluations of both sheets, their residues and quotients, and the orientation and boundary records. A precision prolongation adds the prefix and the compatible cylinder. Thus two states with the same projection q may retain different records.

Operational dependencies have three levels. At the local level, the selector activates a sheet, transport changes its position, and the update computes the new arithmetic coordinates. At the trajectory level, these transitions compose and retain the records of preceding steps. At the refinement level, extension relations determine which continuation preserves the predecessor's prefix, signature, and boundary. The finite calendar describes the phase of these operations; holonomy describes their composition on the history.

2.3.2 Tritic selection and lifted cardinal transport

Let $\bar{\theta} \in \{0, \dots, 107\}$ be the calendar representative. The operating phase and the dodecaphase sector are

$$r(\theta) = 1 + (\bar{\theta} \bmod 9), \quad \beta(\theta) = \left[\left[\frac{\bar{\theta}}{9} \right] \right]_{12}. \quad (2.14)$$

The previously defined selector may be written $\varepsilon_9 = M_{\text{ph}}$. Its value vector in the ordered chart of D_9 is

$$m_{\text{ph}} = (1, -1, 0, 1, -1, 0, 1, -1, 0).$$

The function $M_{\text{ph}} : D_9 \rightarrow \{-1, 0, +1\}$ and this vector are two presentations of the same selector, with their respective domains.

Define the activation indicators

$$a_+(\theta) = \mathbf{1}_{\{1,4,7\}}(r(\theta)), \quad a_\times(\theta) = \mathbf{1}_{\{2,5,8\}}(r(\theta)).$$

Cursor transport is then

$$x'_+ = x_+ + a_+(\theta)v_{d_+}, \quad x'_\times = x_\times + a_\times(\theta)v_{d_\times} \quad \text{in } X_9. \quad (2.15)$$

In phases 3, 6, 9, both positions remain unchanged; the neutral event forms part of the chronology and its record. Cardinal directions determine the vectors v_d , whereas the phase signature determines which vector is applied.

The integer lift preserves boundary crossing. If $s : X_9 \rightarrow D_9^2$ chooses positive representatives and $x' = x + av_d$, with $a \in \{0, 1\}$, define

$$b_d(x, a) = \frac{s(x) + av_d - s(x')}{9} \in \mathbb{Z}^2. \quad (2.16)$$

Divisibility by 9 follows from equality in X_9 . If $z \in \mathbb{Z}^2$ records previous crossings, the update is $z' = z + b_d(x, a)$. Thus

$$s(x') + 9z' = s(x) + 9z + av_d.$$

The integer transport data remain reconstructible after the position has been reduced on the torus.

Proposition 2.12 (Conservation of lifted displacement). *For a cardinal trajectory of n steps, with activation indicators a_t ,*

$$s(x_n) + 9z_n = s(x_0) + 9z_0 + \sum_{t=0}^{n-1} a_t v_{d_t}.$$

The same equality is preserved under concatenation of two compatible trajectories.

Proof. The one-step identity is (2.16). Summing it over the n steps cancels the intermediate coordinates. Splitting the summation interval into two segments produces exactly the concatenation law. For a closed trajectory, the difference $z_n - z_0$ is the previously defined winding number. $\square$

The same principle governs APP evaluations. If a sheet takes the integer value N_t , its record simultaneously preserves $N_t = R_t + 9q_t$. The difference between two steps satisfies

$$N_{t+1} - N_t = (R_{t+1} - R_t) + 9(q_{t+1} - q_t).$$

Spatial-crossing memory z_t and arithmetic quotient q_t have different definitions; preserving them jointly maintains the geometric and arithmetic provenance of every transition.

2.3.3 Two-cursor chronology and composite returns

The cardinal involution j interchanges N with S and E with O . After applying (2.15), both directions are reversed at the end of a block of 27 steps. Writing

$$h(\theta) = \mathbf{1}_{\{0,27,54,81\}}((\bar{\theta} + 1) \bmod 108),$$

the complete rule on the observable projection is

$$F_{\text{obs}}(q) = (x'_+, j^{h(\theta)} d_+, x'_\times, j^{h(\theta)} d_\times, \theta + 1). \quad (2.17)$$

The reading that forms the emission uses the positions before transport, according to the explicit recurrence in Section 4. The observable update and emission therefore compose in a fixed order.

Proposition 2.13 (Reversibility and period of the observable calendar). *The map F_{obs} is a permutation of $\mathcal{Q}_{\text{obs}}$ and satisfies $F_{\text{obs}}^{108} = I$. Within a block of 54 steps, positions and directions are restored; the coordinate in Θ_{108} advances by 54.*

Proof. Given the final state, the preceding phase is first recovered by subtracting one in Θ_{108} . That phase determines whether directions were reversed and which cursor moved. Applying the corresponding involution and subtracting the cardinal vector uniquely recovers the initial state.

Each interval of 27 steps contains nine activations of each cursor. The following block preserves the activations and reverses the directions, so the integer displacements cancel over 54 steps. The rule has period 54 in its position and orientation increments; the same cancellation holds for any calendar origin. After 108 steps, θ is also restored. $\square$

We reserve

$$\mathcal{K}_{27} = F_{\text{obs}}^{27} : \mathcal{Q}_{\text{obs}} \longrightarrow \mathcal{Q}_{\text{obs}}$$

for the iterate of 27 transitions. Its fourth iterate is the identity on the observable state defined in (2.13). Projecting history onto the calendar makes this finite check possible. The trajectory record, its evaluations, and the new prefixes remain in the dynamic state on which $\mathcal{U}_t$ acts; observable equality does not reinitialize them.

2.3.4 Affine memory transport and composition of states

The observable projection admits fibers of arithmetic and geometric data. Over a configuration q , we write a state in the form

$$x = (q, o, h, c, \sigma, C, b, \lambda).$$

Here o preserves orientation and sheet; h , residues and quotients; c , additive transport memory; σ , the boundary signature; C , the precision cylinder; b , its boundary label; and λ , the recorded trajectory. The signature is determined by a map

$$\text{Sig}_q : H_q \times C_q \times \text{Hist}_q \longrightarrow S_q, \quad \sigma = \text{Sig}_q(h, c, \lambda).$$

The symbols H_q , C_q , Hist_q , and S_q denote, respectively, the sets of residue–quotient data, the abelian transport-memory group, the recorded trajectories, and the boundary signatures. Realizations of this signature are specified component by component in the regional and terminal constructions. This dependence prevents treating the signature as a free parameter once the data producing it are fixed.

The elementary residue–quotient chart is explicit. For $n \in \mathbb{Z}$, let

$$r = 1 + ((n - 1) \bmod 9), \quad u = \frac{n - r}{9}.$$

Then $n = r + 9u$, with $r \in D_9$ and $u \in \mathbb{Z}$. Uniqueness follows because two representatives in D_9 congruent modulo 9 are equal. This chart supplies local integer reconstruction. The precision towers additionally incorporate the truncation maps and cylinders developed in 4 and 4.

Let $\gamma : q \rightarrow q'$ be an observable trajectory. Fiber transport is expressed through maps

$$H(\gamma) : H_q \rightarrow H_{q'}, \quad C(\gamma) : C_q \rightarrow C_{q'}, \quad \text{Hist}(\gamma) : \text{Hist}_q \rightarrow \text{Hist}_{q'}.$$

The maps respect identities and composition, and $C(\gamma)$ is an abelian-group homomorphism. The increment $k_{\text{tr}}(\gamma) \in C_{q'}$ satisfies

$$\begin{aligned} k_{\text{tr}}(\text{id}_q) &= 0, \\ k_{\text{tr}}(\gamma_2 \circ \gamma_1) &= k_{\text{tr}}(\gamma_2) + C(\gamma_2)k_{\text{tr}}(\gamma_1). \end{aligned} \tag{2.18}$$

Here k_{tr} denotes the transport cocycle; the rational coordinate κ of record K retains its notation in the dodecaphase-closure chapter.

Transportable coordinates are updated by

$$\begin{aligned} h' &= H(\gamma)h, & c' &= C(\gamma)c + k_{\text{tr}}(\gamma), \\ \lambda' &= \gamma \circ \lambda, & \sigma' &= \text{Sig}_{q'}(h', c', \lambda'). \end{aligned} \tag{2.19}$$

The second equation is an affine action: the linear term transports previous memory and the cocycle adds the increment produced by the trajectory. In the spatial-crossing realization for each cursor separately, $C(\gamma) = I$ and $k_{\text{tr}}(\gamma)$ is the sum of the vectors in (2.16). This concrete instance connects the abstract law with the cardinal operations already defined.

Proposition 2.14 (Compatibility of the update with concatenation). *Updating by (2.19) along γ_1 , followed by updating along γ_2 , coincides with updating along $\gamma_2 \circ \gamma_1$.*

Proof. For the affine coordinate, two updates give

$$c'' = C(\gamma_2)C(\gamma_1)c + C(\gamma_2)k_{\text{tr}}(\gamma_1) + k_{\text{tr}}(\gamma_2).$$

Functoriality of C and (2.18) turn this expression into $C(\gamma_2 \circ \gamma_1)c + k_{\text{tr}}(\gamma_2 \circ \gamma_1)$. The statement for h follows from composition of H , and that for λ from associativity of trajectories. The final signature agrees because it is evaluated on the same three reconstructed coordinates. $\square$

The cylinder and its boundary complete this law through the relations Ext_γ , defined in 2.3.7. Their elements determine which of the preceding updates preserve an admissible prolongation. States, together with the triples (γ, x, x') admitted by these relations, form a category over observable trajectories:

$$(\gamma_2, x', x'') \circ (\gamma_1, x, x') = (\gamma_2 \circ \gamma_1, x, x'').$$

The identity of x is (id_q, x, x) . Composition preserves arithmetic transport and boundary prolongability simultaneously. This is the structure in which returns with memory and successive refinements take place.

2.3.5 Dodecaphase record, phase, and accumulated memory

The 108 transitions are distributed over the ordered windows

$$E_m = \{9m - 8, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The record preserves the phase origin, orientation, and data produced during each window. Its map is

$$\mathcal{R}_{12}(x) = ((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m))_{m=1}^{12},$$

where τ_m is the window's subroute, σ_m its orientation, κ_m the integer boundary increment, and Λ_m the local incidence record. We adopt the chart developed in 8.3; its window indices correspond to $m = \beta + 1$. The domain comprises compatible TPK histories, and the codomain their oriented temporal records. Composition of segments and their increments is governed by (2.19). By contrast, the group C_{12} describes translation of the window index; Θ_{108} preserves the step calendar. This distinction specifies what is transformed and what is observed when a cycle is completed.

In coordinates $\theta = 9b + r$, with $0 \leq r < 9$, advancing 9 steps preserves r and increases b by one. Addition of phases introduces the quotient

$$c_0(r, r') = \left\lfloor \frac{r + r'}{9} \right\rfloor,$$

whose cocycle identity ensures associativity of composition. The finite calendar preserves $[b]_{12}$; a record of turns preserves $b \in \mathbb{Z}$, or $b \in \mathbb{N}_0$ for a history with fixed origin. Section 4 proves the nonsplit exact sequence (4.13). Its function is to describe the relation between these coordinates before they are used to construct the signed state.

The signed record receives phase and memory data through the composition developed in Section 8:

$$R_{\text{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12}. \quad (2.20)$$

Its domain is the terminal state with memory; its output belongs to $\mathbb{Z}^{12}$. The reversible transformation between this signed state and K , together with the rational reading of K , is proved in the corresponding chapter. Record operations precede these transformations. Reconstructing a record from its cyclic differences verifies its recoverability; its dynamic provenance must be preserved through the events that produced it.

For the encoded trace (d_i) retained by the record, the event reader counts codes $\{2, 4, 6, 8\}$ in each window with orientation $(+, +, +, -, -, -, +, +, +, -, -, -)$. These codes and the cardinal symbols of $\mathcal{D}$ are different record coordinates. Chapter 8 develops this reader, the integral decomposition $1 + 5 + 6$, the congruences characterizing its image, and its exact inverse. The inversion proof establishes which data this chart preserves; the definition of $\mathcal{R}_{12}$ specifies the oriented history on which it acts.

2.3.6 Phase incidence and internal channel construction

The selector determines three subsets of D_9 :

$$H_+ = \{1, 4, 7\}, \quad H_- = \{2, 5, 8\}, \quad H_0 = \{3, 6, 9\}.$$

Let $H_{\text{act}} = H_+ \sqcup H_-$ be the set of active phases and $O_{\text{ph}} = D_9 \setminus \{9\}$ the chart preceding the marked return. Denote by $P_2(H_{\text{act}})$ the set of unordered pairs of active phases. Internal incidence is given by

$$\mathcal{F}_6^{\text{ph}} = H_{\text{act}} \times P_2(H_{\text{act}}), \quad \mathcal{F}_8^{\text{ph}} = O_{\text{ph}} \times P_2(H_{\text{act}}), \quad \Theta_{\text{ph}} = D_9 \times P_2(H_{\text{act}}). \quad (2.21)$$

The indices describe the number of positions in the first component. The construction uses the phases and calendar origin already fixed; subsequent exceptional realizations will transport these incidences to their own domains.

Proposition 2.15 (Cardinalities and oriented phase incidence). *The sets in (2.21) have cardinalities 90, 120, 135, respectively. If*

$$\partial_{\text{ph}} = \Theta_{\text{ph}} \times \{-1, +1\}, \quad E_{\text{ph}} = \partial_{\text{ph}} \sqcup \{\infty\}, \quad L_{\text{ph}} = H_0 \times P_2(H_{\text{act}}),$$

then

$$|\partial_{\text{ph}}| = 270, \quad |E_{\text{ph}}| = 271, \quad |L_{\text{ph}}| = 45, \quad |\partial_{\text{ph}} \sqcup L_{\text{ph}}| = 315.$$

Simultaneously translating the phase origin and marked return preserves all these cardinalities and inclusions.

Proof. There are six active phases, so $|P_2(H_{\text{act}})| = \binom{6}{2} = 15$. Multiplying by 6, 8, 9 gives 90, 120, 135. Orientation doubles 135; the return mark adds one element; the three neutral phases give $3 \cdot 15 = 45$. A calendar translation transports each factor by a bijection and carries the marked point to its image. Products and disjoint unions thus preserve the incidence. $\square$

Here the cardinality 271 has a phase-incidence realization. The cubic completion in Section 4 has its own construction of the corona 1000 – 729. The correspondence between the two realizations must preserve the maps and marks identifying them, in addition to their cardinality. Likewise, the selector ε_9 , the incidence (90, 120), and the extractor $\mathcal{E}_{108}^{90,120}$ retain the domains and operations just distinguished.

2.3.7 Trajectory reversal and conjugate prolongations

Bidirectionality begins with an operation on trajectories before scalar values are assigned to them. For $\gamma = (x_0; d_1, \dots, d_n)$, with endpoint x_n , define

$$\text{rev } \gamma = (x_n; jd_n, \dots, jd_1).$$

The reversed trajectory starts at the endpoint of the original. Composition satisfies

$$\text{rev}^2 \gamma = \gamma, \quad \text{rev}(\gamma_2 \circ \gamma_1) = \text{rev } \gamma_1 \circ \text{rev } \gamma_2,$$

and integer displacement changes sign. These identities follow by reversing the order of steps and replacing each cardinal vector with its opposite. For a loop, it follows that $\text{wind}(\text{rev } \gamma) = -\text{wind}(\gamma)$.

On records, reversal also requires transporting the order of components and the orientation of events. Route reversal and signature conjugation are distinguishable operations: their composition is specified when applied to a concrete record. In particular, the regional involution interchanging closure and propagation components preserves the self-scaling axis; its formula is given in Section 4.4.1. This regional operation acts on a different domain from cardinal trajectory reversal.

Extensions over a route $r : q \rightarrow q'$ are expressed by relations between fibers of compatible states. If $\mathcal{F}_q$ denotes the fiber of complete states over q , we require

$$\text{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}, \quad \Delta_{\mathcal{F}_q} \subseteq \text{Ext}_{\text{id}_q},$$

$$\text{Ext}_{r_2} \circ \text{Ext}_{r_1} \subseteq \text{Ext}_{r_2 \circ r_1}.$$

Their composition preserves the evaluations of both sheets, orientation, quotients, prefix, and boundary. Route reversal does not by itself imply that every extension relation is a bijection. Conjugate prolongation is defined on the domain where transport of these data satisfies the corresponding compatibilities.

2.3.8 Nonadic holonomy and continuity of refinement

A phase turn composes transitions on the state with memory:

$$\Gamma_{9,k} = \mathcal{U}_{k+8} \circ \dots \circ \mathcal{U}_k. \quad (2.22)$$

On an individualized history, this functional composition realizes the holonomy relation; on the full domain we retain $\Gamma_{9,k} \subseteq \hat{X}_k \times \hat{X}_{k+9}$, with the admissible prolongations developed later. The cyclic base has nine phases, while the fiber preserves trajectory, quotients, and boundary. Monodromy is the action of one turn of this base on the fibers; holonomy $\Gamma_{9,k}$ expresses its realization through the effective TPK transitions.

If g observes phase and M the number of retained turns,

$$g(\Gamma_{9,k}x) = g(x), \quad M(\Gamma_{9,k}x) = M(x) + 1.$$

The first identity expresses phase closure and the second identifies the new temporal depth. Coinductive prolongation also preserves its cylinders and prefixes. Restrictions between depths compose an inverse system; a compatible section gathers its antecedents without replacing them by the latest observation. This arithmetic of histories and its unity are developed in 5.1–5.4; the regions and their readers are constructed in Section 4.

Phase transfer on emissions already constructed has a different domain. Let $\mathcal{A}_+$ and $\mathcal{A}_\times$ be the additive and multiplicative signature alphabets, with cardinalities 18 and 26. Their product identifies the catalog $\mathcal{U}_6$. For $f : \mathcal{A}_+ \times \mathcal{A}_\times \rightarrow \mathbb{R}$, the sheet averages are

$$(E_+f)(s, e) = \frac{1}{18} \sum_{s' \in \mathcal{A}_+} f(s', e), \quad (E_\times f)(s, e) = \frac{1}{26} \sum_{e' \in \mathcal{A}_\times} f(s, e').$$

These emissions and their cardinalities are constructed through the recurrence in Section 4. Once they have been obtained, the transfer operator is defined on $\mathcal{U}_6 \times C_9$ by

$$P_+ = E_+, \quad P_- = E_\times, \quad P_0 = I, \quad \mathcal{K}_{\text{ph}}((u, r), (v, r+1)) = P_{M_{\text{ph}}(r)}(u, v),$$

with zero weight for all other phase transitions. The index r uses the positive representative of the cyclic class. The phase order $(+1, -1, 0)^3$ then produces the renewal

$$\mathcal{K}_{\text{ph}}^9|_r = E_+ E_\times.$$

Indeed, both averages are idempotent, commute, and average different factors; their composition assigns weight $1/(18 \cdot 26) = 1/468$ to each emission. This equality characterizes a subsequent transfer observable. History transport remains given by (2.22), with its memory and boundary.

The resulting hierarchy establishes the function of the following chapters. Finite emission constructs the domain and regions; extension relations preserve their prefixes; the nonadic connection prolongs them; the dodecaphase record determines the data entering K and the closure of α . Numerical and incidence realizations receive this preceding structure. This order allows every consequence to be presented in its own chapter while preserving, from the formal kernel onwards, the objects that make it possible.

2.4 Accumulated memory and transport resolvents

Calendar return does not remove increments produced during the journey. To express this distinction in operator form, an accumulated trajectory-event record is constructed first, followed by its generating series. Reduction of memory variables will be performed on this transport, also preserving the source and output reader.

2.4.1 Oriented events and record update

Let (d_t) be the sequence of integer direction codes retained by a TPK trajectory. The code d_t and the cursor's cardinal direction are different coordinates; no identification of their alphabets is assumed. We number the windows starting from zero:

$$I_m = \{9m + 1, \dots, 9m + 9\}, \quad m \geq 0.$$

In the first cycle, I_m is window E_{m+1} of the dodecaphase record. Its orientation is

$$\sigma = (1, 1, 1, -1, -1, -1, 1, 1, 1, -1, -1, -1).$$

For a prolonged history, $\sigma_m \in \{-1, 1\}$ is the sign retained in the corresponding window. The signed event is

$$a_m = \sigma_m \sum_{t \in I_m} \mathbf{1}_{\{2,4,6,8\}}(d_t), \quad |a_m| \leq 9. \quad (2.23)$$

Each summand is incorporated when its transition occurs. This definition counts trace events; it does not select directions through a value that is to be obtained later.

Let $\mathbf{e}_0, \dots, \mathbf{e}_{11}$ be the basis of $\mathbb{Z}^{12}$ and $\mathbf{S}\mathbf{e}_j = \mathbf{e}_{j+1 \bmod 12}$. Write $R = S^{-1}$. The accumulated record in fixed coordinates satisfies

$$z_{m+1} = z_m + a_m \mathbf{e}_{m \bmod 12}.$$

A prefix with no accumulated history starts at $z_0 = 0$; at a different origin, z_0 retains the preceding record. The frame accompanying the calendar is $y_m = S^{-m} z_m$.

Proposition 2.16 (Update and return with memory). *The record in the moving frame satisfies*

$$y_{m+1} = R y_m + \eta_m, \quad \eta_m = a_m R \mathbf{e}_0. \quad (2.24)$$

For every $n \geq 1$,

$$y_n = R^n y_0 + \sum_{j=0}^{n-1} R^{n-1-j} \eta_j. \quad (2.25)$$

In particular,

$$y_{12} = y_0 + \sum_{j=0}^{11} a_j \mathbf{e}_j. \quad (2.26)$$

Proof. Multiplying the update of z_m by $S^{-(m+1)}$ gives $y_{m+1} = S^{-1}y_m + a_m S^{-1}\mathbf{e}_0$, since $S^{-m}\mathbf{e}_{m \bmod 12} = \mathbf{e}_0$. Formula (2.25) follows by induction. For $n = 12$, $R^{12} = I$ and $R^{11-j}\eta_j = a_j R^{12-j}\mathbf{e}_0 = a_j \mathbf{e}_j$, which proves (2.26). $\square$

Thus twelve windows restore the frame and preserve the event vector. The count alone does not recover the order of all steps within each window; that order remains in the recorded trajectory.

To specify the balance, define

$$D_3 = I - S^3, \quad D_4 = I - S^4, \quad B = D_3^\top D_3 + D_4^\top D_4,$$

$$\mathcal{M}(y) = \frac{1}{2} y^\top B y, \quad q(y) = \sum_{j=0}^{11} y_j.$$

The first is a quadratic form of the record; the second records the sum of its coordinates. They are not identified here with physical energy or charge.

Proposition 2.17 (Event-induced balance). *Update (2.24) satisfies*

$$\begin{aligned} \mathcal{M}(y_{m+1}) - \mathcal{M}(y_m) &= a_m (By_m)_0 + 2a_m^2, \\ q(y_{m+1}) - q(y_m) &= a_m. \end{aligned} \quad (2.27)$$

Proof. Expanding the products gives $B = 4I - S^3 - S^{-3} - S^4 - S^{-4}$; hence $R^\top B R = B$ and $B_{00} = 4$. Since $y_{m+1} = R(y_m + a_m \mathbf{e}_0)$, the quadratic difference is $a_m \mathbf{e}_0^\top B y_m + \frac{1}{2} a_m^2 B_{00}$. The coordinate sum is invariant under permutation R , giving the second equality. $\square$

2.4.2 Generating series and uniform memory control

The series preserves all increments, not only their sum at the end of a cycle. With a formal variable u , let

$$Y(u) = \sum_{m \geq 0} y_m u^m, \quad H_\eta(u) = \sum_{m \geq 0} \eta_m u^m.$$

Proposition 2.18 (Resolvent and tail bound). *The formal identity*

$$Y(u) = (I - uR)^{-1}(y_0 + uH_\eta(u)) \quad (2.28)$$

holds. Both series converge for $|u| < 1$. If $\ell : \mathbb{R}^{12} \rightarrow \mathbb{R}$ is linear, $L = \|\ell\|_{1 \rightarrow \mathbb{R}}$, $M_0 = \|y_0\|_1$, and $0 < r < 1$, then, for every integer $N \geq 0$ and $|u| \leq r$,

$$\left| \sum_{m > N} \ell(y_m) u^m \right| \leq L r^{N+1} \left(\frac{M_0 + 9(N+1)}{1-r} + \frac{9r}{(1-r)^2} \right). \quad (2.29)$$

The derivative series converges uniformly on every disk $|u| \leq r < 1$.

Proof. Multiplying (2.24) by u^{m+1} and summing gives $Y - y_0 = uRY + uH_\eta$. The constant term of $I - uR$ is invertible, so its formal inverse is $\sum_{k \geq 0} u^k R^k$. Moreover, R preserves the ℓ^1 norm and $\|\eta_m\|_1 \leq 9$, whence $\|y_m\|_1 \leq M_0 + 9m$. For the tail, sum the majorants $L(M_0 + 9m)r^m$, using

$$\sum_{m > N} r^m = \frac{r^{N+1}}{1-r}, \quad \sum_{m > N} mr^m = r^{N+1} \left(\frac{N+1}{1-r} + \frac{r}{(1-r)^2} \right).$$

Finally, $\sum_{m \geq 1} Lm(M_0 + 9m)r^{m-1} < \infty$ majorizes the derivative series and proves its uniform convergence. $\square$

The variable u counts windows of nine transitions. Identifying it with another reader's variable requires preserving that clock and the map relating the two readings.

2.4.3 Memory elimination: operator, source, and reader

On a domain where the update is fixed as $U : X \rightarrow X$, the free vector space $V = \mathbb{Q}^{(X)}$ on the states admits the operator $T\delta_x = \delta_{U(x)}$. The clock is included in x if the transition depends on it. This linear extension preserves distinct states even when they share an observable phase.

Consider a decomposition $V = V_0 \oplus V_1$, where V_1 is the auxiliary sector to be eliminated, and write

$$T = \begin{pmatrix} A & B_1 \\ C & D \end{pmatrix}, \quad v = \begin{pmatrix} v_0 \\ v_1 \end{pmatrix}, \quad \ell = (\ell_0, \ell_1).$$

Here $A : V_0 \rightarrow V_0$, $B_1 : V_1 \rightarrow V_0$, $C : V_0 \rightarrow V_1$, and $D : V_1 \rightarrow V_1$. The following identities are formal; they also hold wherever the resolvents converge.

Proposition 2.19 (Schur reduction with source and reader). *Let*

$$D_u = (I - uD)^{-1}, \quad \mathcal{F}(u) = I - uA - u^2 B_1 D_u C.$$

The retained component of $(I - uT)^{-1}v$ is

$$w_0 = \mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1), \quad w_1 = D_u(v_1 + uCw_0). \quad (2.30)$$

Its complete reading satisfies

$$\begin{aligned} \ell(I - uT)^{-1}v &= (\ell_0 + u\ell_1 D_u C) \mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1) \\ &\quad + \ell_1 D_u v_1. \end{aligned} \quad (2.31)$$

Proof. The two equations of $(I - uT)w = v$ are $w_0 = v_0 + uAw_0 + uB_1 w_1$ and $w_1 = v_1 + uCw_0 + uDw_1$. Solving the second and substituting it into the first gives (2.30); applying ℓ_0 and ℓ_1 to the two components gives (2.31). All formal inverses exist because their constant terms are identities. $\square$

The term $u^2 B_1 D_u C = \sum_{k \geq 0} u^{k+2} B_1 D^k C$ describes departure into memory, k steps within it, and a return. By contrast, $u\ell_1 D_u C$ reads memory before that return. Thus reducing only the operator does not necessarily preserve the observation: the source and reader must also be transformed.

The distinction is visible in the cycle already constructed. If P retains $\mathbf{e}_0$, then $PRP = 0$, but

$$\langle \mathbf{e}_0, (I - uR)^{-1} \mathbf{e}_0 \rangle = \frac{1}{1 - u^{12}}, \quad \langle \mathbf{e}_{11}, (I - uR)^{-1} \mathbf{e}_0 \rangle = \frac{u}{1 - u^{12}}. \quad (2.32)$$

Indeed, $R^n \mathbf{e}_0 = \mathbf{e}_{-n \bmod 12}$: the two series collect, respectively, $n \equiv 0$ and $n \equiv 1 \pmod{12}$. The resolvent of the compression is I , whereas compression of the resolvent preserves the omitted returns. This example concerns the twelve-window record, not the entire TPK state.

2.4.4 Segment composition and coherence of reduction

Let three composable affine transports be $F_i(y) = R_i y + b_i$, with compatible domains and codomains. The memory produced by the complete journey is

$$F_3 F_2 F_1(y) = R_3 R_2 R_1 y + b_3 + R_3 b_2 + R_3 R_2 b_1. \quad (2.33)$$

The proof is successive substitution of the three affine formulas. Grouping the first two segments first, or the last two, produces the same expression; the factors transport each increment from its point of origin. In (2.24), $R_i = R$ and b_i is the event of its window.

Eliminating several layers preserves a similar coherence. To make it explicit, now decompose $V = V_0 \oplus V_1 \oplus V_2$ and write

$$T = \begin{pmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{pmatrix}, \quad Q_2 = (I - u D_{22})^{-1}.$$

Eliminating the third component leaves the four blocks

$$\begin{aligned} A' &= A + u B_2 Q_2 C_2, & B'_1 &= B_1 + u B_2 Q_2 D_{21}, \\ C'_1 &= C_1 + u D_{12} Q_2 C_2, & D'_{11} &= D_{11} + u D_{12} Q_2 D_{21}. \end{aligned} \quad (2.34)$$

Proposition 2.20 (Transitivity of elimination). *If $D_* = (D_{ij})_{i,j=1}^2$, the complement obtained in two stages coincides with that obtained by eliminating both memories together:*

$$\begin{aligned} I - u A' - u^2 B'_1 (I - u D'_{11})^{-1} C'_1 &= I - u A \\ &\quad - u^2 (B_1 \ B_2) (I - u D_*)^{-1} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}. \end{aligned} \quad (2.35)$$

Proof. Solving the third equation of $(I - uT)w = v$ and substituting it into the first two produces (2.34). Eliminating the second component then gives the left-hand side of (2.35). Solving the two memory equations together gives the right-hand side, by Proposition 2.19. The formal solution is unique because $I - uT$ has constant term I . Both procedures also transform the source and reader according to (2.30)–(2.31). $\square$

Balance (2.27) describes record increments. The preceding identities allow them to be transported when memory is reused in dodecaphase closure; they do not by themselves determine additional analytic coefficients.

3 Joint construction of the continuum through compatible histories and prolongations

The arithmetic expressed by a normal form preserves a precise reading of the state that produces it. To situate that reading within HMT, it is also necessary to retain the histories, the sheets, and the operations acting upon them. This chapter brings together that common substrate. Its different constructions are obtained from the same tree of APP-TRIT-TPK executions; the expository divisions allow their properties to be studied without turning them into independent domains chosen by a subsequent application.

The starting point is the formal kernel presented above. An admissible emission contains the APP operation, the TRIT orientation, the change of sheet, the residue and the quotient, together with the effective TPK memory update. When only a residue is expressed, the execution history still belongs to the source object. Two executions with an identical final reading are not identified for that reason alone.

3.1 Admissible histories and memory by composition

Let $H_0 := \{*\}$. The level H_1 is the finite nonempty set of admissible APP-TRIT seeds, and, for $k \geq 1$, H_k is the set of enriched histories of length k admitting compatible TPK prolongations at every finite horizon. Its elements are constructed through successive emissions from the initial domain of the kernel. We write $\zeta\varepsilon$ for the prolongation of ζ by an admissible emission ε , and $\rho_k(\zeta\varepsilon) = \zeta$ for deletion of the final emission. The label produced by the emission is retained, even when different labels express the same scalar state. The space of complete histories is defined by

$$H_\infty^{\text{enr}} := \varprojlim_k (H_k, \rho_k).$$

The cofinal subsystem $X_N := H_{6N}$, with $X_0 = H_0$, coincides with the sextet clock used in the cardinal proof; by cofinality there is a canonical identification $H_\infty^{\text{enr}} \cong \varprojlim_N (X_N, \rho_{6(N+1), 6N})$.

On the two arithmetic sheets, the local data satisfy the exact divisions

$$\delta_\diamond = r_\diamond + 9q_\diamond, \quad 0 \leq r_\diamond < 9, \quad \diamond \in \{\Sigma, \Pi\}.$$

The oriented TRIT coordinate similarly admits an expression $\Delta = o_r + 3\kappa$, with the oriented representative declared by the kernel. Preservation of the quotient allows these representations to be composed without replacing an emission by its isolated residue.

In a memory chart, the update of an emission is written

$$U_\varepsilon(m) = C_\varepsilon m + \kappa_\varepsilon.$$

For a path $\alpha = (\varepsilon_1, \dots, \varepsilon_k)$, the effective operation is the ordered composition of those updates. Its linear part and its translation are

$$C_\alpha = C_{\varepsilon_k} \cdots C_{\varepsilon_1}, \quad \kappa_\alpha = \sum_{j=1}^k C_{\varepsilon_k} \cdots C_{\varepsilon_{j+1}} \kappa_{\varepsilon_j}.$$

The empty product in the last sum is the identity.

Proposition 3.1 (Affine compatibility of concatenation). *If α precedes β , then*

$$C_{\beta\alpha} = C_\beta C_\alpha, \quad \kappa_{\beta\alpha} = C_\beta \kappa_\alpha + \kappa_\beta, \quad U_{\beta\alpha} = U_\beta U_\alpha.$$

Proof. Substituting $U_\alpha(m)$ into U_β yields $C_\beta C_\alpha m + C_\beta \kappa_\alpha + \kappa_\beta$. Decomposition into linear part and translation gives the three identities. Associativity of composition proves that the result does not depend on an auxiliary subdivision of the path. $\square$

This cocycle identity is the precise content of the accumulated memory. The quantity transported by a second path depends on the chart update it performs; in general, it is not the untransported sum of two records.

The nonadic return preserves a second distinction. In the phase and turn coordinates, with $\bar{r} \in \{0, \dots, 8\}$, its representation is

$$\sigma_9(w, r) = \left(w + \left\lfloor \frac{\bar{r} + 1}{9} \right\rfloor, r + 1 \pmod{9} \right), \quad \sigma_9^9(w, r) = (w + 1, r).$$

Over nine steps, exactly one crossing from 8 to 0 occurs. The phase returns and the turn coordinate increases. The enriched holonomy also retains the memory updates accompanying that traversal.

3.2 Tree refinement and measure of histories

The initial measure is the canonical one from the census: $\mu(*) = 1$ and $\mu(c) = 1/|H_1|$ for each seed $c \in H_1$. In this subsection, each history has a finite positive number of prolongations. Deletion of emissions defines the refinement maps. Admissibility of the kernel provides the prolongations; when a residence uses a neutral emission, that emission explicitly provides a section of ρ_k . The existence of this section concerns the tree that includes that emission, not an arbitrary restriction of routes.

Let $d(\zeta) \geq 1$ be the number of admissible emissions at ζ . The APP order and the census of prolongations determine the canonical law $p_{\text{can}}(\varepsilon \mid \zeta) = 1/d(\zeta)$. Without additional parameters, this law defines

$$\mu(\zeta\varepsilon) = \frac{\mu(\zeta)}{d(\zeta)}.$$

A nonuniform weighting may replace p_{can} only when it has already been produced and added to the enriched record. The correlative constructor of this chapter uses p_{can} exclusively; therefore, its measure is a function of the APP-TRIT-TPK tree and not of an external choice.

Proposition 3.2 (Cylindrical compatibility and conditional expectation). *At every level,*

$$\sum_{\rho_k(y)=x} \mu(y) = \mu(x).$$

In the spaces $L^2(H_k, \mu_k)$, the maps

$$(J_k f)(y) = f(\rho_k(y)), \quad (E_k g)(x) = \sum_{\rho_k(y)=x} \frac{\mu(y)}{\mu(x)} g(y)$$

satisfy $E_k = J_k^$, $E_k J_k = I$, and $\|J_k f\| = \|f\|$.*

Proof. The first identity is the normalization of the local probabilities. Applying it to $|f(x)|^2$ yields the isometry. Grouping the sum by fibers of ρ_k yields the formula for E_k ; on a function constant on each fiber, that formula returns its value. $\square$

The compatible masses define a probability measure on the space of infinite histories: the cylinder of a finite history receives its mass $\mu(\zeta)$. The measure extends from the algebra of cylinders, and its uniqueness follows because they generate the σ -algebra. If the tree is finitely branching and every history has a prolongation, the inverse limit of its levels under deletion of emissions is nonempty. In the cylinder topology, every nonempty cylinder has positive measure.

This measure distinguishes histories converging to the same representation. The uniform measure on terminal values would be a different reading, since it would eliminate the multiplicities that the construction has just preserved.

3.2.1 Enriched readers and fiber memory

The difference between history and value admits an operational criterion. At a finite level H_k , let $q_k : H_k \rightarrow Z_k$ be a reader and let $M_k : H_k \rightarrow \mathcal{M}_k$ be the preserved record. The enriched reader is $\hat{q}_k = (q_k, M_k)$. Memory distinguishes states within a single fiber of the reader; its sufficiency is verified on the domain under consideration. This formulation incorporates into the article itself the complete development of the fiber information used below.

Proposition 3.3 (Recoverability at a finite level). *An inverse of $\hat{q}_k$ exists on its image if and only if $\hat{q}_k$ is injective. For a random variable Z taking values in H_k with distribution μ_k ,*

$$H(Z) = H(q_k(Z)) + H(Z \mid q_k(Z)).$$

If $\hat{q}_k$ is injective on the support of Z , then

$$H(Z \mid q_k(Z), M_k(Z)) = 0.$$

Here $H(Z) = -\sum_z \Pr(Z = z) \log \Pr(Z = z)$, with $0 \log 0 = 0$, is finite entropy in nats; conditional entropy is the average entropy of the conditional distributions.

Proof. An inverse recovers a unique antecedent, which requires injectivity; conversely, injectivity makes it possible to assign to each expressed pair its unique antecedent. Since $q_k(Z)$ is a deterministic function of Z , grouping the sum defining $H(Z)$ by fibers of q_k yields the chain rule. If the pair $(q_k(Z), M_k(Z))$ distinguishes the support, each corresponding conditional distribution is concentrated on a single state, and its entropy is zero. $\square$

Not publishing a coordinate and erasing the record that permits its recovery are distinct operations. This result does not identify entropy with the cardinality of the continuum or introduce a thermal interpretation: its domain, reader, and distribution are precisely those just specified.

3.2.2 Equivariance of the measure under tree symmetries

Proposition 3.4 (Joint transport of the tree and its measure). *Let $g = (g_k)_{k \geq 0}$ be a collection of bijections $g_k : H_k \rightarrow H_k$ that commutes with truncations, fixes the root, and bijectively transports admissible emissions while preserving their multiplicities. Then*

$$d(g_k \zeta) = d(\zeta), \quad (g_k)_* \mu_k = \mu_k, \quad (g_\infty)_* \mu = \mu.$$

More generally, if μ_ζ is the law of continuations of a history $\zeta \in H_k$, then

$$(g_\infty)_* \mu_\zeta = \mu_{g_k \zeta}.$$

Proof. The bijection of emissions preserves their number with multiplicities. The uniform seed distribution is likewise invariant. Each path and its image therefore receive the same product of local weights $1/d(\zeta)$. Equality of measures follows at every level and on every cylinder. Compatibility with truncations defines g_∞ , and uniqueness of the measure determined by cylinders proves equality in the limit. The same argument, beginning at ζ , proves the assertion about continuations. $\square$

This establishes the naturality of the tree and its measure: a coherent change of representation transports the weights and does not require choosing them anew. Preservation of emissions and multiplicities is an essential hypothesis of the proposition, not a datum added after the construction.

3.3 Transport by sheets and exact accounting of outputs

First consider the space with orthonormal basis formed by the histories at level k . The transport preserving the label of each prolongation is

$$\mathcal{U}_k e_\zeta = \sum_{\varepsilon \text{ admissible at } \zeta} \sqrt{p_{\text{can}}(\varepsilon \mid \zeta)} e_{\zeta\varepsilon}.$$

The successor fibers of distinct antecedents are disjoint. The normalization of p_{can} therefore proves $\mathcal{U}_k^* \mathcal{U}_k = I$. The representation by masses of histories is related to this one by the diagonal change of basis incorporating $\sqrt{\mu(\zeta)}$.

The sheet label is part of the complete record of a history. We write

$$\hat{\zeta} = ((\zeta_0, s_0), \dots, (\zeta_k, s_k)), \quad s_j \in \{\Sigma, \Pi\},$$

where s_j is the active sheet at step j . The prolongation adds the emission and the sheet s_{k+1} determined by the operational TPK selector, retaining the prefix in full, including the initial sheet. In the basis of these lifted histories,

$$\mathcal{U}_k e_{\hat{\zeta}} = \sum_{\varepsilon \text{ admissible at } \hat{\zeta}} \sqrt{p_{\text{can}}(\varepsilon \mid \hat{\zeta})} e_{\hat{\zeta}\varepsilon}, \quad P_k e_{\hat{\zeta}} = \mathbf{1}_{s_k=\Sigma} e_{\hat{\zeta}}, \quad Q_k = I - P_k.$$

Thus, P_k reads the current sheet, while the distinction between histories is maintained by their complete prefixes. Two initial histories that reach the same active sheet retain distinct successor fibers. The proof of isometry by disjoint collections of prolongations applies verbatim to this basis. The tritic orientation is another coordinate of the record: the value 0 preserves the threshold reading on each sheet and does not add a third sheet to this decomposition. We now separate sheet preservation and sheet exchange:

$$A_k = P_{k+1} \mathcal{U}_k P_k + Q_{k+1} \mathcal{U}_k Q_k, \quad \eta_k = Q_{k+1} \mathcal{U}_k P_k + P_{k+1} \mathcal{U}_k Q_k.$$

Proposition 3.5 (Gram identity by sheets). *For a linear transport U , whether or not it is isometric, the preceding decomposition satisfies*

$$A^* A + \eta^* \eta = P U^* U P + Q U^* U Q.$$

In particular, for $U = \mathcal{U}_k$,

$$A_k^* A_k + \eta_k^* \eta_k = I.$$

Proof. In $A^* A$ and $\eta^* \eta$, the mixed terms containing $P_{k+1} Q_{k+1}$ vanish. Summing the remaining terms produces $P_k U^* (P_{k+1} + Q_{k+1}) U P_k$, and its analogue with Q_k . The first identity follows from complementarity of the projectors; the second also uses $U^* U = I$. $\square$

The first identity is the general formula. Replacing its right-hand side by $U^* U$ requires the cross-blocks of that Gram matrix to vanish. Isometric preservation of history transport provides that condition here; it must not be attributed automatically to a subsequent compression.

Write $F_{0,0} = I$, $F_{0,n} = A_{n-1} \cdots A_0$, and $T_N = F_{0,N}^* F_{0,N}$.

Theorem 3.6 (Finite decomposition with terminal residue). *For each horizon N ,*

$$I = \sum_{n=0}^{N-1} F_{0,n}^* \eta_n^* \eta_n F_{0,n} + T_N.$$

The sequence T_N is positive and decreasing, and has a positive strong limit T_∞ . The sum of outputs converges strongly to $I - T_\infty$.

Proof. Proposition 3.5 gives $T_n - T_{n+1} = F_{0,n}^* \eta_n^* \eta_n F_{0,n}$. The telescoping sum proves the finite equality and monotonicity. For each vector, the quadratic forms decrease to a limit; polarization defines a positive bounded form and, by the representation theorem for bounded forms, an operator T_∞ . Convergence is strong because, for a positive operator $S \leq I$, $\|Sv\|^2 \leq \langle v, Sv \rangle$; this is applied to $S = T_N - T_\infty$. $\square$

The terminal residue is retained until its extinction in the domain under consideration is proved. This distinction prevents a phase return, by itself, from becoming an assumption that all histories are exhausted.

3.4 Oriented balance and cancellation under refinement

A history carries its accumulated orientation $o(h)$ and the counts of its visits to the two sheets. The corresponding local balance is

$$b(h) = o(h)(N_\Sigma(h) - N_\Pi(h)), \quad b^+ = \frac{|b| + b}{2}, \quad b^- = \frac{|b| - b}{2}.$$

When a reader expresses the mean over the prolongations, $b_c = Eb_f$ is defined. The orientation remains in the argument of b_f ; it is not removed before averaging.

Proposition 3.7 (Exact cancellation defect). *The quantity*

$$\kappa_{\text{can}} = \frac{E|b_f| - |Eb_f|}{2}$$

is nonnegative and satisfies

$$E(b_f^+) = (b_c)^+ + \kappa_{\text{can}}, \quad E(b_f^-) = (b_c)^- + \kappa_{\text{can}}.$$

Proof. The weighted triangle inequality gives $|Eb_f| \leq E|b_f|$. Writing the positive and negative parts as $(|b| \pm b)/2$ yields both equalities. $\square$

The information omitted when only the balance is expressed is the common cancellation of the opposing contributions. This quantity is determined from the same prolongations used by the preceding transport. The factor 1/9 appears only when the fiber contains nine prolongations with equal weights; conditional expectation is the rule that remains valid for general branching and weightings.

3.5 Ternary incidence of the sheets and rectangular complex

The two sheets determine the module $V = \mathbb{F}_3 e_\Sigma \oplus \mathbb{F}_3 e_\Pi$. Its four projective directions are the lines generated by $e_\Sigma, e_\Pi, e_\Sigma + e_\Pi, e_\Sigma - e_\Pi$. They give rise to six unordered pairs. The eight nonzero vectors are their oriented representatives. Thus, the counts 4, 6, 8 belong to distinct operations on the same sheet module.

On the six pair positions, define

$$(Ba)_{\{L,L'\}} = a_L + a_{L'}, \quad s(q) = \sum_{\{L,L'\}} q_{\{L,L'\}}.$$

Each direction occurs in three pairs; hence $sB = 0$ in $\mathbb{F}_3$. The reader $W(q) = \mathbf{1}_{s(q)=0} - 1/3$ is invariant under $q \mapsto q + Ba$. On the uniform ambient space $\mathbb{F}_3^6$, its moments are

$$\mathbb{E}W = 0, \quad \mathbb{E}W^2 = \frac{2}{9}, \quad \mathbb{E}W^3 = \frac{2}{27}.$$

Indeed, s is surjective and each fiber contains 3^5 elements. The reader equals 2/3 with probability 1/3 and $-1/3$ with probability 2/3. These means describe the uniform ambient space; the distribution induced by a subset of histories must be calculated with its own multiplicities, already retained in H_k .

To fix the ports without losing multiplicities, at depth k the labeled readings are retained: $\lambda_\Sigma(\zeta) = (\zeta, \Sigma)$ and $\lambda_\Pi(\zeta) = (\zeta, \Pi)$. Their sets are

$$\mathcal{P}_k^\Sigma = \{\lambda_\Sigma(\zeta) : \zeta \in H_k\}, \quad \mathcal{P}_k^\Pi = \{\lambda_\Pi(\zeta) : \zeta \in H_k\}.$$

They are finite and nonempty because H_k is. Values, quotients, and residues are expressed on these ports; equality of values does not identify their labels. The product $\mathcal{P}_k^\Sigma \times \mathcal{P}_k^\Pi$ is the domain of the rectangular reading, within which the support of the admissible incidences is recorded.

With this level fixed, write $I = \mathcal{P}_k^\Sigma$, $J = \mathcal{P}_k^\Pi$, and $A_m = \mathbb{Z}/3^m\mathbb{Z}$. The lexicographic APP order determines the minimal surviving branch and, within it, the reference ports i_0, j_0 . No external choice is added: changing coordinates produces a canonically isomorphic complex. The maps

$$\kappa(u, v)_{ij} = u_i - v_j, \quad (\delta w)_{ij} = w_{ij} - w_{ij_0} - w_{i_0j} + w_{i_0j_0}$$

define the following sequence, valid over the finite ring A_m :

$$0 \longrightarrow A_m \longrightarrow A_m^I \oplus A_m^J \xrightarrow{\kappa} A_m^{I \times J} \xrightarrow{\delta} A_m^{(I \setminus \{i_0\}) \times (J \setminus \{j_0\})} \longrightarrow 0.$$

Proposition 3.8 (Exactness of rectangular incidence). *The preceding sequence is exact, with its first arrow given by the constant diagonal.*

Proof. The equality $u_i - v_j = 0$ for every pair forces all coordinates of u and v to be the same constant. Direct substitution gives $\delta\kappa = 0$. If $\delta w = 0$, take $u_i = w_{ij_0}$ and $v_j = w_{i_0j_0} - w_{i_0j}$. Then $u_i - v_j = w_{ij}$. Finally, any matrix in the last term is prolonged with zero reference row and column; its image under δ is the initial matrix. The proof uses only additions and subtractions, and therefore does not require division by a unit of the ring. $\square$

Reduction from A_{m+1} to A_m , while keeping $k, \mathcal{P}_k^\Sigma, \mathcal{P}_k^\Pi, i_0, j_0$ fixed, commutes with both maps. The reconstruction formulas are the same at every modular depth. This is the arithmetic compatibility proved in this subsection. The prolongation of histories $k \rightarrow k+1$ preserves the prefix labels; its maps between sets of ports are data distinct from the change of ring $m \rightarrow m+1$.

Those maps are obtained from deletion of emissions:

$$\rho_k^\Sigma(\zeta\varepsilon, \Sigma) = (\zeta, \Sigma), \quad \rho_k^\Pi(\zeta\varepsilon, \Pi) = (\zeta, \Pi).$$

The minimal surviving branch of the APP order just constructed has compatible prolongations. Its two labels define compatible reference ports at every level. Coordinate transport is precomposition:

$$(L_k u)_{i'} = u_{\rho_k^\Sigma(i')}, \quad (L_k v)_{j'} = v_{\rho_k^\Pi(j')}, \quad (L_k w)_{i'j'} = w_{\rho_k^\Sigma(i'), \rho_k^\Pi(j')}.$$

For the last module in the sequence, the matrix is first prolonged by zero on the reference row and column, this same formula is applied, and the result is restricted to the new complement of the reference ports. This rule also defines L_k when a new port has the marked port as its antecedent.

Proposition 3.9 (Naturality of the rectangular complex under prolongation). *The preceding maps satisfy $L_k \kappa_k = \kappa_{k+1} L_k$ and $L_k \delta_k = \delta_{k+1} L_k$. They also commute with coefficient reduction $A_{m+1} \rightarrow A_m$ and compose in accordance with prefix deletion.*

Proof. The first identity follows from $u_{\rho_k^\Sigma(i')} - v_{\rho_k^\Pi(j')}$. In the second, each of the four terms defining δ is transported by the same precomposition. Compatibility of the marked ports identifies their rows and columns; when an antecedent is the marked port, the four terms cancel and agree with prolongation by zero. All formulas are sums and differences with integer coefficients, so they commute with modular reduction. The composition of precompositions is precomposition with the composite prefix, proving the final assertion. $\square$

3.6 Memory complex and compatibility of incidences

The same accumulated memory coordinate $c_y \in \mathbb{Z}$, at a residence y , defines a free integral complex. Its reduction to A_m uses the same integer record before reduction; therefore, successive depths are compatible. Its generators are f_y in degree 2, e_y, b_y, r_y in degree 1, and g_y in degree 0:

$$\partial f_y = 3c_y e_y + b_y, \quad \partial e_y = g_y, \quad \partial b_y = -3c_y g_y, \quad \partial r_y = 0.$$

The identity $\partial^2 = 0$ is verified on f_y by $3c_y g_y - 3c_y g_y = 0$, and is immediate on the other generators.

If a path $\alpha : y \rightarrow y'$ updates the memory, its transport in the complex sends the like-named generators to their new representatives, except for

$$\Psi_\alpha(b_y) = b_{y'} + 3(c_{y'} - c_y)e_{y'}.$$

Proposition 3.10 (Naturality of the memory complex). *The identities $\partial\Psi_\alpha = \Psi_\alpha\partial$ and $\Psi_{\beta\alpha} = \Psi_\beta\Psi_\alpha$ hold. The maps extend linearly over the chosen coefficient ring.*

Proof. On f_y , the image of its boundary is $3c_y e_{y'} + b_{y'} + 3(c_{y'} - c_y)e_{y'} = 3c_{y'} e_{y'} + b_{y'}$. On b_y , the boundary of its image is $-3c_{y'} g_{y'} + 3(c_{y'} - c_y)g_{y'} = -3c_y g_{y'}$. The other two cases are immediate. Under concatenation, the memory differences sum telescopically, proving the second identity. $\square$

The relation between two representatives of the same residence can also be written as an explicit boundary. For an integer coordinate v , reduced in the same ring when appropriate, let

$$z^- = r_y, \quad z^+ = r_y + 3c_y v e_y + v b_y, \quad h_* = -v f_y.$$

Then $\partial h_* = z^- - z^+$. Under a transport that also updates v , the correction $K_\alpha = (v_{y'} - v_y)f_{y'}$ satisfies $K_{\beta\alpha} = K_\beta + \Psi_\beta K_\alpha$. The symbols K_α in this local homotopy do not denote the dodecaphase seal K : they belong to another domain, and their function is specified by the preceding identity. Their function can also be checked in the equality

$$z_{y'}^+ - \Psi_\alpha z_y^+ = \partial K_\alpha.$$

Both sides equal $(v_{y'} - v_y)(3c_{y'} e_{y'} + b_{y'})$.

Proposition 3.11 (Homology and memory of the local complex). *Over $\mathbb{Z}$, or after reduction to A_m , one has*

$$H_0(\Gamma) = H_2(\Gamma) = 0, \quad H_1(\Gamma) \cong A_m[r_y]$$

in the modular realization, and $H_1(\Gamma) \cong \mathbb{Z}[r_y]$ in the integral realization.

Proof. Define the degree-1 homotopy by $h(g_y) = e_y$, $h(b_y) = f_y$, and zero on the other generators. Let p be the projection onto the summand generated by r_y , and ι its inclusion. Verification on each generator gives

$$\partial h + h\partial = I - \iota p.$$

In particular, on b_y the terms $3c_y e_y$ and $-3c_y e_y$ cancel. The complex deformation retracts onto the module generated by r_y , concentrated in degree 1, proving the assertion. $\square$

The transport Ψ_α is a shear of determinant 1. Thus, this complex explicitly preserves how c_y changes at chain level, but its local homology does not distinguish the values of that coordinate. The complete memory remains in the labels and transports of the source state; it is not recovered from H_1 alone. A memory-dependent numerical torsion requires the specified assembly and determinant reading, in addition to this local complex.

3.7 Assembly with terminal paths

For a finite horizon, the residences and their prolongations form a path category. Each residence ν is assigned the free module $W(\nu) = A_m[\text{Path}(\nu, \text{Term})]$, which retains each terminal continuation as a generator. An initial path acts on W by precomposition. The complex $\Gamma(\nu)$ of the preceding subsection acts covariantly through Ψ .

The assembly uses both transports in the same bigraded module:

$$\mathcal{B}_{q,r} = \bigoplus_{\nu_0 \rightarrow \dots \rightarrow \nu_q} W(\nu_q) \otimes_{A_m} \Gamma_r(\nu_0).$$

The normalized path complex is used: degenerate chains containing identities are omitted. The finite horizon then bounds the number of strict prolongations and the horizontal degree. The horizontal boundary deletes an intermediate residence by composing the adjacent arrows. At the first endpoint it transports Γ ; at the last it precomposes W . Alternating signs give the path boundary d_{bar} . The total differential is

$$D = d_{\text{bar}} + (-1)^q \partial_\Gamma.$$

The face identities cancel the terms of d_{bar}^2 pairwise. Naturality in Proposition 3.10 implies that d_{bar} commutes with ∂_Γ ; the factor $(-1)^q$ cancels the cross terms in the square. Therefore, $D^2 = 0$. Compatibility is proved on path generators and is not introduced by a label asserting membership in a complex.

Terminal evaluation also does not require choosing a particular history. For each continuation h , the effective composition of its updates is applied. The result retains the pair formed by h and its reading, with its multiplicity. If $k < \ell < N$, the continuations decompose as a disjoint union of prefixes up to ℓ and their continuations up to N . Proposition 3.1 proves that direct evaluation and evaluation of those two parts produce the same reading. This yields the terminal naturality square.

The path modules, incidences, memory transport, and their differentials arise from the same prolongations. Their assembly does not consist in adding to a state five coordinates that would be retained by definition: their relations are constructed through sums, boundaries, and effective compositions, and are verified on their generators.

3.8 Determinants, limit, and arithmetic reading

At each residence, the free local complex Γ admits its graded determinant line

$$\text{Det } \Gamma = \bigotimes_r \left(\bigwedge^{\text{rk } \Gamma_r} \Gamma_r \right)^{(-1)^r}.$$

The total assembly of the preceding subsection, in turn, has the modules $\text{Tot}_d \mathcal{B} = \bigoplus_{q+r=d} \mathcal{B}_{q,r}$. At finite horizon and over the same ring, they are free of finite rank and only finitely many of them are nonzero. Its line is

$$\text{Det}(\text{Tot } \mathcal{B}) = \bigotimes_{q,r} \left(\bigwedge^{\text{rk } \mathcal{B}_{q,r}} \mathcal{B}_{q,r} \right)^{(-1)^{q+r}}.$$

This formula follows from the direct-sum decomposition in each degree. It distinguishes the line of the local complex from that of the assembly with all paths; their bases are ordered by means of the retained labels. Modular reduction commutes with these exterior powers because the modules and their bases arise from the same free integral modules at fixed horizon. The line is defined by the modules and their degrees, both over $\mathbb{Z}$ and after reduction to A_m ; a negative exponent denotes the dual module. A nonzero numerical torsion additionally requires specifying the complex over a field, the bases, and the relevant homology identifications. To calculate Smith factors, one uses the integer matrix produced

before reduction, not an arbitrary lift of a modular matrix. An integer square block with nonzero determinant is invertible over $\mathbb{Q}$; its reduction is invertible over A_m only when that determinant is a unit, that is, when it is not divisible by 3. Refinement of memories and incidences makes it possible to formulate and verify the stability of the relevant invariant factors; these conditions do not follow from the existence of the determinant line.

Passage to the limit preserves the compatible projective systems of residues, the histories, and their multiplicities. The squares proved in the preceding subsections remain commutative because their maps are given by the same formulas at each level. For an Archimedean coordinate, the additional step consists in publishing nonempty cylinder intervals, compatible under inclusion and with diameter tending to zero. Their nested compact closures have a one-point intersection: existence follows from compactness and uniqueness from the diameter bound. That point is the value of the limit reading. If the intervals are half-open, the point may belong only to their closures; the boundary representation rule then determines its positional notation. The value does not replace the history that produced it. The other representation preserves the incidences of the same state. The two targets share their origin without thereby receiving the same information.

The coinductive regions of Section 4, the dodecaphase record of Section 8, and the exceptional incidence of Section 7 are expressed from that substrate with their respective maps. The Hilbert-space realization subsequently retains the phase labels and specifies its own unital inclusion; duality and the screens act on the codomains constructed there. The joint operations assembled here transmit histories, sheets, orientation, and memory to those representations. Identifying a specific functional with an external analytic form in turn requires writing its action on the generators and verifying their inner products: the conservation identities of the present chapter do not replace that composition.

3.9 Mathematical scope of the joint construction

The constructions of state, tree, and measure; the correlative operations; and the terminal assembly with naturality and limit are developed in full in this article. In this chapter their balance, incidence, and memory identities have been formulated and proved on the same generators. The five constructions thus remain operations of a single tree. Each specialization keeps its exact hypotheses visible: isometry of transport, canonical weighting law, treatment of the terminal residue, and nonsingularity of the determinant block, where applicable.

The preceding equations prove the memory cocycles, weighted cancellation, rectangular exactness, and compatibility of the memory complex at every finite horizon, as well as its compatible passage to the limit. A subsequent analytic identification must compose its operator on these generators and verify the additional properties of its codomain; it does not alter or replace the joint construction proved here.

3.10 Joint emergence of the five constructions

For $0 \leq k \leq N$, let

$$\rho_{N,k} := \rho_k \circ \rho_{k+1} \circ \cdots \circ \rho_{N-1}, \quad \rho_{k,k} := \text{id}_{H_k},$$

and, for a surviving history $\zeta \in H_k$, define

$$\text{Desc}_{k,N}(\zeta) := \{x \in H_N : \rho_{N,k}(x) = \zeta\}.$$

The finite tree $\mathcal{T}_{k,N}(\zeta)$ has as vertices

$$\coprod_{j=k}^N \{y \in H_j : \rho_{j,k}(y) = \zeta\}$$

and as edges only the TPK prolongations $a : y \rightarrow z$ with $z \in \rho_j^{-1}(y)$. Therefore, neither its vertices nor its prolongations arise from an auxiliary tree.

For $x \in \text{Desc}_{k,N}(\zeta)$, the marked branch $\lambda_{\zeta,x}$ is the sequence

$$\zeta = x_k \longrightarrow x_{k+1} \longrightarrow \cdots \longrightarrow x_N = x$$

determined by the truncations of x . The common base record is

$$\begin{aligned} \mathcal{R}_{k,N}(\zeta; x) := & (\mathcal{T}_{k,N}(\zeta), \lambda_{\zeta,x}; \\ & (\varepsilon_a, s_a, o_a, (\delta_{\circ,a}, r_{\circ,a}, q_{\circ,a})_{\circ \in \{\Sigma, \Pi\}}, \text{Carr}_a, C_a, t_a, \text{Fr}_a, \text{Route}_a, \text{Mem}_a, \text{Surv}_a)_a; \\ & \text{src}, \text{tgt}, \circ, (\rho_j)_{k \leq j < N}), \end{aligned} \quad (\text{C1})$$

where

$$U_a(m_0) = C_a m_0 + t_a, \quad \delta_{\circ,a} = r_{\circ,a} + 9q_{\circ,a}.$$

The symbol m_0 in this formula is a memory coordinate; it is not the modular depth m used below. The true base space is

$$\mathfrak{R}_{k,N}(\zeta) := \{\mathcal{R}_{k,N}(\zeta; x) : x \in \text{Desc}_{k,N}(\zeta)\}.$$

The marked branch is not an added input: it is the sequence of edges actually produced that ends at x . The remainder of the record simultaneously preserves the lateral fibers of the tree; therefore, the mark does not replace the branching with an isolated trajectory.

Set $R_m := \mathbb{Z}/3^m\mathbb{Z}$, and let $V_j(\zeta) := \text{Desc}_{k,j}(\zeta)$. The transport, the sheet projectors, and the terminal residue are restricted to the subtree by

$$\begin{aligned} \mathcal{U}_j^\zeta &:= \mathcal{U}_j \upharpoonright_{\ell^2(V_j(\zeta), \mu_j)}, & P_j^\zeta &:= P_j \upharpoonright_{V_j(\zeta)}, & Q_j^\zeta &:= I - P_j^\zeta, \\ A_j^\zeta &:= P_{j+1}^\zeta \mathcal{U}_j^\zeta P_j^\zeta + Q_{j+1}^\zeta \mathcal{U}_j^\zeta Q_j^\zeta, & \eta_j^\zeta &:= Q_{j+1}^\zeta \mathcal{U}_j^\zeta P_j^\zeta + P_{j+1}^\zeta \mathcal{U}_j^\zeta Q_j^\zeta. \end{aligned}$$

For $k \leq n \leq N$, let

$$F_{k,k}^\zeta := I, \quad F_{k,n}^\zeta := A_{n-1}^\zeta \cdots A_k^\zeta, \quad T_{k,n}^\zeta := (F_{k,n}^\zeta)^* F_{k,n}^\zeta.$$

The proper ports of the same subtree are

$$\mathcal{P}_j^\Sigma(\zeta) := \{(y, \Sigma) : y \in V_j(\zeta)\}, \quad \mathcal{P}_j^\Pi(\zeta) := \{(y, \Pi) : y \in V_j(\zeta)\};$$

we denote by $B_j^\zeta, \kappa_j^\zeta, \delta_j^\zeta$ the restrictions of the incidence and of the rectangular complex to those ports, and by L_j^ζ the precomposition induced by prefix deletion.

To define the path reader without abbreviations, let

$$W_\ell^\zeta(y) := R_m[\text{Path}_{\mathcal{T}_{k,\ell}(\zeta)}(y, \text{Term}_\ell)]$$

and set

$$\mathcal{D}_{q,r;k,\ell}^{(m)}[\mathcal{R}] := \bigoplus_{y_0 \rightarrow \cdots \rightarrow y_q \text{ in } \mathcal{T}_{k,\ell}(\zeta)} W_\ell^\zeta(y_q) \otimes_{R_m} (\Gamma_r(y_0) \otimes_{\mathbb{Z}} R_m), \quad \text{Tot}_d \mathcal{D} := \bigoplus_{q+r=d} \mathcal{D}_{q,r;k,\ell}^{(m)}[\mathcal{R}].$$

The formulas already proved produce, on each $\mathcal{R} \in \mathfrak{R}_{k,N}(\zeta)$, the five objects

$$\begin{aligned} \mathcal{F}_{\text{sp}}[\mathcal{R}] &:= (\ell^2(V_j(\zeta), \mu_j), \mathcal{U}_j^\zeta, A_j^\zeta, \eta_j^\zeta, T_{k,j}^\zeta)_{k \leq j \leq N}, \\ \mathcal{F}_{\text{sol}}[\mathcal{R}] &:= (b, \kappa_{\text{can}}, C_\alpha, \kappa_\alpha)_{\alpha \in \mathcal{T}_{k,N}(\zeta)}, \\ \mathcal{F}_{\text{gau}}^{(m)}[\mathcal{R}] &:= (\mathcal{P}_j^\Sigma(\zeta), \mathcal{P}_j^\Pi(\zeta), B_j^\zeta, \kappa_j^\zeta, \delta_j^\zeta; R_m)_{k \leq j \leq N}, \\ \mathcal{F}_{\text{coh}}^{(m)}[\mathcal{R}] &:= (\Gamma_y \otimes R_m, \partial, \Psi_\alpha, H_\bullet)_{y, \alpha \in \mathcal{T}_{k,N}(\zeta)}, \\ \mathcal{F}_{\text{det}}^{(m)}[\mathcal{R}] &:= (\text{Det}(\text{Tot } \mathcal{D}_{\bullet, \bullet; k, \ell}^{(m)}[\mathcal{R}]))_{k \leq \ell \leq N}. \end{aligned} \quad (\text{C2})$$

Here $\mathcal{D}_{\bullet, \bullet; k, \ell}^{(m)}[\mathcal{R}]$ is the normalized bicomplex of paths and memory from the preceding sections, restricted to the subtree of horizon ℓ and reduced over R_m . The last reader preserves the ordered collection of determinant lines up to horizon N ; in this way, its truncation is a canonical projection and

does not presuppose a nonexistent determinant factorization. The law μ is the canonical law of the census, and the ports are fixed by the lexicographically minimal branch of the APP order. Therefore, none of the five readers receives an external weighting or an external port.

For $T \in \mathfrak{T} := \{\text{sp}, \text{sol}, \text{gau}, \text{coh}, \text{det}\}$, define the typed total space

$$\tilde{\mathcal{F}}_{T;k,N}^{(m)}(\zeta) := \coprod_{\mathcal{R} \in \mathfrak{R}_{k,N}(\zeta)} \{\mathcal{F}_T^{(m)}[\mathcal{R}]\}_{\mathcal{R}}, \quad p_T : \tilde{\mathcal{F}}_{T;k,N}^{(m)}(\zeta) \longrightarrow \mathfrak{R}_{k,N}(\zeta),$$

where $\mathcal{F}_{\text{sp}}^{(m)} := \mathcal{F}_{\text{sp}}$ and $\mathcal{F}_{\text{sol}}^{(m)} := \mathcal{F}_{\text{sol}}$; that is, the superscript m is inert for those two readers. The projection p_T forgets only the reader structure and preserves its source record. The correct fiber product is

$$\begin{aligned} \mathcal{J}_{k,N}^{(m)}(\zeta) := & \tilde{\mathcal{F}}_{\text{sp};k,N}^{(m)}(\zeta) \times_{\mathfrak{R}_{k,N}(\zeta)} \tilde{\mathcal{F}}_{\text{sol};k,N}^{(m)}(\zeta) \times_{\mathfrak{R}_{k,N}(\zeta)} \tilde{\mathcal{F}}_{\text{gau};k,N}^{(m)}(\zeta) \\ & \times_{\mathfrak{R}_{k,N}(\zeta)} \tilde{\mathcal{F}}_{\text{coh};k,N}^{(m)}(\zeta) \times_{\mathfrak{R}_{k,N}(\zeta)} \tilde{\mathcal{F}}_{\text{det};k,N}^{(m)}(\zeta). \end{aligned} \quad (\text{C3})$$

The equality of the five projections forces preservation of the same tree, the same edges, the same produced branch, and the same record. For each $\mathcal{R}$, the five constructions determine the element

$$s_{k,N}^{(m)}(\mathcal{R}) := (\mathcal{F}_T^{(m)}[\mathcal{R}])_{T \in \mathfrak{T}} \in \mathcal{J}_{k,N}^{(m)}(\zeta).$$

The existence of this element arises from the operations of (C2), not from the fiber-product notation.

Deletion of the last layer induces

$$\text{res}_{N+1,N}^{\mathfrak{R}}(\mathcal{R}_{k,N+1}(\zeta; x)) := \mathcal{R}_{k,N}(\zeta; \rho_N(x)).$$

The restriction of each operator to the retained generators defines

$$r_{N+1,N}^m : \mathcal{J}_{k,N+1}^{(m)}(\zeta) \longrightarrow \mathcal{J}_{k,N}^{(m)}(\zeta).$$

In the determinantal reader, $r_{N+1,N}^m$ deletes only the last coordinate of the collection of lines in (C2). The reduction $R_{m+1} \rightarrow R_m$ defines, by base change,

$$q_{m+1,m}^N : \mathcal{J}_{k,N}^{(m+1)}(\zeta) \longrightarrow \mathcal{J}_{k,N}^{(m)}(\zeta).$$

On each generator, the following identities hold:

$$\begin{aligned} r_{N+1,N}^m q_{m+1,m}^{N+1} &= q_{m+1,m}^N r_{N+1,N}^{m+1}, \\ \partial \Psi_\alpha &= \Psi_\alpha \partial, \quad L_j^\zeta \kappa_j^\zeta = \kappa_{j+1}^\zeta L_j^\zeta, \quad L_j^\zeta \delta_j^\zeta = \delta_{j+1}^\zeta L_j^\zeta, \end{aligned} \quad (\text{C4})$$

and the compositions of the maps r and q satisfy the functorial identities. The first equality holds because deleting the last layer and reducing the coefficients act on distinct coordinates. The other three are the naturality identities proved above. A single pro-system in (N, m) is obtained, not five juxtaposed limits.

At the root, define

$$\mathfrak{F}_{N,\cdot}^{(m)} := \mathcal{J}_{0,N}^{(m)}(*). \quad (\text{C5})$$

The joint discrete structure of the continuum is the limit of those correlative objects:

$$\boxed{\mathcal{C}_{\text{cont}}^{\text{disc}} := \lim_{\substack{\leftarrow \\ N \geq 1 \\ m \geq 1}} (\mathfrak{F}_{N,\cdot}^{(m)}, r_{N+1,N}^m, q_{m+1,m}^N).} \quad (\text{C6})$$

This is a definition by the pro-system generated by the full tree; it is not the image or the graph of a map that retains the input state as its first coordinate.

The records possess a produced terminal mark. Their compatible projections induce

$$\Pi : \mathcal{C}_{\text{cont}}^{\text{disc}} \longrightarrow H_\infty^{\text{enr}}.$$

Conversely, a complete history determines, through its already emitted prefixes, the compatible sections $s_{0,N}^{(m)}$. Thus, after having defined the continuum, one obtains

$$\text{Gen}_{\text{cont}} : H_{\infty}^{\text{enr}} \longrightarrow \mathcal{C}_{\text{cont}}^{\text{disc}}, \quad \Pi \circ \text{Gen}_{\text{cont}} = \text{id}_{H_{\infty}^{\text{enr}}}. \quad (\text{C7})$$

This section proves recoverability; it is not used to define $\mathcal{C}_{\text{cont}}^{\text{disc}}$.

Theorem 3.12 (Correlative emergence of the five constructions). *For every surviving history $\zeta \in H_k$, every $N > k$, every $m \geq 1$, and every $\mathcal{R} \in \mathfrak{R}_{k,N}(\zeta)$, the five constructions of (C2) are defined on the same record, and (C3) contains $s_{k,N}^{(m)}(\mathcal{R})$. Their truncations, modular reductions, and transports satisfy (C4). Furthermore, $\mathcal{C}_{\text{cont}}^{\text{disc}} \neq \emptyset$, the map Π is surjective and is split by Gen_{cont} . The five components are inseparable aspects of the same continuum, not five domains or five independent maps.*

Proof. The survival that defines H_k and the TPK prolongation law give $\rho_k^{-1}(\zeta) \neq \emptyset$ for each vertex. Because the fibers are finite, König's lemma yields $H_{\infty}^{\text{enr}} \neq \emptyset$.

For a fixed record $\mathcal{R}$, the transport of histories and sheets produces $\mathcal{F}_{\text{sp}}[\mathcal{R}]$; affine composition and the exact cancellation defect produce $\mathcal{F}_{\text{sol}}[\mathcal{R}]$; the incidences and the rectangular sequence produce $\mathcal{F}_{\text{gau}}^{(m)}[\mathcal{R}]$; the memory complex and its transport produce $\mathcal{F}_{\text{coh}}^{(m)}[\mathcal{R}]$; and the lines of the normalized path assembly produce $\mathcal{F}_{\text{det}}^{(m)}[\mathcal{R}]$. Each construction acts on the same generated edges, not on an assumed subdomain or on the abstract ambient space of 729 words.

The affine, transport, incidence, and memory compatibilities prove (C4). Truncating the determinant collection of (C2) commutes by definition with r ; its base change commutes with q . The sections $s_{0,N}^{(m)}$ are compatible and thus produce Gen_{cont} . The identity (C7) proves simultaneously the nonemptiness of the limit, the surjectivity of Π , and the recoverability of each history without introducing it as a defining coordinate.

The fiber, relation, number, orientation, carry, memory, boundary, and route remain in the common record; the multisection, the terminals, and the linear and probabilistic readers remain in the five correlative constructions. Therefore, passage to the limit preserves the closure of dependencies and does not disaggregate the continuum. $\square$

The subsequent recognition of the single continuum constructed through spectral, solenoidal, gauge, cohomological, or determinant language belongs to the subsequent conventional comparison. It neither selects its states nor intervenes in its APP–TRIT–TPK generation.

4 Initial conditions and regional generation

An APP position does not by itself contain the initial condition of its dynamics. The latter includes the direction of transport and distinguishes the cursors of the two sheets. Before classifying particles, this domain must be constructed: it produces the regional records that will enter the electronic coordinate, action, and mass readers. The operations in this section receive neither a physical species nor a metrological value as an initial condition.

4.1 Two oriented cursors on the complete support

We use indices $I_9 = \{0, \dots, 8\}$, whose positive label is $i + 1$, and the cardinal directions $\mathcal{D} = \{N, E, S, O\}$. The domain of one cursor and the joint domain are

$$\mathcal{S} = I_9^2 \times \mathcal{D}, \quad \Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_-, \quad |\mathcal{S}| = 324, \quad |\Omega_{\text{seed}}| = 104\,976. \quad (4.1)$$

The subscript seed retains the domain designation used in the antecedents; mathematically, these are oriented initial conditions. The Cartesian product includes all ordered pairs. The additive sheet has one cursor and the multiplicative sheet another; a subsequent coincidence between their readings is a result of the traversal.

During the 54 instants of the initial emission, the phase selector repeats $(+1, -1, 0)$. In the positive regime, $\rho_9(i + j + 2)$ is first evaluated at the additive cursor, and that cursor is then transported by one position. In the negative regime, $\rho_9((i + 1)(j + 1))$ is evaluated at the multiplicative cursor, after which the second cursor is transported. The null regime records a neutral event. The directions are reversed after instants 27 and 54. The latter reversal belongs to the terminal state although it does not alter readings that have already been taken.

The multiplicative observable uses the exponent chart of the group of units of $\mathbb{Z}/9\mathbb{Z}$:

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5.$$

In the sector $\{3, 6, 9\}$, this observable takes the value zero. Membership in the radical sector remains present in the state; assigning that value to a scalar reading does not identify the radical with the multiplicative identity.

4.2 Finite formula for the window signatures

The six windows each contain nine instants. Each sheet is evaluated three times per window. A closed formula reconstructs the signatures without consulting a prior list. For a cursor with origin $p = (i, j)$ and direction d , let

$$f(k) = \begin{cases} k, & 0 \leq k < 9, \\ 18 - k, & 9 \leq k < 18, \end{cases} \quad p_k = p + f(k)v_d \pmod{9}.$$

The index k counts only the evaluations of that sheet. The reversal after nine of its evaluations explains the second expression for f . Writing $p_k = (i_k, j_k)$, for $m = 0, \dots, 5$ we obtain

$$\begin{aligned} S_m(p, d) &= \sum_{k=3m}^{3m+2} \rho_9(i_k + j_k + 2), \\ E_m(p, d) &= \sum_{k=3m}^{3m+2} \ell_9(\rho_9((i_k + 1)(j_k + 1))), \\ s_m &= [S_m]_{10}, \quad e_m = [E_m]_{10}. \end{aligned} \quad (4.2)$$

The integer evaluation and its division quotient remain stored in the record. The preceding signatures are the decimal projections specified by the emitter.

Since each window has $Z_m = 3$ neutral events, the rule coupling the two signatures is

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, & d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10}. \end{aligned} \quad (4.3)$$

The final term 1 is the reduction of $7Z_m = 21$ modulo ten. The coefficients and calendar are stated in the emission law; physical magnitudes do not appear among its arguments. The oriented ternary projection is

$$\Psi(U_0, \dots, U_5) = ([-U_0]_3, \dots, [-U_5]_3) \in \mathbb{F}_3^6. \quad (4.4)$$

The balanced identification of TRIT is then applied. The minus sign belongs to this chart and must be transported when the chart is changed.

Proposition 4.1 (Factorisation and census of the emission). *The initial conditions produce 18 additive signatures and 26 multiplicative signatures. Their coupling is injective and has 468 emissions. The image of Ψ contains 243 words in the ambient space of 729 elements.*

Proof. The hundreds digits of U_m recover s_m . If b_m is the tens digit, $e_m = [b_m - s_m]_{10}$; the units digit checks the third congruence. Hence no distinct pair of signatures has the same emission.

To obtain the census, (4.2) is evaluated at $i, j = 0, \dots, 8$ and the four directions. Appendix A contains all the resulting signatures. Each additive signature has 18 preimages. Among the multiplicative signatures, 24 have 8, one has 24, and one has 108. These sums are respectively $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$. Since the two cursors are independent, the multiplicities multiply. This gives 432 emissions of weight 144, 18 of weight 432, and 18 of weight 1944, with sum

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

The six reductions in (4.4) are applied to each pair of the 18 and 26 signatures expressed in the appendix. The complete orbit decomposition included there gives $38 \cdot 6 + 5 \cdot 3 = 243$ distinct images. All operations in this enumeration are the finite sums, products, and remainders of the preceding formulas. $\square$

The finite proof does not identify exhaustiveness of the domain with unlimited depth. The initial conditions have been enumerated on a fixed support. Depth is constructed through compatible prolongations preserving those conditions and their transports.

4.3 Typed cubic completion of the regional ambient space

The completion admits a set-theoretic description making its operation explicit. Let E_9 be the set of nine residue classes, represented positively by $1, \dots, 9$, and let $*$ be an additional element. The symbol $*$ differs from the zero class, already represented by 9. Consequently, extending E_9 to $\hat{E} = E_9 \sqcup \{*\}$ increases the cardinality from nine to ten without changing the internal modular identification. The cubic completion is $\hat{C} = \hat{E}^3$, and the initial cube is $C = E_9^3$.

The regional ambient space already constructed by Ψ in (4.4), whose exact census is established in Proposition 4.1, is $\mathbb{F}_3^6$. To type its residence in the initial cube, write $[a]_3 \in \{0, 1, 2\}$ for the representative of $a \in \mathbb{F}_3$ and fix the bijection

$$\eta : \mathbb{F}_3^2 \longrightarrow E_9, \quad \eta(a, b) = 1 + 3[a]_3 + [b]_3.$$

Pairwise grouping gives

$$\beta(w_1, \dots, w_6) = (\eta(w_1, w_2), \eta(w_3, w_4), \eta(w_5, w_6)) \quad (4.5)$$

and is therefore a bijection $\mathbb{F}_3^6 \rightarrow E_9^3 = S_0$. In particular, the 243 words produced by the preceding census reside in the initial body of 729 positions; neither η nor β selects a region, a physical species, or a metrological value. Regional selection is a later and distinct stage, proved in Proposition 4.8. The numerical equality $|\text{im } \Psi| = |S_1| = 243$ does not identify the two sets: $\beta(\text{im } \Psi)$ is a subset of S_0 , whereas S_1 belongs to the expansion shell.

4.3.1 Threshold difference and composition of shells

The enlargement of a discrete cell preserves every stratum appearing when its size increases by one unit. The threshold difference expresses this operation on the cardinality reader and tracks its behaviour under addition and multiplication before a metric realization is introduced.

Definition 4.2 (Threshold difference). Let R be a commutative ring with identity and let $f : \mathbb{Z}_{\geq 0} \rightarrow R$. Define

$$(D_{\#}f)(n) = f(n+1) - f(n). \quad (4.6)$$

For an integer reader of nested finite cells, this difference records the cardinality of the shell added in the step $n \rightarrow n+1$.

Proposition 4.3 (Additivity and exact product law). For $f, g : \mathbb{Z}_{\geq 0} \rightarrow R$,

$$D_{\#}(f+g)(n) = D_{\#}f(n) + D_{\#}g(n), \quad (4.7)$$

$$D_{\#}(fg)(n) = f(n+1)D_{\#}g(n) + g(n)D_{\#}f(n) \quad (4.8)$$

$$= f(n)D_{\#}g(n) + g(n)D_{\#}f(n) + D_{\#}f(n)D_{\#}g(n). \quad (4.9)$$

Proof. The difference of a sum separates term by term. For the product, adding and subtracting $f(n+1)g(n)$ gives

$$\begin{aligned} f(n+1)g(n+1) - f(n)g(n) &= f(n+1)(g(n+1) - g(n)) \\ &\quad + g(n)(f(n+1) - f(n)). \end{aligned}$$

This proves (4.8). Substitution of $f(n+1) = f(n) + D_{\#}f(n)$ yields (4.9), including its complete bilinear term. $\square$

If $A_n \subseteq A_{n+1}$ and $B_n \subseteq B_{n+1}$ are finite sets, the shell of the product decomposes into three disjoint parts:

$$\begin{aligned} (A_{n+1} \times B_{n+1}) \setminus (A_n \times B_n) &= (A_n \times (B_{n+1} \setminus B_n)) \\ &\sqcup ((A_{n+1} \setminus A_n) \times B_n) \\ &\sqcup ((A_{n+1} \setminus A_n) \times (B_{n+1} \setminus B_n)). \end{aligned}$$

Taking cardinalities gives exactly (4.9). The bilinear term counts positions new in both factors; it resides in the intersection of the two enlargements, with incidence preserved.

Proposition 4.4 (Binomial law for the dimensional shell). For $d \geq 1$, let $f_d(n) = n^d$. Then

$$D_{\#}f_d(n) = (n+1)^d - n^d = \sum_{k=0}^{d-1} \binom{d}{k} n^k = \sum_{r=1}^d \binom{d}{r} n^{d-r}. \quad (4.10)$$

Proof. The binomial expansion of $(n+1)^d$ contains n^d and the remaining d terms in the first sum. Substituting $r = d - k$ gives the second sum. In the support $\{1, \dots, n+1\}^d$, each added position has a nonempty set of coordinates equal to $n+1$. If that set has r elements, there are $\binom{d}{r}$ choices of positions and n^{d-r} choices of previous coordinate values. These strata are disjoint and cover the shell, also proving the formula by finite incidence. $\square$

For $d = 3$, the three strata have cardinalities $3n^2$, $3n$, and 1 . At the nonadic-decimal step, $n = 9$, the operation gives $243 + 27 + 1 = 271$; excluding the final vertex leaves 270 positions. The following cubic completion makes explicit the union of this shell with the 729 previous positions. The threshold law thus preserves both the total and its distribution by the number of new coordinates. Subsequent lattice realizations transport their incidences through the maps specific to each construction.

Definition 4.5 (Strata of combinatorial codimension). For $0 \leq r \leq 3$, define

$$S_r = \{(x_1, x_2, x_3) \in \hat{E}^3 : \#\{j : x_j = *\} = r\}.$$

The apex of the completion is $a_* = (*, *, *)$, and the punctured completion is $\hat{C}^\circ = \hat{C} \setminus \{a_*\}$.

Proposition 4.6 (Exact decomposition of the completion). *The strata are disjoint and satisfy*

$$\hat{C} = S_0 \sqcup S_1 \sqcup S_2 \sqcup S_3, \quad |S_r| = \binom{3}{r} 9^{3-r}, \quad (4.11)$$

$$|\hat{C}^\circ| = 729 + 243 + 27 = 999, \quad |\hat{C}| = 999 + 1 = 729 + 271 = 1000. \quad (4.12)$$

The punctured expansion contains 270 elements in addition to the initial cube; the complete expansion contains 271.

Proof. An element of S_r is obtained by choosing the r coordinates containing $*$, in $\binom{3}{r}$ ways, and assigning one of the nine classes of E_9 to each remaining coordinate. Classification by the number of additional coordinates is exhaustive and its classes are disjoint. The sole element of S_3 is a_* . Removing it subtracts one from the total cardinality; restoring it recovers the complete Cartesian product. $\square$

The expression *holographic extreme and mean ratio*, abbreviated EMRH in the corpus, here denotes this articulation of the initial body, expansion strata, and unit restoration. Its immediate content is the exact partition of a finite product and its normalization. The number 999 represents a definite object, the punctured completion; 1000 represents its closure through restoration of the apex. The added unit therefore has an explicit incidence location.

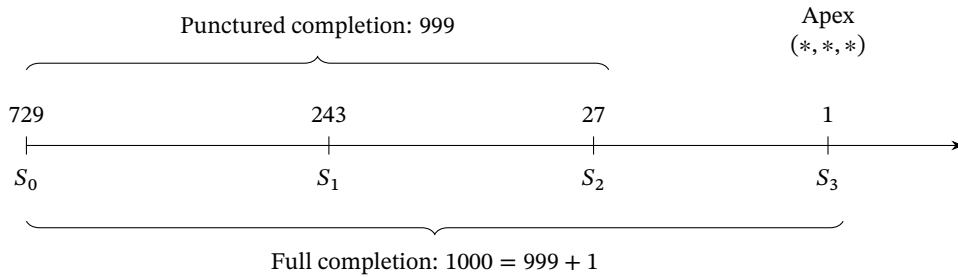


Figure 3: Strata of the cubic completion. The arrangement represents the set-theoretic decomposition, not a metric scale. The 243 elements of S_1 form three coordinate strata of 81 elements; the 27 elements of S_2 form three strata of 9. The apex completes the partition.

4.3.2 Conservation of unity and dimensional compatibility

Normalization is not limited to dividing an isolated identity. In dimension d , the product $\hat{E}^d$ carries the uniform measure $\mu_d(\{x\}) = 10^{-d}$. The coordinate projection $p_d : \hat{E}^{d+1} \rightarrow \hat{E}^d$ deletes the last coordinate.

Proposition 4.7 (Projective compatibility of normalized measures). *For every $d \geq 1$,*

$$(p_d)_* \mu_{d+1} = \mu_d, \quad \mu_d(\hat{E}^d) = 1.$$

The mass of the stratum with r additional coordinates is $\binom{d}{r} 9^{d-r} / 10^d$, and the masses of all strata sum to one.

Proof. Each point of $\hat{E}^d$ has ten preimages. Their total mass is $10 \cdot 10^{-(d+1)} = 10^{-d}$. Equality for any subset follows by summing over its points. The stratified formula follows from the same count as (4.11); the binomial $(9 + 1)^d$ gives total normalization. $\square$

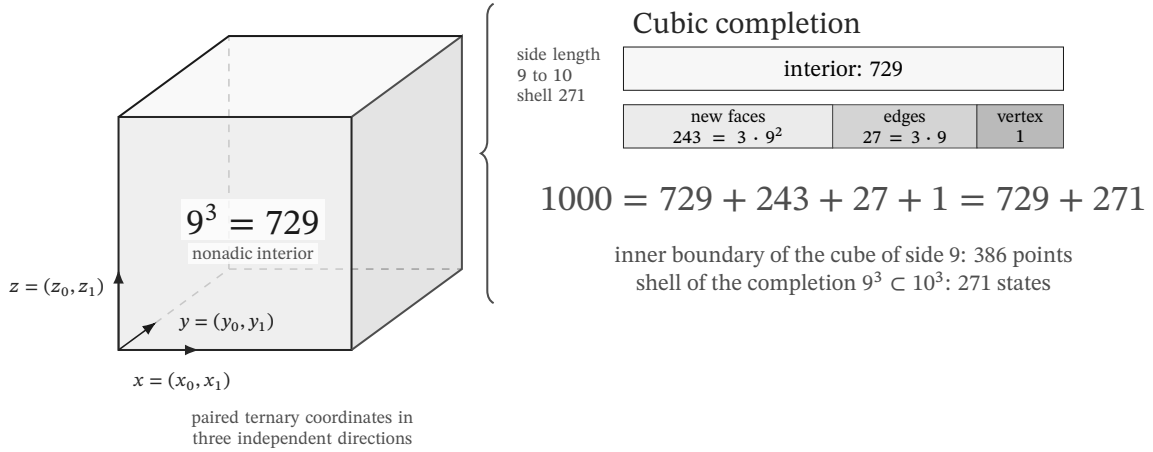


Figure 4: Cubic representation of the expansion from nine to ten classes per coordinate. Strata S_1, S_2, S_3 locate the added incidences on faces, edges, and the apex; their cardinalities form the 271-element shell. The 386-element boundary belongs to the initial cube. This geometric perspective complements the partition in Figure 3; graphical spacing is schematic.

In dimension three, removing the apex leaves mass 999/1000, and restoring it adds 1/1000. Renormalizing before restoration would define a different measure: the distinction is operationally relevant. In particular, conservation of probability mass, conservation of transport records, and thermodynamic dissipation concern different objects. The proposition proves the first and provides the compatible support for studying the second; the third requires a specified dynamic and energetic realization.

The internal geometric boundary also remains distinct. On the Cartesian representative $\{1, \dots, 9\}^3$, removing $\{2, \dots, 8\}^3$ leaves 386 points. This boundary belongs to the initial cube, whereas $\hat{C} \setminus C$ belongs to its expansion. Cubic completion, base-thousand positional expansion, and exceptional-incidence reading share support data; their respective maps determine which information from these data each representation uses.

4.4 Selection of regions through oriented fibres

For $w = (w_1, \dots, w_6)$, define

$$r(w) = (w_3, w_1, w_2, w_5, w_6, w_4), \quad s(w) = (w_4, w_5, w_6, w_1, w_2, w_3).$$

The identities $r^3 = s^2 = I$, $srs = r^{-1}$ realise D_3 . Permuting the six components of U in the same way commutes with Ψ . The preceding enumeration verifies that the orbits preserve the multiplicities of their fibres.

Let $f(w)$ be the number of emissions over w , $m(w)$ its total multiplicity, and $\nu(w) = (n_0, n_1, n_2)$ its ternary census. The additive signature preserves the diagonal class $d = [i + j]_9$ of the origin and the sense ES or NO . We write $\mathcal{D}(\mathcal{O})$ for the set of these classes over an orbit.

Proposition 4.8 (Distinguished regions of the catalogue). *There is a unique orbit of six words with five emissions per word and multiplicities $432 + 4 \cdot 144$. Among the orbits of six words satisfying*

$$f(w) = 1, \quad m(w) = 144 \quad (w \in \mathcal{O}), \quad \mathcal{D}(\mathcal{O}) = \{0, 3, 6\},$$

the census $(2, 3, 1)$ and the census $(2, 2, 2)$ each select a unique orbit. The mark $d = 6$ with orientation ES determines the representatives

$$w^{\text{cl}} = 010211, \quad w^{\text{pr}} = 201101, \quad w^{\text{au}} = 121200.$$

Proof. The complete table of the 43 orbits in A allows the predicates to be applied. The closure orbit is the only one with a fibre of size five and weight 1008; its five weights are those stated. The diagonal condition and the simple fibre of weight 144 leave six orbits. Their censuses select respectively one with (2, 3, 1) and one with (2, 2, 2). Applying the six permutations of D_3 and checking the diagonal class and orientation in (4.2) fixes the three representatives displayed. Uniqueness uses all the conditions: if the set of diagonals is omitted, four balanced orbits with simple fibre and weight 144 appear. $\square$

The designations closure, propagation, and self-scaling apply first to these regions. The next section introduces their evaluation characters. In particular, the multiplicative signature 55555 distinguishes a transverse closure emission, whereas (5, 5) is a position of the electronic support. The two constructions are related through subsequent operations, not through identity of their digits.

4.4.1 Mirror involution of the regional components

On the regional block $B = (u^{\text{cl}}, u^{\text{pr}}, u^{\text{au}}) \in (\mathbb{F}_3^6)^3$, the interchange is

$$cB = (u^{\text{pr}}, u^{\text{cl}}, u^{\text{au}}).$$

Its square is the identity and it preserves the self-scaling component. If $q = u^{\text{cl}} + u^{\text{pr}} + u^{\text{au}}$, $a = u^{\text{cl}} + u^{\text{pr}} - u^{\text{au}}$, and $d = u^{\text{cl}} - u^{\text{pr}}$, the block is recovered by

$$u^{\text{au}} = 2(q - a), \quad s = q - u^{\text{au}}, \quad u^{\text{cl}} = 2(s + d), \quad u^{\text{pr}} = 2(s - d).$$

The proof uses $2^{-1} = 2$ in $\mathbb{F}_3$. The interchange preserves q, a, s and reverses the sign of d . Thus the symmetric aggregate and the oriented difference jointly preserve the two components that the aggregate alone does not distinguish. This regional involution and the reversal of a cardinal route act on different domains. The fixed component may evolve during transport even though it is fixed by the interchange.

4.5 Calendar, monodromy, and memory

The finite calendar consists of twelve nonadic windows. Its group sequence is

$$0 \longrightarrow C_{12} \xrightarrow{[b] \mapsto [9b]} C_{108} \xrightarrow{[t] \mapsto [t]_9} C_9 \longrightarrow 0. \quad (4.13)$$

For $t = 9b + r$, $0 \leq r < 9$, composition contains the transport quotient

$$(b, r) + (b', r') = \left(b + b' + \left\lfloor \frac{r + r'}{9} \right\rfloor, [r + r']_9 \right),$$

with the first coordinate reduced modulo twelve. The projection is exact because its kernel consists of the multiples of nine. It admits no homomorphic section: such a section would give $C_{108} \cong C_{12} \times C_9$, but the respective exponents are 108 and 36.

Monodromy describes the action of one circuit of the phase base on its fibre. In the dynamics with memory, the nonadic holonomy Γ_9 preserves phase and advances depth. Its projection onto phase and counter satisfies

$$(r, n) \longmapsto (r, n + 1).$$

This formula is a reduced representation of the action, not a definition eliminating the rest of the state. On bidirectional histories the counter can be extended to $\mathbb{Z}$; on histories starting at depth zero, backward motion is defined only where the predecessor is preserved. The calendar can return after 108 transitions while preserving a memory different from the initial one. This distinction will be essential when the particle is constructed as persistence rather than repetition of a visible coordinate.

4.6 From records to prolongation operators

The linear representation operators are determined on the transport transitions. If x_j^χ is a regional state and Π_6 its six-component representation, the dependency is

$$b_j^\chi = \Pi_6(x_j^\chi), \quad x_{j+1}^\chi = U_{9j+9} \cdots U_{9j+1}(x_j^\chi).$$

The origin matrices X_a and destination matrices Y_a , in row vectors of $\mathbb{F}_3^6$, are

X_0	Y_0	X_1	Y_1
010211	012222	010211	002111
012222	010211	002111	110221
201101	121221	102011	012222
121221	102011	012222	102011
121200	112202	121020	010210
112202	121020	010210	010200

We have $\det X_0 = 1$, $\det X_1 = -1$ in $\mathbb{F}_3$. The unique solution of $X_a L_a = Y_a$ is $L_a = X_a^{-1} Y_a$. Concatenating the blocks expressed in the calendar (L_0, L_0, L_1, L_1) produces lengths 6, 12, 18, 24, 30. The boundary of length 36 preserves the terminal state and the prolongation relation. These matrices represent transitions: linear inversion proves their uniqueness for the given record, whereas generation of the states belongs to the effective composition U_t of TPK.

4.7 Scope of the regional construction in the study of masses

The census has separated initial conditions, emissions, regions, and operators representing their transitions. The next stage evaluates the regional coordinates and composes the dodecaphase record. Later, the particle atlas will be constructed on the same support but will use another classification. This continuity preserves a common provenance without turning a numerical fibre into a species through mere coincidence of cardinalities or labels.

5 Arithmetic of histories, composition, and refinement

The persistence of a particle and the evaluation of its character require a distinction between the transported history and the coordinates read from it. The APP-TRIT-TPK kernel provides the admissible states, their routes, and the operations preserving their provenance. This section specifies three structures used in passing from those data to the atlas: compatibility between depths, composition of histories with memory, and the transport of observables in the direction opposite to refinement. Algebraic coefficients act on states and routes already specified; the relations proved below preserve that order.

5.1 Compatible sections and evaluation of their readers

Let X_n be the set of admissible states at depth n produced by the kernel. A state records the APP positions, the additive and multiplicative sheets, their remainders and quotients, the tritic regime, orientation, phase, boundary, route, carry, and memory pertinent to survival. These data satisfy the relations of the operations that produced them; they are not regarded as a product of independent coordinates.

For $m \geq n$, the restriction $r_{m,n} : X_m \longrightarrow X_n$ preserves the antecedent at depth n . The restrictions satisfy

$$r_{n,n} = \text{id}_{X_n}, \quad r_{\ell,n} = r_{m,n} \circ r_{\ell,m} \quad (\ell \geq m \geq n). \quad (5.1)$$

New memory belongs to the prolongation; restriction preserves the memory already determined in the prefix.

Definition 5.1 (Compatible numerical section). An HMT numerical section is a sequence of states of the same genealogy

$$x = (x_n)_{n \geq 0} \in X_\infty := \varprojlim_n X_n, \quad r_{m,n}(x_m) = x_n \quad (m \geq n). \quad (5.2)$$

A reader is an operation defined on a specified domain of these sections. A finite observation belongs to a prefix; a scalar value is the evaluation of a reader. The section and the presentation of that section through a reader are therefore objects of different types.

Inverse-limit notation gathers together the compatibility of the states already constructed. The existence of a particular prolongation is established through the admissibility rules of that construction, not merely by writing X_∞ .

In the Archimedean domain of a reader, each prefix determines a nonempty closed interval $I_n(x) = [a_n(x), b_n(x)]$, with

$$I_{n+1}(x) \subseteq I_n(x), \quad b_n(x) - a_n(x) \longrightarrow 0. \quad (5.3)$$

Regional interval evaluation and the determination of its prefixes have been assembled in Proposition 8.1.

Proposition 5.2 (Value determined by compatible cylinders). *For every section in that domain there is a unique real number $\text{ev}(x)$ such that*

$$\{\text{ev}(x)\} = \bigcap_{n \geq 0} I_n(x). \quad (5.4)$$

Proof. Nestedness makes the sequence $a_n(x)$ increasing and $b_n(x)$ decreasing. Both remain bounded in the initial interval. By completeness of the line, $a = \sup_n a_n(x)$ and $b = \inf_n b_n(x)$ exist and satisfy $a \leq b$. The vanishing width forces $a = b$. This point belongs to every interval, and any other point of the intersection would have distance from it less than or equal to every width, and would therefore coincide with it. $\square$

This proposition determines the value of each section in the stated domain. Coverage of all of $\mathbb{R}$ requires a proof that each value belongs to the image of the reader; the inverse-limit definition does not by itself provide that proof. Likewise, identifying two histories from their value requires an injectivity result for the reader in question. Evaluation preserves the genealogy as an antecedent and specifies exactly which part of it is expressed.

5.2 Sectorial route algebra and memory multipliers

Fix a sector of states and the category $\mathcal{C}$ of its admissible histories. We write $vu = v \circ u$: first u is traversed, then v . The elementary steps are TPK selection, transport, and update operations. The category preserves their endpoints and the conditions allowing two routes to be concatenated.

The local tritic fibre can be represented by

$$\mathbb{R}[J_x]/(J_x^2 + \tau(x)1_x), \quad \tau(x) \in \{+1, 0, -1\}.$$

The representation preserves the regime and its orientation. To handle richer coefficients, we shall allow an associative unital real algebra A_x at each object x . A transition between regimes must specify its map on these algebras; it does not amount to identifying them.

The following construction applies to sectors possessing unital transports $\rho_u : A_x \rightarrow A_y$, for $u : x \rightarrow y$, with

$$\rho_{\text{id}_x} = \text{id}_{A_x}, \quad \rho_v \circ \rho_u = \rho_{vu}. \quad (5.5)$$

For each composable pair $u : x \rightarrow y$, $v : y \rightarrow z$, let $\Omega(v, u) \in Z(A_z)^\times$ be an invertible central multiplier. Its rules are

$$\Omega(\text{id}_y, u) = \Omega(u, \text{id}_x) = 1_y \quad (5.6)$$

and, for $w : z \rightarrow t$,

$$\rho_w(\Omega(v, u)) \Omega(w, vu) = \Omega(w, v) \Omega(wv, u). \quad (5.7)$$

These conditions describe the starting structure of the sector: every specific application must provide its transports and multipliers. A realisation through bimodules preserves its own compositions and is not automatically replaced by the strict action (5.5).

Let $t(u)$ be the terminal endpoint of the arrow u . On sums of finite support

$$\mathcal{H} = \bigoplus_{u \in \text{Mor}(\mathcal{C})} A_{t(u)} T_u$$

we define, by bilinearity,

$$(aT_v)(bT_u) = a\rho_v(b)\Omega(v, u)T_{vu} \quad (5.8)$$

when u, v are composable, and zero otherwise. Addition gathers histories with their coefficients; multiplication concatenates the histories and preserves the memory multiplier. Finite support ensures that each product is defined without requiring additional convergence.

5.3 Associativity, local identities, and calendar memory

Proposition 5.3 (Associative composition of the sector). *Under (5.5)–(5.7), the product (5.8) is associative. If there are finitely many objects, its identity is $\sum_x 1_x T_{\text{id}_x}$. For an arbitrary set of objects, the same expressions over finite sets of endpoints give local identities for every finite collection of histories.*

Proof. Let $u : x \rightarrow y$, $v : y \rightarrow z$, $w : z \rightarrow t$, with coefficients $a \in A_y$, $b \in A_z$, $c \in A_t$. The two associations produce, as coefficients of T_{wvu} ,

$$\begin{aligned} ((cT_w)(bT_v))(aT_u) &: c\rho_w(b)\Omega(w, v)\rho_{wv}(a)\Omega(wv, u), \\ (cT_w)((bT_v)(aT_u)) &: c\rho_w(b)\rho_w\rho_v(a)\rho_w(\Omega(v, u))\Omega(w, vu). \end{aligned}$$

The strict action identifies $\rho_w \rho_v(a)$ with $\rho_{wv}(a)$. Centrality allows $\Omega(w, v)$ to be moved to the right of this factor, and the cocycle identity equates the remaining factors. The general coefficients a, b, c , which need not commute, are not permuted. If the three arrows do not form a composable chain, both associations are zero by their endpoints. Bilinearity extends the equality to finite sums. Finally, (5.6) and the unitality of each ρ_u verify the identity. For a finite collection it suffices to add the identities of all its initial and terminal endpoints, proving the last assertion. $\square$

The nonadic calendar provides an effective realisation of these conditions. Take the one-object category with arrows $C_9 = \mathbb{Z}/9\mathbb{Z}$, coefficients $A = \mathbb{R}[z, z^{-1}]$, and identity action. For representatives $0 \leq a, b < 9$, define

$$c_0(a, b) = \left\lfloor \frac{a+b}{9} \right\rfloor, \quad \Omega(b, a) = z^{c_0(a,b)}. \quad (5.9)$$

The formal variable z preserves the number of circuits. Normalisation follows from $c_0(0, a) = c_0(a, 0) = 0$. If $[a+b]_9$ denotes the reduced representative, Euclidean division gives

$$c_0(a, b) + c_0([a+b]_9, c) = \left\lfloor \frac{a+b+c}{9} \right\rfloor = c_0(b, c) + c_0(a, [b+c]_9).$$

This equality proves the cocycle (5.7) for the multipliers, and hence the associativity of the calendar with memory.

If $T = T_{[1]_9}$, the product rules give $T^j = T_{[j]_9}$ for $0 \leq j < 9$ and $T^9 = zT_{[0]_9}$. Subsequently imposing $z^{12} = 1$ yields $T^{108} = 1$. The expressions $z^k T_{[j]_9}$, $0 \leq k < 12$, $0 \leq j < 9$, form the 108 positions of the resulting calendar. This recovers the cyclic group C_{108} and its projection onto C_9 ; retaining z without that relation distinguishes successive circuits. This is a realisation of calendar memory: the remaining coordinates of the TPK histories remain in their own domain.

5.4 Double variance of refinement and preservation of unity

Restriction of states induces a prolongation of observables in the opposite direction. For the function algebras with pointwise multiplication $D_n = \text{Fun}(X_n, \mathbb{R})$, precomposition defines

$$r_{m,n}^* : D_n \longrightarrow D_m, \quad r_{m,n}^* f = f \circ r_{m,n}. \quad (5.10)$$

The contravariant order is preserved: $r_{\ell,n}^* = r_{\ell,m}^* \circ r_{m,n}^*$. These maps are unital homomorphisms. If $r_{m,n}$ is surjective, then $r_{m,n}^*$ is also injective: a difference between two functions can be evaluated at a preimage of a point where they differ.

Lemma 5.4 (Indicators, unity, and compatible evaluation). *For $x \in X_n$, let e_x be its indicator function. Then*

$$r_{m,n}^* e_x = \mathbf{1}_{r_{m,n}^{-1}(x)}, \quad r_{m,n}^* \mathbf{1}_n = \mathbf{1}_m. \quad (5.11)$$

When the fibre $r_{m,n}^{-1}(x)$ is finite, the first equality is the finite sum

$$r_{m,n}^* e_x = \sum_{y: r_{m,n}(y)=x} e_y.$$

For every section $x = (x_n) \in X_\infty$, the rule $[f] \mapsto f(x_n)$, with $f \in D_n$, defines a unital homomorphism

$$\text{Ev}_x : D_{\text{cyl}} \longrightarrow \mathbb{R}, \quad D_{\text{cyl}} = \varinjlim_n (D_n, r_{n+1,n}^*). \quad (5.12)$$

Proof. For $y \in X_m$, we have $(r_{m,n}^* e_x)(y) = 1$ exactly when $r_{m,n}(y) = x$. This is the first equality of (5.11). If the fibre is finite, its point indicators are orthogonal and sum to its indicator. The second equality follows from $\mathbf{1}_n(r_{m,n}(y)) = 1$, without any assumption on the cardinality of the fibre.

For every $f \in D_n$, the naturality of evaluation is

$$(r_{m,n}^* f)(y) = f(r_{m,n}(y)).$$

In a compatible section, $r_{m,n}(x_m) = x_n$, hence $(r_{m,n}^* f)(x_m) = f(x_n)$. Thus two representatives identified at a common depth have the same evaluation. Point evaluation preserves addition, multiplication, and unity, completing the proof of (5.12). $\square$

For infinite fibres the formulation through $\mathbf{1}_{r_{m,n}^{-1}(x)}$ remains valid; there is no need to interpret an infinite sum as a finite algebraic sum. The direct limit gathers the finite-depth observables with the identifications induced by the restrictions. Prolonging route operators additionally requires the compatibility squares of their transports with those restrictions.

The preserved unity is the constant function one on the domain of possibilities. An individual section selects a compatible history in that domain. Refinement may increase depth and distinguish more prolongations while preserving unity and the evaluation of each antecedent. This unital preservation provides the precise algebraic content used here; a probability measure, when it enters another reader, additionally requires its normalisation and its own rules.

6 Integral compatibility of regional generation, cylindrical refinement, and incidence

Regional prolongation produces ordered sequences of blocks together with their reading depths. The Archimedean projection and ternary incidence are calculated on these same sequences. The connection between the two realizations can be formulated entirely through integer operations: selection by rational intervals, Euclidean division, lifting of six trits, carry reading, and the Witt transformation. This formulation makes it possible to trace each coefficient from its generating block to its subsequent use.

The construction uses the three regional readers already obtained from the APP-TRIT-TPK state. We denote them by the indices $c \in \{P, E, F\}$, corresponding, respectively, to the closure, propagation, and autoscaling regions. Their real realization will be written as v_c . The subsequent identifications $v_P = \pi_{\text{HMT}}$, $v_E = e_{\text{HMT}}$, and $v_F = \varphi_{\text{HMT}}$ preserve this causal order. The operation that emits a block inspects the rational intervals of the regional reader; the real value enters the proof of correctness of that operation.

6.1 Positional emission through rational separation of cells

For each region, two rational sequences $\ell_c(j), u_c(j)$, constructed by its reader, are available and satisfy

$$\ell_c(j) \leq v_c \leq u_c(j).$$

The cell-separation property proved for these readers states that, for each integer $s > 0$, there exists J such that, for every $j \geq J$,

$$\lfloor s\ell_c(j) \rfloor = \lfloor su_c(j) \rfloor - 1 = \lfloor sv_c \rfloor. \quad (6.1)$$

Avoidance of rational boundaries by the regional values is the hypothesis of the preceding theorems that guarantees this separation. Here we use their operational conclusion together with the readers and intervals that produce it.

Definition 6.1 (Separation index and emitted prefix). For $s > 0$, let

$$j_c(s) = \min\{j \in \mathbb{N} : \lfloor s\ell_c(j) \rfloor = \lfloor su_c(j) \rfloor - 1\}.$$

For an integer base $b \geq 2$ and a depth $n \geq 0$, define

$$P_c(b, n) = \lfloor b^n \ell_c(j_c(b^n)) \rfloor. \quad (6.2)$$

The prefix contains the integer part and the first n positions in base b ; consequently, $P_c(b, 0)$ is the regional integer part.

Theorem 6.2 (Termination and correctness of emission). *The index $j_c(s)$ exists for every $s > 0$. The emitted prefixes satisfy*

$$P_c(b, n) = \lfloor b^n v_c \rfloor, \quad P_c(b, n) \leq b^n v_c < P_c(b, n) + 1. \quad (6.3)$$

For each scale s , a single sufficiency index $J(s)$ can also be chosen that is valid simultaneously for all three regions.

Proof. The eventual equality in (6.1) makes the set defining $j_c(s)$ nonempty. Its minimum exists by the well-ordering of $\mathbb{N}$. At any index where the equality holds, the rational lower bound gives $\lfloor s\ell_c(j) \rfloor \leq \lfloor sv_c \rfloor$. The upper bound, together with avoidance of a cell boundary, gives $\lfloor sv_c \rfloor \leq \lfloor su_c(j) \rfloor - 1$. When the two integer endpoints coincide, the intervening integer coincides with them. This proves (6.3). For the joint statement, it suffices to take the maximum of the three sufficiency indices obtained at the same scale. The joint index controls all three emissions simultaneously, while each region retains its own intervals. $\square$

Proposition 6.3 (Exact compatibility of prefixes). *For $n, m \geq 0$,*

$$\left\lfloor \frac{P_c(b, n+m)}{b^m} \right\rfloor = P_c(b, n). \quad (6.4)$$

In particular, the next block $d_c(b, n) = P_c(b, n+1) \bmod b$ satisfies

$$P_c(b, n+1) = bP_c(b, n) + d_c(b, n), \quad 0 \leq d_c(b, n) < b. \quad (6.5)$$

Proof. Write $b^n v_c = N + \theta$, with $N = P_c(b, n)$ and $0 \leq \theta < 1$. Then $P_c(b, n+m) = b^m N + \lfloor b^m \theta \rfloor$, and the second summand belongs to $\{0, \dots, b^m - 1\}$. Euclidean division by b^m recovers N . The specialization $m = 1$ yields (6.5) and its residual bound. $\square$

6.2 Regional blocks of six trits at arbitrary depth

The integer equality $729 = 3^6$ determines the grouping of six ternary positions into a block. For $t \geq 0$, define

$$u_c(t) = P_c(729, t+1) \bmod 729, \quad B_{c,j}(t) = \left\lfloor \frac{u_c(t)}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6. \quad (6.6)$$

Thus $B_c(t)$ is an ordered word of six trits, and

$$u_c(t) = \sum_{j=0}^5 B_{c,j}(t) 3^{5-j}. \quad (6.7)$$

The word and its integer lift are mutually recoverable representations of the same emitted block.

Theorem 6.4 (Stability of extraction at any depth). *For every n ,*

$$P_c(729, n) = P_c(3, 6n).$$

If $t < n$, then

$$u_c(t) = \left\lfloor \frac{P_c(729, n)}{729^{n-t-1}} \right\rfloor \bmod 729. \quad (6.8)$$

Consequently, every extension of the prefix preserves all previously emitted blocks and all signatures calculated from them.

Proof. The first equality follows because both prefixes are separated at the same scale, $729^n = 3^{6n}$, and from Theorem 6.2. For the second, apply (6.4) at depths n and $t+1$, and then take the residue modulo 729. The ternary extraction in (6.6) is deterministic. Any explicit function of those eighteen coordinates therefore retains the same result when the prefix is extended. $\square$

The construction is quantified over $t \in \mathbb{N}$. A table materializing one hundred blocks is a finite evaluation of this construction; formula (6.8) determines its compatibility with every subsequent extension exactly.

Example 6.5 (First joint block). Evaluation of the selected regional readers gives the words

$$B_P(0) = 010211, \quad B_E(0) = 201101, \quad B_F(0) = 121200.$$

Their lifting by (6.7) produces

$$u_P(0) = 103, \quad u_E(0) = 523, \quad u_F(0) = 450.$$

For example, $201101_3 = 2 \cdot 243 + 27 + 9 + 1 = 523$. All six positions, including leading zeros, are retained in the band representation. The leading zero in 010211 has a defined position even though the decimal notation for 103 does not contain it.

6.3 Transition registers and reconstruction of linear lifts

Over $\mathbb{F}_3$, collect two consecutive emissions from each region in the matrix

$$R(t) = \begin{pmatrix} B_P(t) \\ B_P(t+1) \\ B_E(t) \\ B_E(t+1) \\ B_F(t) \\ B_F(t+1) \end{pmatrix}. \quad (6.9)$$

This definition fixes both the regional order and the temporal order. The first four registers are

$$X_0 = R(0), \quad Y_0 = R(1), \quad X_1 = R(2), \quad Y_1 = R(3).$$

Their values, obtained by extraction from the regional prefixes, are

$$\begin{aligned} X_0 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 2 & 0 & 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 0 \\ 1 & 1 & 2 & 2 & 0 & 2 \end{pmatrix}, & Y_0 &= \begin{pmatrix} 0 & 1 & 2 & 2 & 2 & 2 \\ 0 & 1 & 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \end{pmatrix}, \\ X_1 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 & 1 & 0 \end{pmatrix}, & Y_1 &= \begin{pmatrix} 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 1 & 0 & 2 & 2 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 & 0 \end{pmatrix}. \end{aligned}$$

Theorem 6.6 (Unique reconstruction of the two initial links). *The equations $X_0 L_0 = Y_0$ and $X_1 L_1 = Y_1$ each have a unique solution in $M_6(\mathbb{F}_3)$. These solutions are*

$$\begin{aligned} L_0 &= \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, & L_1 &= \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix}. \end{aligned}$$

Both matrices are invertible.

Proof. The matrices

$$\begin{aligned} A_0 &= \begin{pmatrix} 1 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 2 & 1 & 2 & 1 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 2 \\ 2 & 1 & 1 & 0 & 2 & 2 \\ 2 & 1 & 1 & 1 & 1 & 2 \end{pmatrix}, & A_1 &= \begin{pmatrix} 1 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 1 \\ 1 & 1 & 0 & 2 & 0 & 0 \\ 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 0 & 2 \end{pmatrix} \end{aligned}$$

satisfy $A_i X_i = X_i A_i = I$, by multiplication in $\mathbb{F}_3$. The solutions are therefore forced by $L_i = A_i Y_i$; multiplication produces the two displayed matrices. Their inverses are

$$L_0^{-1} = \begin{pmatrix} 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 2 & 0 & 0 \\ 2 & 1 & 2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 \\ 0 & 2 & 1 & 1 & 2 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 \end{pmatrix}, \quad L_1^{-1} = \begin{pmatrix} 2 & 0 & 2 & 1 & 2 & 1 \\ 2 & 1 & 0 & 0 & 2 & 0 \\ 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 2 & 2 & 0 \end{pmatrix}.$$

The bilateral verification $L_i L_i^{-1} = L_i^{-1} L_i = I$ proves exact recovery of each transformed word. All products use residues 0, 1, 2 with their interpretation in $\mathbb{F}_3$. $\square$

The causal reading of this result is precise: the regional emissions produce $R(t)$; the registers $R(0), \dots, R(3)$ determine the two lifts above; their inverses recover the transformed words. Continuity for every t belongs to the coinductive readers and Theorem 6.4. The matrix reconstruction proved here has the two links specified in its statement as its domain.

6.4 Simultaneous refinement in ternary and decimal bases

A ternary emission of six trits and a decimal emission of three digits are readings in different bases. To preserve their relationship, record the state

$$s = (n, k, N, D),$$

where N is a prefix at depth n in base 729, and D is a prefix at depth k in base 1000. Their Archimedean cylinders are

$$I_{729}(s) = \left[\frac{N}{729^n}, \frac{N+1}{729^n} \right), \quad I_{1000}(s) = \left[\frac{D}{1000^k}, \frac{D+1}{1000^k} \right). \quad (6.10)$$

The half-open upper endpoint assigns each value to a unique cell at a given base and depth.

Definition 6.7 (Integer compatibility defects). Define

$$A(s) = N1000^k - D729^n, \quad (6.11)$$

$$B(s) = (N+1)1000^k - D729^n = A(s) + 1000^k. \quad (6.12)$$

The state is compatible when

$$A(s) < 729^n, \quad B(s) > 0. \quad (6.13)$$

Lemma 6.8 (Integer criterion for intersection). *The cylinders in (6.10) have a nonempty intersection if and only if (6.13) holds.*

Proof. Two half-open intervals $[a, b)$ and $[c, d)$, with $a < b$ and $c < d$, intersect exactly when $a < d$ and $c < b$. Applying this condition to (6.10) and multiplying by the positive denominators gives $N1000^k < (D+1)729^n$ and $D729^n < (N+1)1000^k$. These are the two inequalities in (6.13). $\square$

Definition 6.9 (Update of depths and prefixes). For $0 \leq u < 729$ and $0 \leq d < 1000$, define

$$T_u(n, k, N, D) = (n+1, k, 729N+u, D),$$

$$T_{u,d}(n, k, N, D) = (n+1, k+1, 729N+u, 1000D+d).$$

The first operation refines only the ternary cylinder. The second refines both readings and preserves both depths in the state.

Proposition 6.10 (Exact boundary recurrences). *The updates above satisfy*

$$A(T_u s) = 729A(s) + u1000^k, \quad (6.14)$$

$$A(T_{u,d} s) = 729000A(s) + u1000^{k+1} - d729^{n+1}. \quad (6.15)$$

Proof. For the first identity,

$$(729N + u)1000^k - D729^{n+1} = 729(N1000^k - D729^n) + u1000^k.$$

For the second,

$$\begin{aligned} (729N + u)1000^{k+1} - (1000D + d)729^{n+1} \\ = 729000(N1000^k - D729^n) + u1000^{k+1} - d729^{n+1}. \end{aligned}$$

Both are integer identities preceding any real approximation. $\square$

Theorem 6.11 (Refinement of the generated regional cylinders). *Let*

$$s_c(n, k) = (n, k, P_c(729, n), P_c(1000, k)).$$

Then $s_c(n, k)$ is compatible for all n, k . Moreover,

$$\begin{aligned} T_{u_c(n)}s_c(n, k) &= s_c(n + 1, k), \\ T_{u_c(n), d_c(1000, k)}s_c(n, k) &= s_c(n + 1, k + 1). \end{aligned}$$

Euclidean division of the refined prefixes recovers the preceding prefixes exactly. For every $a, b \geq 0$,

$$\frac{P_c(729, n + a) - P_c(729, n + a) \bmod 729^a}{729^a} = P_c(729, n),$$

and the analogous identity holds in base 1000.

Proof. Theorem 6.2 places v_c simultaneously in both intervals in (6.10); the intersection criterion gives compatibility. The update identities are (6.5) in each base. Recovery follows from (6.4). Thus the defect recurrence and prefix recovery are consequences of the same update. $\square$

Example 6.12 (First pair of regional cylinders). For $n = k = 1$, evaluation of the three regions gives

c	N	D	A	B
P	2290	3141	211	1211
E	1981	2718	-422	578
F	1179	1618	-522	478

In all three rows, $A < 729$ and $B > 0$. The first prefix column decomposes as $2290 = 3 \cdot 729 + 103$, $1981 = 2 \cdot 729 + 523$, and $1179 = 729 + 450$: exactly the previously emitted six-trit blocks reappear. The different signs of A express the relative positions of the two lower endpoints without altering the regional value's membership in both intervals.

6.5 Selection of the next block by the regional reader

Compatibility of two intervals expresses a relationship between readings. Determining the next block additionally uses the rational interval produced by the region. This operation can be formulated without discarding the depth history.

Definition 6.13 (Admissibility of regional prolongation). For fixed c, n , let $s = 729^{n+1}$ and $j = j_c(s)$. A residue $u \in \{0, \dots, 728\}$ is admissible when

$$[s\ell_c(j)] \leq 729P_c(729, n) + u \leq [su_c(j)] - 1. \quad (6.16)$$

Theorem 6.14 (Uniqueness of the prolongation compatible with the reader). *Condition (6.16) is equivalent to $u = u_c(n)$. There is therefore exactly one admissible prolongation of each region at each depth.*

Proof. At the separation index, both integer endpoints in (6.16) coincide with $P_c(729, n + 1)$. The condition is therefore equivalent to $729P_c(729, n) + u = P_c(729, n + 1)$. By (6.5), the right-hand side is $729P_c(729, n) + u_c(n)$. Cancelling the common term gives equality of the residues. Conversely, this equality satisfies the condition. $\square$

The effective dependence of this selection is fully specified: selected region, rational reader, depth, separation index, preceding prefix, and next residue. The defect A summarizes a boundary relationship; the state (n, k, N, D) and the regional intervals retain the data used by the update.

6.6 Ternary signature, carry, Witt incidence, and reading clock

At a fixed depth t , write $B_{c,j} = B_{c,j}(t)$. For each column j , calculate the following integers and residues:

$$S_j = B_{P,j} + B_{E,j} + B_{F,j}, \quad q_j = S_j \bmod 3, \quad c_j = \lfloor S_j/3 \rfloor, \quad (6.17)$$

$$a_j = (B_{P,j} + B_{E,j} - B_{F,j}) \bmod 3, \quad w_j = \sum_c \mathbf{1}_{B_{c,j} \neq 0}, \quad \bar{w}_j = w_j \bmod 3. \quad (6.18)$$

The residue of the axial expression is taken in the integers. An implementation using natural numbers calculates it as $(B_{P,j} + B_{E,j} + 3 - B_{F,j}) \bmod 3$, which preserves that residue exactly.

The ternary incidence matrix used by the reader is

$$W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}. \quad (6.19)$$

Two residual charges are preserved for each region:

$$v_c^{\text{res}} = \sum_j B_{c,j} \bmod 3, \quad v_c^W = \sum_j (B_c W)_j \bmod 3. \quad (6.20)$$

The reader's complete signature is $\mathcal{S}(B) = (q, a, c, w, \bar{w}, v^{\text{res}}, v^W)$. Its structure separately preserves the column residue, carry, axial contrast, support weight, and the two charge readings.

Theorem 6.15 (Exact recovery and bounds of the signature). *For each column and each region,*

$$S_j = q_j + 3c_j, \quad 0 \leq q_j, a_j < 3, \quad 0 \leq c_j \leq 2, \quad (6.21)$$

$$B_{F,j} = 2(q_j + 3 - a_j) \bmod 3, \quad 0 \leq w_j \leq 3, \quad 0 \leq \bar{w}_j < 3. \quad (6.22)$$

Moreover,

$$v_c^W = \left(\sum_j ((B_c W)_j \bmod 3) \right) \bmod 3. \quad (6.23)$$

The stated recoveries determine each column's total and its autoscaling coordinate; the regional register additionally preserves the three ordered bands.

Proof. Identity (6.21) is Euclidean division of S_j by 3. Each summand of S_j lies between 0 and 2, so $0 \leq S_j \leq 6$ and $c_j \leq 2$. In $\mathbb{F}_3$, subtracting the two congruences defining q_j and a_j gives $q_j - a_j = 2B_{F,j}$. Since the inverse of 2 is 2, this yields (6.22); the summand 3 permits an expression using natural numbers. The bounds on w_j follow by summing three indicators. Finally, reducing an integer sum modulo 3 is equivalent to reducing each summand and then reducing their sum, which proves (6.23). $\square$

Example 6.16 (Signature of the first joint block). The three initial bands produce

$$\begin{aligned} q &= (0, 0, 2, 2, 1, 2), & a &= (1, 2, 0, 1, 1, 2), \\ c &= (1, 1, 0, 1, 0, 0), & w &= (2, 2, 2, 3, 1, 2), \\ v^{\text{res}} &= (2, 2, 0), & v^{\text{W}} &= (2, 1, 1). \end{aligned}$$

The fourth column has total $2 + 1 + 2 = 5$; its signature records $q_3 = 2$ and $c_3 = 1$, so $q_3 + 3c_3 = 5$. Axial recovery gives $2(2 + 3 - 1) \bmod 3 = 2$, precisely the autoscaling coordinate. The Witt images, reduced componentwise, are

$$(2, 0, 2, 1, 1, 2), \quad (0, 0, 0, 2, 0, 2), \quad (2, 1, 1, 2, 1, 0).$$

Their residual sums by row yield the three charges $(2, 1, 1)$.

Definition 6.17 (Integer clock of decimal reading). The transition index t and decimal representation depth are related by

$$\kappa(t) = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\}. \quad (6.24)$$

Proposition 6.18 (Characterization and monotonicity of the clock). *The integer $\kappa(t)$ is uniquely determined by*

$$1000^{\kappa(t)} \leq 729^{t+1} < 1000^{\kappa(t)+1}. \quad (6.25)$$

The function κ is monotone and satisfies $\kappa(20) = \kappa(21) = 20$.

Proof. The powers of 1000 are strictly increasing and unbounded; there is therefore a unique interval between consecutive powers containing 729^{t+1} . Monotonicity of the latter expression proves that of κ . The final two evaluations are the integer comparisons

$$1000^{20} \leq 729^{21} < 1000^{21}, \quad 1000^{20} \leq 729^{22} < 1000^{21}.$$

These comparisons use exact integer powers. □

Equality of decimal depth at two distinct transitions explains why both indices must be retained: the reading clock can remain constant while the ternary band advances and its signature is updated. Chronological information resides in the ordered register of transitions.

6.7 Composition theorem for the block, cylinder, and signature

Theorem 6.19 (Joint compatibility of regional readings). *For each depth n , there exists a unique triple of blocks (u_P, u_E, u_F) satisfying regional admissibility (6.16) in all three regions. This triple is $(u_P(n), u_E(n), u_F(n))$. Simultaneously:*

1. *each block updates its cylinder through recurrences (6.14) and (6.15);*
2. *its six trits produce the signature $S(B_P(n), B_E(n), B_F(n))$;*
3. *the column totals are recovered as $q + 3c$, and the Witt charges are obtained from the same bands through (6.23);*
4. *every deeper prefix returns exactly these blocks, truncated cylinders, and signatures.*

Proof. Apply Theorem 6.14 to each of the three regional indices. Coordinatewise uniqueness gives joint uniqueness. Substitute those same residues into the operations $T_u, T_{u,d}$, which allows Theorem 6.11 to be applied. The deterministic application of (6.6) produces the bands used by (6.17)–(6.20). The recovery identities are those of Theorem 6.15. Finally, Theorem 6.4 and truncation in each base prove stability at greater depth. At every step the identity of the emitted block has been preserved; neither reading requires the selection of a different block. □

The conclusion is an operational compatibility between cylindrical contraction and preservation of incidence. Both use the same generated register, retain the depths, and admit integer recovery. This result is the explicit form of the connection

$$\text{emitted regional block} \longrightarrow \begin{cases} \text{cylinder and integer boundary,} \\ \text{band, signature, carry, and Witt charge.} \end{cases}$$

The subsequent development of incidence and fields uses the second reading while preserving its common provenance with the first.

6.8 Calendar census and simultaneous compatibility of the regional biographies

The preceding rows are assembled, without yet applying any decimal reader, into the three block biographies produced by the transitions:

$$\begin{aligned} B^{\text{cl}} &= 010211|012222|010211|002111|110221, \\ B^{\text{pr}} &= 201101|121221|102011|012222|102011, \\ B^{\text{au}} &= 121200|112202|121020|010210|010200. \end{aligned} \tag{6.26}$$

Indeed, the consecutive pairs of the first two segments are the rows of X_0, Y_0 , and those of the last two are the rows of X_1, Y_1 . Thus, the biographies are another representation of the record of twelve transitions, not a collection of conventional prefixes.

To verify the operative word of the calendar, let $c = c_1c_2c_3c_4 \in \{0, 1\}^4$ and, for each regime $\chi \in \{\text{cl, pr, au}\}$, define

$$u_0^\chi = b_0^\chi, \quad u_j^\chi = u_{j-1}^\chi L_{c_j} \quad (1 \leq j \leq 4).$$

To condense the census, denote by

$$N_{\text{ar}}(c) := \#\{(j, \chi) : u_j^\chi = b_j^\chi, 1 \leq j \leq 4\}, \quad N_{\text{bio}}(c) := \#\{\chi : (u_0^\chi, \dots, u_4^\chi) = B^\chi\}.$$

Exact multiplication in $\mathbb{F}_3$ for the sixteen calendars gives:

c	$N_{\text{ar}}(c)$	$N_{\text{bio}}(c)$
0000, 0001	6	0
0010	9	0
0011	12	3
0100, 0101, 0110, 0111	3	0
1000, 1001, 1010, 1011	0	0
1100, 1101, 1110, 1111	0	0

(D)

The table exhausts the $2^4 = 16$ possibilities, and each of its entries is obtained by four multiplications of a row by one of the two displayed matrices. Consequently, 0011 is the unique calendar that simultaneously reconstructs the three complete biographies. In operator notation,

$$(L_0, L_0, L_1, L_1) \tag{6.27}$$

is not an independent prescription: it is the unique binary reading compatible with the twelve edges already produced by U_t .

The calendar (6.27) produces four new six-component blocks. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate cumulative length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions are fixed; it does not replace their production by U_t . Conversely, (D) proves within the article the uniqueness of the ordering of the four transports. The causal chain is thereby closed: the APP census produces the initial regions, TRIT fixes regime and orientation, TPK produces the transition record, and that record determines L_0, L_1 and 0011 before any Archimedean realization.

7 Regional determination of the dodecaphasic descriptor and selection of the coupling register

The determination of the coupling register combines two constructions that should be developed in their actual order. The first produces the inputs to the selector from the regional representations of the TPK: six-trit bands, column signatures, a conversion calendar, an oriented descriptor, and a functional panel. The second operates on these inputs and on the terminal eight-block repertoire, enumerates its admissible orderings, and selects a unique register through two simultaneous conditions: exceptional incidence and reading heterotypy. The value of the register appears at the end of this composition.

The present development presupposes the regional representations obtained in the preceding chapters from APP-TRIT-TPK. Its purpose is to make explicit every intermediate operation leading from those representations to the finite selector. The realization of the same register through signed incidences, Hadamard inversion, and the representation of the fine-structure constant are then incorporated. The three constructions retain their respective domains: selection of the register, recovery from its channels, and arithmetic evaluation of its representations.

7.1 Regional bands, encoding, and integer signatures

7.1.1 Domains and encoding of the regional bands

The channel order is fixed as

$$\chi \in \{\pi, e, \varphi\}, \quad B_\chi(t) \in \mathbb{F}_3^6, \quad t \in \mathbb{N}.$$

The channel symbols identify the regions already generated. Each $B_\chi(t)$ retains its place in its sequence of representations; the index t counts ternary blocks. The natural-number evaluation of a band $b = (b_0, \dots, b_5)$ is

$$\text{enc}_6(b) = 243b_0 + 81b_1 + 27b_2 + 9b_3 + 3b_4 + b_5, \quad 0 \leq \text{enc}_6(b) < 729. \quad (7.1)$$

Its fixed-width inverse is

$$\text{dig}_j(n) = \left\lfloor \frac{n}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6.$$

Leading zeros are thus retained: the words 002001 and 2001 could share an abbreviated integer notation, but the type of the former contains six positions. All subsequent concatenations use this width.

Within the hundred-block window, the six-hundred-trit representation of channel χ determines an integer $P_\chi^{(600)}$. Extraction is performed by

$$b_\chi(t) = \left\lfloor \frac{P_\chi^{(600)}}{729^{99-t}} \right\rfloor \bmod 729, \quad 0 \leq t < 100, \quad B_\chi(t) = \text{enc}_6^{-1}(b_\chi(t)). \quad (7.2)$$

The six-hundred-trit integer is a representation of the regional producer; the hundred-row table is the evaluation of (7.2). This separation allows a historical table to be replaced by its producer without changing the selector that consumes it.

7.1.2 Residues, quotients, occupancy, and orientation

For each column j and time t , introduce

$$T_j(t) = B_\pi(t)_j + B_e(t)_j + B_\varphi(t)_j, \quad (7.3)$$

$$q_j(t) = T_j(t) \bmod 3, \quad c_j(t) = \lfloor T_j(t)/3 \rfloor, \quad (7.4)$$

$$\alpha_j(t) = (B_\pi(t)_j + B_e(t)_j - B_\varphi(t)_j) \bmod 3, \quad (7.5)$$

$$w_j(t) = \mathbf{1}_{B_\pi(t)_j \neq 0} + \mathbf{1}_{B_e(t)_j \neq 0} + \mathbf{1}_{B_\varphi(t)_j \neq 0}, \quad \bar{w}_j(t) = w_j(t) \bmod 3. \quad (7.6)$$

The subtraction in the third expression is performed in the integers and then reduced. In an implementation over natural numbers, the equivalent expression is $(B_\pi + B_e + 3 - B_\varphi) \bmod 3$; truncated subtraction would change the reader.

Proposition 7.1 (Reconstruction of the column sum). *For all admissible bands,*

$$T_j(t) = q_j(t) + 3c_j(t), \quad 0 \leq q_j(t) < 3, \quad 0 \leq c_j(t) \leq 2, \quad 0 \leq w_j(t) \leq 3.$$

Proof. Each column sums three representatives from $\{0, 1, 2\}$, so that $0 \leq T_j(t) \leq 6$. Euclidean division by three yields the stated residue and quotient. Occupancy counts three binary conditions; its bound follows directly from that definition. $\square$

This identity preserves the quotient of the local sum. Reconstructing the three individual bands requires the bands themselves or an additional reader that distinguishes them: a column sum has fewer coordinates than the triple that produces it. Accordingly, the temporal row used below jointly retains the three bands and the four fields $q, a, c, \bar{w}$.

7.2 Conversion clock and first zero advance

7.2.1 Ternary capacity and decimal depth

A representation of $t + 1$ bands contains $6(t + 1)$ trits. The associated conversion calendar is defined by the integer

$$d_t = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\} = \max\{k \in \mathbb{N} : 1000^k \leq 729^{t+1}\}. \quad (7.7)$$

The definition compares integer capacities. The counter t and the depth d_t record different quantities: a transition adds one six-trit band; the clock counts complete powers of the decimal base one thousand contained in the accumulated ternary capacity.

Lemma 7.2 (Integer characterization of the clock). *For $t, k \in \mathbb{N}$,*

$$d_t = k \iff 1000^k \leq 729^{t+1} < 1000^{k+1}.$$

Moreover, $d_t \leq d_{t+1}$.

Proof. The powers of one thousand form a strictly increasing, unbounded sequence. The power 729^{t+1} lies between two consecutive terms; the lower exponent is precisely the maximum in (7.7). The inequality $729^{t+1} < 729^{t+2}$ implies monotonicity of the maximum. $\square$

Proposition 7.3 (First zero advance). *The least time satisfying $d_{t+1} = d_t$ is*

$$t_* = 20.$$

In particular,

$$d_0 = 0, \quad d_{19} = 19, \quad d_{20} = 20, \quad d_{21} = 20, \quad d_{22} = 21.$$

Proof. The exact integer comparisons

$$1000^{20} \leq 729^{21}, \quad 729^{22} < 1000^{21}, \quad 1000^{21} \leq 729^{23} < 1000^{22}$$

give the values at times twenty, twenty-one, and twenty-two. For $0 \leq t \leq 20$, the expression $729^{t+1}/1000^t = 729(729/1000)^t$ is decreasing and its value at twenty is greater than or equal to one. Hence, $1000^t \leq 729^{t+1} < 1000^{t+1}$ and $d_t = t$. No time less than twenty can satisfy $d_{t+1} = d_t$; at twenty it does hold. All comparisons used involve integer powers. $\square$

The number twenty is therefore an output of the minimality criterion. The rule selecting the instant is “the first zero advance of the clock”; the algorithm and its proof can use this rule without receiving twenty as an external parameter.

7.2.2 Nonadic return and temporal antecedent

The length of the phase return determines

$$r = t_* - 9 = 11. \quad (7.8)$$

The descriptor queries are made at $t_* + 1$, $r + 1$, and r . This distribution preserves the relation between the pause instant, the nonadic translation, and the immediately associated phase reading. Times eleven, twelve, and twenty-one designate positions in the regional calendar; their temporal difference remains recorded even when two times produce the same decimal depth.

7.3 Oriented construction of the twenty-four-trit descriptor

7.3.1 Opposite half-band defects

Take the oriented defects of the regional chart

$$\delta_+ = (0, 0, 0, 1, 1, 1), \quad \delta_- = (0, 0, 0, 2, 2, 2) \in \mathbb{F}_3^6.$$

Their opposition is verified componentwise:

$$\delta_+ + \delta_- = 0 \quad \text{in } \mathbb{F}_3^6.$$

The representative 2 realizes -1 in the ternary chart. Concatenating the three-position tails in negative–positive order produces the word 222111. Adding the defects and concatenating their tails are different operations: the former has codomain $\mathbb{F}_3^6$, whereas the latter preserves the order of two segments of length three.

7.3.2 Dual charge and phase selection

In the regional chart, the six-coordinate transformation uses

$$A_{\text{reg}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

With row vectors, the dual charge of $b \in \mathbb{F}_3^6$ is the sum of the coordinates of bA_{reg} . The integer row sums of A_{reg} are (5, 7, 7, 7, 7, 7); this gives the local expression

$$\eta^\vee(b) = 2b_0 + b_1 + b_2 + b_3 + b_4 + b_5 \pmod{3}. \quad (7.9)$$

The reading phase at instant r is $\theta_r = (r - 1) \bmod 3$. For each channel, define

$$\varepsilon_\chi(r) = \begin{cases} 1, & \eta^\vee(B_\chi(r)) = \theta_r, \\ 2, & \eta^\vee(B_\chi(r)) \neq \theta_r. \end{cases}$$

The phase–charge representation is

$$D(r) = \varepsilon_\pi(r)B_\pi(r) + \varepsilon_e(r)B_e(r) + \varepsilon_\varphi(r)B_\varphi(r) \quad \text{in } \mathbb{F}_3^6. \quad (7.10)$$

The coefficient is determined by an equality of charges before evaluating the word that will result from the combination.

7.3.3 Composition of the descriptor

Let $\parallel$ denote the concatenation of fixed-width words. The complete construction is

$$\Omega_{24} = (B_e(t_* + 1) + \delta_-) \parallel (\delta_-|_{3,4,5} \parallel \delta_+|_{3,4,5}) \parallel (B_\pi(r + 1) + B_e(r + 1)) \parallel D(r). \quad (7.11)$$

Each length-six summand is evaluated in $\mathbb{F}_3^6$; the final concatenation has length $6 + 6 + 6 + 6 = 24$. This construction distinguishes the descriptor Ω_{24} from the regional stages w_{24}^χ : the two objects share a length, but their producers and their places in the composition differ.

Proposition 7.4 (Regional evaluation of the descriptor). *The regional representations and the first-zero-advance criterion determine*

$$\Omega_{24} = 020022\ 222111\ 211201\ 021101, \quad [\Omega_{24}]_3 = 66241910521.$$

Proof. The required bands, preserving the order (π, e, φ) , are

t	$B_\pi(t)$	$B_e(t)$	$B_\varphi(t)$
11	022212	110001	100021
12	012012	202222	000120
21	221022	020100	211021

and come from extraction (7.2). At time twenty-one, $020100 + 000222 = 020022$ in $\mathbb{F}_3^6$. The defect tails provide 222111. At time twelve, $012012 + 202222 = 211201$.

At time eleven, the dual charges are 0, 1, 2, respectively; the phase is $(11 - 1) \bmod 3 = 1$. The resulting coefficients are (2, 1, 2) and

$$2(022212) + (110001) + 2(100021) = 021101 \quad \text{in } \mathbb{F}_3^6.$$

Concatenating these four results gives the stated word. Its integer evaluation is obtained by Horner's method in base three, or by three successive concatenations in base 729. $\square$

The preceding table serves a verification purpose: each of its entries is obtained from the regional producer. Construction rule (7.11) makes it possible to reproduce the descriptor without introducing the final twenty-four-trit string as an input.

7.4 Functional panel and terminal repertoire

7.4.1 Nine regional positions

The functional panel retains the first three bands of each of the three channels, in the declared regional order:

$$\mathcal{P}_9 = (b_\pi(0), b_\pi(1), b_\pi(2), b_e(0), b_e(1), b_e(2), b_\varphi(0), b_\varphi(1), b_\varphi(2)). \quad (7.12)$$

Its evaluation is

$$\mathcal{P}_9 = (103, 161, 103, 523, 457, 301, 450, 398, 438).$$

The nine positions retain their regional provenance, although the value 103 occurs twice. The panel has nine ordered incidences and eight distinct values. The subsequent enumeration uses the incidences; candidate deduplication takes place at a later stage.

7.4.2 Unordered eight-block repertoire

The terminal repertoire involved in this construction is

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\} \subset \mathbb{F}_3^6. \quad (7.13)$$

Its eight elements are distinct. The integer encoding, sorted by value solely to fix a computational representation, is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

The documentary provenance of this repertoire is the terminal segmentation of the seventy-two-trit word and its position-wise recovery through incidences of the phase-charge and affine readers. The selector developed below receives the repertoire without its historical ordering and tests its $8!$ orderings.

The exact formulation of the reassembled implementation maintains an explicit interface here: the regional producers materially replace the descriptor, the panel, and the temporal rows; repertoire (7.13) retains its earlier documentary producer. This distinction identifies where each input enters. Membership of eight words in a temporal reader repertoire, selection of those eight words, and determination of their terminal order are three different statements. The terminal-selection proof establishes the third for the declared repertoire.

Documentary recovery by position can be expressed exactly. Let $\mathcal{I}$ be the phase-charge incidence register and let $\mathcal{J}$ be the register of the affine extension. Each incidence retains the position ℓ in the seventy-two-trit word and the emitted word u . For $5 \leq \ell \leq 12$, form

$$\mathcal{S}_\ell = \{u : (\ell, u) \in \mathcal{I} \cup \mathcal{J}\}.$$

The recovery procedure checks that all eight positions occur and that each $\mathcal{S}_\ell$ is a singleton. If s_ℓ is its unique element, the output delivered to the selector is

$$\mathcal{S}_{\text{term}} = \{s_5, s_6, \dots, s_{12}\}.$$

The singleton proof for each position preserves the coherence of the historical witnesses; the subsequent export retains the eight words and removes only the historical ordering. The selector reconstructs the order through complete enumeration and the terminal conditions, instead of receiving it from the provenance register.

7.4.3 Temporal incidences of the terminal repertoire and recovery by position

The representation of the terminal repertoire uses the same regional blocks, the same carries, and the same calendar that produce the initial twenty-four-trit descriptor. Two operations in this composition should be made explicit separately. The first recovers eight words from recorded temporal incidences. The second presents those eight words to the dodecaphasic selector as an unordered set. The selector re-establishes the order through its incidence, cylinder, and closure conditions. The distinction preserves the provenance witnesses without turning the order of a historical segmentation into an input to the selector.

Let $B_t = (b_{t,0}, b_{t,1}, b_{t,2}) \in (\mathbb{F}_3^6)^3$ be the regional boundary already generated at time t , with $0 \leq t < 100$. In the chart used by the historical register, the incidence matrix is

$$A_{\text{hist}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

The words are regarded as row vectors. Their two charges and the chronological phase are

$$r_{t,i} = \sum_{j=1}^6 b_{t,i,j}, \quad d_{t,i} = \sum_{j=1}^6 (b_{t,i} A_{\text{hist}})_j, \quad \vartheta_t = t - 1 \pmod{3}.$$

All operations in these three expressions take place in $\mathbb{F}_3$. The historical subscript identifies a specific chart of the Witt incidence; it keeps the coordinate transformation explicit when another presentation of that same incidence is used.

For $c \in \{r_t, d_t\}$, $\delta \in \mathbb{F}_3$ and $\epsilon \in \{1, -1\}$, define

$$\eta_i(c, \delta, \epsilon) = \begin{cases} \epsilon, & c_i = \vartheta_t + \delta, \\ -\epsilon, & c_i \neq \vartheta_t + \delta, \end{cases} \quad L_{t,c,\delta,\epsilon} = \sum_{i=0}^2 \eta_i(c, \delta, \epsilon) b_{t,i}.$$

The orientation -1 is written as 2 when printing the ternary coefficients. Alongside these twelve comparative readers per boundary, the register has the affine reader

$$L_t^{\text{af}} = \sum_{i=0}^2 (d_{t,i} + \vartheta_t) b_{t,i}.$$

Adding the phase in the affine reader and comparing charges in the preceding reader are different operations, both acting on data already produced by the TPK boundary.

t	$b_{t,0}$	$b_{t,1}$	$b_{t,2}$	r_t	d_t
8	200121	212212	110110	011	202
11	022212	110001	100021	001	012
30	210112	110202	211110	100	012
34	121022	000011	112110	220	021
37	120121	011202	112221	100	201
49	110212	200212	100122	110	201
58	111121	022121	122201	122	220
67	010012	001220	011120	122	122
75	220000	020111	100002	120	021
81	122201	020202	201122	202	001
90	022112	022110	121010	202	200

Table 1: Temporal boundaries realizing the printed incidences of the terminal repertoire. Each six-digit entry preserves the order of the six coordinates of the regional boundary.

The following table displays all comparative incidences of the register on the twelve blocks of the segmentation under consideration. The letters r and d denote the visible charge and the dual charge, respectively. Column j retains the position index of the historical witness; the operation delivering the data to the selector forgets this index after forming the set of words.

The tenth block has an affine witness. At $t = 30$, $d_{30} = (0, 1, 2)$ and $\vartheta_{30} = 2$, whence

$$(d_{30,0} + \vartheta_{30}, d_{30,1} + \vartheta_{30}, d_{30,2} + \vartheta_{30}) = (2, 0, 1)$$

and, using the bands in Table 1,

$$L_{30}^{\text{af}} = 2(210112) + 0(110202) + (211110) = (001001) \quad \text{in } \mathbb{F}_3^6.$$

The equality holds coordinatewise: before reduction modulo three, the sum vector is $(6, 3, 1, 3, 3, 4)$.

Proposition 7.5 (Complete recovery of the repertoire through incidences). *Consider the comparative-incidence register in Table 2, augmented by the affine witness $(j, t, L) = (10, 30, 001001)$. For each position $5 \leq j \leq 12$, the set of output words of its incidences is a singleton. Extracting these eight words and subsequently forgetting their order produces exactly*

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\}.$$

j	t	charge	δ	ϵ	$(\eta_0\eta_1\eta_2)$	$L_{t,c,\delta,\epsilon}$
4	11	d	0	1	212	021101
5	67	d	1	-1	211	002001
5	67	d	2	1	211	002001
5	67	r	1	-1	211	002001
5	67	r	2	1	211	002001
6	81	d	0	1	222	020111
6	81	r	2	1	222	020111
7	34	r	1	-1	111	200110
7	75	d	1	-1	211	200110
7	75	r	2	-1	211	200110
8	90	r	0	1	121	121012
8	90	r	1	-1	121	121012
9	49	d	0	-1	121	010122
11	11	d	2	-1	211	221110
11	37	d	0	-1	121	221110
12	8	d	0	-1	111	222110
12	8	r	1	-1	111	222110
12	58	d	1	-1	111	222110
12	58	r	0	-1	111	222110

Table 2: Complete phase-charge incidences on the terminal segmentation within the hundred boundaries of the register. Word matches between rows retain their times, orientations, and charge types.

Their encoding as base-three integers is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

Proof. The incidence multiplicities at positions $j = 5, 6, 7, 8, 9, 10, 11, 12$ are, respectively,

$$(4, 2, 3, 2, 1, 1, 2, 4).$$

At each position, all rows print the same word. Each row is verified by substituting its three bands and three coefficients into the formula for L ; the tenth case has been calculated explicitly through L_{30}^{af} . For example, at $t = 67$ and with coefficients $(2, 1, 1)$, one obtains

$$2(010012) + (001220) + (011120) = (002001) \pmod{3}.$$

The eight results are distinct as words of length six. Their encoding $\sum_{k=1}^6 w_k 3^{6-k}$ gives, in position order, $(55, 175, 498, 437, 98, 28, 714, 687)$. Forgetting the order yields the stated set of integers. Thus, extraction preserves each complete word, including its zero positions, and delivers eight distinct elements to the selector. $\square$

The preceding proof is an exact recovery from identified incidences. Its input comprises the incidence register with its positions and witnesses. The subsequent orbital selection receives only $\mathcal{S}_{\text{term}}$ and the initial regional descriptor; it traverses the $8!$ orderings and determines their compatibility with the functional boundaries and the oriented incidences of the dodecaphasic wheel. Both operations are thus presented contiguously, with explicit domains: temporal provenance of the eight words and constructive determination of their order.

The comparative table contains the nineteen incidences that its formula produces on the twelve-block segmentation over the hundred declared times. This finite exhaustiveness is checked by traversing $100 \cdot 2 \cdot 3 \cdot 2 = 1200$ evaluations. The affine reader extends coverage of the terminal positions from seven to eight. This count characterizes the local reader repertoire just defined; the other TPK operations retain the domains and constructive functions established in their respective derivations.

7.5 Comparative readers, affine reader, and temporal incidence

7.5.1 Witt transformation of the selector

The selector uses the six-coordinate chart

$$A_{\text{ter}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}.$$

Define $W(b) = bA_{\text{ter}}$ in $\mathbb{F}_3^6$, together with the charges

$$\eta(b) = \sum_{j=0}^5 b_j \pmod{3}, \quad \eta_{\text{ter}}^{\vee}(b) = \eta(W(b)).$$

The descriptor chart and the selector chart are related by the permutation $\sigma = (0, 1, 2, 4, 5, 3)$, written as a list of images:

$$(A_{\text{ter}})_{ij} = (A_{\text{reg}})_{\sigma(i), \sigma(j)}. \quad (7.14)$$

Lemma 7.6 (Compatibility of the charge functional). *For every band b ,*

$$\eta_{\text{ter}}^{\vee}(b) = \eta^{\vee}(b).$$

Proof. The integer sum of each corresponding row of the two matrices agrees: it is five in the first row and seven in the other five. By distributivity,

$$\sum_j \sum_i b_i (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{reg}})_{ij}.$$

Reduction modulo three gives the equality of charges. Matrix identity (7.14) is verified by substituting its thirty-six entries and also preserves the coordinate change. $\square$

Equality of the total charge functional allows its representations to be compared. The vector transformations remain identified by their matrices and the explicit permutation; the preceding proof preserves that coordinate information.

7.5.2 Twelve comparative readers per instant

Define the temporal phase

$$\theta(t) = (t + 2) \pmod{3}.$$

This expression realizes $(t - 1) \pmod{3}$ also at $t = 0$, where the phase equals two. For each charge type $\nu \in \{\eta, \eta_{\text{ter}}^{\vee}\}$, offset $o \in \{0, 1, 2\}$ and orientation $s \in \{1, 2\}$, take

$$\epsilon_{\chi}^{\nu, o, s}(t) = s \begin{cases} 1, & \nu(B_{\chi}(t)) = \theta(t) + o \pmod{3}, \\ 2, & \nu(B_{\chi}(t)) \neq \theta(t) + o \pmod{3}. \end{cases}$$

The corresponding comparative word is

$$C_{\nu, o, s}(t) = \sum_{\chi \in \{\pi, e, \varphi\}} \epsilon_{\chi}^{\nu, o, s}(t) B_{\chi}(t) \quad \text{in } \mathbb{F}_3^6. \quad (7.15)$$

The parameters generate twelve labelled representations per row. Two labels may produce the same word; the temporal witnesses and labels remain available in the complete calendar.

7.5.3 Affine phase–charge reader

The affine reader uses coefficients

$$a_{\chi}^{\text{af}}(t) = \eta_{\text{ter}}^{\vee}(B_{\chi}(t)) + \theta(t) \pmod{3},$$

and produces

$$F(t) = \sum_{\chi \in \{\pi, e, \varphi\}} a_{\chi}^{\text{af}}(t) B_{\chi}(t) \text{ in } \mathbb{F}_3^6. \quad (7.16)$$

The comparative rule distinguishes agreement and difference of charges; the affine rule adds charge and phase before combining the bands. This operational difference is what heterotypic selection uses.

7.5.4 Word–residue incidence

Let R be the cyclic rotation of the six positions of a band. In the row at time t , form the residual repertoire

$$\mathcal{R}(t) = \{R^j v : v \in \{q(t), a(t), c(t), \bar{w}(t)\}, 0 \leq j < 6\}.$$

The temporal incidence relations are

$$\mathcal{A}_{\text{cmp}} = \{(C_{\nu, o, s}(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t), \nu, o, s \text{ admissible}\}, \quad (7.17)$$

$$\mathcal{A}_{\text{af}} = \{(F(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t)\}. \quad (7.18)$$

Each pair relates a read word to a residue of the same row. Time is quantified jointly across both coordinates of the pair. In the calendar used, times $0, \dots, 99$ are distinct; identifying the same row and identifying the same time are therefore equivalent operations.

Proposition 7.7 (Existential compression of the temporal repertoire). *The function*

$$M(v, u) = \mathbf{1}_{(v, u) \in \mathcal{A}_{\text{cmp}}} + 2\mathbf{1}_{(v, u) \in \mathcal{A}_{\text{af}}} \in \{0, 1, 2, 3\}$$

preserves exactly the existence of each reading type for the pair (v, u) .

Proof. The first bit equals one exactly when there exist a time, a residual field, a rotation, and a comparative label that produce the pair. The second bit expresses the corresponding condition for the affine reader. Union by bitwise disjunction preserves each existence condition as all rows are traversed. A pair may admit both types, in which case the mask equals three. $\square$

The function M is an existence reader. The witness times, their multiplicities, and the individual labels remain in (7.18) and in the calendar that produces them. The finite implementation uses M to decide heterotypy, whereas the genealogical archive retains the richer provenance data.

7.6 Enumeration of cylinders and decimal candidates

7.6.1 Concatenation of seventy-two and seventy-eight trits

For an ordering $\tau = (s_1, \dots, s_8)$ of $\mathcal{S}_{\text{term}}$, define

$$P_{72}(\tau) = [\Omega_{24} \| s_1 \| \cdots \| s_8 \|]_3.$$

Recursively, start from $p_0 = [\Omega_{24}]_3$ and compute $p_i = 729p_{i-1} + [s_i]_3$ for $1 \leq i \leq 8$. This recurrence preserves the leading zeros of each block through explicit multiplication by 729. For each position b of the functional panel,

$$P_{78}(\tau, b) = 729P_{72}(\tau) + b.$$

The lengths are $24 + 8 \cdot 6 = 72$ and $72 + 6 = 78$, respectively. The corresponding ternary cylinder is

$$I_{\tau, b} = \left[\frac{P_{78}(\tau, b)}{3^{78}}, \frac{P_{78}(\tau, b) + 1}{3^{78}} \right). \quad (7.19)$$

7.6.2 Exact intersection with the twelve-triad grid

The decimal reading uses twelve triads, that is, the grid $N/10^{36}$. Write $D = 10^{36}$ and $Q = 3^{78}$ for brevity.

Lemma 7.8 (Integer endpoints of the cylinder). *For an integer prefix p of length seventy-eight, the numerators of the grid points belonging to its cylinder are*

$$\left\{ N \in \mathbb{Z} : \left\lceil \frac{pD}{Q} \right\rceil \leq N < \left\lceil \frac{(p+1)D}{Q} \right\rceil \right\}. \quad (7.20)$$

Each cylinder contains at most one point of this grid.

Proof. The condition $N/D \in [p/Q, (p+1)/Q]$ is equivalent to $pD/Q \leq N < (p+1)D/Q$. For integer N , the first inequality is equivalent to $\lceil pD/Q \rceil \leq N$; the second is equivalent to $N < \lceil (p+1)D/Q \rceil$. The interval length in coordinate N is $D/Q < 1$, since $10^{36} < 3^{78}$. It can therefore contain at most one integer. $\square$

Ceilings are evaluated by integer division: $\lceil a/Q \rceil = \lfloor (a+Q-1)/Q \rfloor$ for $a \geq 0$. The entire enumeration uses exact integers; the choice of boundary points is fixed by the half-open right endpoint.

First retain the indexed list of incidences

$$\mathcal{C}_{\text{ind}} = \{(\tau, j, N) : N/D \in I_{\tau, \mathcal{P}_9(j)}\}, \quad 1 \leq j \leq 9,$$

and then obtain the set of numerators

$$\mathcal{C} = \{N : \exists \tau, j (\tau, j, N) \in \mathcal{C}_{\text{ind}}\}.$$

Passing to $\mathcal{C}$ removes repetitions of a numerator, retained in $\mathcal{C}_{\text{ind}}$ together with their provenance.

Lemma 7.9 (Recovery of the ordering from the cylinder point). *If $(\tau, j, N) \in \mathcal{C}_{\text{ind}}$, block number i of the seventy-eight-trit word is recovered by*

$$\left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13.$$

In particular, blocks five through twelve recover τ .

Proof. Membership in the cylinder means $p \leq QN/D < p+1$, where $p = P_8(\tau, \mathcal{P}_9(j))$. Hence, $\lfloor QN/D \rfloor = p$. Extraction of the earlier blocks agrees with successive division of the prefix p by powers of 729: truncating a positional expansion before extracting a block preserves that block. The displayed formula performs exactly this extraction. The first four blocks form Ω_{24} and the next eight are the ordering τ . $\square$

Proposition 7.10 (Exact candidate count). *For Ω_{24} , $\mathcal{S}_{\text{term}}$ and $\mathcal{P}_9$ defined above,*

$$\# \text{Ord}(\mathcal{S}_{\text{term}}) = 40320, \quad \# \mathcal{C}_{\text{ind}} = 21902, \quad \# \mathcal{C} = 19446.$$

Proof. The eight blocks are distinct, so their orderings are the $8! = 40320$ permutations. For each permutation and each of the nine panel positions, compute $p = P_8(\tau, b)$, apply endpoints (7.20), and add the unique integer when the integer interval is nonempty. The sum of the cardinalities of these intervals is 21902. The union of their integers contains 19446 distinct elements. The exhaustive evaluation of these operations is certified by the theorems `indexed_cylinder_count` and `candidate_count` of the finite selector. The algorithm performs exactly the enumeration described: permutations, nine panel incidences, bounds by integer division, and duplicate removal. $\square$

The numbers 21902 and 19446 count different objects. The former counts indexed incidences containing a decimal point; the latter counts distinct numerators. The total number of cylinder queries is $40320 \cdot 9 = 362880$, including cylinders whose intersection with the grid is empty.

7.7 Exceptional incidence and heterotypy condition

7.7.1 Exceptional face produced by the panel

A band is lifted to twelve coordinates by

$$\Gamma(b) = (b, W(b)) \in \mathbb{F}_3^{12}.$$

The support of a word is the set of positions of its nonzero coordinates, with human-readable indices $1, \dots, 12$. The first regional bands of the panel produce

$$\mathcal{B} = \text{supp } \Gamma(B_\pi(0)) \cap \text{supp } \Gamma(B_e(0)) = \{4, 6, 9, 10\}. \quad (7.21)$$

The operation uses the regional words 103 and 523 and the selector's Witt matrix. The face is computed before querying the candidate coordinates.

For $N \in \mathcal{C}$, its twelve triads are

$$K_i(N) = \left\lfloor \frac{N}{1000^{12-i}} \right\rfloor \bmod 1000, \quad 1 \leq i \leq 12. \quad (7.22)$$

The crown support is

$$\mathcal{H}(N) = \{i : K_i(N) \geq 729\}, \quad \mathcal{C}_{\mathcal{B}} = \{N \in \mathcal{C} : \mathcal{H}(N) = \mathcal{B}\}.$$

The boundary $729 = 3^6$ relates the six-trit band to the decimal triad. Exhaustive evaluation of the equality of supports gives

$$\#\mathcal{C}_{\mathcal{B}} = 79. \quad (7.23)$$

The support preserves positions, while the complete register also preserves the twelve magnitudes. The next stage uses this positional information together with relations between magnitudes.

7.7.2 Ternary words and oriented differences of the candidate

The i th ternary band of the point N/D is obtained by

$$T_i(N) = \left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13. \quad (7.24)$$

The oriented edge $a \rightarrow b$ has residue

$$R_{a,b}(N) = (K_a(N) - K_b(N)) \bmod 729. \quad (7.25)$$

Subtraction is performed in $\mathbb{Z}$ before reduction. For this edge, the pair

$$(\text{enc}_6^{-1}(T_b(N)), \text{enc}_6^{-1}(R_{a,b}(N))).$$

is queried in relations (7.18). It is therefore a joint query of the destination word and the oriented difference of the triads.

Write $\text{Cmp}(a, b; N)$ when the pair belongs to $\mathcal{A}_{\text{cmp}}$ and $\text{Af}(a, b; N)$ when it belongs to $\mathcal{A}_{\text{af}}$.

7.7.3 Determination of the transverse orbit

The twenty-four-trit descriptor determines the first three decimal triads before the terminal filtrations. Indeed, its cylinder satisfies the exact inclusion

$$\left[\frac{66241910521}{3^{24}}, \frac{66241910522}{3^{24}} \right) \subseteq \left[\frac{234543140}{10^9}, \frac{234543141}{10^9} \right).$$

The two endpoint inequalities are verified by integer cross-products. Thus, every candidate shares the three reference positions 1, 2, 3 and their representations (234, 543, 140).

The step-four translation on $\mathbb{Z}/12\mathbb{Z}$ has the four orbits

$$(1, 5, 9), \quad (2, 6, 10), \quad (3, 7, 11), \quad (4, 8, 12).$$

The first three each contain one of the reference positions; the fourth is the only one disjoint from $\{1, 2, 3\}$. The oriented forest of steps three and four is fixed by the edges

$$\begin{aligned} 1 &\rightarrow 4, & 1 &\rightarrow 5, & 2 &\rightarrow 6, \\ 3 &\rightarrow 7, & 4 &\rightarrow 8, & 5 &\rightarrow 9, \\ 6 &\rightarrow 10, & 7 &\rightarrow 11, & 8 &\rightarrow 12. \end{aligned}$$

Its first edge has step three and the remaining edges have step four. The two internal edges of the distinguished orbit are precisely $4 \rightarrow 8$ and $8 \rightarrow 12$. The location of the reading condition is thus fixed by the geometry of the positions and by the initial prefix, before examining the candidates of the exceptional face.

7.7.4 Terminal heterotypy condition

The terminal condition is

$$\begin{aligned} \mathcal{J}(N) = & [\text{Cmp}(4, 8; N) \wedge \text{Af}(8, 12; N)] \\ & \vee [\text{Af}(4, 8; N) \wedge \text{Cmp}(8, 12; N)]. \end{aligned} \tag{7.26}$$

Both assignments of types to the two edges are admitted. The selection requires the existence of a realization with distinct types; the individual labels and their times remain available in the incidence witnesses.

Theorem 7.11 (Terminal selection of the coupling register). *On the candidate domain produced by Ω_{24} , $\mathcal{S}_{\text{term}}$, $\mathcal{P}_9$ and the hundred-row regional calendar, there exists a unique numerator simultaneously satisfying exceptional incidence and heterotypy:*

$$\{N \in \mathcal{C}_{\mathcal{B}} : \mathcal{J}(N)\} = \{234543140729659824621058914794146601\}.$$

Its twelve-triad register is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \tag{7.27}$$

Proof. The preceding count produces the 79 elements of $\mathcal{C}_{\mathcal{B}}$. For each one, its twelve coordinates are extracted by (7.22); the destination bands T_8, T_{12} and residues $R_{4,8}, R_{8,12}$ are computed; the two masks of the temporal repertoire are queried; and disjunction (7.26) is applied. The resulting list contains exactly the stated integer, according to the exhaustive evaluation certified by `terminal_selection`. Equality of the list with a singleton implies existence and uniqueness in the declared domain. Base-one-thousand extraction from that integer gives (7.27).

The selection procedure receives the enumerated producers and rules. The final integer appears in the statement of the verified equality after selection has been executed. This logical position allows the result to be checked without using it as a selection criterion. $\square$

7.7.5 Arithmetic witness of membership in a cylinder

One of the incidences of the selected numerator uses the ordering

$$(002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110)$$

and the boundary $438 = 121020_3$. For this incidence,

$$P_{72} = 5283881584882673136514102926090957,$$

$$P_{78} = 3851949675379468716518781033120308091.$$

The integer-endpoint formula yields

$$\left\lceil \frac{P_{78} 10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146601,$$

$$\left\lceil \frac{(P_{78} + 1) 10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146602.$$

The unique integer in the interval is therefore N_K . The thirteen ternary bands of its decimal point are

020022	222111	211201	021101
002001	020111	200110	121012
010122	001001	221110	222110
121020			

This calculation exhibits a concrete witness of membership in the enumerated domain. Uniqueness of the register follows from the exhaustive examination of all candidates, in addition to this particular witness.

7.8 Regional composition and continuity of the realizations

7.8.1 Composition of the regional generators

Let $\text{Sel}(\Omega, \mathcal{S}, \mathcal{P}, \mathcal{N})$ be the selector defined by the enumeration and the two preceding filters. The regional producers satisfy the equalities

$$\Omega_{\text{reg}} = \Omega_{24}, \quad \mathcal{P}_{\text{reg}} = \mathcal{P}_9, \quad \mathcal{N}_{\text{reg}} = \mathcal{N}_{100},$$

where $\mathcal{N}_{\text{reg}}$ is computed by band extraction and signature reading. By congruence of functions,

$$\begin{aligned} \text{Sel}(\Omega_{\text{reg}}, \mathcal{S}_{\text{term}}, \mathcal{P}_{\text{reg}}, \mathcal{N}_{\text{reg}}) \\ = \text{Sel}(\Omega_{24}, \mathcal{S}_{\text{term}}, \mathcal{P}_9, \mathcal{N}_{100}) = \{N_K\}. \end{aligned} \tag{7.28}$$

The equality replaces three documentary interfaces with their actual regional construction. The terminal repertoire retains the provenance specified in (7.13).

The register is then formed by the twelve Euclidean divisions in (7.22). Each coordinate belongs to $\{0, \dots, 999\}$ and the length is twelve by construction. The implementation that takes the first element of the selected list is justified by equality with the singleton: the default value of a computational operation on empty lists never enters this result.

7.8.2 Relation to signed incidences and inversion

The present construction provides a prospective selection in an explicit finite domain. The subsequent development of the dodecaphasic register produces its oriented channels from the enriched terminal state and proves an integral inversion through Hadamard. That inversion recovers a register from compatible channels; here a register is selected from its regional and incidence-based inputs. Equality of their results allows the two realizations to be composed while preserving the origin of each operation.

Uniqueness of selection and uniqueness of inversion are distinct properties. The former quantifies over the candidates constructed in this chapter. The latter quantifies over the registers sharing a compatible signature. Their conjunction justifies transporting the selected register to the signed realization and recovering it from that realization, without turning a retrospective recovery into the producer of the register.

7.8.3 Relation between the terminal counts

The sequence

$$40320 \longrightarrow 21902 \longrightarrow 19446 \longrightarrow 79 \longrightarrow 1$$

records, successively, orderings, indexed incidences with the grid, distinct numerators, candidates of the exceptional face, and heterotypic survivors. The arrow notation indicates the order of the operations; the second number counts a different class of objects from the first.

The regional-catalogue count $196 \cdot 197 \cdot 196 \longrightarrow 331 \longrightarrow 2 \longrightarrow 1$, developed in its corresponding place, starts from another population: regional blocks from three catalogues and terminal conditions of Gram, occupancy, distance, and orientation. Both counts retain their domains. Their convergence to the same representation is expressed through the identification of their readers and outputs; the intermediate numbers describe different operations and remain documented separately.

7.8.4 Regional composition of the fine-structure constant

The output K subsequently enters the ordered regional composition and the base-one-thousand carry. Its later use preserves the twelve positions and the reading boundary. Coupling to the analytic root and representations at arbitrary depth uses that same output, together with the regional coordinates already generated and the rules of the analytic chart. This dependency prepares the chapter devoted to the two correlated realizations of the fine-structure constant.

The scope of this chapter is precise: production of the descriptor and its regional inputs, determination of the finite domain, unique selection on that domain, and material composition with the producers. The analytic recurrence at all orders and the compatibility of infinite carries are developed subsequently with their own proofs.

7.9 Executable certification and traceability of the operations

The formal development preserves the correspondence between the mathematical definitions and their executable realizations. The signature and clock proofs are found in `N69RegionalSignature.lean`; band extraction and reproduction of the hundred rows, in `GeneratedN69Rows.lean`; the minimality criterion and construction of Ω_{24} , in `RegionalW24.lean`. Compatibility of the two Witt matrices is assembled in `WittReaderCompatibility.lean`.

The temporal readers, rotations, oriented residues, and heterotypic condition are formalized in the module

`TerminalReaders.lean`.

Enumeration and counts are formalized in the module

`TerminalOrbitalSelection.lean`.

The regional panel is linked through `RegionalPanelBridge.lean`, and composition (7.28) is expressed in `SelectedFromRegionalInputs.lean`.

The finite enumeration checks use Lean’s native evaluation; their reports record `Lean.ofReduce Bool`, in addition to the library’s customary logical axioms. The regional reconstruction of the descriptor uses kernel proofs and ordinary reduction. This distinction describes the verification mechanisms actually employed. The compilation receipts attest to their declared modules and dependencies; the mathematical proofs retain the domains, producers, and hypotheses made explicit in the body of the chapter.

7.10 Complete census of cylinders and selection of the terminal boundary

The terminal readout jointly uses two kinds of information: compatibility of the regional cylinders and an incidence profile with labelled channels and positions. This section constructs the complete catalogues, enumerates their triples, and proves the effect of each reader condition. The starting point consists of the regional outputs already produced in Section 4, not values introduced to define the APP-TRIT-TPK transport. Reduction to a matrix belongs to a subsequent readout of these outputs and preserves the order of the three channels.

The result is a census theorem with an explicit starting structure. The profile of Gram products, weights, distances, and marginals is declared as a datum of the terminal reader. The exhaustiveness of the enumeration proves what this profile selects on the specified catalogues; it does not, by itself, prove that the profile is a universal law or that it can be reconstructed from the unique triple that ultimately satisfies it. This distinction allows the complete finite proof to be incorporated without reversing its genealogy.

7.10.1 From regional outputs to ternary catalogues

Let $x_\pi, x_e, x_\varphi \in [0, 1)$ be the fractional parts of the three regional coordinates already constructed. We use their expressed cylinders of one thousand decimal places,

$$J_\chi^{(1000)} = \left[\frac{d_\chi}{10^{1000}}, \frac{d_\chi + 1}{10^{1000}} \right), \quad \chi \in \{\pi, e, \varphi\}.$$

The integers d_χ are finite output readouts. A length of one thousand is sufficient for the integers that follow, as verified from the two bounds of each interval; no infinite depth is assumed. We define

$$p = 30, \quad r = 1050, \quad L = p + r = 1080, \quad k = 171, \quad D = 3^L, \quad Q = 1000^k, \quad (7.29)$$

and

$$P_\chi = \lfloor 3^{30} x_\chi \rfloor, \quad T_\chi = \lfloor Q x_\chi \rfloor.$$

Here k counts decimal triads and does not denote the dodecaphasic register K . We have $1000^{171} \leq 3^{1080} < 1000^{172}$.

Lemma 7.12 (Stable extraction of an integer). *For $d, m, h \in \mathbb{Z}$, with $m, h > 0$, the value of $\lfloor mx \rfloor$ is constant on $[d/h, (d+1)/h)$ if and only if*

$$\left\lfloor \frac{md}{h} \right\rfloor = \left\lfloor \frac{m(d+1)-1}{h} \right\rfloor. \quad (7.30)$$

When this holds, either side gives that value.

Proof. The minimum of mx is attained at the left endpoint. Its supremum is $m(d+1)/h$ and is not attained. The largest integer value that $\lfloor mx \rfloor$ can take is $\lceil m(d+1)/h \rceil - 1$, which equals the second side because the numerator and denominator are integers. Equality of the endpoints of the range of values proves the assertion. $\square$

The stability test holds in all three channels for $m = 3^{30}, Q, 3^{1080}$. The last choice will be used solely for a subsequent comparison, not to select the profile. The first thirty recovered positions are precisely the five six-trit stages of the regional transport:

Channel	Thirty-trit prefix	P_χ
π	010211012222010211002111110221	29152671743887
e	201101121221102011012222102011	147887858824447
φ	121200112202121020010210010200	127247717616687

The agreement preserves the prolongation already constructed; the decimal readout is not used to replace its transport matrices retrospectively.

For a fixed channel, a tail of r trits is represented by $0 \leq R < 3^r$. Its cylinder and the cylinder of the first k triads are

$$I_{\chi,R} = \left[\frac{P_\chi 3^r + R}{D}, \frac{P_\chi 3^r + R + 1}{D} \right), \quad J_{\chi,T} = \left[\frac{T_\chi}{Q}, \frac{T_\chi + 1}{Q} \right). \quad (7.31)$$

The catalogue includes all tails for which these two cylinders have a nonempty intersection.

Proposition 7.13 (Catalogue by exact intersection). *The compatible tails are the integers in the interval $[a_\chi, b_\chi]$, where*

$$a_\chi = \max \left(0, \left\lfloor \frac{T_\chi D}{Q} \right\rfloor - P_\chi 3^r \right), \quad (7.32)$$

$$b_\chi = \min \left(3^r - 1, \left\lfloor \frac{(T_\chi + 1)D - 1}{Q} \right\rfloor - P_\chi 3^r \right). \quad (7.33)$$

For the specified regional cylinders, no clipping applies and we obtain

χ	$a_\chi \bmod 729$	$b_\chi \bmod 729$	$b_\chi - a_\chi + 1$
π	67	262	196
e	584	51	197
φ	117	312	196

(7.34)

Proof. Write $N = P_\chi 3^r + R$. The two half-open intervals in (7.31) intersect exactly when

$$NQ < (T_\chi + 1)D, \quad (N + 1)Q > T_\chi D.$$

The second inequality is equivalent to $N \geq \lfloor T_\chi D / Q \rfloor$; the first, to $N \leq \lfloor ((T_\chi + 1)D - 1) / Q \rfloor$. Subtracting $P_\chi 3^r$ and retaining $0 \leq R < 3^r$ proves the formulae. Evaluation by integer division of the T_χ extracted using (7.30) gives the table. Since $r \geq 6$, $P_\chi 3^r \equiv 0 \pmod{729}$, so the residue of the first endpoint can also be obtained directly from $\lfloor T_\chi D / Q \rfloor \bmod 729$. $\square$

The last six trits of a tail form the word $\nu_6(R \bmod 729)$, where ν_6 denotes the ternary representation of length six, including its leading zeros. By the table, each catalogue contains fewer than 729 consecutive elements, so this reduction is injective on it. Thus, the three boundary catalogues are completely described, without choosing the final triples:

$$\begin{aligned} \mathcal{B}_\pi &= \{\nu_6(67 + i) : 0 \leq i < 196\}, \\ \mathcal{B}_e &= \{\nu_6((584 + j) \bmod 729) : 0 \leq j < 197\}, \\ \mathcal{B}_\varphi &= \{\nu_6(117 + \ell) : 0 \leq \ell < 196\}. \end{aligned} \quad (7.35)$$

Requiring the left endpoint of $I_{\chi,R}$ to belong to $J_{\chi,T}$ would impose a different condition: it would lose the first cylinder that intersects the boundary and would give 195, 196, 195. The census uses the intersection of the complete cylinders, in accordance with its definition.

7.10.2 Declared profile and compatibility indicators

For $u, v \in \mathbb{F}_3^6$, let $g(u, v) = \sum u_i v_i \in \mathbb{F}_3$, $n(u) = g(u, u)$, $w(u) = \#\{i : u_i \neq 0\}$, and $h(u, v) = \#\{i : u_i \neq v_i\}$. The last two functions take integer values; the first two take values in $\mathbb{F}_3$. The reader's local profile, in the order (π, e, φ) , is

$$\begin{aligned} (g_{\pi e}, g_{\pi \varphi}, g_{e \varphi}) &= (2, 2, 0), & (n_\pi, n_e, n_\varphi) &= (0, 0, 0), \\ (w_\pi, w_e, w_\varphi) &= (3, 3, 3), & (h_{\pi e}, h_{\pi \varphi}, h_{e \varphi}) &= (2, 5, 3). \end{aligned} \quad (7.36)$$

The words are ordered as in (7.35). We use $[E] \in \{0, 1\}$ for the indicator of a condition E . For each pair of channels χ, ψ , the rectangular matrix $A_{\chi\psi}$ has entry $[g(u, v) = g_{\chi\psi}]$. The matrix $A_{\chi\psi}^h$ multiplies that entry by $[h(u, v) = h_{\chi\psi}]$. Let D_χ^n be the diagonal matrix of $[n(u) = 0]$, and D_χ^{nw} that of $[n(u) = 0][w(u) = 3]$.

Theorem 7.14 (Exhaustive enumeration by finite sums). *The product of the three catalogues contains 7 567 952 triples. The successive conditions in (7.36) leave*

$$7\,567\,952 \longrightarrow 275\,794 \longrightarrow 6235 \longrightarrow 3127 \longrightarrow 331. \quad (7.37)$$

These four counts are, respectively,

$$\begin{aligned} N_g &= \text{tr}(A_{\pi e} A_{e\varphi} A_{\varphi\pi}), \\ N_{gn} &= \text{tr}(D_{\pi}^n A_{\pi e} D_e^n A_{e\varphi} D_{\varphi}^n A_{\varphi\pi}), \\ N_{gnw} &= \text{tr}(D_{\pi}^{nw} A_{\pi e} D_e^{nw} A_{e\varphi} D_{\varphi}^{nw} A_{\varphi\pi}), \\ N_{gnwh} &= \text{tr}(D_{\pi}^{nw} A_{\pi e}^h D_e^{nw} A_{e\varphi}^h D_{\varphi}^{nw} A_{\varphi\pi}^h). \end{aligned} \quad (7.38)$$

All products and traces in this formula are calculated over the integers, not over the field with three elements.

Proof. The size of the product is $196 \cdot 197 \cdot 196$. When a trace in (7.38) is expanded, its indices run through exactly one triple of words. The summand equals one if it satisfies all the represented conditions and zero otherwise. Each triple is counted once. The diagonal matrices add the individual conditions; the rectangular factors add the pairwise conditions. This proves that the traces are precisely the stated cardinalities.

To make their evaluation explicit, fix the indices i, j of the first two channels and form the sets of indices ℓ of the third that satisfy each pairwise condition. Their indicators are intersected, the relevant individual indicators are added, and the cardinality is summed. The final sum runs over $0 \leq i < 196, 0 \leq j < 197$. Substitution of the words in (7.35) and the profile (7.36) gives, in that order, 275794, 6235, 3127, 331. The supplement records the four contributions of each of the 196 indices i and the complete list of 331 triples; its program evaluates these sums with exact binary indicators and checks each survivor by scalar evaluation. No sample or tolerance is involved. $\square$

7.10.3 Labelled incidence and orientation charge

For a surviving triple, let $B \in \mathbb{F}_3^{3 \times 6}$ be its matrix of ordered rows. The second profile requires

$$\text{rank}_{\mathbb{F}_3} B = 3, \quad w_{\text{col}}(B) = (1, 1, 2, 2, 3, 0), \quad q_{\partial}(B) = (1, 2, 2, 1, 2, 0), \quad (7.39)$$

where w_{col} counts the nonzero elements in each column and q_{∂} sums its entries modulo three. This is a profile of labelled columns; permuting them without transporting the profile defines a different reader. Denote by $\mathcal{F}$ the set of 331 triples from the preceding theorem.

Proposition 7.15 (Reduction of the census to the terminal pair). *The sums of indicators over $\mathcal{F}$, successively imposing rank, column weights, and column sums, are*

$$\sum_{B \in \mathcal{F}} [\text{rank } B = 3] = 331, \quad N_{\text{rango,pesos}} = 18, \quad N_{\text{rango,pesos,sumas}} = 2. \quad (7.40)$$

With catalogue indices starting at one, the two final triples are (120, 197, 157) and (129, 197, 148), and their matrices are

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

Proof. For each triple in the already enumerated set, the rank is computed by elimination over $\mathbb{F}_3$, the nonzero elements of the six columns are counted, and their entries are summed modulo three. The products of the indicators in (7.39) give (7.40). Substituting the two final indices into (7.35) gives the displayed matrices. The complete list of survivors and the exact filters in the supplement allow the elimination to be reproduced without presupposing these two matrices as the domain. $\square$

The orientation $\sigma = (1, 1, -1) = (1, 1, 2) \in \mathbb{F}_3^3$ selects the line $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$. The row charges of B_0, B_1 are $(0, 2, 0)$ and $(2, 2, 1)$; only the latter belongs to this line. This recovers the last step of the selector in Section 7.15, now preceded by the construction and enumeration of its complete domain. The stable readout of $[3^{1080}x_\chi]$, performed after the filters, gives the boundaries 021020, 001220, 100210, with the same ranks (129, 197, 148). This subsequent comparison does not define the profile or enter the sums of indicators.

7.10.4 Preserved information and scope of the result

The procedure preserves the thirty-trit prefix, the compatible tail, the regional order, the position of each trit, and the orientation. Reduction modulo 729 preserves the identity of each candidate here because the interval of tails has length less than 729; this injectivity is not asserted for an arbitrary history. The census of 331 also proves that the local profile alone does not select a single matrix: the labelled marginals and the charge supply additional effective conditions.

Thus, a complete finite enumeration and its selector have been recovered on output cylinders, with an explicit readout profile. Construction of the signed register additionally uses its memory reader and its channels, as specified below; the visible matrix is not identified with that register by equality of the number of components. Nor does this count prove the universality of the profile, generate the regional coordinates anew, or decide a claim about alpha or about the complete continuum.

8 Generated coordinates, action, and electronic normalisation

The construction of the collections uses results from the preceding stages: the regional characters, the dodecaphase record, the action line, and the electronic section. We assemble here the operations connecting them and the proofs needed to compose them. Their order is causal: the regions and their records are produced on APP through TRIT and TPK; real evaluations are performed afterwards; the choice of units enters when dimensional magnitudes are realised.

Initial conditions and regional selection were developed in the section preceding the arithmetic of histories. The domain of the incidence ledger, its counts, and the reconstruction of the record are defined explicitly below. Selecting a specific collection of visits and traces is an additional operation beyond evaluating a ledger: it requires specifying its multiplicities, boundary transport, and coverage of the admitted traces. The reconstruction formulas proved here allow that record to be evaluated and checked; their invertibility does not by itself select the incidence collection.

8.1 Regional readers and compatibility of evaluation

Closure, propagation, and self-scaling are three regions of the same generation. The closure region has five emissions and multiplicity $432 + 4 \cdot 144 = 1008$. In the oriented gauge of the catalogue, its emissions are

(501, 614, 498, 169, 272, 272)	432
(810, 923, 870, 169, 272, 272)	144
(810, 983, 810, 169, 272, 272)	144
(870, 923, 810, 169, 272, 272)	144
(870, 923, 810, 418, 674, 521)	144.

The common ternary word is 010211; the representatives of propagation and self-scaling are 201101 and 121200. The census and orientation precede the evaluations that follow.

At the five positions of the closure reader, the uniform projection $P_5 = \mathbf{11}^T/5$, together with $P_5\lambda = \lambda$ and $\mathbf{1}^T\lambda = 1$, determines $\lambda = \mathbf{1}/5$. This normalisation of the reader positions differs from the seed distribution given in the table. In the elliptic regime $J^2 = -I$, multiplication gives

$$(I + aJ)(I + bJ) = (1 - ab)I + (a + b)J.$$

Whenever $1 - ab \neq 0$, the composite slope is $a \oplus b = (a + b)/(1 - ab)$. Two doublings of $1/5$ give $5/12$ and $120/119$. The positive compensator $1/q$, $q > 1$, that returns the slope to one satisfies

$$\frac{120/119 - 1/q}{1 + (120/119)/q} = 1, \quad q = 239.$$

Thus the closure character of this reader is

$$\Pi_{cl} = 4 \left(4 \arctan \frac{1}{5} - \arctan \frac{1}{239} \right). \quad (8.1)$$

The composition rule, its compensation, and the choice of angular branch fix this value; the conventional angular identity recognises it as π .

The propagation character is realised in a commutative formal collection P_t , normalised by $P_0 = 1$, $P'_0 = 1$, and $P_{t+s} = P_t P_s$. If $P_t = \sum_{n \geq 0} a_n t^n$, comparison of the coefficient of $s^n t$ gives

$$(n + 1)a_{n+1} = a_n, \quad a_0 = 1.$$

Hence $a_n = 1/n!$; representation at one unit of the parameter is $\Pi_{pr} = \sum_{n \geq 0} 1/n!$, recognised as e . The unit and the sign of the generator belong to the reader: the semigroup law alone would admit other generators.

For self-scaling, TRIT selects the modes (Σ, Π, Π) . Their three cyclic transitions are $\Sigma \rightarrow \Pi$, $\Pi \rightarrow \Pi$, and $\Pi \rightarrow \Sigma$. Their incidence matrix, in areal-volumetric order, is

$$F = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (8.2)$$

The positive eigenvector normalised as $(1, x)^\top$ satisfies $x = \lambda$, $1 + x = \lambda x$; hence $x^2 - x - 1 = 0$. Its positive root is the self-scaling coordinate φ . This equation recognises the character of the incidence already produced by the calendar.

Proposition 8.1 (Regional evaluation through compatible intervals). *The three preceding readers admit rational intervals of arbitrarily small width. Their finite decimal prefixes are determined by refinement and are compatible under truncation.*

Proof. For $0 < x < 1$, two consecutive alternating partial sums of $\sum_{j \geq 0} (-1)^j x^{2j+1}/(2j+1)$ enclose $\arctan x$. The first omitted term bounds the error. This provides rational intervals for (8.1). If $E_n = \sum_{j=0}^n 1/j!$, then, for $n \geq 1$,

$$E_n < \Pi_{\text{pr}} < E_n + \frac{n+2}{(n+1)(n+1)!}.$$

The bound follows by bounding successive ratios of the tail by $1/(n+2)$. Finally, F^n has entries F_{n-1}, F_n, F_{n+1} , where $F_0 = 0, F_1 = 1$ and $F_{n+1} = F_n + F_{n-1}$. The identity $F_{n+1}^2 - F_n F_{n+2} = (-1)^n$ shows that two consecutive ratios enclose φ with width $1/(F_n F_{n+1})$. Successive intervals may be intersected to preserve nestedness. After conventional recognition, the irrationality of π, e, φ ensures that no coordinate lies on a rational decimal boundary. At each finite depth, a sufficiently narrow interval lies in a unique decimal cell. Its index determines the prefix, and nestedness proves compatibility. $\square$

8.2 Signed dodecaphase chart of the full TPK state

The evolution producing the record is formulated on the complete state of the Topological Phase Kernel. Denote by TPKFullState the state space jointly preserving APP position, TRIT regime, route, sheet, orientation, carry, boundary, memory, incidence ledger, and terminal coordinates. In a canonical chart,

$$X = (X_{\text{APP}}, X_{\text{TRIT}}, X_{\text{ruta}}, X_{\text{mem}}, C_{\text{term}}, U_{12}^{\text{sgn}}, X_{\partial}, \dots).$$

The dynamics of one hundred and eight steps acts through the composition

$$X_{\text{term}} = (\mathcal{U}_{108} \circ \dots \circ \mathcal{U}_1)(X_{\text{seed}}), \quad \mathcal{U}_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t, \quad (8.3)$$

where Sel_t applies the tritic phase selector, Tra_t transports sheet, route, and orientation, and Upd_t incorporates the carry and memory produced at instant t . Each factor therefore acts on the full state, and the composition preserves the genealogy of the twelve nonadic windows.

The incidence ledger of the following section is the coordinate $\mathcal{R}_{12}(X_{\text{term}})$. Its evaluation by the 90/120 channels defines

$$\mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12} : \text{TPKFullState} \longrightarrow \mathbb{Z}^{25}, \quad X \longmapsto C(X) = (b^{90}(X), b^{120}(X), Q(X)).$$

The signed dodecaphase projection is

$$\mathcal{S}_{12} := \text{pr}_{U_{12}^{\text{sgn}}} : \text{TPKFullState} \longrightarrow \mathbb{Z}^{12}.$$

On the compatible sublattice produced by the dynamics, the two charts satisfy the operator identity

$$\mathcal{S}_{12} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12} = \Pi_H \circ \text{pr}_C. \quad (8.4)$$

Thus the dodecaphase signature is obtained by projection and evaluation of the same state that preserves the ledger; its domain includes the orientation and memory entering each count. The raw projection $\rho_{108} : \text{TPKFullState} \rightarrow \text{CT108RawProjection}$, by contrast, denotes the chart that forgets those coordinates after production. This separation of types fixes the causal order between generation, signed representation, and reduced reading.

8.3 From temporal incidences to the dodecaphase record

Definition 8.2 (Domain of the oriented incidence ledger). A ledger $\mathcal{L}$ contains a nonempty finite collection of routes, their identified visits, and a finite collection of traces. Each route preserves one visit per instant during the first 108 instants, and the later visits used by its traces. A visit preserves

$$v = (\gamma, t, i, j, s, \tau, w, \varepsilon_\partial),$$

where γ identifies the route, t its instant, $1 \leq i, j \leq 9$ its APP position, $s \in \{\Sigma, \Pi, 0\}$ its sheet, $\tau \in \{-1, 0, 1\}$ its tritic orientation, and w its positive integer multiplicity. The incidence chart evaluates

$$n_v = \begin{cases} i + j, & s = \Sigma, \\ ij, & s = \Pi, \\ 9, & s = 0. \end{cases} \quad r_v = 1 + ((n_v - 1) \bmod 9), \quad q_v = (n_v - r_v)/9.$$

The reading $n_v = 9$ fixes the threshold convention of this chart. For $r_v = 9$, the label ε_∂ distinguishes corona from interior; it is not obtained from the remainder alone.

A trace $\theta = (v_1, \dots, v_N)$ traverses consecutive instants of the same route. It preserves a multiplicity $u(\theta) \in \mathbb{N}_{>0}$ and a reduced length unit $\lambda(\theta) \in \mathbb{N}_{>0}$, with

$$N\lambda(\theta) = 120(1 + 3j(\theta)), \quad j(\theta) \in \mathbb{N}_0.$$

Its channel is $\sigma(\theta) = \text{sgn}(\sum_{a=1}^N \tau(v_a))$. The collection counted in the 120 channels comprises traces of nonzero sign. Reduced length and number of instants are related through λ ; they are not identified with one another.

The definition specifies the evaluation domain, including the data that must be transmitted to the reader. Its multiplicities and trace collection belong to the dynamic selection of the ledger. An enumeration of the domain or a format check does not replace that selection.

Generation preserves a ledger $\mathcal{L}$ of visits and traces. The temporal chart divides it into windows $E_m = \{9m - 8, \dots, 9m\}$, $1 \leq m \leq 12$, equivalent to positions $m = \beta + 1$ of the record $\mathcal{R}_{12}$. Each window retains its subroute, orientation, boundary increment, and local incidences. Each visit records APP position, arithmetic sheet, instant, multiplicity, and boundary orientation. Each evaluation n_v preserves the division $n_v = r_v + 9q_v$, with $1 \leq r_v \leq 9$. In window m , the incidence reader expresses

$$\begin{aligned} A_m &= \sum_{v \in \mathcal{L}_m} w(v) \mathbf{1}_{\{1,3,5,7\}}(r_v), \\ C_m &= \sum_{j \geq 0} (\nu_{120,m}^-(j) - \nu_{120,m}^+(j)), \\ V_m &= \sum_{v \in \mathcal{L}_m} w(v) \mathbf{1}_{\{9\}}(r_v) \varepsilon_\partial(v). \end{aligned} \tag{8.5}$$

Here the traces in each window have finite support, $\varepsilon_\partial = +1$ on the corona and -1 in the interior, and the labels $\pm, 120$ belong to the preserved transport. The multiplicity w and the traces ν are those obtained from that transport; they are not reconstructed by selecting a value of α .

In terms of the identified traces, the second sum has the finite expression

$$C_m = - \sum_{\theta: t(v_1) \in E_m} u(\theta) \sigma(\theta).$$

Indeed, grouping by $j(\theta)$ and by sign produces $\nu_{120,m}^-(j) - \nu_{120,m}^+(j)$. This equality specifies the operation actually applied to each record; it keeps channel, multiplicity, and length separate.

Proposition 8.3 (Updating and concatenation of the ledger). *Once the collection of visits and traces admitted by transport is fixed, its counts are updated through integer increments. When a visit v in window m is added, the increment of (A_m, C_m, V_m) is*

$$(w(v) \mathbf{1}_{\{1,3,5,7\}}(r_v), 0, w(v) \mathbf{1}_{\{9\}}(r_v) \varepsilon_\partial(v)).$$

When a trace θ whose origin belongs to E_m is completed, the increment is $(0, -u(\theta)\sigma(\theta), 0)$. For two consecutive segments, the count of their concatenation is the sum of the counts of the visits and traces contained in each segment, plus that of the traces crossing their junction.

Proof. Each sum in (8.5) is indexed by visit or trace identifiers, so adding an identifier adds exactly the summand displayed. For concatenation, visits are distributed between the two segments. Traces are partitioned into those contained in the first, those contained in the second, and those crossing the junction. Additivity of integer sums proves the formula. Preserving the identifiers and pending prolongations prevents a trace from being counted twice or one that has not yet ended from being lost. Decimal reduction is applied after counting; the quotients of that reduction remain in memory. $\square$

Decimal reduction of these three counts produces

$$z_m = 100[A_m]_{10} + 10[C_m]_{10} + [V_m]_{10}, \quad 0 \leq z_m < 1000. \quad (8.6)$$

Let $(D_3 z)_i = z_i - z_{i+3}$, $(D_4 z)_i = z_i - z_{i+4}$, with indices in $\mathbb{Z}/12\mathbb{Z}$, and $Q(z) = \sum_i z_i$. The reader preserves $\mathcal{D}(z) = (D_3 z, D_4 z, Q(z))$. Decimal reduction forgets multiples of ten in each count, whereas the original ledger preserves the visits and transports that generated them. The reading of even-code events on that same history, its decomposition $12 = 1 + 5 + 6$, and its integral inversion are developed in Proposition 12.1. That event vector does not replace the present incidence record: the domains and preserved coordinates remain distinct.

Proposition 8.4 (Incidence conjugation and decimal reduction). *Let $\varrho(m) = 13 - m$. Consider a conjugation of the ledger that maps the visits and traces of window m to those of $\varrho(m)$ through bijections preserving their multiplicities. Suppose it preserves the arithmetic class and boundary label of the visits, and interchanges the positive and negative channels of the traces. Then the counts in (8.5) satisfy*

$$(A_m^\dagger, C_m^\dagger, V_m^\dagger) = (A_{\varrho(m)}, -C_{\varrho(m)}, V_{\varrho(m)}). \quad (8.7)$$

Writing $u_m = [C_m]_{10}$, their decimal blocks satisfy

$$z_m^\dagger = z_{\varrho(m)} + 100\mathbf{1}_{\{u_{\varrho(m)} \neq 0\}} - 20u_{\varrho(m)}. \quad (8.8)$$

Proof. The bijections preserve every summand of A_m and V_m . Interchanging the channels permutes ν^- and ν^+ , thereby reversing the sign of C_m . This yields (8.7). If $C = 10q + u$, $0 \leq u < 10$, Euclidean division of $-C$ gives

$$[-C]_{10} = 10\mathbf{1}_{\{u \neq 0\}} - u, \quad \left\lfloor \frac{-C}{10} \right\rfloor = -q - \mathbf{1}_{\{u \neq 0\}}.$$

In (8.6), only the middle digit changes. Its difference is $10([-C]_{10} - [C]_{10}) = 100\mathbf{1}_{\{u \neq 0\}} - 20u$, which proves (8.8). The second identity simultaneously preserves the integer quotient of the reduction. $\square$

The proposition specifies the action on the ledger before transporting its decimal representation. Its application to route reversal also preserves the convention used to read the endpoints. Let $\gamma = (x_0, \dots, x_n)$ be an enriched route and let f be an integer observable evaluated before each advance. The reverse route $\gamma^{-1} = (x_n, \dots, x_0)$ satisfies

$$\sum_{k=0}^{n-1} f(x_{n-k}) = \sum_{k=0}^{n-1} f(x_k) + f(x_n) - f(x_0). \quad (8.9)$$

Indeed, the left-hand side sums indices $1, \dots, n$; the first term on the right sums $0, \dots, n-1$. Their difference is exactly the stated boundary term. The identity applies to each incidence observable with its preserved data. The endpoints are accounted for in the integer counts before reduction modulo ten; on a closed route the term vanishes, whereas in an open window it can change the block. Traces crossing a junction are preserved through the partition in Proposition 8.3.

This connects the counts to route conjugation in Section 10, with the specific action on each incidence kept explicit. Negating a channel and reversing a route are operations on the ledger; their decimal reduction is the representation just calculated.

Proposition 8.5 (Integral image and reconstruction of the record). *Let $b, c \in \mathbb{Z}^{12}$, $q \in \mathbb{Z}$, with indices from zero to eleven. Define*

$$w_i = c_i - b_{i+1}, \quad P_0 = 0, \quad P_i = \sum_{j=0}^{i-1} w_j, \quad T_w = \sum_{i=0}^{11} P_i.$$

There exists a unique $k \in \mathbb{Z}^{12}$ such that $D_3 k = b$, $D_4 k = c$, $Q(k) = q$, if and only if

$$b_i = w_i + w_{i+1} + w_{i+2}, \quad \sum_i w_i = 0, \quad q + T_w \equiv 0 \pmod{12}. \quad (8.10)$$

In that case,

$$k_i = \frac{q + T_w}{12} - P_i. \quad (8.11)$$

Proof. If k exists, then $c_i - b_{i+1} = k_i - k_{i+1}$. The cyclic sum of these differences is zero, their sum over length three is b_i , and $k_i = k_0 - P_i$. Summing the twelve coordinates gives $q = 12k_0 - T_w$, proving the necessary conditions and the formula. Conversely, the conditions make the formula integral and cyclic. Its differences over length three are b_i , and those over length four are $w_i + b_{i+1} = c_i$. Its sum is q . The difference between two solutions has all consecutive differences zero and sum zero; it is therefore zero. $\square$

Reconstruction is an operation on the typed image, not a retrospective selection of its arguments. In particular, $\mathcal{D}(z) = \mathcal{D}(z')$ implies $z = z'$. Since the three-digit expression of an integer in $[0, 999]$ is unique, two triples of counts express the same record if and only if they coincide componentwise modulo ten.

The signed change of representation uses the block

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad H_4^2 = 4I_4. \quad (8.12)$$

The operator H_{12} applies H_4 to the three ordered orbits $(1, 4, 7, 10)$, $(2, 5, 8, 11)$, $(3, 6, 9, 12)$. On the sublattice $\mathcal{L}_H = \{U \in \mathbb{Z}^{12} : H_{12}U \equiv 0 \pmod{4}\}$, the maps

$$U = H_{12}K, \quad K = \frac{1}{4}H_{12}U \quad (8.13)$$

are inverses. The congruence expresses exactly the integrality of the second map, and $H_{12}^2 = 4I$ proves both compositions.

The signed record is also calculated directly from the transverse coordinates before reconstructing K . With indices from zero to eleven, let, for $r = 0, 1, 2$,

$$S_r = \sum_{j=0}^3 c_{r+3j}, \quad a_0 = \frac{q + 2S_0 + S_1}{3}, \quad a_1 = a_0 - S_0, \quad a_2 = a_1 - S_1, \quad d_{r,j} = b_{r+3j}.$$

The harmonic map expresses by orbits

$$\Pi_H^{(r)}(b, c, q) = (a_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}). \quad (8.14)$$

To prove the formula on the integral image, let k be its preimage. If $s_r = \sum_j k_{r+3j}$, then $S_r = s_r - s_{r+1}$ and $q = s_0 + s_1 + s_2$, with orbit indices modulo three. Solving gives $s_r = a_r$. For the four coordinates (x_0, x_1, x_2, x_3) of an orbit, $d_j = x_j - x_{j+1}$. Their three combinations in (8.14) are respectively $x_0 + x_1 - x_2 - x_3$, $x_0 - x_1 + x_2 - x_3$, and $x_0 - x_1 - x_2 + x_3$. Together with $a_r = \sum_j x_j$, they are exactly $H_4(x_0, x_1, x_2, x_3)^\top$. The effective composition is thus $\mathcal{L} \rightarrow (b, c, q) \rightarrow \Pi_H(b, c, q) = U \rightarrow H_{12}U/4 = K$. The preimage used in this proof verifies the identity; it is not an argument of the map Π_H .

For the terminal state of (8.3), the coordinate produced before signed representation is

$$\begin{aligned} b^{90} &= (-495, -116, -684, 108, 601, -90, -173, -88, 313, 560, -397, 461), \\ b^{120} &= (-425, -281, -481, 671, -255, 30, 475, -543, 680, 251, 6, -128), \\ Q_{\text{TPK}} &= 6263. \end{aligned} \quad (8.15)$$

Substituting these twenty-five fields into (8.14), the three signed orbits are

$$\begin{aligned} U_{\text{term}}^{(1)} &= (2378, -452, -668, -322), \\ U_{\text{term}}^{(2)} &= (1406, 998, -204, -28), \\ U_{\text{term}}^{(3)} &= (2479, -551, -371, -997). \end{aligned}$$

Interleaving by columns produces

$$U_{\text{term}} = \mathcal{S}_{12}(X_{\text{term}}) = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997). \quad (8.16)$$

The orbital transform $H_{12}/4$ is defined on this sublattice by the exact divisibility of every block of four. Consequently, the chain

$$X_{\text{term}} \xrightarrow{\mathcal{R}_{12}} \mathcal{L}(X_{\text{term}}) \xrightarrow{\mathcal{C}_{108}^{90,120}} C_{\text{term}} \xrightarrow{\Pi_H} U_{\text{term}} \xrightarrow{H_{12}/4} K \quad (8.17)$$

is a typed composition from the full state to its periodic seal. Propositions 8.3 and 8.5 provide, respectively, the finite update of the ledger and the integral inverse of the last representation.

The generated record provides the ordered witness

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \quad (8.18)$$

The leading zero in 058 forms part of the block reading. To check it through Proposition 8.5, its differences and sum are

$$\begin{aligned} b &= (-495, -116, -684, 108, 601, -90, -173, -88, 313, 560, -397, 461), \\ c &= (-425, -281, -481, 671, -255, 30, 475, -543, 680, 251, 6, -128), \\ q &= 6263. \end{aligned}$$

In the preceding three orbits, the signed record is

$$\begin{aligned} U^{(1)} &= (2378, -452, -668, -322), \\ U^{(2)} &= (1406, 998, -204, -28), \\ U^{(3)} &= (2479, -551, -371, -997). \end{aligned}$$

Applying $H_4/4$ to these three vectors reproduces (8.18). These identities verify the representation of the incidences; their genesis remains that of the ledger (8.5).

8.4 Periodic representation of K and normalisation of α

Fix $B = 1000$, the phase origin, and the twelve positions. Write

$$N_K = \sum_{j=1}^{12} K_j B^{12-j}, \quad \kappa = \frac{N_K}{B^{12} - 1}.$$

This is the exact periodic evaluation of the record. Multiplication by $B^{12} - 1$ and successive Euclidean division recover its twelve blocks, including any leading zeros. For $\rho K = (K_2, \dots, K_{12}, K_1)$, the same concatenation gives $\kappa(\rho K) = B\kappa(K) - K_1$. Thus reversibility of the record preserves a specified phase origin.

Proposition 8.6 (Minimal period of the rational representation). *The periodic prolongation of the record (8.18) has minimal period 12 blocks in base 1000 and 36 digits in base 10.*

Proof. The twelve components of K are distinct. Any nonzero cyclic translation smaller than twelve would identify two components if it were a period. The minimal block period is therefore twelve.

Concatenating its three-digit blocks, including 058, produces a decimal period of length 36. This expansion is canonical: it is not eventually constant at the digit 9. Let d be its minimal decimal period. Then $d \mid 36$, because the shifts preserving a periodic word form a cyclic subgroup. Grouping the digits in threes, the minimal block period is the least $k > 0$ such that $d \mid 3k$, namely $d/\gcd(d, 3)$. No proper divisor of 36 satisfies $d/\gcd(d, 3) = 12$. It follows that $d = 36$. $\square$

This period belongs to the rational coordinate of the record, with the phase origin fixed. The following composition also preserves the regional coordinates and the normalisation quotients of its output.

Let P_j, E_j, Φ_j be the fractional blocks of the three regional characters already constructed, and extend K_j with period twelve. The composition of records is

$$Z_j = P_j + E_j - \Phi_j - K_j, \quad A_j = Z_j + c_{j+1} - Bc_j. \quad (8.19)$$

The integer quotient c_j transports the continuation towards the observed position. In a finite window, c_{N+1} is fixed first, and then

$$c_j = \left\lfloor \frac{Z_j + c_{j+1}}{B} \right\rfloor, \quad A_j = Z_j + c_{j+1} - Bc_j.$$

Euclidean division uniquely determines $0 \leq A_j < B$. Since $-1998 \leq Z_j \leq 1998$, the set $\{-2, -1, 0, 1, 2\}$ is stable for these quotients.

Theorem 8.7 (Exact composition at unlimited depth). *The complete evaluation of the normalisation, in the canonical decimal convention, is*

$$\alpha_A = \pi + e - \varphi - 4 - \kappa. \quad (8.20)$$

There is a unique canonical expansion (A_j) , with blocks between zero and $B-1$, and a bounded sequence of integer quotients satisfying (8.19) and $c_1 = 0$.

Proof. The regional integer parts are 3, 2, 1. By absolute convergence,

$$y := \sum_{j \geq 1} Z_j B^{-j} = (\pi - 3) + (e - 2) - (\varphi - 1) - \kappa.$$

The first blocks are 141, 718, 618, 234, so that $Z_1 = 7$. The absolute value of the tail sum is at most $1998 \sum_{j \geq 2} B^{-j} = 0.002$; in particular, $0 < y < 1$. Take the canonical expansion $y = \sum A_j B^{-j}$, choosing the terminating expansion rather than the eventually constant $B-1$ expansion if a boundary occurs. Set

$$c_j = \sum_{r \geq 0} (Z_{j+r} - A_{j+r}) B^{-r-1}.$$

Equality of the complete values shows that each c_j is the negative of a finite sum of integers weighted by powers of B ; it is therefore an integer, and $c_1 = 0$. Separating the first term gives $Bc_j = Z_j - A_j + c_{j+1}$. The tail bounds make the sequence bounded. More precisely, the tail of Z lies in $[-2, 2]$ and the canonical tail of A in $[0, 1)$, so $-3 < c_j \leq 2$. Being integral, c_j belongs to the stated set $\{-2, -1, 0, 1, 2\}$. Conversely, summing the recurrence through N gives

$$\sum_{j=1}^N A_j B^{-j} = \sum_{j=1}^N Z_j B^{-j} - c_1 + c_{N+1} B^{-N}.$$

Boundedness makes the final term vanish in the limit. Uniqueness of the canonical expansion of y determines A , and the tail formula then determines every c_j . $\square$

The finite evaluation $\kappa_{36} = N_K/B^{12}$ differs from κ by $B^{-12}\kappa$. Moreover, imposing $c_{13} = 0$ in a twelve-block window does not replace the quotient of the complete continuation. In the expressed record, the continuation gives $c_{13} = -1$: the twelfth block is 662, whereas the truncated window with zero terminal quotient gives 663. The complete evaluation begins

$$\alpha_A = 0.007297352569283800997285105472380662999 \dots$$

The period of K does not impose a period on (A_j) , since (8.19) also involves the three regional records and the integer quotients.

In the following formulas, α denotes this complete positional output α_A . The analytic vacancy route has its own operator; a polynomial truncated at order nine and the root of that polynomial are not identified here with the complete operator. The mass compositions that follow do not require that approximation to be promoted to an equality between the two infinite routes.

8.5 Three-dimensional construction of the electronic prefactor

Let $\mathcal{A}_3 = \{1, \dots, 9\}^3$, with uniform measure, and $\rho_9(n) = 1 + ((n - 1) \bmod 9)$. On this domain act

$$\Sigma(x) = \rho_9(x_1 + x_2 + x_3), \quad \Pi(x) = \rho_9(x_1 x_2 x_3).$$

On $\ell^2(\mathcal{A}_3)$, let P and Q be the conditional averages over the fibres of Σ and Π , respectively. Each is idempotent and self-adjoint: within each fibre it replaces a vector by its constant component and preserves that component orthogonally. They are therefore orthogonal projectors.

The additive fibres have cardinality 81. The multiplicative fibres have cardinalities

$$m = (36, 36, 108, 36, 36, 108, 36, 36, 297).$$

The joint incidence $N_{ab} = \#\{x : \Sigma(x) = a, \Pi(x) = b\}$ is $9N_0$, where

$$N_0 = \begin{pmatrix} 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 1 & 0 & 2 & 0 & 1 & 2 & 0 & 0 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 0 & 2 & 1 & 0 & 2 & 0 & 1 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 1 & 2 & 0 & 0 & 2 & 1 & 0 & 3 \end{pmatrix}. \quad (8.21)$$

This matrix is obtained by counting the 729 triples. It is also reproduced without rounding by $N_{ab} = \sum_{i,j,k=1}^9 \mathbf{1}_{\{a\}}(\rho_9(i+j+k)) \mathbf{1}_{\{b\}}(\rho_9(ijk))$. In the normalised bases of functions constant on each fibre, the intertwining has matrix $C_{ab} = N_{ab}/\sqrt{81m_b}$.

Proposition 8.8 (Norm of the noncommutativity of the two sheets). *The preceding projectors satisfy*

$$\|[P, Q]\| = \frac{\sqrt{3}}{4}. \quad (8.22)$$

Proof. On the space of the nine additive fibres, PQP has matrix $M = CC^\top$. Multiplication of the matrices in (8.21) gives eigenvalue one on the constant vector, $1/4$ on $(1, -1, 0, 1, -1, 0, 1, -1, 0)$, and $7/132$ on $(1, 1, -2, 1, 1, -2, 1, 1, -2)$. On coordinates $\{3, 6, 9\}$ with zero sum it acts by $1/18$, with multiplicity two. On each of the sets $\{1, 4, 7\}$ and $\{2, 5, 8\}$, the zero-sum vectors belong to the kernel; they give four dimensions. These orthogonal subspaces add up to the nine dimensions.

For a unit eigenvector $u \in \text{ran } P$ of PQP with eigenvalue $0 < \lambda < 1$, the vector $v = (Qu - \lambda u)/\sqrt{\lambda(1-\lambda)}$ is a unit vector orthogonal to $\text{ran } P$. In the basis (u, v) ,

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} \lambda & \sqrt{\lambda(1-\lambda)} \\ \sqrt{\lambda(1-\lambda)} & 1-\lambda \end{pmatrix}.$$

The commutator has norm $\sqrt{\lambda(1-\lambda)}$. The blocks corresponding to $\lambda = 0, 1$ commute. Among the calculated eigenvalues, the maximum is attained at $\lambda = 1/4$, since $\lambda(1-\lambda)$ is increasing on $[0, 1/2]$. Its square root is $\sqrt{3}/4$. $\square$

In the extremal block, $S = 2P - I$ and $J = 4[P, Q]/\sqrt{3}$ satisfy

$$S^2 = I, \quad J^2 = -I, \quad SJ = -JS.$$

The four matrices I, S, J, SJ are linearly independent, so they realise $\text{Cl}_{1,1} \simeq M_2(\mathbb{R})$. The elements $n_{\pm} = (S \pm J)/2$ satisfy $n_{\pm}^2 = 0$ and $n_+ n_- + n_- n_+ = I$, by direct expansion. The prefactor of the electronic equation thus arises from two evaluations of APP in dimension three and their commutator, before any reference mass.

8.6 Boundary reading and basal electronic composition

The incidence (8.2) has positive eigenvector $r = (1, \varphi)^T$. The normalisation $p_{ij} = F_{ij}r_j/(\varphi r_i)$ produces

$$P_{\text{av}} = \begin{pmatrix} 0 & 1 \\ \varphi^{-2} & \varphi^{-1} \end{pmatrix}, \quad \mu = \frac{(1, \varphi^2)}{1 + \varphi^2}.$$

The identity $\varphi^{-2} + \varphi^{-1} = 1$ proves that the rows sum to one. Since F is symmetric, weights proportional to r_i^2 satisfy detailed balance; hence $\mu P_{\text{av}} = \mu$. The incidence is irreducible and has a self-loop, so this stationary distribution is unique. The chronological frequency $(1/3, 2/3)$ counts the three instants of the calendar; μ weights its admissible trajectories with memory one.

The operator preserving the sheets, fixing the volumetric normalisation at one, and marking the areal sheet by the closure character is $D_{\pi} = \text{diag}(\pi, 1)$. By induction,

$$F^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}.$$

The ratio of marked diagonal returns is

$$\Theta_n = \pi \frac{F_{n-1}}{F_{n+1}} \longrightarrow \frac{\pi}{\varphi^2}.$$

Convergence follows from the dominant eigenvalue φ , or from the consecutive ratios already bounded. Under interchange of its arguments, the reader $\mathcal{R}(x, y) = x/y^2$ also expresses

$$\rho_A = \frac{\pi}{\varphi^2}, \quad \rho_E = \frac{\varphi}{\pi^2}, \quad \varphi^3 \rho_A^2 \rho_E = 1. \quad (8.23)$$

The last equality is checked by multiplying the fractions. It relates two orientations of the same reader and does not select the remaining coefficients of the electronic equation.

The typed central return has length 27, with support $\mathbb{Z}/9\mathbb{Z} \times \mathbb{F}_3$. In the fixed regional gauge, the five microfibre positions are inserted as $(4, 0), (5, 0), (6, 0), (6, 1), (6, 2)$. The complement has $27 - 5 = 22$ positions. Its incidence with the three tritic phase states has $5 \cdot 3 = 15$ pairs. This yields the integers -22 and 15 used by the electronic reader; orientation fixes the sign of the first term. The cubic composition uses three copies of the closure coordinate, so its evaluation is α^3 . The support, regional insertion, and this composition rule are distinct operational data; an isolated count does not replace their specification.

The global defect and its nonadic return are expressed through

$$d_4 = \frac{\pi - e \log \pi}{270}, \quad \Delta_4 = d_4 \left(1 + \frac{\pi}{729}\right). \quad (8.24)$$

The denominator 270 belongs to the corona without its closure unit; $729 = 9^3$ belongs to the cubic support. The function $f(x) = x - e \log x$ attains its minimum zero at $x = e$, since $f'(x) = 1 - e/x$ changes sign there. As $\pi \neq e$, we obtain $d_4 > 0$ and $\Delta_4 > 0$.

Definition 8.9 (Basal electronic coordinate). The composition of the preceding readers is

$$\varepsilon_e^{(0)} = \frac{\sqrt{3}}{4} \left[\exp\left(\frac{\varphi}{\pi^2}\right) - 22\alpha^3 \right] (1 + 15\Delta_4). \quad (8.25)$$

The exponential acts on orientation ρ_E of (8.23); the vacancy reading corrects that component, and the final factor preserves the return of the global defect. For $0 < \alpha < 0.01$, all factors are positive: $\exp(\varphi/\pi^2) > 1$, $22\alpha^3 < 22 \cdot 10^{-6} < 1$, and $\Delta_4 > 0$. This is a dimensionless coordinate. Its presence in a mass chart is defined after construction of the scale.

8.7 Determinantal scale and the reduced Planck constant

The local block (8.12) has determinant -16 . Its Smith normal form is $\text{diag}(1, 2, 2, 4)$. Indeed, the greatest common divisor of the entries is one; the minors of order two are even, and one has absolute value two. By the adjugate identity, the minors of order three have absolute value four, and the determinant has absolute value sixteen. The determinantal divisors are 1, 2, 4, 16, whose successive ratios give the four Smith factors. Consequently,

$$G_H = \text{coker } H_4 \simeq \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}, \quad |G_H| = 16. \quad (8.26)$$

The local normed reader multiplies the same coordinate α once for each class of this cokernel and applies the self-scaling coordinate once. Its definition produces

$$\Sigma_H = \varphi \prod_{g \in G_H} \alpha = \varphi \alpha^{16}. \quad (8.27)$$

The order of the cokernel determines the exponent of this reader. The Smith normal form and the choice of normed reader serve distinct functions: the former does not turn an arbitrary functional of the block into expression (8.27).

Proposition 8.10 (Decade of action). *The output (8.27) satisfies $10^{-34} < \Sigma_H < 10^{-33}$; its decimal valuation is -34 .*

Proof. The regional and positional evaluations give the intervals $1.618 < \varphi < 1.619$ and $0.007297 < \alpha < 0.007298$. By positivity,

$$\frac{1618\,7297^{16}}{10^{99}} < \Sigma_H < \frac{1619\,7298^{16}}{10^{99}}.$$

The integer inequalities $1618\,7297^{16} > 10^{65}$ and $1619\,7298^{16} < 10^{66}$ prove the result without using an external action constant. $\square$

The decade and the mantissa are representations of different stages. Before constructing the latter, the regional reader selects the persistent return of the closure microfibre. The projection $p_{\text{ret}}(u_1, \dots, u_6) = (u_4, u_5, u_6)$, applied to the five emissions of 8.1, produces the block $b_\pi = (169, 272, 272)$ four times and $b_\pi^- = (418, 674, 521)$ once. These multiplicities count regional rows, not the seed multiplicities 432, 144.

Proposition 8.11 (Equivariant selector of the regional return). *Among rules preserving permutations of the five regions, selecting the terminal block of maximal multiplicity, equivariant under permutations of its three positions, and retaining a position fixed by the stabiliser of the block, the selected coordinate is unique and equals $\rho_\pi = 169$.*

Proof. Projection of the five rows shows that the block of maximal multiplicity is unique and is b_π . Its stabiliser in $\mathfrak{S}_3$ interchanges the two positions of value 272 and fixes only the position of value 169. Retaining that position is compatible with any simultaneous permutation of coordinates. Equivalently, the value is expressed without choosing an initial position as $\text{sing}(b_\pi) = \sum_j (b_\pi)_j - 2 \bmod a(b_\pi) = 169$. $\square$

Grading by length expresses a word of length n through the character $\chi_z([w]) = z^{|w|}$. Concatenation adds lengths, so $\chi_z([uv]) = \chi_z([u])\chi_z([v])$. Since the regional emissions belong to $\mathcal{U}_6$, the return impulse appears as $\rho_\pi z^6$. Degree six corresponds to the length of the emission, not the length of a closed walk traversing several anchors of the graph.

The prolongation weight is obtained on the determinant line of bilayer transport. For a real involution R in dimension two, with $\text{tr } R = 0$, let

$$x_A = \frac{50\pi}{9}\alpha, \quad \mathcal{T}(x_A, y) = \exp(-x_A I_2 - yR).$$

The oriented coordinate y is still formal. In an eigenbasis of R , the two entries are $e^{-x_A - y}$ and $e^{-x_A + y}$. Therefore,

$$\bigwedge^2 \mathcal{T} = \det \mathcal{T} = e^{-2x_A} = D_A.$$

Dependence on y cancels. Multiplicativity under concatenation determines the prolongation character $k \mapsto D_A^k$: any character with elementary value D_A must have that value at $k = 1 + \dots + 1$. This construction precedes evaluation of the conjugate angular coordinate.

The fifth-order regional reader uses eigenvalues 9, 5 of the central block $\begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$, the orbit of four positions, and the calendar of twelve windows. Its graded rule fixes

$$\begin{aligned} H_5 &= \frac{1}{2} \sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5, \\ D_A &= \exp\left(-\frac{100\pi\alpha}{9}\right), \\ \mathcal{C}_\pi &= 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}, \\ \eta_{\text{ret}} &= H_5 - \frac{90}{\pi}\mathcal{C}_\pi. \end{aligned} \tag{8.28}$$

In this rule, the ratios are $9/16 = 9/4^2$, $5/9$, $7/48 = (\text{tr } B_c/2)/(4 \cdot 12)$, and $1/54 = 1944/104976$. Their assignment to the degrees and signs forms part of the graded reader. The integer 169 is the representation of the selector proved in Proposition 8.11. Its regional datum precedes the evaluation of α ; the analytic reader subsequently composes it with the closure coordinate.

The tail of $\mathcal{C}_\pi$ has a renewal construction. The impulse $\rho_\pi z^6$ and the strict continuation belong to different summands. After recording the impulse once, the continuation is normalised by the unit of the determinant line; each new step multiplies it by D_A . The orientation of the regional gauge places this return in the compensating sector subtracted from the fifth-degree character. With formal parameter z , the rule

$$R(z) = D_A z^7 + D_A z R(z) \tag{8.29}$$

uniquely determines $R(z) = D_A z^7/(1 - D_A z)$. This can be proved by comparing coefficients or by inverting the formal series $1 - D_A z$. Normalisation of the first return, in addition to the recurrent ratio, is essential to this uniqueness. Since $0 < D_A < 1$ and $\alpha < 1$, evaluation at $z = \alpha$ converges absolutely.

Proposition 8.12 (Positivity and the action return ratio). *On the generated branch, $0 < \eta_{\text{ret}} < H_5$. The ratio*

$$R_{\text{act}} = \frac{H_5}{\eta_{\text{ret}}} \tag{8.30}$$

is positive, greater than one, and dimensionless.

Proof. The bounds $0.007 < \alpha < 0.008$ and $\varphi < 1.619$ give $\frac{1}{2}\sqrt{1000\alpha/\varphi} > 1$. The sum of the absolute values of the remaining negative terms of H_5 is less than 0.008001, so $H_5 > 0.99$. Since $D_A < 1$, $\pi > 3$, and $\alpha < 0.008$,

$$0 < \frac{90}{\pi}\mathcal{C}_\pi < 30 \left(169(0.008)^6 + \frac{(0.008)^7}{1 - 0.008} \right) < 10^{-6}.$$

It follows that $0 < \eta_{\text{ret}} < H_5$. The ratio of two action coordinates in the same basis is dimensionless and does not change when that basis is rescaled. $\square$

Let $\mathcal{U}_S$ be a basis of the action line. The two sections, before and after the return, are

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S, \quad \hbar_{\text{ret}} = \eta_{\text{ret}} 10^{-34} \mathcal{U}_S. \quad (8.31)$$

Their physical reading is the proposed construction of the reduced Planck constant, with the return section made explicit. The unit $\mathcal{U}_S \mapsto \text{J s}$ is a subsequent metrological realisation. The coefficient of Σ_H within its decade is not H_5 : the determinant expresses the scale, and the regional reader expresses the coordinate occupying that scale. For each section, $h = 2\pi\hbar$ expresses the action of a complete circuit.

8.8 Angular coordinates and the clock of the absolute realisation

The angular pair uses the dimensionless coordinates already generated:

$$A_{\text{deg}} = 1000\alpha, \quad C_{\text{deg}}^* = 2(\eta_{\text{ret}} + \alpha), \quad x = \frac{\pi}{180} A_{\text{deg}}, \quad y = \frac{\pi}{180} C_{\text{deg}}^*. \quad (8.32)$$

The first operation shifts the base-thousand expansion by one block. The second is the bidirectional representation of the regional coordinate: equivalently, $y = \pi(H_5 + \alpha)/90 - \mathcal{C}_\pi$. This equality follows by substituting (8.28); it introduces no additional rule for selecting the counter-angle. The chart of one circuit has 360 angular units or 2π radians. Thus, for a lifted coordinate,

$$\hbar\theta_{\text{rad}} = h \frac{\theta_{\text{deg}}}{360}.$$

The factor 10^{-34} remains on the action line and is not inserted into the angular coordinates.

The positive channels are

$$q_+ = e^{-x-y}, \quad q_- = e^{-x+y}.$$

The pair is recovered through $x = -\frac{1}{2} \log(q_+ q_-)$, $y = \frac{1}{2} \log(q_- / q_+)$. Channel interchange preserves x and reverses the sign of y . The operator with oriented projectors $P_\pm$, $T = q_+ P_+ + q_- P_-$, preserves those labels when publishing its two eigenvalues.

For an internal time unit $t_0 > 0$, the closure of 108 steps determines

$$\omega_0 = \frac{2\pi}{108t_0}, \quad E_{\text{clk}} = \hbar_{\text{ret}} \omega_0, \quad m_{\text{clk}} = \frac{E_{\text{clk}}}{c_{\text{int}}^2}. \quad (8.33)$$

These formulas produce dimensions of frequency, energy, and mass. Velocity and length are obtained through the response of the same channels, as specified by the following identities. Expressing them in seconds, joules, or mass units requires the corresponding charts; those charts do not alter the channels or the electronic section normalised relative to the clock.

8.8.1 Identity between constitutive velocity and clock velocity

In the domain $x > |y|$, the two channels satisfy $0 < q_\pm < 1$. On the two-sheet realisation, with rank-one projectors P_+, P_- , the temporal response is defined by

$$\mathcal{V} = (I - T^{90})(I - T^{120})^{-1} = r_+ P_+ + r_- P_-, \quad r_\pm = \frac{1 - q_\pm^{90}}{1 - q_\pm^{120}} > 0.$$

The inverse exists because both eigenvalues of T^{120} are less than one. The symmetric reading of the response is

$$\mathcal{E} = \log[(1 - q_+^{90})(1 - q_-^{90})] - \log[(1 - q_+^{120})(1 - q_-^{120})] = \log(r_+ r_-).$$

Let u_S, u_C, u_T be positive bases of action, velocity, and time, and $u_L = u_C u_T$ the length basis. The kinematic and constitutive readings are

$$c_{\text{int}} = e^{-\mathcal{E}} u_C, \quad c_{\text{vac}} = (r_+ r_-)^{-1} u_C.$$

Proposition 8.13 (Common realisation of propagation and absolute mass). *In the same dimensional chart,*

$$c_{\text{int}} = c_{\text{vac}} = \hat{c} u_C, \quad \hat{c} = (\det V)^{-1} = (r_+ r_-)^{-1}.$$

For $t_0 = \tau_0 u_T$, with $\tau_0 > 0$, the elementary length is

$$\ell_0 = c_{\text{int}} t_0 = \tau_0 \hat{c} u_L.$$

The chronological mass of (8.33) therefore retains the form

$$m_{\text{clk}} = \frac{2\pi \eta_{\text{ret}} 10^{-34}}{108 \tau_0 \hat{c}^2} \frac{\mathcal{U}_S}{u_T u_C^2}.$$

Proof. The identity $\mathcal{E} = \log(r_+ r_-)$ gives $e^{-\mathcal{E}} = (r_+ r_-)^{-1}$; the rank-one property of each sheet gives $\det V = r_+ r_-$. Sheet interchange permutes the factors and preserves their product. Length is obtained by multiplying the same velocity by t_0 . Substituting these expressions and (8.31) into (8.33) produces the mass formula. The factor $\mathcal{U}_S/(u_T u_C^2)$ has the dimension of mass; it does not enter the selection of the channels. $\square$

For a homogeneous route of $n > 0$ transitions, the additive readings $T(\gamma) = nt_0$ and $L(\gamma) = n\ell_0$ give $L(\gamma)/T(\gamma) = c_{\text{vac}}$. If the channel varies from edge to edge, their lengths are added; the ratio is then the mean velocity of those edges, not necessarily a common local velocity.

Equality of sections persists under changes of chart. If $u'_C = du_C$, $u'_T = \eta u_T$, with $d, \eta > 0$, then

$$\hat{c}' = \hat{c}/d, \quad \tau'_0 = \tau_0/\eta, \quad u'_L = d\eta u_L, \quad \tau'_0 \hat{c}' u'_L = \ell_0.$$

In particular, choosing the adapted basis $u'_C = c_{\text{vac}}$ makes $\hat{c}' = 1$, but the reader coordinate in the internal chart remains $(r_+ r_-)^{-1}$.

8.9 Two readers of the electronic trajectory and beta evaluation

The central trajectory starts at $(r, c) = (5, 5)$, with orientation $(h, v) = (1, 1)$. TRIT repeats the cycle $(+1, -1, 0)$. In the first regime, $r + c$ is read and the advance $c \mapsto 1 + ((c - 1 + h) \bmod 9)$ follows; in the second, rc is read and the advance $r \mapsto 1 + ((r - 1 + v) \bmod 9)$ follows; in the zero regime, 9 is read without displacement. The class modulo three of the digital root of the reading is expressed. After steps 27, 54, 81, h, v are simultaneously reversed. The representation of 108 steps preserves the complete trajectory and gives

$$a = 100000220, \quad b = 120200000, \quad \gamma_e = (a^3 b^3)^2, \quad \tau_e = (+, +, +, -, -, -, +, +, +, -, -, -). \quad (8.34)$$

The identity is checked by traversing the nine instants of each window; at the third block, reversal changes a to b , and the second traversal of 54 steps repeats the representation. The state also preserves its memory coordinates.

The first reader measures cyclic disagreements of the ternary representation:

$$D_k = \#\{t \in \mathbb{Z}/108\mathbb{Z} : \gamma_e(t) \neq \gamma_e(t + k)\}.$$

Since the expressed word has period 54, every disagreement appears together with its translate by 54; all D_k are even. In the half $a^3 b^3$, the disagreement count for $k = 1, 3, 4, 27, 36$ gives respectively 27, 28, 34, 24, 16. Doubling it yields

$$(D_1, D_3, D_4, D_{27}, D_{36}) = (54, 56, 68, 48, 32).$$

Hence

$$K_D = \left(D_{27} + D_{36}, D_1, \frac{D_4 - D_3}{2} \right) = (80, 54, 6). \quad (8.35)$$

The second reader acts on the orientation of the twelve windows. Define, with cyclic indices,

$$C_k = \sum_{r=0}^{11} \tau_r \tau_{r+k}, \quad A_\ell = \sum_{r=0}^{11} \left(\sum_{j=0}^{\ell-1} \tau_{r+j} \right)^2, \quad P_6 = \sum_{r=0}^5 \tau_r \tau_{r+6}.$$

The oriented primitive integer combination cancelling the diagonal part of A_3, A_4 is $\Omega = 4A_3 - 3A_4$: its coefficients satisfy $3u + 4v = 0$, $\gcd(u, v) = 1$, and $u > 0$ is fixed. Expanding the square proves

$$A_\ell = \ell C_0 + 2 \sum_{k=1}^{\ell-1} (\ell - k) C_k, \quad \Omega = -2C_1 - 4C_2 - 6C_3.$$

For τ_e , we obtain $C_0 = 12, C_1 = 4, C_2 = -4, C_3 = -12, A_3 = 44, A_4 = 32, P_6 = 6$. Therefore,

$$K_\Omega = (\Omega, 54, P_6) = (80, 54, 6) = K_D. \quad (8.36)$$

The second component preserves the half-return of 108 steps; it is not introduced by diagonal cancellation. Equality of the two readers has been proved on the specified central trajectory; it is not extended by definition to all collections of the atlas.

Definition 8.14 (Electronic coordinate after the return). The complete electronic composition is

$$\varepsilon_e^\beta = \varepsilon_e^{(0)} R_{\text{act}}^{80-54\alpha+6\Delta_4}. \quad (8.37)$$

Positivity of (8.25) and (8.30) allows this power to be defined through the real logarithm. The integer triple arises equally from the two central readers just calculated. Relative evaluation preserves the reduced Planck constant through the ratio of its sections, and absolute realisation additionally uses (8.33) and the stated dimensional chart.

In the inherited tables, $\mathbf{e}_0$ denotes the basal coordinate $\varepsilon_e^{(0)}$ in the chart specified there. The section $\hbar_{\text{int}}$ used for a survival evaluation must be stated in the same chart: for the post-return realisation of the present development, it is $\hbar_{\text{ret}}$. Historical rows retain the convention stated in their source table and are not recalculated through a silent change of this section.

If $\mathcal{U}_E$ is the energy basis of the electronic evaluation, $E_e = \varepsilon_e^\beta \mathcal{U}_E$. When the chronological quantum is used as the basis, the coefficient is

$$\hat{\varepsilon}_e = \varepsilon_e^\beta \frac{\mathcal{U}_E}{E_{\text{clk}}}, \quad E_e = \hat{\varepsilon}_e E_{\text{clk}}.$$

The equality preserves the object and transforms its coordinate. Hence a number presented in an MeV chart cannot be multiplied without conversion by an arbitrary different energy quantum. The electron retains its structural construction when expressed relative to a chronological scale: it is not replaced by the mass of that quantum.

Metrological comparison applies to the result of this composition, with the version of α , the action section, the Δ_4 reader, and the unit chart stated. In particular, d_4 and Δ_4 are not interchangeable: replacing them simultaneously in the basal factor and in the exponent changes the result by the positive ratio

$$\frac{1 + 15\Delta_4}{1 + 15d_4} R_{\text{act}}^{6(\Delta_4 - d_4)} > 1.$$

The preserved formulas allow that difference to be located without attributing it to an external mass or selecting the trajectory anew.

9 The particle as a class of persistent sections

The material construction begins with the state produced by APP, TRIT, and TPK. APP supplies the positions and the two arithmetic operations; TRIT distinguishes the regime and orientation of their transitions; TPK transports this information together with the quotients, boundary incidences, and memory. As the state is prolonged, a finite evaluation remains linked to the evaluations preceding it. The notion of a particle rests on this compatibility and on its material representation. Its mass will be an observable of the section thus constructed.

This sequence determines the meaning of the three objects to be used: the genealogical state preserves the complete extension; the route records its observable history; the material representation makes that history act on a state space and permits the evaluation of observables. The distinction is essential even when the three objects share the same electronic label or the same atlas coordinate.

9.1 Extension, observable history, and compatible section

Let $\mathcal{S}_m$ be the admissible states surviving through depth m , obtained by TPK prolongation, and let $\rho_{n,m} : \mathcal{S}_n \rightarrow \mathcal{S}_m$, for $n \geq m$, be their restriction maps. Compatibility of refinement means

$$\rho_{m,m} = I, \quad \rho_{n,m} \rho_{p,n} = \rho_{p,m} \quad (p \geq n \geq m). \quad (9.1)$$

A surviving extension is an element

$$x = (x_m)_{m \geq 0} \in \Omega_{\text{HMT}}^{\text{surv}} := \varprojlim_m \mathcal{S}_m, \quad \rho_{n,m}(x_n) = x_m. \quad (9.2)$$

The existence and admissibility of these states arise from the construction of the formal kernel. The inverse-limit expression describes the compatibilities preserved by a state already generated.

Let $\mathcal{R}_m$ be the space of observable histories at depth m . The route reader $\text{sh}_m : \mathcal{S}_m \rightarrow \mathcal{R}_m$ and the restriction $\text{tr}_{n,m}$ satisfy

$$\text{tr}_{n,m} \text{sh}_n = \text{sh}_m \rho_{n,m}. \quad (9.3)$$

Thus the reader transmits the restrictions of the state to its histories. The route of x is the collection $\gamma_x = (\gamma_x^{[m]})_m$, where $\gamma_x^{[m]} = \text{sh}_m(x_m)$.

Proposition 9.1 (Compatibility of the expressed history). *For every extension (9.2), its finite histories satisfy $\text{tr}_{n,m}(\gamma_x^{[n]}) = \gamma_x^{[m]}$.*

Proof. Applying (9.3) to the state x_n gives

$$\text{tr}_{n,m}(\gamma_x^{[n]}) = \text{sh}_m(\rho_{n,m} x_n) = \text{sh}_m(x_m) = \gamma_x^{[m]}.$$

The second equality uses precisely the inverse-limit condition. $\square$

The proposition establishes descent from the extension to the history. Its content does not include injectivity of the reader: two extensions can express the same finite history while preserving different memories. Hence a reused route retains its state of provenance, instead of reconstructing it from a numerical value or a visible word.

On each history $\gamma_x^{[m]}$, consider the discrete bundle $\mathcal{B}_{\gamma_x^{[m]}}$ of material states of the stated representation. Let $\mathcal{E}_m(x) = \Gamma(\mathcal{B}_{\gamma_x^{[m]}})$ be its spaces of sections, with restrictions $p_{n,m} : \mathcal{E}_n(x) \rightarrow \mathcal{E}_m(x)$ compatible with those of the state. A persistent section is a collection

$$s = (s_m)_{m \geq m_0}, \quad s_m \in \mathcal{E}_m(x), \quad p_{n,m} s_n = s_m \quad (n \geq m \geq m_0). \quad (9.4)$$

The initial depth m_0 allows a state to be described from any cut beyond which its representation is defined. Prior information entering its observables remains in the record transported by the extension.

9.2 Identity under refinement and equivalence of representations

Fix the internal symmetry groups G_m acting on the material fibres and preserving the stated observables. Two persistent sections s and s' represent the same material identity when, from some level m_1 onwards, there is a collection $g_m \in G_m$ such that

$$s'_m = g_m s_m, \quad p_{n,m} g_n = g_m p_{n,m}, \quad (9.5)$$

and the observable characters of the genealogical record coincide. The admissible symmetries are fixed before forming the quotient; their action transports sheet, orientation, memory, and representation according to the corresponding laws.

Proposition 9.2 (Compatible equivalence of sections). *Relation (9.5), on collections carrying the same types of representation and observables, is an equivalence relation. Its classes are unaffected by removing a finite number of initial levels, provided their transported genealogical information is preserved.*

Proof. The identity of each G_m gives reflexivity. The inverse of a compatible collection is again compatible: $p_{n,m} g_n = g_m p_{n,m}$ implies $p_{n,m} g_n^{-1} = g_m^{-1} p_{n,m}$. This gives symmetry. If $s' = g s$ and $s'' = h s'$, then from the larger of the two initial levels we have $s''_m = h_m g_m s_m$ and $p_{n,m} h_n g_n = h_m g_m p_{n,m}$. Composition also preserves the observables and proves transitivity. All these relations are formulated on a common tail; changing the initial point within that tail preserves the class. The last assertion concerns a change of presentation, not deletion of memory from the state. $\square$

Definition 9.3 (Persistent material particle). An HMT–MD particle is a class $[s]_{\text{pers}}$ of sections (9.4) under (9.5), accompanied by its extension and provenance record, admitting an irreducible material representation or a minimal invariant block and a stable spectral output under the specified coupling.

The definition gathers conditions at different levels. Prolongation and the record belong to the HMT state. Representation and coupling belong to its material realisation. Spectral stability is predicated of the operator and domain used by that coupling. This organisation permits identity, charge, spin, and mass to be studied together without turning one particular observable into the definition of all the others.

The nonadic return illustrates the distinction between identity and instantaneous state. For the phase reader g and memory record $\mathcal{B}$, the following can coexist:

$$g(k+9) = g(k), \quad p_{k+9,k}(s_{k+9}) = s_k, \quad \mathcal{B}_{k+9} \neq \mathcal{B}_k. \quad (9.6)$$

The first equality is periodic; the second expresses persistence; the third records advancement. The phase return belongs to the history of the same material identity, while the state preserves the fact that an additional circuit has been completed.

9.3 The species as a fibre: scope of the finite atlas

The atlas constructed by the APP–TRIT–TPK operations has support $\mathcal{C}_{81} = \{1, \dots, 9\}^2$. Its four orientations per cell form the domain of 324 oriented routes. The record map $\mathcal{R}$ distinguishes 56 route configurations, while the internal classifier

$$q : \mathcal{C}_{81} \longrightarrow \Sigma_{13} \quad (9.7)$$

gathers 13 types of collection and level. These censuses and their rules are developed in the part devoted to the atlas; here we specify which object they transmit to the definition of a particle.

For $y \in \Sigma_{13}$, the fibre $q^{-1}(y)$ preserves all its cells and records. Once the internal ordering of each fibre of $\mathcal{R}$ is fixed, let $k(c)$ be the rank of the cell within it. Its finite presentation is

$$\mathfrak{C}_y^{\text{atlas}} = \{(\mathcal{R}(c), k(c)) : q(c) = y\}. \quad (9.8)$$

Rank is added to separate cells possessing the same collection record; it does not select a mass. The pair $(\mathcal{R}(c), k(c))$ recovers the cell within the ordered chart because the second datum individualises its position in the fibre of the first.

If $\pi_{81} : \Omega_{\text{HMT}}^{\text{surv}} \rightarrow \mathcal{C}_{81}$ is the cellular reader, the lift of the species is

$$\tilde{\mathcal{E}}_y = \{x \in \Omega_{\text{HMT}}^{\text{surv}} : q(\pi_{81}x) = y\}. \quad (9.9)$$

This preimage contains extensions, whereas (9.8) contains finite presentations. The construction and domain of the reader π_{81} form part of the passage between them. A particular particle is realised as a persistent section within the corresponding enriched multisection.

Theorem 9.4 (Central fixed point and equivariant selection). *Let $\rho_c(r, s) = (10 - r, 10 - s)$ be cellular inversion. In the ρ_c -stable fibres of the classifier (9.7), with trivial action on the label y , an equivariant point selection can take values only in cell $(5, 5)$. The electronic fibre at level zero contains this unique fixed point; no other fibre admits such a selection.*

Proof. Solving $\rho_c(r, s) = (r, s)$ gives $r = s = 5$. If $a(y) \in q^{-1}(y)$ is an equivariant selection and y is fixed, then $a(y) = a(\rho_c y) = \rho_c a(y)$. Its value must be the fixed point $(5, 5)$. The atlas places that cell in the electronic fibre at level zero, whose cardinality is one. For any other fibre, the selection condition would produce a fixed point not belonging to it. $\square$

The consequence is constructive: the other species are presented by orbits or multisections, and a realisation requiring a particular line must specify its representation and coupling. The electron provides the central rank-one case without turning all collections into copies of that case. Moreover, cell $(5, 5)$, the raw central signature, the regional signature 555555, and the relative origin of an action line retain their own domains. Uniqueness of the point does not identify those records with one another.

9.4 From the material record to the mass observable

The chain used by the mass observable preserves the order

$$\text{extension} \longrightarrow \text{route} \longrightarrow \text{record} \longrightarrow \text{integer signature} \longrightarrow \text{character} \longrightarrow \text{towers} \longrightarrow \text{material operator}. \quad (9.10)$$

Each arrow retains the data required by the next. The signature is a reading of the record; the character evaluates that signature with the internal coefficients already constructed; the towers incorporate its refinement; finally, the operator acts in the representation of the fibre selected by the coupling. Its output can be a scalar, a multiplet, or a spectral distribution, according to that representation. Metrological comparison takes place after this construction.

A species is described by charge, spin representation, colour, flavour, generation, route configuration, and material operator. These data explain why two preparations with coinciding masses can preserve different information, and why the same identity can express different observables under different couplings. Identity resides in the persistent class and its admissible transports; measurement is a reading of that identity in the stated codomain.

The same hierarchy separates a persistent extension from a section with a finite horizon. If $\text{Ext}_L(s) \neq \emptyset$ and $\text{Ext}_{L+1}(s) = \emptyset$, the horizon L records an obstruction to prolongation. An intermediate participation of this kind in a composition does not by itself create a new asymptotic species. Similarly, a resonance requires the survival operator and its pole, as well as the identity of the section it transports. The distinction between persistence, finite prolongation, and spectral evaluation is thus preserved.

10 Conjugation, orientation, and return of material sections

The persistent identity of Section 9 admits inversion operations at different levels of its structure. Cellular inversion acts on the atlas; dodecaphase conjugation acts on the orientation record; lifting a loop acts on a fibre; a spinor rotation acts on a representation space. Specifying these domains allows the operations to be composed and explains what information is preserved in each return.

10.1 Dodecaphase involution and parity of observables

Let $\mathcal{J}_{12} = \{-1, 0, 1\}^{12}$ be the ambient space of tritic records with twelve positions. The admissible TPK domain is the subset satisfying the route compatibilities already constructed. The operation on the ambient space is

$$C_M : \mathcal{J}_{12} \longrightarrow \mathcal{J}_{12}, \quad (C_M \tau)_j = -\tau_{11-j}, \quad 0 \leq j < 12. \quad (10.1)$$

Restriction to admissible records uses stability of that domain under conjugation. Reversed order and sign reversal are simultaneous components of the operation.

Proposition 10.1 (Linear and quadratic parity of the record). *We have $C_M^2 = I$. If $a_j = a_{11-j}$ and $L_a(\tau) = \sum_j a_j \tau_j$, then $L_a(C_M \tau) = -L_a(\tau)$. If a quadratic form $Q_b(\tau) = \sum_{j,k} b_{jk} \tau_j \tau_k$ satisfies $b_{jk} = b_{11-j, 11-k}$, then $Q_b(C_M \tau) = Q_b(\tau)$.*

Proof. At each position, $[C_M(C_M \tau)]_j = -(-\tau_{11-(11-j)}) = \tau_j$. For the linear functional, substituting $k = 11 - j$ in the sum gives $L_a(C_M \tau) = -\sum_k a_{11-k} \tau_k = -L_a(\tau)$. In the quadratic expression the two sign reversals cancel, and the substitution $(r, s) = (11 - j, 11 - k)$ recovers Q_b through the symmetry of its coefficients. $\square$

Linear orientation and memory correlations therefore have different laws. Even a composition of these observables must preserve their parities: if L_a is odd, its exponential character satisfies

$$\exp(L_a(C_M \tau)) = \exp(L_a(\tau))^{-1}. \quad (10.2)$$

The equality expresses multiplicative inversion. Invariance of a complete material operator is studied on that operator, with its channels and representation, and is established by the corresponding covariance.

Cellular inversion in Theorem 9.4 is $\rho_c : \mathcal{C}_{81} \rightarrow \mathcal{C}_{81}$, whereas C_M has domain (10.1). Both are involutions, but comparing them requires the route record map and its transport law. This datum prevents conjugation of a history from being replaced by inversion of an isolated atlas coordinate.

10.2 Lifting to antisections and preservation of mass

Conjugation is applied to the particle through two compatible maps. The base map C_m^b transports state x_m to its conjugate state $\bar{x}_m$; the lift $\tilde{C}_m : \mathcal{E}_m(x) \rightarrow \mathcal{E}_m(\bar{x})$ transports its material sections. Let t_m be the reader of the dodecaphase record and $\bar{p}_{n,m}$ the restrictions in the conjugate collection. Their conditions are preservation of admissibility, involutivity, and compatibility:

$$\begin{aligned} \rho_{n,m} C_n^b &= C_m^b \rho_{n,m}, & t_m(C_m^b x_m) &= C_M t_m(x_m), \\ \bar{p}_{n,m} \tilde{C}_n &= \tilde{C}_m p_{n,m}, & \tilde{C}_{m,\bar{x}} \tilde{C}_{m,x} &= I. \end{aligned} \quad (10.3)$$

In the first row, the reader expresses the orientation record of the history. The remaining transport of the state accompanies that representation. The second row gives the laws of the lift; the subscripts $x, \bar{x}$ make its two source fibres explicit. If s is persistent, its antisection is $\bar{s} = (\tilde{C}_m s_m)_m$.

Proposition 10.2 (Persistence of the antisection). *Conditions (10.3) transport persistent sections to persistent sections and restore the section upon conjugating twice. If they also transport the internal symmetries by $g_m \mapsto \tilde{C}_m g_m \tilde{C}_m^{-1}$, they induce an involution on the material classes of Definition 9.3.*

Proof. Compatibility gives $\bar{p}_{n,m}\bar{s}_n = \tilde{C}_m p_{n,m} s_n = \tilde{C}_m s_m = \bar{s}_m$. Involutivity restores s . For $s'_m = g_m s_m$, their antisections are related through $\tilde{C}_m g_m \tilde{C}_m^{-1}$. The restriction law in (10.3) preserves compatibility of this collection, which belongs to the internal groups by the stated condition. Thus the construction descends to the classes. $\square$

The charge representation uses the odd part of the material record. If q is the stated charge character, its law is $q(\bar{s}) = -q(s)$. The elementary charge entering the expression of that coordinate on a dimensional line is the internal output already constructed; the representation is subsequently applied to the state without using a target charge to select its record.

For mass, covariance of its operator is required. Let $\hat{M}_s$ and $\hat{M}_{\bar{s}}$ be the self-adjoint operators of the two realisations, and let $\mathcal{C}$ be their unitary or antiunitary intertwiner. Its domain is transported by $\mathcal{C} \text{Dom}(\hat{M}_s) = \text{Dom}(\hat{M}_{\bar{s}})$. The operator condition is

$$\mathcal{C}\hat{M}_s\mathcal{C}^{-1} = \hat{M}_{\bar{s}}. \quad (10.4)$$

Proposition 10.3 (Spectral preservation under conjugation). *Under (10.4), transport of an eigensection with mass $m \in \mathbb{R}$ preserves that eigenvalue. The spectral measure is transported by $E_{\bar{s}}(B) = \mathcal{C}E_s(B)\mathcal{C}^{-1}$ for every real Borel set B . In particular, a scalar realisation preserves mass on passage to the antisection.*

Proof. If $\hat{M}_s\psi = m\psi$, covariance gives $\hat{M}_{\bar{s}}\mathcal{C}\psi = \mathcal{C}(m\psi) = m\mathcal{C}\psi$; the last equality also holds in the antiunitary case because m is real. The real functional calculus of self-adjoint operators transports their spectral projectors and proves the second assertion. $\square$

In a band record the same condition can be seen in elementary form. Given a real reading $E_0(\tau)$, let $E_{\pm} = (E_0 \pm E_0 \circ C_M)/2$ and let $\xi \in \{+1, -1\}$ be the sheet orientation. Then

$$E_{\text{band}}(\tau, \xi) = E_+(\tau) + \xi E_-(\tau), \quad E_{\text{band}}(C_M\tau, -\xi) = E_{\text{band}}(\tau, \xi). \quad (10.5)$$

The equality follows from the evenness of E_+ and oddness of E_- . It explains how band energy can be preserved by transporting record and sheet simultaneously; its identification with the mass of a particular realisation uses (10.4).

The electronic record $\tau_e = (+, +, +, -, -, -, +, +, +, -, -, -)$ satisfies $C_M\tau_e = \tau_e$. This equality concerns the twelve-position word. The pair (τ_e, ξ) preserves an additional orientation changed by conjugation. The distinction allows the unique electronic centre and the conjugate charge realisations to be retained together; a fixed word does not become a proof of neutrality.

10.3 Sign holonomy and the persistence band

Let ℓ be a return loop of the material reader of a surviving extension. Its circuit group is $\langle \ell \rangle \simeq \mathbb{Z}$. On this group, the sign character is specified by

$$\varepsilon : \langle \ell \rangle \longrightarrow \{+1, -1\}, \quad \varepsilon(\ell) = -1. \quad (10.6)$$

The circuit group records how many times the loop is traversed; its type differs from the finite phase C_9 . The interval bundle associated with the character is

$$\mathcal{M}_{\ell} = ([0, 1] \times [-1, 1]) / ((0, u) \sim (1, -u)). \quad (10.7)$$

Theorem 10.4 (Oriented return on the band). *The bundle (10.7) is a Möbius band. Its transport over ℓ reverses the sign of the fibre and over ℓ^2 restores it. Thus a nonzero vector of that line requires two circuits for its oriented return. This preserves the distinction between return of orientation and return of the complete state.*

Proof. The endpoint relation in (10.7) is the linear map $u \mapsto -u$, with negative determinant. It gives the nonorientable real line over the circle and its corresponding interval band. Composing two transports produces $u \mapsto u$. The memory of the extension is not part of this sign quotient; its updating continues to obey TPK transport and (9.6). $\square$

Parametrising one circuit of the base by 2π represents restoration of the cover by 4π . If the temporal reader assigns 108 steps to the relevant circuit, the oriented lift closes at 216 steps. These assignments require the stated loop and reader; they do not divide the history into two successive particles of 108 steps.

Proposition 10.5 (Decomposition of observables on the double cover). *Let $p : \tilde{X} \rightarrow X$ be a principal double cover, with involution ι . For $f \in C^0(\tilde{X}, \mathbb{R})$,*

$$f^+ = \frac{f + f \circ \iota}{2}, \quad f^- = \frac{f - f \circ \iota}{2}$$

give $f = f^+ + f^-$, with $f^+ \circ \iota = f^+$ and $f^- \circ \iota = -f^-$. The first part descends to X ; the second represents a section of the associated sign line. On the Möbius line, every continuous global section has a zero.

Proof. The identities follow by substituting $\iota^2 = I$. The invariant part is constant on each fibre and defines a function on the quotient. The anti-invariant part satisfies the transition law of the sign representation. On the circle, a section of this line is represented by a continuous real function u on $[0, 1]$ with $u(1) = -u(0)$. If an endpoint is zero, a zero already exists; otherwise the endpoints have opposite signs, and the intermediate value theorem provides an interior zero. $\square$

The decomposition distinguishes two classes of observables on the same cover. It represents odd orientation information and preserved even data. Applying it to a material record uses the lift (10.3); the sign character describes the transported line, not all the data of the particle.

10.4 Bidirectional lattice and channel conjugation

The integer angular record expresses two coordinates

$$W_+ = 6n_A + s_C, \quad W_- = 6n_A - s_C, \quad (n_A, s_C) \in \mathbb{Z}^2. \quad (10.8)$$

The coefficients n_A, s_C are internal route readings. The factor 6 belongs to the dodecaphase normalisation of that record.

Proposition 10.6 (Lattice index and interchange of readings). *The image of (10.8) has index 12 in $\mathbb{Z}^2$ and Smith normal form $\text{diag}(1, 12)$. It is characterised by $W_+ + W_- \equiv 0 \pmod{12}$. The operation $s_C \mapsto -s_C$ interchanges W_+ and W_- .*

Proof. The matrix of the map is $\begin{pmatrix} 6 & 1 \\ 6 & -1 \end{pmatrix}$. The greatest common divisor of its entries is 1, and its determinant is -12 , giving the stated invariant factors. The inverse coordinates are $n_A = (W_+ + W_-)/12$ and $s_C = (W_+ - W_-)/2$. If the sum is divisible by 12, the two integers have the same parity and both expressions are integral. The condition is also necessary by summing (10.8). The interchange assertion is checked by substituting $-s_C$. $\square$

In the real angular representation, the coordinates already generated, $a = A_{\text{rad}}$ and $b = C_{\text{rad}}^*$, produce the channels $q_{\pm} = \exp[-(a \pm b)]$. In the domain $a > |b|$, both belong to $(0, 1)$. Let $R^* = R, R^2 = I$ be the sheet grading and C a unitary interchange with $CR C^{-1} = -R$. For $T(b) = \exp[-(aI + bR)]$ we obtain

$$CT(b)C^{-1} = T(-b), \quad CP_{\pm}C^{-1} = P_{\mp}, \quad P_{\pm} = (I \pm R)/2. \quad (10.9)$$

Indeed, conjugation transforms the exponent and every term of its convergent series. This proof transmits the interchange to the channels. The integer lattice and the real operator have distinct domains and express the same interchange operation through their stated reading maps.

10.5 Spinor representation of the electronic block

The three-dimensional APP construction of the addition and multiplication projectors selects a principal electronic plane through the tritic mode $(1, -1, 0)^3$. Its derivation and the evaluation of the projectors are included among the antecedents of the present article. In an orthonormal basis of that plane, their restrictions are

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 1/4 & \sqrt{3}/4 \\ \sqrt{3}/4 & 3/4 \end{pmatrix}, \quad S = 2P - I, \quad J = \frac{4}{\sqrt{3}}[P, Q]. \quad (10.10)$$

This representation receives the two constructed projectors and preserves their orientation. Multiplication gives $S^* = S$, $J^* = -J$, $S^2 = I$, $J^2 = -I$, and $SJ = -JS$. Specifically, $S = \text{diag}(1, -1)$ and $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

Let $\mathcal{S}_e = E_e \otimes_{\mathbb{R}} \mathbb{C}$ be the complexification of the real plane. The scalar unit i belongs to the field of this representation; J remains the real endomorphism arising from the projectors.

Proposition 10.7 (Hermitian triple and representation of rotations). *In $\mathcal{S}_e$, define*

$$\sigma_1 = SJ, \quad \sigma_2 = -iJ, \quad \sigma_3 = S. \quad (10.11)$$

These operators are Hermitian and traceless, and satisfy

$$\sigma_a \sigma_b = \delta_{ab} I + i \sum_c \epsilon_{abc} \sigma_c, \quad L_a = \sigma_a / 2, \quad [L_a, L_b] = i \sum_c \epsilon_{abc} L_c, \quad \sum_a L_a^2 = \frac{3}{4} I. \quad (10.12)$$

They constitute the irreducible spin-1/2 representation on the electronic block.

Proof. The relations of S, J give $(SJ)^* = SJ$, $(-iJ)^* = -iJ$, and $\sigma_a^2 = I$. The cyclic products are $(SJ)(-iJ) = iS$, $(-iJ)S = iSJ$, and $S(SJ) = J = i(-iJ)$; reversing the order reverses the sign. This proves the product identity and the commutators. In the basis of (10.10), the three matrices are

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Their trace is zero and, together with I , they generate $M_2(\mathbb{C})$. A subspace invariant under all three is invariant under that entire algebra and can only be 0 or $\mathcal{S}_e$. Finally, summing $L_a^2 = I/4$ gives the value $3I/4$ of the quadratic operator. $\square$

Recognition of this triple as the Pauli matrices follows (10.10). The norm and product $\langle X, Y \rangle = \frac{1}{2} \text{tr}(XY)$ on the real space of traceless Hermitian matrices give its space of directions. For $n \in \mathbb{R}^3$, $\frac{2}{|n|} = 1$, let

$$H_e(n) = \sum_a n_a \sigma_a, \quad h_e(n) = \frac{1}{2} H_e(n), \quad \Pi_{\pm}(n) = \frac{1}{2} (I \pm H_e(n)). \quad (10.13)$$

Proposition 10.8 (Directional decomposition and double return). *The operators $\Pi_{\pm}(n)$ are rank-one orthogonal projectors and $h_e(n)\Pi_{\pm}(n) = \pm \frac{1}{2}\Pi_{\pm}(n)$. The rotation group around n is*

$$U_n(\theta) = e^{-i\theta H_e(n)/2} = \cos(\theta/2)I - i \sin(\theta/2)H_e(n), \quad U_n(2\pi) = -I, \quad U_n(4\pi) = I. \quad (10.14)$$

Proof. The antisymmetric part of (10.12) vanishes when contracted with $n_a n_b$, so $H_e(n)^2 = I$. Idempotence, orthogonality, and the sum $\Pi_+ + \Pi_- = I$ follow. They are Hermitian with trace one, hence their rank is one. Multiplication by $H_e(n)/2$ gives the eigenvalues. Separating even and odd powers of the exponential produces (10.14) and its group law. Angular normalisation is checked by

$$U_n(\theta)H_e(v)U_n(\theta)^* = H_e(v \cos \theta + (n \times v) \sin \theta + n(n \cdot v)(1 - \cos \theta)),$$

which follows from multiplication using the product identity of the triple. The induced action on directions is a rotation through angle θ ; the spinor lift carries its half-angle. $\square$

If n is realised as the direction of propagation, $h_e(n)$ is dimensionless helicity. Reversal of that direction satisfies $H_e(-n) = -H_e(n)$ and $\Pi_{\pm}(-n) = \Pi_{\mp}(n)$. This operation changes the directional labels; charge conjugation transports the material record through (10.3). Chirality is defined through the relativistic grading appropriate to the representation and its projectors, which differ from (10.13).

The scalar electronic evaluation acts as $\varepsilon_e^\beta I$ on $\mathcal{S}_e$ and commutes with the triple and with $\Pi_{\pm}(n)$. Rotational information preserves the structure of that evaluation without introducing another factor into mass. The Möbius band, in turn, transports a real line around a loop, and nonadic holonomy transports phase and memory. The three returns can enter the same realisation; their domains and intertwiners determine the precise composition.

11 Classification of persistences: atlas, collections, and internal observables

Classification begins with the support on which transport acts. Physical names are interpreted after its invariants have been constructed. This priority distinguishes a position, a history, and a species: the position belongs to the finite atlas; the history preserves the transitions producing it; the persistence type gathers histories compatible with the same classification. An evaluated mass is a subsequent reading of that object.

11.1 Parity, depth, and multisections

On the positional copy of APP, consider

$$I_9 = \{1, \dots, 9\}, \quad \mathcal{C} = I_9 \times I_9.$$

The ordered pair preserves the two axes. Define the parity classifier

$$f(r, c) = \begin{cases} \ell, & r, c \text{ odd}, \\ \nu, & r \text{ odd}, c \text{ even}, \\ d, & r \text{ even}, c \text{ odd}, \\ u, & r, c \text{ even}, \end{cases} \quad g(r, c) = \max\{|r - 5|, |c - 5|\}.$$

The symbols of the first map initially name internal classes. The second measures combinatorial depth relative to the centre. Subsequent association with physical sectors must preserve these definitions.

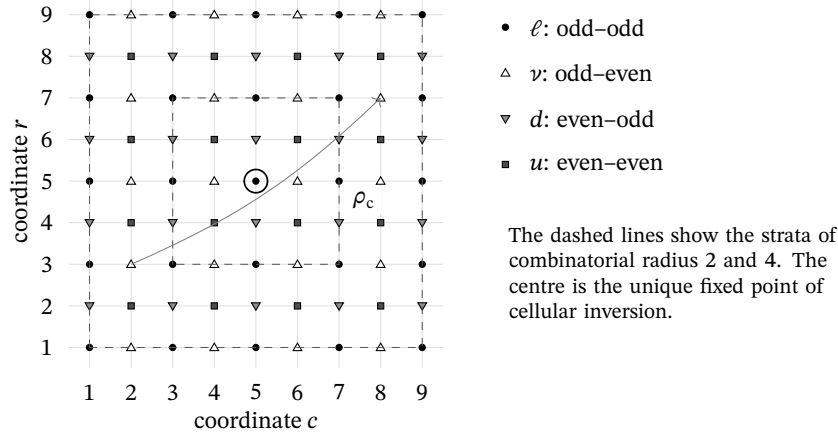


Figure 5: Classification on the positional atlas. The marks distinguish the four internal parity classes, preceding their physical realisations. Depth is measured by $g(r, c) = \max(|r - 5|, |c - 5|)$. The arrow illustrates $(2, 3) \mapsto (8, 7)$ under cellular inversion; it does not represent a dynamic trajectory or a change of flavour. The central circle marks $(5, 5)$, at depth zero.

Definition 11.1 (Family-and-depth multisection). For every realised pair $y = (f, g)$, define

$$\mathfrak{S}_y = \{(r, c) \in \mathcal{C} : (f(r, c), g(r, c)) = y\}.$$

Its lift to persistent states is the preimage under the projection onto the atlas, preserving the sheets, orientations, and memory of the extension.

Proposition 11.2 (Census of the multisections). *The atlas contains 81 cells and exactly 13 nonempty multisections. Their distribution is*

class	depths	cardinalities	total
ℓ	0, 2, 4	1, 8, 16	25
ν	1, 2, 3, 4	2, 4, 6, 8	20
d	1, 2, 3, 4	2, 4, 6, 8	20
u	1, 3	4, 12	16

Proof. In I_9 there are 5 odd and 4 even numbers. The four classes have cardinalities 25, 20, 20, 16. For the odd–odd class, the centred squares of radii 0, 2, 4 contain 1, 9, 25 points; the differences are 1, 8, 16. For odd–even, the cumulative rectangles at radii 1, 2, 3, 4 contain 2, 6, 12, 20 points, whose differences are 2, 4, 6, 8. Interchanging the axes proves the even–odd case. In even–even there are 4 points through radius 1 and 16 through radius 3, so the strata have 4, 12 points. The sets are disjoint, and their union is $\mathcal{C}$. $\square$

This census explains the meaning of each cardinality before any particle table. Depth g is not by itself identified with the index of a physical generation either: that identification uses the realisation maps of the corresponding sector.

11.2 Central conjugation and selection of a cell

Cellular inversion is

$$\rho_c(r, c) = (10 - r, 10 - c).$$

It preserves parity and depth and therefore acts within every multisection. Its unique fixed point satisfies $2r = 2c = 10$: it is $(5, 5)$. The remaining 80 cells are grouped into 40 pairs. There are thus 41 orbits, only one of which is a singleton.

Proposition 11.3 (Obstruction to noncentral point selection). *If a cell selector is required to depend only on the family–depth pair and to be equivariant for ρ_c , it can exist only in the central multisection.*

Proof. Inversion acts trivially on the pair y . Equivariance of $a(y) \in \mathfrak{S}_y$ requires $a(y) = a(\rho_c y) = \rho_c a(y)$. Thus $a(y)$ would have to be the unique fixed point $(5, 5)$. In the central multisection, this is precisely the available choice. $\square$

The consequence is structural. Treating the electron through a rank-1 fibre does not authorise all other collections to be treated in the same way. Outside the centre, the orbit, multisection, and preserved orientations belong to the object being evaluated.

11.3 Route configurations and positional memory

The chronological chart assigns to each cell the sextuple

$$\mathcal{R}(r, c) = (n_A, s_C, k_{\Delta_4}, b_{120}, b_\zeta, \tau_{12})_{r,c}.$$

The five integer coordinates arise from the oriented record; the last preserves the dodecaphase word. Two cells belong to the same route collection when these six readings coincide. This quotient differs from the family–depth quotient.

The documented finite census of this chart contains 56 collections. The size histogram is

$$41 \times 1, \quad 10 \times 2, \quad 2 \times 3, \quad 1 \times 4, \quad 2 \times 5.$$

The identities

$$41 + 10 + 2 + 1 + 2 = 56, \quad 41 + 20 + 6 + 4 + 10 = 81$$

check the number of classes and coverage of the support. Reproducing the census requires the sextuples produced by transport, not merely these two sums. The data dossier preserves the 81 rows and the 56 complete classes.

Fixing the lexicographic order in each fibre, let κ_{pos} be the rank of the cell. Then

$$(r, c) \longmapsto (\mathcal{R}(r, c), \kappa_{\text{pos}}(r, c))$$

is injective. 5 positional-memory values are required because fibres of size 5 exist; the pigeonhole principle rules out a uniform injective lift with fewer values. This memory is neither colour nor the mass exponent, although all subsequently enter the same history.

11.4 Residual colour, flavour, and sheets

In the 36 cells with even first coordinate, the residual-colour observable is defined by

$$\text{col}(r, c) = r - c \pmod{3}.$$

For each even r , the 9 values of c traverse each class 3 times. This gives the exact distribution

$$36 = 12 + 12 + 12.$$

Central inversion transforms this observable into its opposite, since

$$(10 - r) - (10 - c) = -(r - c).$$

It fixes class 0 and interchanges 1 and 2. The colour representation subsequently uses this ternary support and its operators; the number of residues does not replace construction of the representation.

Flavour distinguishes classes and realisations of the sector, whereas residual colour distinguishes positions within the corresponding ternary support. The sheet, by contrast, identifies the arithmetic evaluation entering a transition. Sheet interchange acts on a history; it is not an operation that re-names flavour. This distinction allows mixing, conjugation, and mass evaluations to be formulated separately.

11.5 The domain of orientations

Each cell admits $(h, v) \in \{\pm 1\}^2$, so that

$$\mathcal{C}^{\text{or}} = \mathcal{C} \times \{\pm 1\}^2, \quad |\mathcal{C}^{\text{or}}| = 324.$$

The canonical orientation of 81 routes and the complete domain of 324 are two distinct domains. A reader may descend to a collection in the first chart and cease to be constant when all four orientations are considered. Every assertion about coincidence of readers therefore identifies its domain before publishing the census.

The finite structure thus combines positions, route configurations, multisections, conjugation orbits, and residual observables. None of these partitions replaces the others. The subsequent mass development acts on their persistent lifts, retaining the information required by each evaluation.

12 From the trajectory record to the mass operator

The construction distinguishes two operations. First, the dynamics produces a trajectory record. Then a functional extracts integer coordinates and evaluates them through the internal constants already obtained. Preserving that order explains how mass can be a magnitude of persistence without exhausting its structure.

12.1 Chronological reading and oriented events

Let $d_1, \dots, d_{108}$ be the digital reading of an enriched route. In the stated chart, the action support is $\{1, 3, 5, 7\}$, and

$$n_A = \#\{t : d_t \in \{1, 3, 5, 7\}\}.$$

The 12 events of the 9-step blocks are also preserved:

$$e_m = \sigma_m \#\{t : 9m + 1 \leq t \leq 9m + 9, d_t \in \{2, 4, 6, 8\}\}, \quad 0 \leq m < 12,$$

where $\sigma = (+, +, +, -, -, -, +, +, +, -, -, -)$. The support of this extractor is an explicit part of the reading chart; its function must not be confused with the ternary phase selector of TPK.

The distribution of events matters even when their sums coincide. To see this, introduce

$$p_i = e_i + e_{i+6}, \quad a_i = e_i - e_{i+6}, \quad s_C = \sum_{i=0}^5 p_i, \quad M_i = p_i - p_5 \quad (0 \leq i < 5).$$

Proposition 12.1 (Integral reconstruction of events). *The transformation*

$$(e_0, \dots, e_{11}) \mapsto (s_C, M_0, \dots, M_4, a_0, \dots, a_5)$$

is injective on $\mathbb{Z}^{12}$. Its image is characterised by

$$s_C - \sum_{i=0}^4 M_i \equiv 0 \pmod{6}, \quad p_i \equiv a_i \pmod{2} \quad (0 \leq i < 6),$$

where $p_5 = (s_C - \sum M_i)/6$ and $p_i = p_5 + M_i$ for $i < 5$.

Proof. Adding the six expressions for p_i gives $s_C = 6p_5 + \sum_{i<5} M_i$. This recovers the p_i , and then

$$e_i = \frac{p_i + a_i}{2}, \quad e_{i+6} = \frac{p_i - a_i}{2}.$$

The congruences are necessary for these quantities to be integers and sufficient for the formulas to define an integral preimage. Substitution into the forward transformation recovers every coordinate. $\square$

The decomposition $12 = 1 + 5 + 6$ thus acquires an operational meaning: the total, differences, and orientations recover the events. Division by 6 belongs to this reconstruction. When the character uses $s_C/6$, s_C remains the integer sum; a coordinate already divided is not divided a second time.

12.2 Torsional memory and signature

For every occurrence of 9, the reading preserves its predecessor. Define, with cyclic indices,

$$j_2(d) = \begin{cases} 0, & d = 9, \\ 1, & d \in \{2, 4, 6, 8\}, \\ -1, & d \in \{1, 3, 5, 7\}, \end{cases} \quad z_t = \mathbf{1}_{\{d_t=9\}} j_2(d_{t-1}).$$

Then

$$T_z = \sum_{t=1}^{108} z_t, \quad C_z = \sum_{t \in \{27, 54, 81, 108\}} z_t, \quad k_{\Delta_4} = T_z - 2C_z.$$

The first sum preserves accumulated memory, and the second distinguishes corona closures. Removing the predecessors before evaluating these expressions changes the object being read.

With $\tau_m = \text{sgn}(e_m)$, the remaining coordinates are

$$b_{120} = \sum_{a=0}^3 \text{sgn}(e_a + e_{a+4} + e_{a+8}), \quad b_\zeta = \sum_{m=0}^{11} \tau_m \tau_{m+3},$$

where the second index is modulo 12. The signature obtained is

$$L_{\text{tick}}(\gamma) = \sigma(\gamma) = (n_A, s_C, k_{\Delta_4}, b_{120}, b_\zeta) \in \mathbb{Z}^5.$$

This map is deterministic in the stated chart. The signs and correlations explain why composition must take place on the record before extracting the 5 integers: adding signatures does not, in general, represent the composition of two histories.

12.2.1 The seam term in the $108 \rightarrow 1080$ concatenation

Prolongation of the chronological reading must retain the effective predecessor of every event. Let $d^{(b)} = (d_1^{(b)}, \dots, d_{108}^{(b)})$, $0 \leq b < 10$, be ten consecutive words, and define

$$a_b = d_1^{(b)}, \quad q_b = d_{108}^{(b)}, \quad q_{-1} = q_9.$$

Let $T_z^\sqcup$ be the sum of the ten readers evaluated separately, using q_b as the cyclic predecessor of a_b , and let $T_z^\parallel$ be the reader of the concatenated word, which uses q_{b-1} as the effective predecessor of a_b .

Lemma 12.2 (Exact seam correction). *With $\Delta T_z = T_z^\parallel - T_z^\sqcup$,*

$$\Delta T_z = \sum_{b=0}^9 \mathbf{1}_{\{a_b=9\}} (j_2(q_{b-1}) - j_2(q_b)). \quad (12.1)$$

If crowns occur at $t \equiv 0 \pmod{27}$, then

$$\Delta C_z = 0, \quad \Delta k_{\Delta_4} = \Delta T_z. \quad (12.2)$$

Proof. Away from the ten block beginnings, every event has the same predecessor under both evaluations. If $a_b \neq 9$, the indicator defining z_t also annihilates the possible predecessor difference. If $a_b = 9$, the separate term is $j_2(q_b)$, whereas the concatenated term is $j_2(q_{b-1})$; their difference is the corresponding summand in (12.1). Summing the ten blocks proves the first identity. The new block beginnings occupy positions $108b + 1 \equiv 1 \pmod{27}$, never a crown. Hence the seam does not change C_z , and the definition $k_{\Delta_4} = T_z - 2C_z$ gives (12.2). $\square$

The correction genuinely attains -2 . Take a single beginning $a_b = 9$, with $q_{b-1} = 1$, $q_b = 2$, and require $a_r \neq 9$ at every other beginning. Then $j_2(q_{b-1}) - j_2(q_b) = -1 - 1 = -2$. This word is a seam-reader control; it is not promoted to a surviving route, particle, or physical signature.

Length 1080 repeats ten phase stations, not ten copies of an unchanging depth. The correct transport is the ordered composition

$$\mathcal{H}_9 \circ \mathcal{H}_8 \circ \dots \circ \mathcal{H}_0.$$

Only an additional stationarity proof would justify replacing it by $\mathcal{H}_{108}^{10}$. The seam formula retains exactly the memory that such a replacement could erase.

12.3 Character and internal specialisation

Let $X = (X_A, X_C, X_4, X_{120}, X_{270})$. The formal character is

$$\chi(\sigma; X) = \exp(n_A X_A + s_C X_C + k_{\Delta_4} X_4 + b_{120} X_{120} + b_{\zeta} X_{270}).$$

For each X , it satisfies $\chi(\sigma + \sigma'; X) = \chi(\sigma; X)\chi(\sigma'; X)$ because the exponent is linear in the signature. This is a law on signatures; it does not assert that the extractor is additive on routes without preserving the interface.

Specialisation uses the internal coordinates constructed previously:

$$X_A = A_{\text{rad}}, \quad X_C = C_{\text{rad}}^*/6, \quad X_4 = \Delta_4, \quad X_{120} = s_{120}, \quad X_{270} = \zeta_{270} := 135\Delta_4^2.$$

The character is dimensionless and positive. Its effect is a homothety on the mass line. The exponential form determines how the coefficients act; their provenance is fixed by the preceding records and constructions.

12.4 Bilayer hierarchy and preservation of the two genealogies

Write $x = A_{\text{rad}}, y = C_{\text{rad}}^*$. In the chamber $x > y > 0$, the channels are

$$q_+ = e^{-x-y}, \quad q_- = e^{-x+y}, \quad 0 < q_+ < q_- < 1.$$

For an involution R , with $P_{\pm} = (I \pm R)/2$, the two-channel operator is

$$T = q_+ P_+ + q_- P_-.$$

Its scalar and oriented moments are expressed as

$$c_n = q_-^n + q_+^n, \quad s_n = q_-^n - q_+^n.$$

Multiplying these expressions proves

$$c_n^2 - s_n^2 = 4e^{-2nx}, \quad c_{m+n} = \frac{c_m c_n + s_m s_n}{2}, \quad s_{m+n} = \frac{s_m c_n + c_m s_n}{2}.$$

Both sequences satisfy

$$X_{n+2} = c_1 X_{n+1} - e^{-2x} X_n, \quad X \in \{c, s\}.$$

These relations show that the heights share the same operator. A different collection of moments is not defined for every species.

Proposition 12.3 (Semigroup of heights).

$$\langle 90, 120 \rangle = \{90, 120\} \cup \{30r : r \geq 6\}.$$

Proof. After division by 30, it suffices to study $\langle 3, 4 \rangle$. The integers 6, 7, 8 are $3 + 3, 3 + 4, 4 + 4$; adding 3 produces every subsequent integer. Below 6, the only positive elements are 3, 4. $\square$

The heights $360 = 4 \cdot 90 = 3 \cdot 120$ illustrate the distinction between equality of evaluation and memory of composition. The algebraic quotient identifies the two exponents; the record preserves whether transport used four stages of one type or three of the other.

The refined specialisation has exponent

$$\log \chi_{\text{ref}} = n_A x + \frac{s_C}{6} y + k_{\Delta_4} \Delta_4 + b_{\zeta} \zeta_{270} + \sum_{h \in \langle 90, 120 \rangle} N_h s_h, \quad N_h = \eta_h + b_{120} \mathbf{1}_{\{h=120\}}.$$

The pair (η_{120}, b_{120}) is preserved even though evaluation uses its sum. s_{270} and ζ_{270} also remain separate: the former is a bilayer moment, the latter a quadratic expression of refinement.

Proposition 12.4 (Convergence of the refined evaluation). *If $\sum_h |N_h s_h| < \infty$, the refined character defines a finite positive scalar, and its truncated evaluations converge to it.*

Proof. The assumption gives absolute convergence of the exponent. Continuity of the exponential permits passage to the limit; the exponential of a finite real number is strictly positive. $\square$

Convergence evaluates a sequence of coefficients already specified. Selection of those coefficients by the route is a distinct operation and must accompany every evaluation attributed to a species.

The notation χ_{full} retained in some sector evaluation formulas denotes this same character χ_{ref} , with the same arguments and coefficients. The change of symbol adds no further specialisation or tower.

12.5 Displacement and autocorrelation readers

The same route admits two readings of the return:

$$K_D = \left(D_{27} + D_{36}, D_1, \frac{D_4 - D_3}{2} \right), \quad K_\Omega = (\Omega, 54, P_6).$$

The first map preserves displacement defects; the second, return memory and autocorrelation. In the complete census of 324 orientations there are 14 profiles of the first, 9 of the second, and 40 joint pairs. The 18 coincidences have value (80, 54, 6). Equality of the two operators throughout the atlas is not inferred from this.

For the electronic route, the displacements are

$$(D_1, D_3, D_4, D_{27}, D_{36}) = (54, 56, 68, 48, 32).$$

Substitution produces $K_D = (80, 54, 6)$, coinciding with K_Ω . Its exponent is

$$\kappa_e = 80 - 54\alpha + 6\Delta_4.$$

The electronic trajectory is constructed in (8.34); its displacement and autocorrelation readers are evaluated, respectively, in (8.35) and (8.36). The coincidence $K_D = K_\Omega = (80, 54, 6)$ belongs to the beta-closure theorem in Subsection 8.9. Here that central exception is preserved, and the operation is extended to the atlas fibres.

12.6 Operator on a fibre and dimensional section

Let $\mathcal{H}_M$ be the finite Hilbert space with orthonormal basis (e_γ) labelled by the routes of a multisection. For $r \in \{D, \Omega\}$, define the return operator by $B_M^r e_\gamma = \kappa_\gamma^r e_\gamma$, with real exponents; it is therefore self-adjoint. Let $M_M^{(0)}$ be the positive basal operator, whose connection with the character is fixed below. The dimensionless ratio of the action sections is $R_{\text{act}} > 0$. Define

$$M_M^{\beta,r} = R_{\text{act}}^{B_M^r/2} M_M^{(0)} R_{\text{act}}^{B_M^r/2}.$$

Proposition 12.5 (Positivity and scalar restriction). *The operator $M_M^{\beta,r}$ is positive. If both operators are diagonal in the route basis, each basal value $m_\gamma^{(0)}$ is transformed into*

$$m_\gamma^\beta = m_\gamma^{(0)} R_{\text{act}}^{\kappa_\gamma}.$$

Proof. Let $S = R_{\text{act}}^{B_M^r/2}$, self-adjoint and invertible. For every v , $\langle v, S M_M^{(0)} S v \rangle = \langle S v, M_M^{(0)} S v \rangle \geq 0$. In the diagonal basis, the two factors S multiply each value by $R_{\text{act}}^{\kappa_\gamma/2}$; their product gives the formula. $\square$

The absolute mass section is obtained from action and the clock:

$$\omega_0 = \frac{2\pi}{108t_0}, \quad m_{\text{clk}} = \hbar_{\text{ret}} \omega_0 c_{\text{int}}^{-2}.$$

In the chart normalised relative to the electron, the preceding basal operator is

$$M_M^{(0)} e_\gamma = m_e^{(0)} \chi_{\text{ref}}(\sigma_\gamma - \sigma_e, \eta_\gamma - \eta_e) e_\gamma.$$

Here $m_e^{(0)}$ is the realisation of the basal electronic coordinate, with its prefactor, refinement, and dimensional basis. If $E_e^{(0)} = \varepsilon_e^{(0)} \mathcal{U}_E$, then

$$m_e^{(0)} = \varepsilon_e^{(0)} \mathcal{U}_E c_{\text{int}}^{-2} = b_e m_{\text{clk}}, \quad b_e = \varepsilon_e^{(0)} \frac{\mathcal{U}_E}{E_{\text{clk}}}.$$

In the chart $\mathcal{U}_E = E_{\text{clk}}$, we obtain $b_e = \varepsilon_e^{(0)}$; in another chart, the stated ratio of bases is retained. If the tower coordinates are already relative to the electron, a second subtraction of η_e is omitted. This equality links the operator to the signature of each route and explicitly preserves electronic normalisation. A different choice of reference section must also transport that normalisation. The scale of the reduced Planck constant belongs to this dimensional construction. The character and return factor subsequently act as scalars. The electron preserves its structure: it is not identified with the chronological quantum m_{clk} .

For two routes in the same absolute chart,

$$\frac{m_Y^\beta}{m_X^\beta} = \chi_{\text{ref}}(\sigma_Y - \sigma_X, \eta_Y - \eta_X) R_{\text{act}}^{\kappa_Y - \kappa_X}.$$

The common dimensional section cancels. The difference of the return exponents does not cancel unless it is zero. This identity preserves the distinction between the absolute scale and the relative structure of a collection.

12.7 The null propagation sector

The exponential takes strictly positive values. The null chart is constructed before that character, in the split realisation

$$\mathcal{A}_{\text{sp}} = \mathbb{R}[R]/(R^2 - 1), \quad z = r_+ P_+ + r_- P_-, \quad N(z) = r_+ r_-.$$

The directions $\mathbb{R}P_+$ and $\mathbb{R}P_-$ have zero norm. Propagation transported in one of them belongs to that sector; the bilayer material reading uses the coupling of both. In this way, the photon and the gluonic sector retain their place in the theory even though they do not form a row of the positive character. The realisation of their representations is presented alongside the corresponding sectors.

13 Composition of action, proper frequency, and radial reciprocity

The preceding mass operator receives the signature, character, and towers of the enriched history. The action section fixes its absolute scale and also participates in the counterangle and return correction. The following composition makes that concurrence explicit: one structural route coordinate determines mass, proper frequency, two reciprocal radial readings, and, in the specified canonical realization, thermal weighting. The order APP-TRIT-TPK, enriched state, and joint discrete structure of the continuum is retained; the regional coordinates and α are previously generated outputs. The selection of K , incidence, and orientation remain in their antecedents. The operations in this section are subsequent compositions of the readers already constructed.

13.1 Radial recovery of the action section

With the angular pair in the same unit, write

$$A = 1000\alpha, \quad C^* = 2(\eta_{\text{ret}} + \alpha), \quad \xi = C^*/A, \quad R_\star = \left(\frac{1 - \xi}{1 + \xi} \right)^{1/4}, \quad q = R_\star^4.$$

On the positive branch $0 < \eta_{\text{ret}} < 499\alpha$, one has $0 < q < 499/501$. The explicit inverse is

$$\eta_{\text{ret}} = \alpha \frac{499 - 501q}{1 + q}, \quad \hbar_{\text{ret}} = 10^{-34} \mathcal{U}_S \eta_{\text{ret}}, \quad R_{\text{act}} = \frac{H_5(1 + q)}{\alpha(499 - 501q)}. \quad (13.1)$$

Proof. The identity $q = (1 - \xi)/(1 + \xi)$ gives $\xi = (1 - q)/(1 + q)$. Substitution into $\eta_{\text{ret}} = \alpha(500\xi - 1)$ proves the first formula; the other two use the definitions of the action section and correction $R_{\text{act}} = H_5/\eta_{\text{ret}}$. For both orientations, let $\varepsilon = \pm 1$ and retain $q_\varepsilon = (1 - \varepsilon\xi)/(1 + \varepsilon\xi)$. Then

$$\eta_{\text{ret}} = \alpha \left[500\varepsilon \frac{1 - q_\varepsilon}{1 + q_\varepsilon} - 1 \right].$$

The transformation $(\varepsilon, q_\varepsilon) \mapsto (-\varepsilon, q_\varepsilon^{-1})$ preserves the expression and hence $\hbar_{\text{ret}}$ and R_{act} . $\square$

This recovery retains the orientation mark, α , H_5 , and the dimensional basis. The signature and towers continue to arise from the route record.

13.2 Longitudinal section, induced clock, and electronic cotransformation

Let $L_\star > 0$ be the reference longitudinal section and $c_{\text{int}} > 0$ the constitutive velocity in the chosen chart. We write α_{HMT} and π_{HMT} for the coordinates already generated by the enriched state; they are not metrological data selecting this construction. In the previously established formulae below, $\alpha := \alpha_{\text{HMT}}$ and $\pi := \pi_{\text{HMT}}$. The marked incidence contains

$$N = 12 \cdot 4 \cdot 120 = 5760, \quad r_N = \frac{N - 1}{4N} = \frac{5759}{23040}, \quad L_\alpha = \alpha_{\text{HMT}}^{16} r_N L_\star. \quad (13.2)$$

The coefficient r_N is the diagonal expectation of the residual Hadamard return, $\Delta(CC^*) = r_N I$. The complementary channel $1/(4N)$ is retained as a complement but is not added to the longitudinal observable. Hence L_α is constructed before the clock and before any gravitational reading.

The 108-step clock is the subsequent circular realization of that length:

$$t_0 = \frac{\pi_{\text{HMT}} L_\alpha}{54c_{\text{int}}}, \quad \omega_0 = \frac{2\pi_{\text{HMT}}}{108t_0} = \frac{c_{\text{int}}}{L_\alpha}, \quad R_{108} = \frac{c_{\text{int}}}{\omega_0} = L_\alpha. \quad (13.3)$$

For a fixed returned action section $\hbar_{\text{ret}} > 0$, the clock energy and mass are defined first, and only then the electronic factor:

$$Q_L = \frac{\hbar_{\text{ret}} c_{\text{int}}}{L_\alpha}, \quad E_{\text{clk}} = Q_L, \quad m_{\text{clk}} = \frac{Q_L}{c_{\text{int}}^2}, \quad E_e^{(0)} = \varepsilon_e^{(0)} U_E, \quad b_e(Q_L) = \frac{E_e^{(0)}}{Q_L}. \quad (13.4)$$

Hereafter, b_e abbreviates the evaluated function $b_e(Q_L)$. One of the return readers $r \in \{D, \Omega\}$ is retained. The refined character is the one constructed in the preceding section, retaining convergence and towers relative to the electron. The basal electronic coordinate and its exponent, derived above, are

$$d_4 = \frac{\pi - e \log \pi}{270}, \quad \Delta_4 = d_4 \left(1 + \frac{\pi}{729}\right), \quad \varepsilon_e^{(0)} = \frac{\sqrt{3}}{4} \left[e^{\varphi/\pi^2} - 22\alpha^3\right] (1 + 15\Delta_4), \quad \kappa_e = 80 - 54\alpha + 6\Delta_4.$$

For a route with a positive scalar realization,

$$\Xi_\Gamma = b_e \chi_{\text{ref}}(\sigma_\Gamma - \sigma_e, \eta_\Gamma - \eta_e) R_{\text{act}}^{\kappa_\Gamma^r}, \quad m_\Gamma^\beta = m_{\text{clk}} \Xi_\Gamma. \quad (13.5)$$

This formula uses absolute towers before taking their difference; when the record already provides relative towers, that difference is taken only once. The height-120 term likewise retains its single inclusion in the character. At the central state,

$$m_e^\beta = \frac{\varepsilon_e^{(0)} U_E}{c_{\text{int}}^2} R_{\text{act}}^{\kappa_e}.$$

The equality follows by cancelling Q_L between m_{clk} and $b_e(Q_L)$. This retains the complete electronic normalization. Action enters the absolute scale, R_{act} , and the counterangle determining the constitutive channels. No species-dependent decimal correction occurs: the decade of the action section remains exclusively in the absolute action-clock-mass branch.

Chart covariance prevents b_e from being frozen. With c_{int} , $E_e^{(0)}$, and all dimensionless route data fixed—the refined character, exponents, and R_{act} —, the cotransformation of the longitudinal and action sections is written

$$L_\star \mapsto \lambda L_\star, \quad \hbar_{\text{ret}} \mapsto \mu \hbar_{\text{ret}}, \quad \lambda, \mu > 0,$$

then

$$L_\alpha \mapsto \lambda L_\alpha, \quad Q_L, m_{\text{clk}} \mapsto \frac{\mu}{\lambda} (Q_L, m_{\text{clk}}), \quad b_e \mapsto \frac{\lambda}{\mu} b_e, \quad m_{\text{clk}} b_e = \frac{E_e^{(0)}}{c_{\text{int}}^2}. \quad (13.6)$$

Consequently, varying the sections cotransforms the electronic factor and does not alter the already constructed masses or their tables.

In the following identities, $m_\Gamma = m_\Gamma^\beta$ is used as an abbreviation, retaining the complete electronic normalization.

Proposition 13.1 (Proper frequency and reduced wavelength). *With $E_\Gamma = m_\Gamma c_{\text{int}}^2$, $\Omega_\Gamma = E_\Gamma / \hbar_{\text{ret}}$, $\nu_\Gamma = \Omega_\Gamma / (2\pi_{\text{HMT}})$, and $R_{108} = L_\alpha$, one has*

$$E_\Gamma = E_{\text{clk}} \Xi_\Gamma, \quad \Omega_\Gamma = \omega_0 \Xi_\Gamma, \quad \nu_\Gamma = \frac{\Xi_\Gamma}{108 t_0}, \quad \bar{\lambda}_\Gamma = \frac{\hbar_{\text{ret}}}{m_\Gamma c_{\text{int}}} = \frac{R_{108}}{\Xi_\Gamma}.$$

Proof. Substituting (13.5) and the definitions of m_{clk} and E_{clk} gives all four identities. For two positive routes, the frequency ratio equals the mass ratio; the multiplicative property of the character retains the relative signature and towers together with $R_{\text{act}}^{\kappa_Y^r - \kappa_X^r}$. $\square$

13.3 Joint rigidity of action and radial reading

In this subsection we abbreviate $\hbar = \hbar_{\text{ret}}$. Let $M = M^{\beta, r}$ be the already constructed mass operator on a finite, strictly positive massive fibre. After M , define

$$X = \frac{M}{m_{\text{clk}}}, \quad H = M c_{\text{int}}^2 = Q_L X, \quad \Lambda = L_\alpha X^{-1}, \quad \mathcal{R}_g = L_\alpha X.$$

Theorem 13.2 (Two conserved relations and unique gravitational coefficient). *The preceding readings satisfy*

$$H\Lambda = \hbar c_{\text{int}} I. \quad (\text{R01})$$

$$\mathcal{R}_g \Lambda = \Lambda \mathcal{R}_g = L_\alpha^2 I. \quad (\text{R02})$$

If the physical realization subsequently identifies the reduced radial reading by $\mathcal{R}_g = (G/c_{\text{int}}^2)M$, then there is a unique compatible coefficient:

$$G = \frac{c_{\text{int}}^3 L_\alpha^2}{\hbar}. \quad (13.7)$$

Proof. The definitions and $Q_L = \hbar c_{\text{int}}/L_\alpha$ give (R01) and (R02) directly. They imply $\mathcal{R}_g = L_\alpha^2 H/(\hbar c_{\text{int}})$. Substitution of $H = M c_{\text{int}}^2$ and comparison with $\mathcal{R}_g = (G/c_{\text{int}}^2)M$ yield (13.7). If G' satisfies the same readings, then $(G' - G)M/c_{\text{int}}^2 = 0$; invertibility of M forces $G' = G$. The null sector is treated separately and is not inverted. In particular, G is deduced after the mass operator and does not enter its construction. $\square$

If X is only positive semidefinite, the construction is restricted to $(\ker X)^\perp$. With the pseudoinverse X^+ , $\Lambda^+ = L_\alpha X^+$, and $P = P_{(\ker X)^\perp}$, the relations become $H\Lambda^+ = \hbar c_{\text{int}} P$ and $\mathcal{R}_g \Lambda^+ = L_\alpha^2 P$. The null sector retains its separate treatment and does not yield a vacuous determination of G .

For any positive action section $\hbar_a$, let $Q_{L,a} := \hbar_a c_{\text{int}}/L_\alpha = E_{P,a}$ be the longitudinal Planck energy in that same section. The same proof gives $G_a = c_{\text{int}}^3 L_\alpha^2/\hbar_a$ and preserves $\hbar_a G_a/c_{\text{int}}^3 = L_\alpha^2$. The clock representation is a corollary, not the premise of the deduction:

Corollary 13.3 (Circular form).

$$t_0 = \frac{\pi_{\text{HMT}} L_\alpha}{54 c_{\text{int}}} \implies G_a = \left(\frac{54}{\pi_{\text{HMT}}} \right)^2 \frac{c_{\text{int}}^5 t_0^2}{\hbar_a}. \quad (13.8)$$

Theorem 13.4 (Reciprocity of radial readings). *With the reduced convention $r_{g,\Gamma}^{(a)} = G_a m_\Gamma/c_{\text{int}}^2$,*

$$r_{g,\Gamma}^{(a)} = \frac{\hbar_{\text{ret}}}{\hbar_a} R_{108} \Xi_\Gamma, \quad r_{g,\Gamma}^{(a)} \bar{\lambda}_\Gamma = \frac{\hbar_{\text{ret}}}{\hbar_a} R_{108}^2.$$

In the same section $a = \text{ret}$,

$$r_{g,\Gamma} = R_{108} \Xi_\Gamma, \quad \bar{\lambda}_\Gamma = R_{108}/\Xi_\Gamma, \quad r_{g,\Gamma} \bar{\lambda}_\Gamma = R_{108}^2. \quad (13.9)$$

Proof. Substituting G_a and $m_\Gamma = \hbar_{\text{ret}} \omega_0 \Xi_\Gamma/c_{\text{int}}^2$ gives $r_g^{(a)} = (\hbar_{\text{ret}}/\hbar_a) R_{108}^2 \omega_0 \Xi_\Gamma/c_{\text{int}}$. The identity $R_{108} \omega_0 = c_{\text{int}}$ proves the first formula; multiplication by R_{108}/Ξ_Γ gives the second. Choosing the same section cancels the action ratio. $\square$

The convention $2Gm/c^2$ would introduce a factor of two into the product and would constitute a different reader. Here the lengths are readings of the same state: in logarithmic coordinates their displacements are $\log \Xi_\Gamma$ and $-\log \Xi_\Gamma$. Their geometric mean is $R_{108} = L_\alpha$. The physical identification of $\mathcal{R}_g$ is stated explicitly in the preceding theorem; it does not redefine the mass operator.

13.4 Thermal reading of the spectrum and joint compatibility

Thermal transduction is a positive linear map between the oriented dimensional lines of temperature and energy: $k_B : \mathcal{L}_\Theta \rightarrow \mathcal{L}_E$. The uniform distribution over the three local states has information $-3(1/3)\log(1/3) = \log 3$. A positive temperature section Θ_{108} , normalized by the energy of one chronological decision, uniquely determines the transducer through

$$k_B \Theta_{108} \log 3 = E_{\text{clk}}.$$

In positive bases u_E, u_Θ , let $E_{\text{clk}} = \epsilon u_E$, $\Theta_{108} = \theta u_\Theta$, and $k_B(u_\Theta) = b u_E$. The equation gives $b = \epsilon/(\theta \log 3) > 0$, proving existence and uniqueness. Under $u'_E = s_E u_E$, $u'_\Theta = s_\Theta u_\Theta$, one has $b' = (s_\Theta/s_E)b$, $\epsilon' = \epsilon/s_E$, and $\theta' = \theta/s_\Theta$; the identity is preserved. The individual coordinate of the transducer retains those bases and the thermal section as declared data. Action and period previously fix $E_{\text{clk}} = h_{\text{ret}}/(108t_0)$. In the canonical realization at that temperature, with a common energy origin and zero chemical potential,

$$\beta_{108} = (k_B \Theta_{108})^{-1}, \quad \beta_{108} E_\Gamma = (\log 3) \Xi_\Gamma, \quad e^{-\beta_{108} E_\Gamma} = 3^{-\Xi_\Gamma}.$$

The proof substitutes $E_\Gamma = E_{\text{clk}} \Xi_\Gamma$. For the finite fibre, its full operator expression is

$$S(\rho) := -\text{Tr}(\rho \log \rho), \quad Z = \text{Tr}(3^{-X}), \quad \rho_{108} = 3^{-X}/Z.$$

$$S(\rho_{108}) = \log Z + (\log 3) \text{Tr}(\rho_{108} X).$$

The last equality follows from $\log \rho_{108} = -(\log 3)X - (\log Z)I$ and $\text{Tr} \rho_{108} = 1$. If eigenvalues are grouped with physical multiplicities g_Γ , then $Z = \sum_\Gamma g_\Gamma 3^{-\Xi_\Gamma}$. States of the realization are counted; equivalent descriptions of a history retain their identification. In infinite dimension, 3^{-X} must be trace class and finite entropy additionally requires $\text{Tr}(X 3^{-X}) < \infty$.

Here S is dimensionless von Neumann entropy; dimensional entropy is obtained through the transducer k_B .

The full composition, on the same action section and using $R_{108} = L_\alpha$, is

$$\boxed{\Xi_\Gamma = \frac{m_\Gamma}{m_{\text{clk}}} = \frac{\Omega_\Gamma}{\omega_0} = \frac{r_{g,\Gamma}}{L_\alpha} = \frac{L_\alpha}{\bar{\lambda}_\Gamma} = \frac{E_\Gamma}{k_B \Theta_{108} \log 3}.} \quad (13.10)$$

This equality identifies readings of the same section; the genealogy producing it remains in b_e , the refined character, and the return exponent. At $\Xi = 1$, the radial readings coincide and the energy is E_{clk} ; whether a species occupies that point depends on the spectrum actually constructed. The joint identity provides a compatibility check between dimensional realizations and retains the distinction between proper energy, action, temperature, and route memory.

14 Electron trajectory, return spectrum, and mass conservation

This section develops, on the mass operator already constructed, the dynamical structure that preserves the electron trajectory. It introduces neither a second definition of the electron nor an additional calibration. The causal chain is the one established throughout this article:

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow X_{\text{enr}} \longrightarrow \mathcal{C}_{\text{cont}}^{\text{disc}} \longrightarrow (\Gamma_e, \tau_e, \sigma_e) \longrightarrow M^\beta.$$

The coordinates π_{HMT} , φ_{HMT} , e_{HMT} , and α_{HMT} are correlated HMT outputs generated from the enriched state; they are not introduced as conventional input data. The Fourier analysis below is applied after the word has been produced and does not select the route.

14.1 Distinct central objects and fibre memory

The fixed cell $(5, 5)$, the boundary datum 111111, the transverse emission 555555, and the orientation $\tau_e = (+ + + - - -)^2$ do not name one and the same object. The cellular involution

$$\rho_c(r, c) = (10 - r, 10 - c)$$

has the unique fixed point $(5, 5)$. At that point,

$$5 + 5 = 1 + 9, \quad 5 \cdot 5 = 7 + 2 \cdot 9,$$

whereas at $(8, 2)$ and $(2, 8)$ the sum is preserved but the product quotient changes. The fixed cell determines the central exception; the boundary register and the transverse emission preserve other state coordinates.

In particular, the product-channel fibre of 555555 contains 24 oriented positions. After one TPK advance it splits into

$$555177, \quad 555717, \quad 555771,$$

with eight positions in each output. The next value is not a function of the isolated scalar signature: the memory class and orientation are required to reconstruct the evolution. This is why the mass operator acts on route fibres rather than on a list of numbers.

14.2 Exact cellular support of the central trajectory

The trajectory in (8.34) starts at $(r, c, h, v) = (5, 5, 1, 1)$. Each TRIT block $(+1, -1, 0)$ advances each coordinate once and retains one threshold step. At the endpoints of every block one has $r = c$; at the intermediate step,

$$c - r \equiv h \pmod{9}.$$

Nine blocks visit every position on the corresponding diagonal. The simultaneous inversions of (h, v) at steps 27, 54, 81 add the conjugate diagonal without resetting the state. Consequently, the cellular projection of the 108 steps has the exact support

$$\text{Supp}_{\text{cell}}(\Gamma_e) = \{(r, c) \in \{1, \dots, 9\}^2 : c - r \equiv 0, 1, -1 \pmod{9}\}, \quad |\text{Supp}_{\text{cell}}(\Gamma_e)| = 27. \quad (14.1)$$

The equality follows from the update rule, not from a retrospective count. It describes a discrete trajectory morphology on the positional torus. Its realisation as a Möbius band additionally requires the orientation transport and gluing of Section 10; the finite support is not identified with a three-dimensional material shape.

14.3 Two histories, one sector chart, and exact reconstruction

The central biography contains the histories

$$e_{++} = (2, 2, 5, -4, -3, -2)^2, \quad e_{--} = (4, 3, 2, -2, -2, -5)^2. \quad (14.2)$$

Both are expressed in the fixed sector chart $\Sigma_{12} = (+ + + - -)^2$. The absolute orientation of the recurrence is transported separately; the sign of a history is not identified with the sector label. The two histories have zero total and the same absolute-value histogram, but they do not have the same positional memory.

For a history $e = (e_0, \dots, e_{11})$, define

$$p_i = e_i + e_{i+6}, \quad a_i = e_i - e_{i+6}, \quad s_C = \sum_{i=0}^5 p_i, \quad M_i = p_i - p_5 \quad (0 \leq i < 5). \quad (14.3)$$

The decoder is

$$p_5 = \frac{s_C - \sum_{i=0}^4 M_i}{6}, \quad p_i = p_5 + M_i, \quad e_i = \frac{p_i + a_i}{2}, \quad e_{i+6} = \frac{p_i - a_i}{2}. \quad (14.4)$$

The integrality conditions are divisibility by six in the first expression and common parity of p_i and a_i . For (14.2), one obtains

$$M^{++} = (8, 8, 14, -4, -2), \quad M^{--} = (18, 16, 14, 6, 6), \quad a = 0.$$

Substitution in (14.4) restores all twelve components. Thus the histories are demonstrably distinct even when a partial reading coincides. It does not follow that their masses must differ: a mass reader may be invariant under a transport that preserves fewer coordinates than the enriched state.

14.4 Spectral identity of the return reader

Let $\tau = (\tau_0, \dots, \tau_{11}) \in \{-1, +1\}^{12}$ with cyclic indices. Define

$$C_s(\tau) = \sum_{j=0}^{11} \tau_j \tau_{j+s}, \quad \Omega(\tau) = -2C_1 - 4C_2 - 6C_3, \quad P_6(\tau) = \frac{C_6}{2}. \quad (14.5)$$

The first identity is the expanded form of $\Omega = 4A_3 - 3A_4$ already proved in (8.36). To make explicit which modes transport the correction, introduce, after τ has been produced,

$$T_k(\tau) = \sum_{j=0}^{11} \tau_j e^{-2\pi i k j / 12}, \quad 0 \leq k < 12. \quad (14.6)$$

Proposition 14.1 (Exact spectral decomposition). *For every real word of length twelve,*

$$C_s = \frac{1}{12} \sum_{k=0}^{11} |T_k|^2 \cos\left(\frac{2\pi k s}{12}\right), \quad (14.7)$$

$$\Omega = \frac{1}{12} \sum_{k=0}^{11} |T_k|^2 w_k, \quad w_k = -2 \cos \theta_k - 4 \cos 2\theta_k - 6 \cos 3\theta_k, \quad \theta_k = \frac{2\pi k}{12}, \quad (14.8)$$

$$P_6 = \frac{1}{24} \sum_{k=0}^{11} |T_k|^2 (-1)^k. \quad (14.9)$$

Proof. Expanding $|T_k|^2$ yields $\sum_{j,\ell} \tau_j \tau_\ell e^{-2\pi i k(j-\ell)/12}$. The sum over k equals twelve when $j - \ell$ is the prescribed displacement and vanishes otherwise. This is finite character orthogonality and proves (14.7). Inserting successively $s = 1, 2, 3$ in (14.5) yields (14.8); $s = 6$ yields (14.9). $\square$

For $\tau_e = (+ + + - - -)^2$, with $z_k = e^{-2\pi i k/12}$,

$$T_k = (1 + (-1)^k)(1 - z_k^3)(1 + z_k + z_k^2). \quad (14.10)$$

Only $k = 2, 6, 10$ survive, with

$$T_2 = 4 - 4i\sqrt{3}, \quad T_6 = 4, \quad T_{10} = 4 + 4i\sqrt{3},$$

and hence

$$|T_2|^2 = 64, \quad |T_6|^2 = 16, \quad |T_{10}|^2 = 64. \quad (14.11)$$

The corresponding weights w_2, w_6, w_{10} are 7, 4, 7. Therefore,

$$\Omega(\tau_e) = \frac{64 \cdot 7 + 16 \cdot 4 + 64 \cdot 7}{12} = 80, \quad P_6(\tau_e) = 6. \quad (14.12)$$

This recovers the second reader of (8.36) from the trajectory modes. The alternating word $(+-)^6$, which retains the same sign histogram, has $|T_6|^2 = 144$ and gives $\Omega = 48$; it is a negative control showing that order, not merely counting, enters. Cyclic translations multiply T_k by a phase and preserve $|T_k|^2$, so this reader alone does not recover the time origin.

The return contribution to the electron exponent is thus

$$\kappa_e = \frac{1}{12} \sum_{k=0}^{11} |T_k|^2 \left(w_k + \frac{\Delta_4}{2} (-1)^k \right) - 54\alpha_{\text{HMT}} = 80 - 54\alpha_{\text{HMT}} + 6\Delta_4. \quad (14.13)$$

This identity introduces no spatial Hamiltonian: it proves that the already-defined mass correction is an exact spectral functional of the return word.

14.5 Extension of the electron equation to multisections

The register of a route γ supplies $\sigma_\gamma = (n_A, s_C, k_\Delta, b_{120}, b_\zeta)$ and tower coordinates $\eta_\gamma = (\eta_h)_h$. In a common chart, the refined character is

$$\begin{aligned} \log \chi_{\text{ref}}(\sigma_\gamma, \eta_\gamma) &= n_A x + \frac{s_C}{6} y + k_\Delta \Delta_4 + b_\zeta \zeta_{270} + b_{120} s_{120} + \sum_{h \in \langle 90, 120 \rangle} \eta_h s_h \\ &= n_A x + \frac{s_C}{6} y + k_\Delta \Delta_4 + b_\zeta \zeta_{270} + \sum_{h \in \langle 90, 120 \rangle} N_h s_h, \\ N_h &:= \eta_h + b_{120} \mathbf{1}_{\{h=120\}}. \end{aligned} \quad (14.14)$$

where $x = A_{\text{rad}}$, $y = C_{\text{rad}}^*$, $\zeta_{270} = 135\Delta_4^2$, and $s_h = q_-^h - q_+^h$. The two lines are the same evaluation: b_{120} occurs once, either as an explicit term or absorbed into N_{120} . Absolute convergence of the last series is required for an infinite tower. This formula keeps separate the torsion ζ_{270} , the two sheets, and the two compositions of height $360 = 4 \cdot 90 = 3 \cdot 120$.

In the route basis of a fibre, let $B^r e_\gamma = \kappa_\gamma^r e_\gamma$, with $r = D$ or Ω . The operator law of Section 12 gives, in its diagonal chart,

$$\boxed{\frac{m_\gamma^\beta}{m_e^\beta} = \chi_{\text{ref}}(\sigma_\gamma - \sigma_e, \eta_\gamma - \eta_e) R_{\text{act}}^{\kappa_\gamma - \kappa_e}.} \quad (14.15)$$

The electron equation is the reference section of this identity, not an experimental number used to select routes. Physical labels are assigned only after fibres, representations, and channels have been constructed; numerical proximity is not a substitute for that prospective matching.

14.6 Operator criterion for mass-preserving transport

Define the dimensionless coordinate

$$\Lambda_\gamma := \log \frac{m_\gamma^\beta}{m_e^\beta} \quad (14.16)$$

in a common chart with positive masses. This symbol is neither the reciprocal length $\Lambda = L_\alpha X^{-1}$ nor the coordinate $\bar{\lambda}_\Gamma$ used elsewhere.

Let $\mathcal{D}$ be a finite route domain stable under a bijective TPK transport $F : \mathcal{D} \rightarrow \mathcal{D}$, and let $V_F e_\gamma = e_{F\gamma}$ be its permutation operator. Then

$$[M^\beta, V_F]e_\gamma = m_e^\beta (e^{\Lambda_{F\gamma}} - e^{\Lambda_\gamma}) e_{F\gamma}. \quad (14.17)$$

Consequently,

$$[M^\beta, V_F] = 0 \iff \Lambda_{F\gamma} = \Lambda_\gamma \text{ for every } \gamma \in \mathcal{D}. \quad (14.18)$$

Using (14.14), this condition is

$$\delta n_A x + \frac{\delta s_C}{6} y + \delta k_\Delta \Delta_4 + \delta b_\zeta \zeta_{270} + \sum_h \delta N_h s_h + \delta \kappa \log R_{\text{act}} = 0. \quad (14.19)$$

Here $\delta N_h = \delta \eta_h + \delta b_{120} \mathbf{1}_{\{h=120\}}$, so the difference retains the original tower coordinates and adds the b_{120} incidence term exactly once. It is checked after F has been produced by the TPK; F is not selected retrospectively by the mass value. The criterion permits events and memory to change provided the total coordinate is preserved. It does not state that every TPK transport preserves mass, nor that it exhausts bound-state dynamics. Its hypotheses are a stable domain, bijective transport, a fixed chart, positive masses, and convergence of the towers.

14.7 Joint result

Equations (14.1), (14.4), (14.13), (14.15), and (14.18) close one chain. The central trajectory visits exactly twenty-seven cells; its histories preserve reversible memory; the return has three nonzero modes producing $(\Omega, P_6) = (80, 6)$; the mass law extends the electron section to the multisections; and the commutator identifies exactly which transports preserve mass. Cellular morphology, return spectrum, and magnitude are coordinated readers of one enriched state, not interchangeable observables.

15 Graded completion of the channels and the exclusion principle

The construction of a particle provides a space of channels, their labels, and the operators that transport orientation and memory. The composition of several particles adds a different question: how these channels are exchanged and how their occupations are counted. Here this stratum is constructed from the data already produced by APP-TRIT-TPK. Mass and the Hamiltonian can subsequently act on the resulting space; their values do not enter the proof of the occupation rules.

The parity used is that of sheet transport in the one-particle representation. It is not the sign of charge, the parity of a visible integer, or the third coordinate of the TRIT considered in isolation. The different readings retain their maps. Section 10 constructs the oriented return and spinorial representation; this section develops the multiparticle algebra corresponding to a stated grading and exchange rule.

15.1 Finite channel space and parity character

At a fixed depth K , let S_K be the finite set of channels of the realization under consideration. Its elements retain the route and representation data that still distinguish states: sheet, orientation, boundary, memory, flavour, colour, and spin component when they belong to the sector. The quotient defining S_K is taken before constructing occupation; it does not identify two channels solely because their masses are equal.

The one-particle space is

$$\mathcal{H}_K = \bigoplus_{s \in S_K} V_s, \quad \langle e_{s,a}, e_{t,b} \rangle = \delta_{st} \delta_{ab}, \quad (15.1)$$

where V_s carries the stated representation multiplicity. Each $e_{s,a}$ is a complete mode, not an unlabelled atlas cell. Its positive inner product belongs to that finite realization and makes explicit the normalization used below.

The parity datum is a compositional character of transport,

$$p : \text{Mor}(\mathbf{P}_{\text{TPK}}) \longrightarrow \mathbb{Z}/2\mathbb{Z}, \quad p(\gamma\eta) = p(\gamma) + p(\eta). \quad (15.2)$$

On the channels on which this grading is realized, the holonomy involution J_p satisfies $J_p e = (-1)^{p(e)} e$. Thus,

$$\mathcal{H}_K = \mathcal{H}_{K,0} \oplus \mathcal{H}_{K,1}, \quad \Pi_0 = \frac{I + J_p}{2}, \quad \Pi_1 = \frac{I - J_p}{2}.$$

The two projections are orthogonal because $J_p^* = J_p$ and $J_p^2 = I$. Equality (15.2) ensures that composing transports preserves the grading. When a realization has not supplied this character, the completion below is a construction relative to its explicit choice, not a statistical assignment obtained from the appearance of a route.

15.2 Graded exchange and permutation projections

For homogeneous vectors, define the exchange

$$\tau_p(v \otimes w) = (-1)^{p(v)p(w)} w \otimes v. \quad (15.3)$$

The sign is negative exactly when both modes belong to the odd sector. Exchange does not modify a mode's labels: it transports the complete mode from one tensor position to another.

Proposition 15.1 (Graded representation of permutations). *The adjacent exchanges $\tau_{p,j}$ on $\mathcal{H}_K^{\otimes n}$ are unitary and satisfy*

$$\begin{aligned}\tau_{p,j}^2 &= I, & \tau_{p,j}\tau_{p,k} &= \tau_{p,k}\tau_{p,j} \quad (|j-k| > 1), \\ \tau_{p,j}\tau_{p,j+1}\tau_{p,j} &= \tau_{p,j+1}\tau_{p,j}\tau_{p,j+1}.\end{aligned}$$

They therefore define a unitary representation U_p of $\mathfrak{S}_n$.

Proof. Applying an exchange twice gives the sign $(-1)^{2p(v)p(w)} = 1$ and restores both vectors. Operators acting on disjoint pairs commute. On a homogeneous triple of degrees a, b, c , both sides of the three-exchange relation produce the same final order and the sign $(-1)^{ab+ac+bc}$. Each operator permutes an orthonormal basis while multiplying it by signs of modulus one, and is therefore unitary. $\square$

The average

$$P_n^p = \frac{1}{n!} \sum_{\pi \in \mathfrak{S}_n} U_p(\pi)$$

is the orthogonal projection onto tensors invariant under graded exchange. On a homogeneous even space it coincides with the ordinary symmetrizer; on a homogeneous odd space it coincides with the antisymmetrizer. If U denotes the unsigned permutation,

$$S_n = \frac{1}{n!} \sum_{\pi} U(\pi), \quad A_n = \frac{1}{n!} \sum_{\pi} \text{sgn}(\pi) U(\pi). \quad (15.4)$$

Proposition 15.2 (Exact symmetrization and antisymmetrization). *One has $S_n^* = S_n^2 = S_n$ and $A_n^* = A_n^2 = A_n$. For $n \geq 2$, $S_n A_n = 0$. Their images are, respectively, $\text{Sym}^n \mathcal{H}$ and $\bigwedge^n \mathcal{H}$.*

Proof. Taking adjoints replaces each permutation by its inverse and leaves the sums invariant. In the square of each average, every permutation ρ occurs $n!$ times; in the second average its coefficient is $\text{sgn}(\rho)$. This proves idempotence. The product of the two projections includes the sum of the signs of all permutations, which is zero for $n \geq 2$. Finally, their images are the invariant and alternating tensors, by their averaging formulas. $\square$

15.3 Universal algebra and normalization of occupation states

Let $T(\mathcal{H}) = \bigoplus_{n \geq 0} \mathcal{H}^{\otimes n}$ be the tensor algebra. The graded algebraic completion is the quotient by the ideal generated by

$$v \otimes w - (-1)^{p(v)p(w)} w \otimes v. \quad (15.5)$$

The universal property of the quotient means that every graded linear map from the channels to an algebra satisfying these relations extends uniquely to a homomorphism from the quotient. Once stated, the exchange rule thus fixes the product of occupations without recourse to target energies.

Theorem 15.3 (Symmetric and exterior decomposition). *If $d_0 = \dim \mathcal{H}_0$ and $d_1 = \dim \mathcal{H}_1$, the graded completion has the decomposition*

$$\mathcal{F}(\mathcal{H}) = \mathcal{F}_+(\mathcal{H}_0) \otimes \mathcal{F}_-(\mathcal{H}_1), \quad (15.6)$$

$$\mathcal{F}_+(\mathcal{H}_0) = \bigoplus_{m \geq 0} \text{Sym}^m \mathcal{H}_0, \quad \mathcal{F}_-(\mathcal{H}_1) = \bigoplus_{k=0}^{d_1} \bigwedge^k \mathcal{H}_1.$$

The total-number sector n is

$$\mathcal{F}_n(\mathcal{H}) = \bigoplus_{k=0}^{\min(n, d_1)} \text{Sym}^{n-k} \mathcal{H}_0 \otimes \bigwedge^k \mathcal{H}_1.$$

Proof. With ordered homogeneous bases fixed, relations (15.5) allow the even factors to be placed first and the odd factors afterwards. Even factors commute; odd factors anticommute, and the square of an odd factor is zero because we work in characteristic zero. Each monomial is written as

$$e_1^{m_1} \cdots e_{d_0}^{m_{d_0}} f_{i_1} \cdots f_{i_k}, \quad m_j \geq 0, \quad i_1 < \cdots < i_k.$$

The realization by symmetric and alternating tensors makes these monomials linearly independent. It provides the inverse map and proves the decomposition. Grouping them by $m_1 + \cdots + m_{d_0} + k = n$ gives the final assertion. $\square$

The Hilbert completion declares the normalized occupation vectors $|\mathbf{m}; I\rangle$ orthonormal, where $\mathbf{m} \in \mathbb{N}_0^{d_0}$ and $I \subset \{1, \dots, d_1\}$. In the bosonic sector of degree $n = \sum_j m_j$, the tensor normalization is

$$|\mathbf{m}\rangle = \sqrt{\frac{n!}{\prod_j m_j!}} S_n(e_1^{\otimes m_1} \otimes \cdots \otimes e_{d_0}^{\otimes m_{d_0}}).$$

For $I = \{i_1 < \cdots < i_k\}$, the exterior vector is

$$|I\rangle = \frac{1}{\sqrt{k!}} \sum_{\pi \in \mathfrak{S}_k} \text{sgn}(\pi) f_{i_{\pi(1)}} \otimes \cdots \otimes f_{i_{\pi(k)}}.$$

In the first formula there are $n! / \prod m_j!$ distinct tensors; in the second there are $k!$. Counting them proves that both norms are one. Degree zero is $\mathbb{C}|0\rangle$; this occupation vacuum is not identified with the constitutive chart of the electromagnetic vacuum or with a vacancy in the TPK record.

15.4 Fermionic creation and derivation of the CAR

Let I be an ordered subset of odd modes and $r_j(I) = \#\{i \in I : i < j\}$. Define

$$\begin{aligned} a_j^\dagger |I\rangle &= \begin{cases} 0, & j \in I, \\ (-1)^{r_j(I)} |I \cup \{j\}\rangle, & j \notin I, \end{cases} \\ a_j |I\rangle &= \begin{cases} (-1)^{r_j(I)} |I \setminus \{j\}\rangle, & j \in I, \\ 0, & j \notin I. \end{cases} \end{aligned} \quad (15.7)$$

The sign counts the transpositions needed to insert or remove the mode while maintaining the order of the exterior basis. It uses neither its mass, its charge, nor the sign of an expressed scalar value. The matrices of the two maps are adjoints of one another and their norms are 1, except in the trivial case with no odd modes.

Theorem 15.4 (Canonical anticommutation relations). *The operators in (15.7) satisfy*

$$\{a_i, a_j^\dagger\} = \delta_{ij} I, \quad \{a_i, a_j\} = 0, \quad \{a_i^\dagger, a_j^\dagger\} = 0.$$

These are exact relations on the whole of $\mathcal{F}_-(\mathcal{H}_1)$.

Proof. For $i = j$, the product $a_j^\dagger a_j$ is the identity on vectors containing j and zero on the others; $a_j a_j^\dagger$ has the complementary actions. Their sum is I . Two insertions or two removals of the same mode give zero. For $i \neq j$, if an operation is prohibited, both terms of the corresponding anticommutator are zero. If both compositions are admissible, they produce the same set of occupations. Changing the order changes by one the number of modes preceding one of the insertions or removals; the signs are opposite. This proves the three anticommutators on every basis vector. $\square$

Corollary 15.5 (Exclusion and occupation of a complete mode). *For $N_j = a_j^\dagger a_j$,*

$$N_j^* = N_j, \quad N_j^2 = N_j, \quad \text{Spec}(N_j) \subset \{0, 1\}.$$

The same complete mode admits at most one fermionic occupation.

Proof. The CAR give

$$N_j^2 = a_j^\dagger a_j a_j^\dagger a_j = a_j^\dagger (I - a_j^\dagger a_j) a_j = N_j,$$

because $a_j^2 = (a_j^\dagger)^2 = 0$. Self-adjointness is immediate; an eigenvalue of a self-adjoint idempotent is 0 or 1. $\square$

The proof constructs the CAR before deducing Pauli exclusion: it does not introduce exclusion as an additional restriction on a particle table. It also explains the real generators used in Proposition 17.5: that proposition now acts on the explicit exterior algebra. Its Majorana interpretation retains the conditions of Section 17.

15.5 Bosonic creation, CCR, and the finite-particle domain

On the normalized bosonic basis, define

$$b_j^\dagger |\mathbf{m}\rangle = \sqrt{m_j + 1} |\mathbf{m} + \mathbf{e}_j\rangle, \quad b_j |\mathbf{m}\rangle = \sqrt{m_j} |\mathbf{m} - \mathbf{e}_j\rangle, \quad (15.8)$$

with value zero in the second expression when $m_j = 0$. The factors arise from the normalization of symmetric tensors. The common domain $\mathcal{D}_{\text{fin}}$ is the algebraic direct sum of the number sectors: vectors have only finitely many nonzero occupation components. It is dense and invariant under the preceding maps.

Theorem 15.6 (Canonical commutation relations). *On $\mathcal{D}_{\text{fin}}$,*

$$[b_i, b_j^\dagger] = \delta_{ij} I, \quad [b_i, b_j] = [b_i^\dagger, b_j^\dagger] = 0.$$

The number operators $N_j^B = b_j^\dagger b_j$ satisfy $N_j^B |\mathbf{m}\rangle = m_j |\mathbf{m}\rangle$.

Proof. For the same mode, $b_j b_j^\dagger |\mathbf{m}\rangle = (m_j + 1) |\mathbf{m}\rangle$ and $b_j^\dagger b_j |\mathbf{m}\rangle = m_j |\mathbf{m}\rangle$. The difference is the identity. For distinct modes, the factors affect different occupation coordinates and therefore commute. The number formulas follow by applying the two operations in (15.8) consecutively. $\square$

If there is an even mode, b_j and $b_j^\dagger$ are unbounded operators on the Hilbert completion: their norms on $|\mathbf{m}_j\rangle$ grow as square roots of m_j . The preceding equalities therefore have an explicit domain and are not presented as identities between finite matrices on the entire bosonic space. The fermionic operators act on the second factor of (15.6); the bosonic operators act on the first. On the common domain, operators on these two factors commute.

Proposition 15.7 (Exact defect of a hard bosonic cutoff). *In one mode, let P_N be the projection onto $|0\rangle, \dots, |N\rangle$ and let $b^{(N)} = P_N b P_N$. On this finite space,*

$$[b^{(N)}, (b^{(N)})^\dagger] = I - (N + 1) |N\rangle \langle N|.$$

Proof. For $m < N$, creation and annihilation retain the calculation $(m + 1) - m = 1$. On $|N\rangle$, truncated creation is zero and the reverse product equals N ; the commutator equals $-N = 1 - (N + 1)$. These are all the basis vectors. $\square$

The correction resides at the last level of the cutoff, not in the occupation law. A finite calculation can verify the CCR away from that boundary and must retain the boundary term when it includes the terminal level. This distinction is indispensable if a matrix certificate is not to conflate a truncation with the full bosonic completion.

15.6 The meaning of “the same state” in the Pauli principle

Exclusion concerns the same vector of a complete mode. Two channels may share a cell, mass, or reduced signature while remaining linearly independent through their spin, sheet, colour, flavour, or another coordinate preserved by transport. These two vectors admit a nonzero exterior product. If a physical quotient identifies them, that quotient must act before one states how many independent modes there are.

Proposition 15.8 (Independence criterion for double occupation). *For two vectors in a channel space,*

$$\|u \wedge v\|^2 = \|u\|^2 \|v\|^2 - |\langle u, v \rangle|^2.$$

In particular, two orthonormal modes produce an exterior state of norm one, and two proportional vectors produce zero.

Proof. The degree-two exterior realization is $(u \otimes v - v \otimes u)/\sqrt{2}$. Expanding its norm gives the formula. Equality in Cauchy–Schwarz characterizes linear dependence; orthonormality gives squared norm one. $\square$

For an orbital channel u and two orthogonal spin components ξ_+, ξ_- , the modes $u \otimes \xi_+$ and $u \otimes \xi_-$ are distinct and their exterior product can be occupied. There is no double occupation of the same mode: there is occupation of two distinct components of the representation. The same applies to two sheets when these remain independent components of the realization.

The central electronic word may be fixed by C_M , while the charge representation distinguishes electron and positron. This coincidence of words does not identify their complete modes or determine their occupation. The word involution, cellular inversion, and exchange of Fock factors retain the differentiated domains of Section 10.

If r orthonormal fermionic modes have the same value of a partial observable, their total occupation can take the values $0, 1, \dots, r$. Each N_j remains a projection; the sum $\sum_{j=1}^r N_j$ need not be one. This consequence distinguishes spectral degeneracy from exclusion of a mode, two properties that an isolated mass value does not separate.

15.7 Finite dimensions and transport by refinement

Proposition 15.9 (Occupation census). *For d modes,*

$$\dim \bigwedge^n \mathbb{C}^d = \binom{d}{n}, \quad \dim \operatorname{Sym}^n \mathbb{C}^d = \binom{d+n-1}{n} \quad (d \geq 1).$$

The first dimension is zero for $n > d$. In the graded space, the dimensions for total number n are obtained by summing the products of the two dimensions for $k = 0, \dots, \min(n, d_1)$.

Proof. An exterior basis is chosen through subsets of n distinct modes. A symmetric basis is chosen through $(m_1, \dots, m_d) \in \mathbb{N}_0^d$ with sum n . Placing $d - 1$ separators among n occupations counts these weak compositions and gives the second binomial coefficient. The graded formula follows from (15.6) by direct sums and tensor products. $\square$

Let $T : \mathcal{H}_K \rightarrow \mathcal{H}_L$ be the stated refinement intertwiners, preserving the grading and corresponding channel data. Each induces

$$\Gamma(T) = \bigoplus_{n \geq 0} T^{\otimes n}|_{\mathcal{F}_n}, \quad T^{\otimes 0} = I_{\mathbb{C}}. \quad (15.9)$$

Theorem 15.10 (Compatibility of completion with transport). *If T preserves degree, then $T^{\otimes n} P_n^p = P_n^{p'} T^{\otimes n}$. If, in addition, $\|T\| \leq 1$, then $\Gamma(T)$ is a bounded operator of norm at most 1 on Fock space and*

$$\Gamma(T_2 T_1) = \Gamma(T_2) \Gamma(T_1), \quad \Gamma(I) = I.$$

Consequently, compatible refinement squares for the channels induce compatible occupation squares.

Proof. Preservation of degree makes $T^{\otimes n}$ commute with each adjacent exchange; by Proposition 15.1, it commutes with the permutations and their average. The restrictions are therefore well defined. Their norm is bounded by $\|T\|^n \leq 1$, so the direct sum is bounded. Identities and compositions are verified in each degree and, by continuity, on the completion. $\square$

Contractivity is a sufficient condition for treating the bosonic sum without additional restrictions. In finite degrees, the powers exist for any linear map, but their sum is not always bounded: for one bosonic mode and $T = 2I$, $\Gamma(T)|n\rangle = 2^n|n\rangle$. In the finite exterior sector there are only $d_1 + 1$ degrees, so any linear map of that sector has a bounded finite second quantization. The difference concerns domains, not causality.

Proposition 15.11 (Transport of creations and annihilations). *Let $T : \mathcal{H}_K \rightarrow \mathcal{H}_L$ be a bounded degree-preserving transport. For $f \in \mathcal{H}_K$ and $g \in \mathcal{H}_L$, on the algebraic finite-particle domain one has*

$$\Gamma(T)c_K^\dagger(f) = c_L^\dagger(Tf)\Gamma(T), \quad (15.10)$$

$$c_L(g)\Gamma(T) = \Gamma(T)c_K(T^*g). \quad (15.11)$$

The subscripts indicate the fibre on which each operator acts. If T is isometric, the second identity specializes to

$$c_L(Tf)\Gamma(T) = \Gamma(T)c_K(f), \quad N_{L,Tf}\Gamma(T) = \Gamma(T)N_{K,f},$$

where $N_{K,f} = c_K^\dagger(f)c_K(f)$ and, for a unit mode, $N_{L,Tf}$ is its transported occupation operator.

Proof. In a symmetric or exterior product, applying T to every factor commutes with inserting f , which becomes Tf . This proves (15.10). Annihilation contracts each factor with the observed mode. For each factor v ,

$$\langle g, Tv \rangle = \langle T^*g, v \rangle.$$

The exterior or graded signs are preserved because T preserves degree; the symmetric normalizations are the same on both sides. The sum of contractions proves (15.11). If $T^*T = I$, substitute $g = Tf$. Composing the creation and annihilation identities then gives the occupation identity. All operations considered preserve the finite-particle domain; when $\|T\| \leq 1$, $\Gamma(T)$ also has the bounded extension of Theorem 15.10. $\square$

The general identity retains the adjoint T^* and is valid without isometry. The need for the isometric condition arises when transporting the same mode. For example, if $T = tI$, with $0 < t < 1$, for a unit mode f and the vacuum Ω ,

$$c(tf)\Gamma(T)f = t^2\Omega, \quad \Gamma(T)c(f)f = \Omega.$$

The general formula correctly gives $\Gamma(T)c(T^*tf)f = t^2\Omega$. Transport by contractions, with its explicit adjoint, and the isometric specialization of occupation are thereby retained.

15.8 Exchange symmetry and the record of a composite

The composition in Section 16.8 acts first on histories and their boundaries. Fock space organizes channel occupation; $\mathfrak{G}_{12}$ organizes the linkage record. These are coordinated operations that do not replace each other. In particular, antisymmetrization does not remove the memory of the coupling order.

Let $\mathcal{D} \subset \mathcal{H}^{\otimes n}$ be the space generated by constituent expressions admissible for an interface. The linearization of composition is written $C_{\mathcal{J}} : \mathcal{D} \rightarrow \mathcal{H}_{\text{comp}}$. When a finite group G of exchanges preserves admissibility and boundary, the specific compatibility is

$$C_{\mathcal{J}}U(g) = V(g)C_{\mathcal{J}}, \quad g \in G,$$

where U, V are unitary representations and V transports the composite records, including their sheet and memory data. The equation is the exchange condition for that realization; it is not inferred from the mere census of expressions.

Proposition 15.12 (Descent of composition to the statistical sector). *Let $\chi : G \rightarrow \{+1, -1\}$ be the character of the exchange sector. Under the preceding compatibility,*

$$P_\chi^{\text{comp}} C_{\mathcal{T}} = C_{\mathcal{T}} P_\chi^{\text{in}},$$

where

$$P_\chi^{\text{in}} = \frac{1}{|G|} \sum_g \chi(g) U(g), \quad P_\chi^{\text{comp}} = \frac{1}{|G|} \sum_g \chi(g) V(g).$$

Proof. Multiplying each intertwining equality by $\chi(g)/|G|$ and summing gives exactly the identity. The averages are projections by the same calculation as in Proposition 15.2. $\square$

The result justifies composing within a symmetric or alternating sector when the boundary has the required transport. If a permutation changes the physical linkage channel, the distinct trees and their interrelations are retained; they are not identified by a symmetrization that forgets that boundary.

15.8.1 Colour antisymmetry and the remainder of the baryonic state

In the baryonic realization, Proposition 16.8 constructs the colour vector $|\epsilon\rangle$, which changes sign under a transposition. After regrouping the factors of the complete state, a factorized vector has the form

$$\Psi = |\epsilon\rangle_{\text{col}} \otimes \Phi_{\text{spin,flavour,orbital,sheet}}.$$

Exchange of two constituents acts simultaneously on both factors. If $U_{\text{rest}}(\pi)\Phi = \Phi$, then the complete state changes sign and belongs to the fermionic sector. Conversely, for a factorized vector with a nonzero colour singlet, total alternation requires the remainder to be symmetric under those same permutations. The proof consists in factoring $U_{\text{tot}}(\pi) = U_{\text{col}}(\pi) \otimes U_{\text{rest}}(\pi)$ and cancelling the colour sign.

The word “remainder” explicitly includes flavour and sheet. Exchanging colours without transporting those components is not the permutation of two complete modes. For a nonfactorizable state, the alternating projection is applied to the complete tensor space, without replacing it by a scalar rule.

15.8.2 Compatibility of physical projections

Colour, channel, and statistics projections can be combined by multiplication when they commute. If P, Q are orthogonal projections and $[P, Q] = 0$, then $(PQ)^* = PQ$, $(PQ)^2 = PQ$, and $\text{im}(PQ) = \text{im } P \cap \text{im } Q$. Inclusion in both images is immediate; for a vector in their intersection, $PQv = v$. If they do not commute, their product does not generally represent the projection for the two simultaneous conditions: the realization must construct the common subspace or the physical order of the operations.

This precision preserves the meaning of the singlet and Pauli projections that occur in the composites of Section 16. Coupling selects a sector; the composite mass is evaluated after constructing that sector’s linkage record.

15.9 Action of the material operator and scope of physical identification

Let $B = B^*$ be a channel operator already obtained from internal dynamics and preserving the grading. In degree n it acts through

$$d\Gamma_n(B) = \sum_{j=1}^n I^{\otimes(j-1)} \otimes B \otimes I^{\otimes(n-j)}.$$

The sum commutes with graded permutations and restricts to $\mathcal{F}_n$. If $B \geq 0$, every restriction is positive. In an eigenbasis, its value is the sum of occupations multiplied by the eigenvalues of B , with the occupations allowed by the preceding proofs. This is the action of an additive observable on independent channels; it does not identify the operator of a bound composite with the sum of its constituents’ isolated masses.

In the bosonic Hilbert sum, $d\Gamma(B)$ has the natural domain

$$\left\{ \psi = (\psi_n) : \sum_n \|d\Gamma_n(B)\psi_n\|^2 < \infty \right\}.$$

Each block is self-adjoint in finite dimension, and its direct sum with this domain is self-adjoint. The linkage or open-channel operator is added only when it has been constructed by the corresponding dynamics. The formation of a complex pole follows the treatment in 16.12.

Complete algebraic results have therefore been obtained: symmetry projections, normalized bases, CAR and CCR on their domains, single-mode exclusion, occupation censuses, and compatibility with refinement intertwiners. The starting point of these results is stated in (15.1), (15.2), and (15.3).

When the realization of rotations satisfies $U(2\pi) = (-1)^p$, the same grading coordinates the restoration of orientation at 4π and the exterior completion. That compatibility is not replaced by a coincidence between two signs. Identification with the relativistic spin–statistics theorem additionally requires the Lorentzian representation, locality, and positivity of the physical realization. Braiding sectors that do not descend to permutations remain in the codomain of Section 17; they are not forced into this bosonic or fermionic completion.

16 Sectorial realizations, material composition, and collection spectra

The atlas of Section 11 and the reader of Section 12 have complementary roles. The former classifies supports, depths, and orbits; the latter preserves the chronology of each trajectory and expresses its mass operator. Applying them to a collection requires both structures to be retained. Equality between mass values does not identify routes; equality between cells does not identify histories, representations, or composite states either.

This section develops the sectorial realizations of that construction. The internal constants, angular pair, absolute scale, and bidirectional readers are those already constructed from APP-TRIT-TPK. The physical representation specifies which information of the state is observed; metrological comparison takes place after that representation has been fixed. In particular, the exposition preserves the distinction between a fiber operator, an exact evaluation for a declared signature, and a spectral line individuated by a coupling.

16.1 Collection descriptors and the spectral criterion for individuation

For an elementary state, write

$$X = (\mathfrak{c}, \rho, g, f, q, \epsilon, \varsigma, \mathfrak{a}).$$

Here $\mathfrak{c}$ is the sectorial class, ρ the representation, g the physical generation, f flavor, q charge, ϵ sheet orientation, ς statistical parity, and $\mathfrak{a}$ the coupling data. The combinatorial depth of the atlas and the physical generation remain coordinates of different types. Flavor distinguishes components of a collection representation; color distinguishes components of the color representation; the sheet and its orientation preserve the operatorial provenance of the trajectory. None of these labels replaces the event register.

Let $\mathcal{P}_{\text{adm}}$ be the set of admissible physical descriptors, let $\mathcal{R}_{\text{adm}}$ be the register of enriched routes produced by the atlas, and let $\mathcal{D}$ be the space of incidence descriptors that preserve sectorial class, representation, depth, charge, color, orientation, and coupling data. Define

$$d_{\mathcal{P}} : \mathcal{P}_{\text{adm}} \longrightarrow \mathcal{D}, \quad d_{\mathcal{R}} : \mathcal{R}_{\text{adm}} \longrightarrow \mathcal{D},$$

where $d_{\mathcal{P}}(X)$ expresses the structural requirements of state X , and $d_{\mathcal{R}}(\gamma)$ expresses the same coordinates preserved by route γ . The fiber, orbit, or multisection produced by the descriptor is the fiber product

$$F_X = d_{\mathcal{R}}^{-1}(d_{\mathcal{P}}(X)) = \mathcal{R}_{\text{adm}} \times_{\mathcal{D}} \{d_{\mathcal{P}}(X)\}. \quad (16.1)$$

For the realizable descriptors in this article, F_X is finite and nonempty. On $\mathcal{H}_X = \ell^2(F_X) \otimes V_{\rho_X}$ act the two operators $M_X^{\beta, D}$ and $M_X^{\beta, \Omega}$ of Section 12. Since they are diagonal in the route basis, their joint spectral measure is

$$E_X(B) = \sum_{\gamma \in F_X} \mathbf{1}_B(m_D(\gamma), m_{\Omega}(\gamma)) |\gamma\rangle\langle\gamma| \otimes I_{V_{\rho_X}}, \quad B \subset \mathbb{R}_{>0}^2. \quad (16.2)$$

Preserving this measure allows two fibers to be distinguished even when their means coincide. A scalar functional of the two readers is applied only when it forms part of the declared representation. If the coupling preserves several values, the output is a spectrum with multiplicities.

Proposition 16.1 (Spectral-line criterion). *Let M be a self-adjoint operator on a finite fiber and P a preparation projector. The channel expresses exclusively the value m if and only if*

$$E_M(\{m\})P = P, \quad \text{equivalently} \quad (M - mI)P = 0.$$

The equality $PMP = mP$ alone determines a scalar compression, but does not guarantee that the prepared spectral support is a single point.

Proof. By the spectral theorem, $\ker(M - mI)$ is the image of $E_M(\{m\})$. The first two conditions state that $\text{im } P$ is contained in that eigenspace. To verify the last distinction, take $M = \text{diag}(1, 3)$ and the projector onto $(1, 1)/\sqrt{2}$. Then $PMP = 2P$, whereas the prepared mass distribution has support $\{1, 3\}$, not $\{2\}$. $\square$

Individuation thus preserves the scope of each result. The fiber and operator are determined objects; passage to a line uses the projector and spectral condition stated above. This formulation will apply equally to charged, neutral, and composite collections.

16.2 Charged leptons: electronic center and generation multisections

The three charged multisections have ranks 1, 8, 16 at depths G_0, G_2, G_4 . They share the charge representation and trivial color, but not the support rank or trajectory register. The electron occupies the center (5, 5); the muonic and tauonic realizations preserve their multisections until spectral coupling is performed. Conjugation of their representations is treated through the operators of Section 10, not through a new choice of masses.

The basal electronic coordinate e_0 provides the intermediate angular chart

$$m_X^{(0)} = e_0 \exp\left(n_A(X)x + \frac{s_C(X)}{6}y\right), \quad x = A_{\text{rad}}, \quad y = C_{\text{rad}}^*. \quad (16.3)$$

In this formula, e_0 is the basal section, not the beta electronic closure. Bidirectional return belongs to the subsequent stage of the construction. The two stages preserve their own normalization.

The recovered nominal registers for the two noncentral collections express

$$(n_A, s_C)_\mu = (42, -3), \quad (n_A, s_C)_\tau = (64, 0),$$

and their declared complete signatures are

$$\sigma_\mu = (42, -3, 8, 0, 2), \quad \sigma_\tau = (64, 0, 25, -1, 6). \quad (16.4)$$

To explain the effect of each coordinate, denote by $\mathcal{E}_j$ the successive readings of a declared signature:

$$\begin{aligned} \mathcal{E}_0(\sigma) &= n_A x + s_C y / 6, \\ \mathcal{E}_1(\sigma) &= \mathcal{E}_0(\sigma) + k_{\Delta_4} \Delta_4, \\ \mathcal{E}_2(\sigma) &= \mathcal{E}_1(\sigma) + b_{120} s_{120}, \\ \mathcal{E}_3(\sigma) &= \mathcal{E}_2(\sigma) + b_\zeta \zeta_{270}, \quad \zeta_{270} = 135 \Delta_4^2. \end{aligned} \quad (16.5)$$

The symbol $\mathcal{E}_3$ here denotes an evaluation exponent, not a species or an additional mode. The beta supplement preserving these signatures uses the explicit normalization $m_X^{[3]} = m_e^\beta \exp \mathcal{E}_3(\sigma_X)$. This is an evaluation relative to the supplement's signatures; it does not replace the route-by-route construction of $M_X^{\beta,D}$ and $M_X^{\beta,\Omega}$.

The difference of signatures gives, without approximation,

$$\log \frac{m_\tau^{[3]}}{m_\mu^{[3]}} = 22x + \frac{1}{2}y + 17\Delta_4 - s_{120} + 4\zeta_{270}. \quad (16.6)$$

The proof consists of subtracting the exponents in (16.5). The angular term, torsional memory, and two return readings remain identified in the result. The preserved evaluations are

$$\begin{aligned} m_\mu^{[3]} &= 105.658360294435829303 \dots \text{ MeV}/c^2, \\ m_\tau^{[3]} &= 1776.860917631432295595 \dots \text{ MeV}/c^2. \end{aligned}$$

Their documentary status is exact evaluation upon fixing the recovered signature. Physical selection of that signature within the multisection and the spectral-line criterion are distinct operations and are preserved as such.

At the electronic center the rank is one, and the readers coincide at $(80, 54, 6)$. The active history $(2, 2, 5, -4, -3, -2)^2$ preserves more information than its historical envelope $(4, 3, 5, -4, -3, -5)^2$, although both share the sign pattern. This distinction links the electronic particularity to the positional memory of events: collections are not defined solely by the visual form of a sign word.

16.3 Neutrinos: neutrality, mass spectrum, and flavor transport

The neutrino support has dimension $20 = 2 + 4 + 6 + 8$, distributed over four depths. Its three flavor labels ν_e, ν_μ, ν_τ and its three possible mass eigenvalues belong to the physical realization of the sector, not to the census of depths. Electrical neutrality preserves orientation, sheet memory, and spinorial holonomy. It therefore does not eliminate the variables involved in mixing.

The neutral register separates the contributions of two half-cycles. If J_m is the block of 9 iterations of phase m , θ_t marks the visit to 9, ϵ_t preserves the sheet, and ω_t its corona orientation, the local reader is

$$T_\nu(m) = \sum_{t \in J_m} \omega_t \epsilon_t \theta_t, \quad e_m^\nu = T_\nu(m) - T_\nu(m+6).$$

16.3.1 Exact cancellations of the antipodal extractor

Indices in this subsection are read in $\mathbb{Z}/12\mathbb{Z}$, and $\text{sgn}(0) = 0$. Let A_r be the phase shift $(A_r T)(m) = T(m+r)$. For the neutral register,

$$e^\nu = (I - A_6)T_\nu, \quad e_{m+6}^\nu = -e_m^\nu. \quad (16.7)$$

Define

$$B_a^\nu = e_a^\nu + e_{a+4}^\nu + e_{a+8}^\nu, \quad \tau_m^\nu = \text{sgn}(e_m^\nu), \quad c_m^\nu = \tau_m^\nu \tau_{m+3}^\nu.$$

Proposition 16.2 (Vanishing of the two antipodal scalar readers). *The neutral extractor satisfies*

$$B_{a+2}^\nu = -B_a^\nu, \quad b_{120}^\nu := \sum_{a=0}^3 \text{sgn}(B_a^\nu) = 0, \quad (16.8)$$

and

$$c_{m+3}^\nu = -c_m^\nu, \quad b_\zeta^\nu := \sum_{m=0}^{11} c_m^\nu = 0. \quad (16.9)$$

Proof. By (16.7),

$$B_{a+2}^\nu = e_{a+2}^\nu + e_{a+6}^\nu + e_{a+10}^\nu = -e_{a+8}^\nu - e_a^\nu - e_{a+4}^\nu = -B_a^\nu.$$

The four summands in b_{120}^ν cancel in the pairs $(0, 2)$ and $(1, 3)$, including when one of them vanishes. Likewise, $\tau_{m+6}^\nu = -\tau_m^\nu$, whence

$$c_{m+3}^\nu = \tau_{m+3}^\nu \tau_{m+6}^\nu = -\tau_{m+3}^\nu \tau_m^\nu = -c_m^\nu.$$

The sum of the twelve terms vanishes in alternating pairs. □

Scalar cancellation does not destroy oriented information. On the anti-invariant subspace

$$S_- := \ker(A_6 + I), \quad J := A_3|_{S_-},$$

one has $J^2 = -I$. Since the shift is unitary, $J^* = J^{-1} = -J$: the register retains an oriented complex structure. Moreover,

$$e^\nu = (I - A_6)T_\nu = 2P_- T_\nu, \quad P_- = (I - A_6)/2,$$

so the complete antipodal vector remains available even though its two scalar contractions vanish.

The cyclic return $A_{12} = I$ concerns this phase representation; it is not the return of the complete TPK state, which retains carry, sheet, boundary, and memory. For the same reason, (16.8)–(16.9) do not annul the general coefficients b_{120} and b_ζ of a non-antipodal route, nor do they modify masses, composite signatures, or PMNS transport. The longitudinal seam used in register prolongation is controlled independently by Lemma 12.2.

This difference retains the oriented information even when its total sum vanishes. The complete operator also uses torsional predecessors, vacancies, and transport between sheets. Passage to the physical subspace requires the rank-3 neutral projector,

$$M_\nu = P_\nu \mathcal{M}_\nu P_\nu \quad \text{on } \text{im } P_\nu.$$

The eigenvalues of that compression are the masses in the internal action–clock–mass scale. Mixing angles act between their frames; they neither replace the scale nor retrospectively generate the eigenvalues. The compression coincides with the restriction of the complete operator when $\text{im } P_\nu$ is a reducing subspace of $\mathcal{M}_\nu$.

The register allows a Gram kernel to be assembled before diagonalization. For each data type $r \in \{\text{car, hoj, ori, vac, hol}\}$, let $\Xi_r(c)$ be the vector produced by cell or route c . Define

$$K_{\text{reg}}(c, d) = \sum_{r: N_r > 0} \frac{\langle \Xi_r(c), \Xi_r(d) \rangle}{N_r}, \quad N_r = \sum_a \|\Xi_r(a)\|^2. \quad (16.10)$$

Omission of a term with zero total norm is fixed before evaluation. It preserves a well-defined domain and introduces no fit.

Proposition 16.3 (Positivity and rank of the neutral register). *The kernel K_{reg} is Hermitian and positive semidefinite. For a three-dimensional aggregation G , the condition $\det G \neq 0$ certifies that the aggregated data separate three directions. For any invertible square matrix G , its polar factor*

$$U_G = G(G^*G)^{-1/2}$$

is unitary. This factor does not by itself determine the spectral frames of the charged and neutral operators.

Proof. For coefficients z_c , the quadratic form of (16.10) is

$$\sum_{c,d} \bar{z}_c K_{\text{reg}}(c, d) z_d = \sum_{r: N_r > 0} N_r^{-1} \left\| \sum_c z_c \Xi_r(c) \right\|^2 \geq 0.$$

The rank is the dimension of the image of the aggregated vectors. For the last statement, $U_G^* U_G = (G^* G)^{-1/2} G^* G (G^* G)^{-1/2} = I$; invertibility and equality of dimensions also give $U_G U_G^* = I$. The formula uses matrix G , whereas a spectral frame requires diagonalization of the corresponding mass operator. $\square$

If $F_\ell, F_\nu : \mathbb{C}^3 \rightarrow \mathcal{H}$ are complete orthonormal frames of the same physical subspace, then $U_{\text{PMNS}} = F_\ell^* F_\nu$ is unitary. If their images are different, the product is only a contraction; unitarity requires equality of the transported subspaces. In the presence of degeneracies, mixing is determined as a double coset under frame changes in the degenerate blocks, not as a numerical matrix chosen by convention.

The minimal phase construction

$$\Phi_K = I + (e^{i\pi/6} - 1)|b_K\rangle\langle b_K|, \quad \|b_K\| = 1,$$

remains a documented negative control: it is unitary, but the best permutation of its angular identification gives $\theta_{13} \simeq 17.02^\circ$, incompatible with the comparison for that reduced model. The result concerns the rank-one phase. It does not replace the complete kernel (16.10), nor permit a mixing matrix that has not passed its own rank and angle checks to be presented as validated.

16.3.2 Neutral self-conjugation and the possibility of a sterile sector

The Dirac–Majorana alternative uses the fermionic representation. A neutral real structure must satisfy

$$\mathcal{C}_\nu^2 = I, \quad \mathcal{C}_\nu M_\nu = M_\nu \mathcal{C}_\nu, \quad \mathcal{C}_\nu H_\nu = H_\nu \mathcal{C}_\nu,$$

with $\mathcal{C}_\nu$ antiunitary and a compatible realification. Commutation with mass preserves its real part. For the self-adjoint Hamiltonian, by contrast, antiunitarity also conjugates the factor i : if $U_\nu(t) = \exp(-itH_\nu/\hbar)$, then

$$\mathcal{C}_\nu U_\nu(t) \mathcal{C}_\nu^{-1} = U_\nu(-t).$$

Consequently, the fixed modes $\psi = \mathcal{C}_\nu \psi$ define a neutral real structure, but the stated conditions are insufficient to restrict the entire Schrödinger evolution to it. Preservation of that fixed part requires $\mathcal{C}_\nu$ to commute with the group $U_\nu(t)$; Proposition 17.3 proves the corresponding condition on its generator. A Majorana realization must also declare the compatibility of self-conjugation with the fermionic representation and its dynamics. A nonzero conserved charge that changes sign under $\mathcal{C}_\nu$ keeps conjugate modes separate in the Dirac realization. The word involution of Section 10 and electrical neutrality, considered in isolation, do not decide this alternative.

A sterile sector requires a nonzero projector P_s with $P_s P_{\text{act}} = 0$ and $P_s J_{\text{weak}} = 0$. If $[P_s, M_\nu] = 0$, it is reducing and does not mix with the active sector. If $P_s M_\nu P_{\text{act}} \neq 0$, mixing exists and the projector ceases to commute with mass. Consequently, an additional atlas depth does not automatically count as a fourth neutrino species.

16.4 Quarks: flavor, color orbit, and scale transport

The quark state preserves the tuple

$$(\sigma_{ud}, g, [\gamma]_{\text{col}}, \mathfrak{M}, \mathcal{L}),$$

which distinguishes up or down type, generation, color orbit, multisection, and register. Section 11 constructs the color residue $\kappa_{\text{col}}(r, c) = r - c \pmod{3}$ and the three classes of 12 cells. The color index acts within the support; flavor preserves generation and signature information that this residue does not identify.

The six nominal angular pairs are

f	n_A	s_C	W_+	W_-	
u	11	7	73	59	
d	17	8	110	94	
c	61	8	374	358	$W_\pm = 6n_A \pm s_C.$
s	41	-3	243	249	
t	100	-1	599	601	
b	71	-5	421	431	

These last integers rewrite the same exponent:

$$n_A x + s_C y / 6 = \frac{W_+(x+y) + W_-(x-y)}{12}.$$

Adding and subtracting $W_\pm$ recovers n_A and s_C , so that this presentation preserves both channels. The six pairs belong to the documented angular chart; the three remaining coordinates are not filled with zeros to manufacture a complete signature. Their color operators and angular evaluations are preserved as results of different scope.

Proposition 16.4 (Color degeneracy of a flavor mass). *Let $V_{\text{col}} \simeq \mathbb{C}^3$ be the fundamental color representation and U its irreducible unitary action. The reduction*

$$\mathbb{E}_{\text{col}}(M) = \int_{SU(3)} U(g) M U(g)^* dg$$

is a multiple of I_3 . If M already commutes with the action, its three color components have the same flavor-mass value.

Proof. Invariance of Haar measure gives $U(h)\mathbb{E}_{\text{col}}(M)U(h)^* = \mathbb{E}_{\text{col}}(M)$. Schur's lemma makes this operator scalar. Since trace is preserved, the scalar is $\text{Tr } M/3$. If M is invariant from the outset, the integral returns M . $\square$

The proof explains what the color quotient eliminates: the dependence of the numerical value on the red, green, or blue representative. It does not eliminate the color routes subsequently involved in a bound state. Nor does it turn this reduction into a flavor selector.

The current quark mass is presented as a section at a declared scale and in a declared scheme, $m_q(\mu, \mathcal{R})$. If $t = \log \mu$ and the connection induces $\gamma_m(g(t))$, transport satisfies

$$\left(\frac{d}{dt} + \gamma_m(g(t))\right)m_q = 0, \quad m_q(t_2) = \mathcal{P} \exp\left(-\int_{t_1}^{t_2} \gamma_m(g(t)) dt\right)m_q(t_1).$$

The connection and its endomorphism are transport data, not numbers chosen to match a table. A physical residual is meaningful only after both sections have been placed at the same scale and in the same scheme.

CKM mixing acts on another coordinate: the flavor frame. Once the internal unitary matrix has been constructed, the down operator, expressed in the up frame, is $B_d^{(u)} = V_{\text{CKM}} B_d V_{\text{CKM}}^*$. Unitary conjugation preserves the characteristic polynomial, since

$$\det(zI - V B_d V^*) = \det(V(zI - B_d)V^*) = \det(zI - B_d).$$

The role of the angles is thus understood: they relate frames without becoming retrospective generators of mass eigenvalues.

16.5 Zero-mass charts: the photon and the adjoint color representation

The null domain is separated before the positive character is applied. In the total realization,

$$\mathcal{H}_{\text{phys}} = \mathcal{H}_0 \oplus \mathcal{H}_+, \quad M|_{\mathcal{H}_0} = 0, \quad M|_{\mathcal{H}_+} > 0.$$

The exponential of a finite signature does not vanish. Thus a null chart is not obtained by assigning an arbitrarily large signature to a positive route and calling its approximation zero.

The photon belongs to the electromagnetic chart, with its gauge representation and its two physical transverse polarizations. Gluons belong to the adjoint color representation, of dimension 8. Equality of their zero masses does not identify the two representations. The fiber, gauge, polarizations, and coupling channels remain part of the state.

Oriented propagation with zero mass is also compatible. In the documented chart,

$$c_{\pm} = \frac{c}{1 \pm \varepsilon}, \quad \varepsilon = q_-^{90} - q_+^{90}, \quad 0 < \varepsilon < 1.$$

Then $c_{\pm} > 0$ and

$$\frac{2}{c_+^{-1} + c_-^{-1}} = c.$$

The identity follows by adding $(1 + \varepsilon)/c$ and $(1 - \varepsilon)/c$. The asymmetry of the two readings preserves its orientation information; it does not by itself express a mass term.

16.6 Massive bosons and the scalar sector

The descriptors of W , Z , and the scalar sector are not forced into one of the 13 fermionic multisections. They preserve their representation and realization domain. The documented nominal readings are

X	nominal route	direction	(n_A, s_C)	(W_+, W_-)
W	15/37	$E-O$	$(94, -1)$	$(563, 565)$
Z	20/23	$E-O$	$(95, -1)$	$(569, 571)$
H	24/29	$N-S$	$(97, 9)$	$(591, 573)$

These route names belong to the preserved nominal register. Their joining to the complete oriented domain must retain the trajectory, representation, and projector data. The expressed cardinal counts, respectively $(23, 44, 11, 30)$, $(33, 45, 10, 20)$, and $(40, 34, 22, 12)$, are preserved together with the direction and are not replaced by a single label.

For W and Z , the complete signatures are recovered:

$$\sigma_W = (94, -1, 0, -2, 4), \quad \sigma_Z = (95, -1, -11, 0, -4).$$

Reading (16.5) produces, for these signatures,

$$m_W^{[3]} = 80378.964909704547024865 \dots \text{ MeV}/c^2,$$

$$m_Z^{[3]} = 91187.533444313407844855 \dots \text{ MeV}/c^2.$$

Their ratio preserves exactly the difference of the evaluated registers:

$$\log \frac{m_Z^{[3]}}{m_W^{[3]}} = x - 11\Delta_4 + 2s_{120} - 8\zeta_{270}.$$

By contrast, the nominal coordinate $(97, 9)$ of the scalar sector belongs to the documented angular chart $\mathbb{Z}^2$: this expression does not attribute to it a $\mathbb{Z}^5$ signature or an order-3 refinement. Its realization as a scalar excitation requires the scalar projector and the corresponding gauge and fermionic couplings.

For an unstable species, the physical result is a pole of the channel operator, not merely the evaluation of a positive character. Subsection 16.12 constructs this reading and separates pole mass and width. In the preserved documentary register, the preceding numbers are signature-conditioned evaluations; the individuation of route, projector, and pole constitutes the sectorial realization that must accompany them.

16.7 Finite grammar of material compositions

Let $\mathcal{Q}_a$, $a \in \mathbb{Z}/3\mathbb{Z}$, be the three color classes of the atlas, each of cardinality 12. The formal dual $q^\vee$ preserves the conjugate support and reverses charge and color orientation. It is not merely the cell $\rho_c(q)$ stripped of these data. Mesonic and baryonic expressions are defined by

$$\begin{aligned} \mathfrak{M} &= \{q \otimes (q')^\vee : q, q' \in \mathcal{Q}_a \text{ for some } a\}, \\ \mathfrak{B} &= \mathcal{Q}_0 \times \mathcal{Q}_1 \times \mathcal{Q}_2. \end{aligned} \tag{16.11}$$

The pairs are ordered and the color positions of the triple are fixed. Residual neutrality is, respectively, $a - a = 0$ and $0 + 1 + 2 = 0 \pmod{3}$.

Proposition 16.5 (Internal compositional census). *The grammar contains 432 mesonic expressions, 1728 baryonic expressions, and 372 compatible charged arrows, divided into 320 leptonic and 52 quark arrows. These cardinalities count internal expressions and arrows, not physical species or independent mass predictions.*

Proof. For each color, an ordered pair of 12 cells is chosen; therefore, $|\mathfrak{M}| = 3 \cdot 12^2 = 432$. The baryon's three positions give $|\mathfrak{B}| = 12^3 = 1728$. Lepton–neutrino transitions of equal depth share only G_2, G_4 ; in both directions this gives $2(8 \cdot 4 + 16 \cdot 8) = 320$.

Up–down transitions additionally require equality of color. In G_1 , the color multiplicities are $(2, 1, 1)$ and $(0, 1, 1)$; in G_3 , they are $(4, 4, 4)$ and $(2, 2, 2)$. There are $2 + 24 = 26$ pairs in one direction and 52 in both. Adding $320 + 52$ completes the census. All factors come from the atlas preceding mass evaluation. $\square$

The 81 cell identities complete the neutral arrows of this grammar, but a compositional identity does not thereby become a neutral boson. The representation and realization channel are the data distinguishing the two notions.

16.8 Composition of the register and bonding memory

A composite contains more information than the list of its constituents. Let L_1, L_2 be enriched registers, g the transport aligning phase, sheet, and orientation, and B the subtrace identified at the common boundary. In action and event coordinates, composition is

$$(n, e)_1 \star_{(g, B)} (n, e)_2 = (n_1 + n_2 - n_B, e_1 + g e_2 - e_B). \quad (16.12)$$

Subtraction prevents the same subtrace from being counted twice; transport preserves orientation at the interface. Torsional predecessors and corona marks are composed at the same level before L_{tick} is applied again. The reversible event decomposition of Proposition 12.1 permits the complete register to be transported without reducing it to its total.

For an ordered tree $\mathcal{T}$ of n constituents, this gives

$$(\Gamma_1, \dots, \Gamma_n; \mathcal{T}, \partial) \xrightarrow{\mathfrak{G}_{12}^{(n)}} L_Y \xrightarrow{L_{\text{tick}}} \sigma_Y \mapsto (\eta_Y, M_Y). \quad (16.13)$$

The boundary ∂ includes the transports and subtraces of each tree edge. Once these admissible data are fixed, each composition is an explicit operation. The character is applied to the final signature; it does not preselect the tree through the destination mass.

Proposition 16.6 (Compatibility of parenthesization). *In a register chart where $L_1 \star L_2 = L_1 + L_2 + \mathfrak{b}(L_1, L_2)$, the two parenthesizations of an admissible triple coincide if and only if*

$$\mathfrak{b}(L_1, L_2) + \mathfrak{b}(L_1 \star L_2, L_3) = \mathfrak{b}(L_2, L_3) + \mathfrak{b}(L_1, L_2 \star L_3).$$

The equality is required on the domain where both trees represent the same oriented interface.

Proof. Expanding $(L_1 \star L_2) \star L_3$ gives $L_1 + L_2 + L_3 + \mathfrak{b}(L_1, L_2) + \mathfrak{b}(L_1 \star L_2, L_3)$. Expanding $L_1 \star (L_2 \star L_3)$ gives the same sum of three registers and the two terms on the right-hand side. Their equality is exactly equivalent to the stated identity. $\square$

The cocycle law ensures independence of an equivalent parenthesization; it does not identify physically different trees. Two bonding structures preserving different channels remain distinct realizations until an intertwiner of registers, projectors, and operators is available.

16.8.1 A finite witness of the information a sum of signatures would lose

The historical reduced event extractor is defined by

$$F_0(n, e) = \left(n, \sum_m e_m, K_0(e), \sum_{a=0}^3 \text{sgn}(e_a + e_{a+4} + e_{a+8}), \sum_m \tau_m \tau_{m+3} \right),$$

where $\tau_m = \text{sgn}(e_m)$ and $K_0(e) = \sum_{\tau_m=0} \Sigma_{12}(m)$, with $\Sigma_{12} = (+, +, +, -, -, -)^2$. Its third coordinate is a reading of null events; it is not the strong torsion $k_{\Delta_4} = T_z - 2C_z$ of Section 12.

Consider the registers of the routes

$$a = (1, 3, -1, +1), \quad b = (6, 7, -1, +1), \quad c = (1, 2, -1, -1),$$

with

$$\begin{aligned} e(a) &= (5, 4, 4, -4, -4, -6)^2, \\ e(b) &= (4, 5, 4, -4, -6, -4)^2, \\ e(c) &= (1, 4, 1, -2, -4, -3)^2. \end{aligned}$$

They have $F_0(a) = F_0(b) = (8, -2, 0, 0, -12)$ and $F_0(c) = (28, -6, 0, -4, -12)$. Superposing the registers gives

$$F_0(a \boxplus c) = (36, -8, 0, 0, -12), \quad F_0(b \boxplus c) = (36, -8, 0, -2, -12).$$

The projected inputs are equal and the composite outputs differ. Consequently, no operation on the two five-tuples of this projection recovers all superpositions. The example proves a precise obstruction for F_0 ; it does not attribute the same result to every possible extension of a signature. The current construction avoids this loss through (16.13): it preserves the events and predecessors, composes them, and only then evaluates the complete extractor.

16.9 Mesons: dual pair, singlet, and differentiation of excitations

The expression $q \otimes (q')^\vee$ establishes the ordered pair and its residual neutrality. The color singlet belongs to the realization $V \otimes \bar{V}$, $V = \mathbb{C}^3$. If $\Omega = \sum_{a=1}^3 |a\bar{a}\rangle$, the projector is

$$P_M^{(1)} = \frac{1}{3} |\Omega\rangle\langle\Omega|. \quad (16.14)$$

Proposition 16.7 (Mesonic color projection). *The map $P_M^{(1)}$ is a rank-one orthogonal projector, and its image is invariant under $U \otimes \bar{U}$, $U \in SU(3)$.*

Proof. We have $\langle\Omega, \Omega\rangle = 3$, hence $(P_M^{(1)})^2 = P_M^{(1)}$. Moreover,

$$(U \otimes \bar{U})\Omega = \sum_{a,b,c} U_{ba} \bar{U}_{ca} |b\bar{c}\rangle = \sum_b |b\bar{b}\rangle = \Omega.$$

The image thus preserves the color contraction without privileging any color. $\square$

The mesonic descriptor combines flavors, spin and parity, isospin and charge, radial or orbital excitation, bonding tree, and boundary. The label J^{PC} is used when C is defined for the channel; charge conjugation is not automatically assigned to a charged state. Two states with the same flavors may have different radial, orbital, or channel projectors and, therefore, different spectra. The orbital construction of Section 18 provides a possible reading when its dynamical hypotheses hold; orbital numbers do not follow from the meson's name alone.

The composite-evaluation register preserves

	n_A	s_C	k_{Δ_4}	b_{120}	b_ζ
π^0	44	-4	-25	1	-2
$\pi^\pm$	44	1	-2	2	-1
$K^\pm$	54	-1	17	-2	2
K^0	54	0	32	3	-2

For these signatures, the same reading (16.5) explains the differences within the pairs:

$$\begin{aligned}\log \frac{m_{\pi^\pm}^{[3]}}{m_{\pi^0}^{[3]}} &= \frac{5}{6}y + 23\Delta_4 + s_{120} + \zeta_{270}, \\ \log \frac{m_{K^0}^{[3]}}{m_{K^\pm}^{[3]}} &= \frac{1}{6}y + 15\Delta_4 + 5s_{120} - 4\zeta_{270}.\end{aligned}\tag{16.15}$$

The action terms $44x$ and $54x$ cancel within each pair. The remaining differences preserve which register coordinate produces each contribution. They are exact consequences of the declared signatures, not a substitute proof of their generation by the bonding tree.

The resulting evaluations are

$$\begin{aligned}m_{\pi^0}^{[3]} &= 134.976724327872125739 \dots \text{ MeV}/c^2, \\ m_{\pi^\pm}^{[3]} &= 139.570485171572191374 \dots \text{ MeV}/c^2, \\ m_{K^\pm}^{[3]} &= 493.677085649530612475 \dots \text{ MeV}/c^2, \\ m_{K^0}^{[3]} &= 497.611186497750457358 \dots \text{ MeV}/c^2.\end{aligned}$$

Each numerical value retains the status of an exact evaluation conditioned on the recovered composite signature. The singlet, grammar, and register extraction are constructed; complete nominal realization requires the binding tree to regenerate that signature before comparison begins.

16.10 Baryons: color triple, exchange, and the neutron–proton difference

The neutrality of $q_0 \otimes q_1 \otimes q_2$ preserves the three color positions. In the tensor realization, introduce the normalized vector

$$|\varepsilon\rangle = \frac{1}{\sqrt{6}} \sum_{a,b,c=1}^3 \varepsilon_{abc} |abc\rangle, \quad P_B^{(1)} = |\varepsilon\rangle\langle\varepsilon|.$$

Proposition 16.8 (Baryonic singlet and exchange rule). *The operator $P_B^{(1)}$ is a rank-one orthogonal projector, invariant under $U^{\otimes 3}$, $U \in SU(3)$. A permutation of two positions reverses the sign of its generating vector and preserves the projector.*

Proof. There are 6 nonzero orthogonal terms, each of modulus $1/\sqrt{6}$; hence the vector has norm one. The determinant identity gives $U^{\otimes 3}|\varepsilon\rangle = \det(U)|\varepsilon\rangle = |\varepsilon\rangle$. A transposition reverses the sign of ε_{abc} . In the outer product of the vector with its adjoint, the two signs cancel. $\square$

Color antisymmetry is one part of the complete state. The physical projector combines color, spin and parity, isospin, charge, and the exchange symmetry of orbital and flavor components. Pauli's principle acts on the complete fermionic product; applying it to a bare list of colors is insufficient. The triple tree records which pair was coupled first and how the third constituent crossed the boundary. Proposition 16.6 controls equivalent parenthesizations.

The recovered proton and neutron signatures are

$$\sigma_p = (59, 0, 9, 1, 0), \quad \sigma_n = (59, 1, -34, 0, 1).$$

They share the action coordinate 59, but differ in sheet, torsion, and returns. Their exponent difference is

$$\log \frac{m_n^{[3]}}{m_p^{[3]}} = \frac{1}{6}y - 43\Delta_4 - s_{120} + \zeta_{270}.\tag{16.16}$$

The equality is proved by subtracting the five coordinates. In this signature, -34 is a torsional integer of the neutron; it is not identified with the decimal exponent -34 of the absolute action scale. Equality of two integers in different maps does not eliminate their type or provenance.

The preserved evaluations are

$$\begin{aligned} m_p^{[3]} &= 938.271751546984898298 \dots \text{ MeV}/c^2, \\ m_n^{[3]} &= 939.565810472507023312 \dots \text{ MeV}/c^2. \end{aligned}$$

Formula (16.16) locates the effect of each component of the nominal signatures. The physical derivation of the difference requires isospin, vacancy orientation, the bonding boundary, and the electromagnetic contribution to be transported into a single chart. Character evaluation preserves its exactness under the fixed signature; that exactness does not replace the prospective construction of the bond.

16.11 Higher-order compositions and channel equivalence

The map $\mathfrak{G}_{12}^{(n)}$ admits an arbitrary finite number of constituents. A tetraquark uses $qq\bar{q}\bar{q}$; a pentaquark, $qqqq\bar{q}$; a gluonic bound state uses adjoint representations projected onto the singlet; a hybrid preserves fermionic and gauge constituents in the same tree. Nuclei, atoms, and molecules use the same precedence of register and reading, with their own couplings and boundaries.

Binding energy belongs to the realization of the composite register. It is not obtained by adding isolated masses and naming the desired residual a binding contribution. Even when a cocycle takes additive values on registers, the sign and correlation extractor is applied afterwards; the multiplicative character of a signature therefore does not turn the composite mass into a sum of constituent masses.

If two trees $\mathcal{T}_1, \mathcal{T}_2$ have a unitary V intertwining their operators and channel projectors,

$$VH_1 = H_2V, \quad VP_1 = P_2V,$$

then $V(z - H_1)^{-1} = (z - H_2)^{-1}V$ on the common resolvent set. Its continuation, when it exists and is transported in the same chart, preserves poles and their multiplicities. This equality gives the operational criterion for identifying two realizations of a composite; sharing flavors or an approximate mass is insufficient.

16.12 Resonances: pole mass, width, and survival memory

Let P be the prepared subspace and $Q = I - P$ that of the eliminated channels. In a region where $z - QHQ$ is invertible, eliminating the component $Q\psi$ from $(z - H)\psi = 0$ gives

$$H_{\text{eff}}(z) = PHP + PHQ(z - QHQ)^{-1}QHP. \quad (16.17)$$

Proposition 16.9 (Reduced reading of a bound or resonant state). *If $P\mathcal{H}$ is finite-dimensional, the admissible spectral values of the reduced problem satisfy $\det(z - H_{\text{eff}}(z)) = 0$. A resonance additionally requires meromorphic continuation to the relevant channel chart and an isolated zero with admissible physical residue.*

Proof. The equation in the complement is $(z - QHQ)Q\psi = QHP\psi$. Solving it and substituting into the equation for $P\psi$ produces $(z - H_{\text{eff}}(z))P\psi = 0$. In finite dimension, a nonzero solution exists exactly when the determinant vanishes. The algebraic calculation is performed on the resolvent domain; passage to a resonance uses the stated continuation and residue, which do not follow merely from a channel being open. $\square$

A stable state is a normalizable spectral line with no open decay channel; a bound state lies below the relevant threshold, although it may decay through another channel; a resonance is a pole of the continuation. If

$$z_X = M_X c_{\text{int}}^2 - \frac{i}{2} \Gamma_X, \quad \Gamma_X > 0,$$

the real and imaginary parts are two coordinated readers of the same pole. Width is not a sixth coordinate of σ .

The discrete reading uses a survival eigenvalue $\lambda_X = re^{-i\theta}$, $0 < r < 1$, and a time step $t_0 > 0$. Once the branch $\theta \in I_\theta$ of the quasienergy chart is fixed,

$$z_X = \frac{\hbar_{\text{int}}}{t_0}(\theta + i \log r), \quad M_X = \frac{\hbar_{\text{int}}\theta}{t_0 c_{\text{int}}^2}, \quad \Gamma_X = -\frac{2\hbar_{\text{int}}}{t_0} \log r. \quad (16.18)$$

Indeed, substitution into $\lambda_X = \exp(-iz_X t_0 / \hbar_{\text{int}})$ recovers $re^{-i\theta}$. Since $\log r < 0$, the pole is retarded and the width positive. The survival probability after N steps is $r^{2N} = \exp(-\Gamma_X N t_0 / \hbar_{\text{int}})$, whence $\tau_X = \hbar_{\text{int}} / \Gamma_X$.

If this reading is obtained from a compression SUS , asymptotic interpretation through a single mode requires a simple dominant eigenvalue, nonzero initial overlap, and separation in modulus from the rest of the spectrum. Normalization on probabilities rather than amplitudes changes the factor 2. The temporal branch and that convention therefore accompany every expressed width.

Equality of mass and width for conjugate states also requires the channels to be transported:

$$U_C H_{\text{eff}}^X(z) U_C^{-1} = H_{\text{eff}}^X(z).$$

The same poles are then transported with the same causal prescription. An antiunitary representation additionally requires that prescription to be transported: bare conjugation of z interchanges retarded and advanced poles.

16.13 Three topological codomains and the virtual sector

Topological classes preserve objects and fusion rules, not a universal scalar mass. The primary datum is $(N_{ab}^c, F_d^{abc}, R_c^{ab})$, with its coherence equations. An energy gap subsequently requires a Hamiltonian realization and a spectral projector. Assigning this output type makes it possible to include these sectors without turning them into elementary-mass rows.

16.13.1 Fibonacci sector

The objects $\mathbf{1}, \tau$ satisfy $\tau \otimes \tau = \mathbf{1} \oplus \tau$. The positive dimension d_τ preserves fusion multiplication: $d_\tau^2 = 1 + d_\tau$. The corpus realization identifies this dimension with the internal coordinate φ already generated; the fusion equation is its recognition in this codomain, not a substitute for its APP-TRIT-TPK genealogy. In the fixed convention,

$$F_\tau^{\tau\tau\tau} = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}, \quad R_1^{\tau\tau} = e^{-4\pi i/5}, \quad R_\tau^{\tau\tau} = e^{3\pi i/5}.$$

The identity $\varphi^{-2} + \varphi^{-1} = 1$ directly proves that $F^* F = I$; off-diagonal terms cancel. The phases of R preserve the braiding orientation.

16.13.2 Ising sector and Majorana realization

The rules are

$$\psi \otimes \psi = \mathbf{1}, \quad \psi \otimes \sigma = \sigma, \quad \sigma \otimes \sigma = \mathbf{1} \oplus \psi,$$

with

$$F_\sigma^{\sigma\sigma\sigma} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad R_1^{\sigma\sigma} = e^{-\pi i/8}, \quad R_\psi^{\sigma\sigma} = e^{3\pi i/8}.$$

Matrix multiplication gives $F^2 = I$. The sector preserves its three fusion types and its representation; its Majorana designation does not automatically turn a neutrino from the preceding sector into a self-conjugate mode. Such a statement uses the specific antiunitary conditions of the neutral realization.

16.13.3 Abelian sector with six classes

For $a, b \in \mathbb{Z}/6\mathbb{Z}$,

$$a \otimes b = a + b \pmod{6}, \quad R_{ab} = \zeta_6^{ab}, \quad \zeta_6 = e^{2\pi i/6}.$$

The identity $R_{a+b,c} = R_{a,c}R_{b,c}$ and its analogue in the second argument express the bilinear character of braiding. The doubled construction $\mathcal{C} \boxtimes \mathcal{C}^{\text{rev}}$ preserves both orientations. Its classes must not be counted as six new elementary masses.

16.13.4 Finite persistence and virtuality

The virtual sector preserves finite extensions, their channel, and their persistence horizon. A finite section with a terminal obstruction does not occupy the same codomain as an asymptotically persistent section. Virtuality is read in an amplitude or resolvent and is not, by definition, equivalent to a universal imaginary mass. Nor does the absence of certification of a global extension in a record suffice to assert that an obstruction exists: the horizon and its proof are different data. The classification precisely preserves this finite information without inventing a stable particle for every internal calculation line.

16.14 Classification of twenty-three particle sectors and scope of their realizations

The public inventory comprises 17 elementary classes, 2 composite collections, 3 topological codomains, and 1 virtual sector: $17 + 2 + 3 + 1 = 23$. The three neutrino rows name flavors, the six quark rows preserve their color triplicity, and the gluon row preserves the adjoint representation. None of these counting conventions modifies the atlas 81/56/13/324.

The following list retains the output type and the scope of the documented realizations. The expressions “declared signature” and “fiber operator” refer to the explicit constructions in this section, not to different degrees of decimal precision. The provenance register distinguishes the electronic scalar closure and the two null charts from physical scalarization still pending in noncentral collections. For neutrinos, it retains as pending the complete realization of mixing and scale; for quarks, flavor selection and its scale chart; for nominal bosons, the joining of route and pole; for named composites, the scalar realization of their bond. These documentary statuses do not alter the operators, grammars, or conditioned evaluations presented above.

The exact documentary statuses are expressed in Appendix B.3: Table 7 preserves the status legend, and Table 8 preserves, for the twenty-three classes, residence, incidence, representation, and scalar status.

Proposition 16.10 (Realization fiber product and observable codomain). *For $X \in \mathcal{P}_{\text{adm}}$, the fiber defined in (16.1) is nonempty if and only if*

$$d_{\mathcal{P}}(X) \in \text{im } d_{\mathcal{R}}.$$

On this fiber, the joint spectral measure (16.2) determines the observable codomain of the readers $(M_X^{\beta,D}, M_X^{\beta,\Omega})$. A preparation P_X expresses a scalar line m exactly when it satisfies the criterion of Proposition 16.1; in all other cases it preserves the multiplet, spectral distribution, or sectorial type produced.

Proof. By the definition of image, the stated condition is equivalent to the existence of a route $\gamma \in \mathcal{R}_{\text{adm}}$ with $d_{\mathcal{R}}(\gamma) = d_{\mathcal{P}}(X)$; this is precisely the condition for membership in F_X . The second statement follows from the spectral theorem applied to the two diagonal operators and the characterization of single-point support proved in Proposition 16.1. $\square$

Class	Representatives	Preserved result and realization operation
1	e^-, e^+	Rank-one scalar beta closure and conjugate representation.
2	μ^-, μ^+	Exact fiber operators; nominal evaluation and order-3 refinement with a declared signature; distinct physical spectral selection.
3	τ^-, τ^+	Exact fiber operators; nominal evaluation and order-3 refinement with a declared signature; distinct physical spectral selection.
4	$\nu_e, \bar{\nu}_e$	Neutral fiber operator; realization of mixing, mass frame, and physical scale through their complete maps.
5	$\nu_\mu, \bar{\nu}_\mu$	Same neutral domain, with a different flavor label; not an individual atlas depth.
6	$\nu_\tau, \bar{\nu}_\tau$	Same neutral domain; preservation of the Dirac–Majorana alternative under its conditions.
7	$u, \bar{u}$	Color-fiber operator and nominal angular reading; scalarization, scheme, and scale preserved.
8	$d, \bar{d}$	Color-fiber operator and nominal angular reading; scalarization, scheme, and scale preserved.
9	$c, \bar{c}$	Color-fiber operator and nominal angular reading; scalarization, scheme, and scale preserved.
10	$s, \bar{s}$	Color-fiber operator and nominal angular reading; scalarization, scheme, and scale preserved.
11	$t, \bar{t}$	Color-fiber operator and nominal angular reading; explicit mass convention and comparison channel.
12	$b, \bar{b}$	Color-fiber operator and nominal angular reading; scalarization, scheme, and scale preserved.
13	Photon	Exactly zero-mass section; electromagnetic representation.
14	Gluons	Exactly zero-mass section; adjoint color representation.
15	W^+, W^-	Complete nominal signature and order-3 evaluation; joining to route, projector, and pole as a distinct physical realization.
16	Z^0	Complete nominal signature and order-3 evaluation; joining to route, projector, and pole as a distinct physical realization.
17	H	Scalar descriptor and nominal angular evaluation; projector, couplings, and pole of the scalar realization.
18	Mesons	Exact grammar of the dual pair and bond; recovered composite signatures; distinct nominal scalarization and pole channels.
19	Baryons	Exact grammar of the color triple and bond; recovered proton and neutron signatures; distinct nominal binding realization.
20	Fibonacci	Fusion and braiding codomain; a gap requires its Hamiltonian realization.
21	Ising/Majorana	Fusion codomain and real structure according to the representation; no assigned universal scalar mass.
22	$\mathbb{Z}/6\mathbb{Z}$	Six braiding classes; a topological codomain, not six universal masses.
23	Virtual sections	Finite extension and obstruction horizon; no universal stable scalar.

The external documentary snapshot contains 471 mass observables and 384 width observables, totaling 855. These are comparison rows, not 855 internal predictions. In particular, the 243 mesonic mass rows and 166 baryonic rows do not coincide with the grammar's cardinalities 432 and 1728: they count objects of different types. The external table is opened after identity, fiber, register, operator, projector, and chart have been fixed.

Reconstruction of a positive ratio through a tower receiving that ratio demonstrates the representational capacity of the evaluation system. The independent output of a species, by contrast, requires the tower's coefficients to arise from its route and boundary. This separation preserves both representational completeness and effective sectorial results. The whole is organized by a common rule: construct the structure and its memory, apply the reader of the corresponding type, and then perform physical recognition.

17 Persistence, fusion structures, and spectral realisations

The atlas and mass operators admit different representations. The classification of these representations preserves the datum characterising each: a multiplication rule, a real conjugation, an exchange phase, or a spectral measure. This section places the topological sectors and neutral realisations within the same architecture. Mass coordinates are obtained once the representation and its dynamics have been determined.

The distinction between these stages is particularly visible in the areal–volumetric sector. An incidence matrix can fix a fusion ring and its positive dimension; the energy of an excitation also depends on the operator acting in the representation. The first determination retains exact content when several dynamic realisations of the same ring are studied. The distinction preserves the structural information preceding the magnitude.

17.1 The based ring of areal–volumetric incidence

The primitive incidence of the sheets, obtained from the TPK calendar, is

$$F_{av} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

In the basis $(1, \tau)$, the first column expresses $\tau \cdot 1 = \tau$; the other expresses $\tau \cdot \tau = 1 + \tau$. The incidence thus induces the based ring

$$\mathcal{F}_{av} = \mathbb{Z}[\tau]/(\tau^2 - \tau - 1), \quad \mathcal{F}_{av}^+ = \{a + b\tau : a, b \in \mathbb{Z}_{\geq 0}\}.$$

The positive cone records multiplicities. The operation is internal to the two types and does not use a mass as a coefficient.

Proposition 17.1 (Product and positive dimension). *We have*

$$(a + b\tau)(c + d\tau) = (ac + bd) + (ad + bc + bd)\tau.$$

There is a unique ring homomorphism $d : \mathcal{F}_{av} \rightarrow \mathbb{R}$ with $d(1) = 1$ and $d(\tau) > 0$. Its value at τ is the golden coordinate recognised in the realisation of F_{av} .

Proof. Expanding the product and substituting $\tau^2 = 1 + \tau$ gives the coefficients. A unital homomorphism is determined by $x = d(\tau)$, with $x^2 = x + 1$. Its roots are $(1 + \sqrt{5})/2$ and $(1 - \sqrt{5})/2$; only the first is positive. Evaluation at that root defines a homomorphism of the quotient. The equation identifies the dimension of this representation; the HMT regional coordinate has been constructed earlier. $\square$

With $F_0 = 0$, $F_1 = 1$, and $F_{n+1} = F_n + F_{n-1}$, the powers satisfy

$$\tau^n = F_{n-1} + F_n \tau \quad (n \geq 1).$$

The initial case is immediate; multiplication by τ reapplies the incidence and produces the induction step. The coefficients thus retain their origin as counts of sheet composition.

A categorical representation adds morphism spaces and an associator to these multiplicities. A braided representation adds compatible exchange operators. The material realisation adds the Hamiltonian, its domains, and the preparation selecting a spectral sector. If it has a gap $\Delta E > 0$ and effective velocity c_{eff} , its gap-mass coordinate is

$$m_{\text{brecha}} = \Delta E c_{\text{eff}}^{-2}.$$

The positive dimension of the ring and the gap are outputs of different operators. The fusion rule preserves its content without by itself fixing a universal mass for all its realisations.

17.2 Dimension and representation type

In a pointed representation, all simple objects are invertible. If X is simple and invertible and d is a positive dimension compatible with duals, then

$$1 = d(X \otimes X^*) = d(X)d(X^*) = d(X)^2,$$

and $d(X) = 1$. Dimension therefore provides a check on changes of representation.

Proposition 17.2 (Dimension obstruction in the pointed sector). *A simple object of dimension φ or $\sqrt{2}$ cannot be identified, preserving dimension, with a simple object of a pointed representation. In particular, the 81 invertible simple objects of the untwisted centre based on $\mathbb{Z}/9\mathbb{Z}$ do not provide that identification.*

Proof. Invertible simple objects have dimension 1, different from both dimensions. In the stated abelian untwisted centre, a simple object is labelled by a group element and a character of its dual. Both sets have 9 elements. The 81 labels are invertible under multiplication of the group and of the character. $\square$

This census of 81 has a domain different from the atlas of 81 cells. An extension or functor to another codomain specifies the actions and maps it transports. Areal–volumetric incidence preserves its exact ring, and the topological realisation preserves the conditions of its own type.

17.3 Neutral self-conjugacy and real generators of CAR

A real structure on a complex space is an antiunitary operator $\mathcal{C}$ with $\mathcal{C}^2 = I$. Its fixed part is a real space. Commutation of the mass operator with $\mathcal{C}$ preserves that real part on its domain. Restricting an evolution to the fixed part is expressed through commutation of its group with $\mathcal{C}$; in a Schrödinger evolution, this condition must be distinguished from commutation of the Hamiltonian. Both conditions belong to the material representation of Section 10.2.

Proposition 17.3 (Antiunitary conjugation and time evolution). *Let H be self-adjoint, $\hbar > 0$, and $\mathcal{C}$ an antiunitary conjugation preserving $\mathcal{D}(H)$. For $U(t) = \exp(-itH/\hbar)$, if $\mathcal{C}H\mathcal{C}^{-1} = \epsilon H$, with $\epsilon \in \{1, -1\}$, then*

$$\mathcal{C}U(t)\mathcal{C}^{-1} = U(-\epsilon t). \quad (17.1)$$

The group $U(t)$ preserves the fixed part of $\mathcal{C}$ for every t if and only if it commutes with $\mathcal{C}$, equivalently if $\mathcal{C}H\mathcal{C}^{-1} = -H$. If instead $\mathcal{C}H = H\mathcal{C}$, a vector fixed by $\mathcal{C}$ remains fixed by it throughout the evolution exactly when it belongs to $\ker H$.

Proof. Antiunitary conjugation sends i to $-i$. Spectral functional calculus gives

$$\mathcal{C} \exp(-itH/\hbar) \mathcal{C}^{-1} = \exp(it\mathcal{C}H\mathcal{C}^{-1}/\hbar),$$

which proves (17.1). If $U(t)$ commutes with $\mathcal{C}$, it transports its fixed vectors to fixed vectors. Conversely, every complex vector is uniquely written as $x + iy$ with x, y fixed by $\mathcal{C}$. Preservation of that real part implies, by linearity of $U(t)$ and antilinearity of $\mathcal{C}$, that the two commute. Uniqueness of the self-adjoint generator of the unitary group then identifies $\mathcal{C}H\mathcal{C}^{-1}$ with $-H$.

In the commuting case $\epsilon = 1$, for $\mathcal{C}\psi = \psi$ we have $\mathcal{C}U(t)\psi = U(-t)\psi$. The condition that $U(t)\psi$ be fixed for every t is equivalent to $U(2t)\psi = \psi$ for every t . By spectral calculus, this is equivalent to the spectral measure of ψ being supported on $\{0\}$, that is, to $\psi \in \ker H$. $\square$

Thus an antiunitary symmetry commuting with H reverses time direction. A realisation through preserved real states uses a different condition on the generator. This distinction preserves the scope of algebraic self-conjugacy and allows the dynamics of a Majorana field to be specified separately.

Proposition 17.4 (Self-conjugacy and charge). *Let Q be a self-adjoint charge operator and $\mathcal{C}$ a conjugation transporting its domain and satisfying $\mathcal{C}Q\mathcal{C}^{-1} = -Q$. If $\psi \neq 0$, $Q\psi = q\psi$, and $\mathcal{C}\psi = \psi$, then $q = 0$.*

Proof. Self-adjointness implies $q \in \mathbb{R}$. Applying $\mathcal{C}$ to the eigenvalue equation gives $-Q\psi = q\psi$. Together with the initial equation this yields $2q\psi = 0$, hence $q = 0$. $\square$

Neutrality is necessary in this self-conjugacy problem; the kernel of Q does not by itself specify an antiunitary operator or its compatibility with the evolution group. The criterion applies to the neutrino blocks after constructing their operators and mixing. The fixed dodecaphase word, the real spinor, and the charge representation thus retain their respective domains.

The fermionic completion provides another realisation. On the exterior space of the stated modes, a_j^* is exterior multiplication by the basis vector and a_j is its adjoint. Exterior antisymmetry gives

$$\{a_j, a_k\} = 0, \quad \{a_j, a_k^*\} = \delta_{jk}I.$$

Proposition 17.5 (Self-adjoint generators of CAR). *The operators*

$$\gamma_{2j-1} = a_j + a_j^*, \quad \gamma_{2j} = -i(a_j - a_j^*)$$

are self-adjoint and satisfy $\{\gamma_r, \gamma_s\} = 2\delta_{rs}I$.

Proof. Adjunction fixes each operator. For the same mode, $a_j^2 = (a_j^*)^2 = 0$ and $a_j a_j^* + a_j^* a_j = I$, so each square equals I . The anticommutator between the two combinations of the same mode vanishes by expansion. For distinct modes, all elementary anticommutators vanish. $\square$

These are real generators of a Clifford structure. A topological zero mode additionally requires the dynamics to realise a localised zero-energy excitation, separated from the continuum by a gap and stable under the perturbations considered. The algebraic and spectral properties are linked through the dynamic realisation.

17.4 Exchange phases and winding of order six

For $n \geq 2$, in the order-six reader the exponent sum defines $w_6 : B_n \rightarrow \mathbb{Z}/6\mathbb{Z}$. The Artin relations preserve this sum: commutation of distant generators retains two terms, and the relation involving three generators retains three.

Proposition 17.6 (Characters of the winding reader). *The unitary characters of the quotient are*

$$\chi_k(b) = \exp\left(\frac{2\pi i k}{6} w_6(b)\right), \quad k = 0, 1, \dots, 5.$$

A scalar representation of this form descends to the permutation group exactly for $k = 0$ or $k = 3$.

Proof. A cyclic character is determined by the image q of 1, with $q^6 = 1$. To descend to permutations, it must satisfy the additional relation $\sigma_j^2 = 1$, which requires $q^2 = 1$. Among the sixth roots, only $q = 1, -1$ satisfy this condition. Both satisfy all relations of the permutation group. $\square$

The other four characters preserve winding information. A material species requires the reader to be realised on a multisection and the dynamics to preserve its structure. Route reversal changes w_6 to $-w_6$ and transports each character to its complex conjugate.

17.5 Stability and visibility of a section

The positivity proved in Section 12 has a spectral consequence. For a positive self-adjoint operator M , functional calculus gives $\text{Spec}(M^2) \subset [0, \infty)$. In a realisation identifying M^2 with the squared-mass observable on a physical space with positive inner product, a persistent eigensection cannot have a negative eigenvalue of that observable. An unstable direction of the linearisation of a background belongs to another spectral problem, studied with its domain and dynamics.

Visibility depends on coupling. If P_X is the support of a section and P_a is the projector of a channel, $P_a P_X = 0$ expresses orthogonality of supports. A nonzero product expresses possible incidence; its intensity is determined by the coupling operator. Two projectors can commute and have nonzero product, so the commutator does not replace that incidence.

In a neutral block, a section sterile with respect to the weak channel belongs to the kernel of its coupling. If that kernel is invariant under the mass operator, the restriction has a spectral measure of its own. Atlas depth, neutrality, and absence of weak coupling are distinct properties verified on the same section. Representational possibilities are retained without turning a collection label into an external mass.

The organisation thus preserves the order of the argument: incidence determines multiplicities; lifts determine conjugations and phases; the representation and material operator determine the spectral output. Comparison is subsequently applied to that output, with its units and coupling stated.

18 Central realisation of trajectories and Sommerfeld holonomy

The particle defined in Section 9 possesses an observable history before receiving an orbital realisation. The question in this section is how a representation of that history preserves flux, area and phase when expressed as central motion. The answer is organised into four operations: cancellation of interior fluxes in a graph cut, normalisation of its boundary in a metric chart, evolution under the declared central operator and phase transport over its cycles.

This order keeps visible which datum enters each assertion. The graph and its records arise from APP-TRIT-TPK; the metric chart specifies how its boundary is measured; the dynamical realisation fixes the Lagrangian acting in that chart; the semiclassical condition uses the reduced Planck constant already constructed and the holonomy of the transported line. The results of orbital geometry are proved with these explicit starting data. Central mechanics is recognised in the resulting realisation, without making its conventional constants inputs to the generator.

18.1 Flux conservation and boundary reading

Let $G = (V, E)$ be a finite cut of the route graph of the realisation under consideration, with an arc in each orientation of its edges. An oriented flux is a function $F : \vec{E} \rightarrow \mathbb{R}$ satisfying $F(y, x) = -F(x, y)$. At a vertex x , its divergence is

$$\operatorname{div} F(x) = \sum_{y: \{x, y\} \in E} F(x, y). \quad (18.1)$$

For $S \subseteq V$, the outward flux is evaluated over the arcs originating in S and ending in $V \setminus S$.

Theorem 18.1 (Flux balance in a finite cut). *For every subset $S \subseteq V$,*

$$\sum_{x \in S} \operatorname{div} F(x) = \sum_{\substack{x \in S, y \notin S \\ \{x, y\} \in E}} F(x, y). \quad (18.2)$$

Proof. An edge with both endpoints in S contributes twice to the sum of divergences, through $F(x, y) + F(y, x) = 0$. Edges with both endpoints outside S do not contribute. Exactly the edges crossing the boundary remain, with the indicated orientation. $\square$

The theorem transmits to the boundary the balance produced in the interior. For a cubic representation with nearest neighbours, consider

$$C_R = \{x \in \mathbb{Z}^3 : \|x\|_\infty \leq R\}, \quad R \in \mathbb{Z}_{\geq 0}.$$

Each of the six faces contains $(2R + 1)^2$ outward links in its normal direction. Links in different directions are counted separately, even when they originate at the same edge or corner vertex. Hence,

$$|\partial C_R| = 6(2R + 1)^2. \quad (18.3)$$

This census counts outward links, not the number of boundary vertices.

If Q_F is the total flux of the enclosed source, its mean per link is $Q_F/[6(2R + 1)^2]$. Under the reader's mean-isotropy condition, this mean is the intensity assigned to the shell. Define its effective radius, in units of the chart spacing, by

$$4\pi r_{\text{eff}}^2 = 6(2R + 1)^2. \quad (18.4)$$

We then obtain

$$E_R = \frac{Q_F}{4\pi r_{\text{eff}}^2}. \quad (18.5)$$

The coordinate π in this normalisation is the already expressed HMT regional output. The identity follows from the census and the definition of the radius; passage from a mean to a central realisation also uses the stated isotropy. The dimensional chart subsequently determines whether Q_F represents electric, gravitational or another observable flux, and how it converts the lattice spacing into length. The finite balance preserves this distinct status of each operation.

18.2 Central dynamics on the Euclidean sheet

Let $\mu > 0$ be the reduced mass of the material realisation and let $\mathcal{K} > 0$ be its central coupling, with the dimension of energy times length. These coordinates are received after the internal constructions of mass, action and coupling; their origin and dimensional line are preserved when carrying out the realisation. The domain of the following theorem consists of all realisations that adopt, on their Euclidean sheet,

$$L(\mathbf{r}, \dot{\mathbf{r}}) = \frac{\mu}{2} \|\dot{\mathbf{r}}\|^2 + \frac{\mathcal{K}}{r}, \quad H(\mathbf{r}, \mathbf{p}) = \frac{\|\mathbf{p}\|^2}{2\mu} - \frac{\mathcal{K}}{r}, \quad r = \|\mathbf{r}\| > 0, \quad \mathbf{p} = \mu \dot{\mathbf{r}}. \quad (18.6)$$

The exclusion of $r = 0$ delimits the regular domain of these coordinates. The trajectory is considered over a collision-free time interval.

Proposition 18.2 (Orbital plane and conic equation). *A non-radial solution of the Euler–Lagrange equation of (18.6) conserves $\ell = \mathbf{r} \times \dot{\mathbf{r}}$, with $\ell = \|\ell\| > 0$. In the plane orthogonal to ℓ , it can be written*

$$r(\theta) = \frac{p}{1 + \epsilon \cos(\theta - \theta_0)}, \quad p = \frac{\mu \ell^2}{\mathcal{K}}, \quad \epsilon \geq 0. \quad (18.7)$$

Negative energy corresponds to the elliptic regime $0 \leq \epsilon < 1$.

Proof. Differentiating the potential gives

$$\mu \ddot{\mathbf{r}} = -\frac{\mathcal{K}}{r^3} \mathbf{r}. \quad (18.8)$$

Hence, $\dot{\ell} = \mathbf{r} \times \ddot{\mathbf{r}} = 0$. Position and velocity remain orthogonal to the non-zero constant vector ℓ , fixing the plane. In its polar coordinates, $r^2 \dot{\theta} = \ell$. Introducing $u = 1/r$ into the radial equation gives Binet's equation

$$u''(\theta) + u(\theta) = \frac{\mathcal{K}}{\mu \ell^2}.$$

Its general solution is $u = \mathcal{K}[1 + \epsilon \cos(\theta - \theta_0)]/(\mu \ell^2)$, and inversion proves (18.7) on the domain where the denominator is positive. Substitution into the energy gives $H = \mathcal{K}(\epsilon^2 - 1)/(2p)$, which is negative exactly when $\epsilon < 1$. $\square$

Here the letter ϵ denotes orbital eccentricity, not Euler's number. The vector ℓ is specific angular momentum, not the Planck constant. These notational choices allow the geometric observables and the internal constants entering their realisation to be compared without conflation.

Proposition 18.3 (Areal conservation and period law). *The orbit's areal velocity is constant:*

$$\frac{d\mathcal{A}_{\text{orb}}}{dt} = \frac{\ell}{2}. \quad (18.9)$$

For an ellipse with semimajor axis a , its period satisfies

$$T^2 = \frac{4\pi^2 \mu}{\mathcal{K}} a^3. \quad (18.10)$$

Proof. The sector swept out in time dt has area $r^2 d\theta/2 = \ell dt/2$. If $b = a\sqrt{1 - \epsilon^2}$ is the semiminor axis, the total area is πab and $T = 2\pi ab/\ell$. The relation $p = a(1 - \epsilon^2) = \mu \ell^2/\mathcal{K}$ implies $\ell^2 = \mathcal{K}a(1 - \epsilon^2)/\mu$. Substitution into $T^2 = 4\pi^2 a^2 b^2/\ell^2$ gives (18.10). $\square$

Areal conservation is determined by the central symmetry of the motion. The ellipse in this proposition lies in configuration space. By contrast, the action ellipse constructed from the angular pair lies in its action chart and transports the corresponding normalisations. The connection between them requires the realisation map identifying their variables; sharing an elliptic shape does not replace that map. The orbital reading thus preserves the dependencies on mass, coupling and geometry that produce it.

18.3 Orbital covering and real sign line

On the complex plane with the origin removed, the map

$$q = z^2 : \mathbb{C} \setminus \{0\} \longrightarrow \mathbb{C} \setminus \{0\} \quad (18.11)$$

is a double covering. If $q(t) = r(t)e^{i\theta(t)}$ and a root is chosen initially, its lift is $z(t) = \sqrt{r(t)}e^{i\theta(t)/2}$. One circuit with an increment 2π in θ ends at $-z(0)$, and two circuits restore the initial point. This is the transport of the two lifts over the punctured plane.

An orbital band requires another object: a real line associated with the orbit and its sign character. Let $\gamma : S^1 \rightarrow \mathbb{R}^2$ be a regular injective parametrisation of an elliptic orbit, and let $\delta > 0$. With holonomy -1 over one circuit, its interval band is represented by

$$M_\gamma = ([0, 2\pi] \times [-\delta, \delta]) / ((0, u) \sim (2\pi, -u)). \quad (18.12)$$

Identifying the base with the image of γ places the zero section over the orbit.

Proposition 18.4 (Oriented transport over the orbital cycle). *The line transport in (18.12) satisfies $\mathcal{H}(\gamma) = -I$ and $\mathcal{H}(\gamma^2) = I$.*

Proof. The terminal fibre is identified with the initial fibre by $u \mapsto -u$; composing this map twice gives the identity. This is the construction in Theorem 10.4, applied to the orbital loop. $\square$

The covering (18.11) has two-point fibres; (18.12) has interval fibres. Monodromy of order two appears in both, while preserving these distinct types. Levi–Civita dynamical regularisation also uses the transformation of momenta and time reparametrisation. The present construction uses only its covering and its lifting law, with the scope just proved.

18.4 Phase condition and half-integer shift

Let γ be a cycle of the Hamiltonian realisation, and let

$$J_\gamma = \oint_\gamma \mathbf{p} \cdot d\mathbf{q} \quad (18.13)$$

be its action. The reduced Planck constant $\hbar_{\text{int}} > 0$ is received from the HMT action construction, in the same dimensional line as J_γ . The quotient $J_\gamma/\hbar_{\text{int}}$ is therefore dimensionless. Suppose that semiclassical transport in the complexified line contributes the phase $\exp(iJ_\gamma/\hbar_{\text{int}})$ and that the real orientation line contributes the character $\varepsilon_\gamma \in \{+1, -1\}$. The closure condition for the transported section is

$$\varepsilon_\gamma \exp\left(\frac{iJ_\gamma}{\hbar_{\text{int}}}\right) = 1. \quad (18.14)$$

The product preserves both contributions: the action of the cycle and the sign of the line. The latter is not absorbed into a modification of the Planck constant.

Theorem 18.5 (Integer and half-integer action sectors). *Under the transport (18.14),*

$$J_\gamma = \begin{cases} 2\pi\hbar_{\text{int}}n, & \varepsilon_\gamma = +1, \\ 2\pi\hbar_{\text{int}}(n + \frac{1}{2}), & \varepsilon_\gamma = -1, \end{cases} \quad n \in \mathbb{Z}. \quad (18.15)$$

Proof. For positive sign, the exponential must equal 1, whose real arguments are $2\pi n$. For negative sign, it must equal -1 , whose arguments are $(2n+1)\pi$. Multiplying by $\hbar_{\text{int}}$ produces the two collections. $\square$

The formula determines the closure constraint on the action of a given cycle. Obtaining an atomic spectrum also uses the specific Hamiltonian, the independent cycles and the quantisation prescription. If this prescription includes Maslov indices, their phases are composed with (18.14) according to the chosen transport, avoiding counting the same topological contribution twice.

Reversing the orientation of the cycle changes J_γ to $-J_\gamma$, whereas the sign character satisfies $\varepsilon_{\gamma^{-1}} = \varepsilon_\gamma^{-1}$. The total phase becomes its inverse, and the closure condition is preserved. This is a precise law of orbital bidirectionality. Its identification with material antisections requires the lift in Section 10.2; equality of two signs does not by itself establish that identification.

The realisation thus brings together three consecutive conservation laws: flux passes from the interior to the boundary; central dynamics preserves angular momentum and the area swept out per unit time; section transport preserves its closure condition through action and holonomy. Each result preserves the equation, domain and data that support it. The material history of the TPK is represented in that dynamics with its memory, rather than being replaced by the orbital figure it expresses.

19 Realisation of observables and metrological comparison

A mass value acquires its meaning within the structure that produces it and the representation in which it is measured. The APP-TRIT-TPK construction first determines admissible trajectories, their oriented records and the signature readers; the towers and material realisation act on these objects. Experimental comparison takes place at the end of this composition. Its function is to compare observables of the same type, preserve the identity of the state and quantify agreement in a common chart. This position makes it possible to study together the mathematical structure, the evaluated coordinate and its physical interpretation.

This section brings together three levels with different functions: the internal domain of routes and their quotients; the classes and individualised evaluations in the mass source owners; and the external inventory of observables used as a reference. The order of exposition preserves the constructive direction of the article. External quantities appear as terms of comparison, while internal values preserve the signatures, operations and conditions fixing their evaluation.

19.1 Structural domain, material classes and external inventory

Let $\mathcal{C}$ be the set of APP cells and $\mathcal{R}$ its domain of oriented routes. The identification of a route contains its cell coordinates and the two initial orientations:

$$\gamma = (r, c, h_0, v_0), \quad r, c \in \{1, \dots, 9\}, \quad h_0, v_0 \in \{-1, +1\}.$$

Hence, $\#\mathcal{C} = 81$ and $\#\mathcal{R} = 81 \cdot 4 = 324$. The 56 trajectory collections and 13 multisections belong to quotients and structural groupings of the same domain, with the relations and readers established earlier. These four cardinalities describe different internal objects. Assigning a physical representation may preserve an entire fibre; a species label does not by itself impose a pointwise choice within it.

The public table of 23 classes organises realisations by sector. Its function is to relate internal residence to representation, charge, spin, colour, flavour and observable type. A class may comprise several species, and a catalogue identity may bring together different charge states. The correspondence is expressed through typed maps and preserves these multiplicities.

The frozen metrological inventory contains 471 mass records and 384 width records. The corpus documents 1170 identities in 438 groups, of which 403 contain at least one of these observables. The sum

$$471 + 384 = 855$$

counts external summary-table observables. The 81/56/13/324 atlas, the 23 classes and the 855 records thus belong to three levels of description. Their association preserves the type of each set, rather than converting their cardinalities into a single number of particles or predictions.

Catalogue collection	Groups	Masses	Widths
General	46	53	9
Quarks	6	8	1
Mesons	201	243	225
Baryons	149	166	149
External gravitational record	1	1	0
Total with observable	403	471	384

The gravitational row preserves the identity appearing in the external source; its presence in the inventory does not define an HMT elementary particle. The remaining 35 groups preserve their catalogue identity, although the snapshot assigns them no summary-table mass or width. Documentary integrity encompasses both facts: the recorded quantities and the identities for which that snapshot does not express one of those quantities.

19.2 The observable before the numerical coordinate

A material output is specified through the state, representation, operator and reading procedure. For a fibre F_X , let $\hat{M}_X$ be the mass operator already constructed, and let $P_{X,a}$ be the projector corresponding to the preparation state or channel a . For Ψ_X normalised in its image, the spectral distribution is

$$\mu_{X,a}(B) = \langle \Psi_X, P_{X,a} E_{\hat{M}_X}(B) P_{X,a} \Psi_X \rangle,$$

where $E_{\hat{M}_X}$ is the operator's spectral measure. This expression preserves the prepared state and the multiplicity of the fibre. A single scalar coordinate for every state prepared in that subspace is obtained under the condition

$$\text{Ran}(P_{X,a}) \subset \text{Dom}(\hat{M}_X), \quad \hat{M}_X P_{X,a} = m_X P_{X,a}.$$

Indeed, every vector in that subspace is then an eigenvector of the mass operator with the same value m_X , and its spectral measure is concentrated at that value. If the compression equality $P_{X,a} \hat{M}_X P_{X,a} = m_X P_{X,a}$ is used, one additionally requires that $P_{X,a}$ reduce $\hat{M}_X$; in finite dimension, this is equivalent to commutation. Without that condition, the compression may fix only the mean value. When the spectral measure retains several values, the output is presented as a distribution or multiplet with its weights.

For a fixed density ρ in a finite fibre, let m_j be the distinct eigenvalues and E_j their spectral projectors. The weights $p_j = \text{tr}(\rho E_j) \geq 0$, with $\sum_j p_j = 1$, determine

$$\bar{m} = \sum_j p_j m_j, \quad \text{Var}_\mu(m) = \sum_j p_j (m_j - \bar{m})^2.$$

Scalar-concentration criterion for the fixed state. The variance vanishes if and only if all values with positive weight coincide. Indeed, every summand is non-negative; the sum is zero precisely when $m_j = \bar{m}$ for every j with $p_j > 0$. This determines when a mass distribution admits a scalar representation without losing the dispersion of the prepared state. The criterion concerns the fixed density and differs from the operator condition on the entire subspace. The electronic case uses its operator on the central fibre, as constructed earlier; the rank-one property of an arbitrary projector is insufficient to concentrate the spectral measure of another operator.

Mass and width use coordinated readers of different types. For a resonant state, a pole convention may be expressed in energy coordinates as

$$z_X = M_X c_{\text{int}}^2 - \frac{i}{2} \Gamma_X.$$

The real part is the mass–energy coordinate; Γ_X has the dimension of energy and measures the width. Once the time chart is fixed, $\tau_X = \hbar_{\text{int}}/\Gamma_X$. The relation assumes the pole and survival regime for which it was constructed; a rate, a width and a lifetime retain their units and conversion rules. This distinction explains why the inventory preserves two observable classes.

The same principle determines comparison by sector. Running quark masses are expressed with a scheme $\mathcal{S}$ and a scale μ ; resonances distinguish pole mass, Breit–Wigner parametrisation and on-shell convention. Neutrinos distinguish mass eigenvalues, mass-squared differences, flavours and mixing parameters. One-sided bounds preserve the documented confidence level and statistical convention. An upper bound is not converted into a positive central value, nor a difference of squared masses into an absolute mass.

19.3 Fixing the construction and matching records

The realisation of a species X transmits the necessary information from its internal antecedents. For comparison purposes, we may collect it in the record

$$\mathfrak{D}_X = (F_X, \rho_X, P_{X,a}, \mathcal{L}_X, \sigma_X, \eta_X, \hat{M}_X, \mathcal{U}_X),$$

where ρ_X identifies the representation, $\mathcal{L}_X$ the genealogical record, $\sigma_X \in \mathbb{Z}^5$ the signature, η_X the tower section and $\mathcal{U}_X$ the unit chart. All these data receive their meaning from the APP-TRIT-TPK operations and the realisations proved in the article. Collecting them in a tuple is a documentary operation: their constructions and proofs remain in the sections that produce them.

The external reference is represented separately by

$$o = (\text{id}, \text{species}, \text{observable}, v, \text{unit}, \mathcal{S}, \mu, \text{uncertainty}, \text{limit}, \text{confidence}, \text{source}, \text{edition}).$$

The association $(\mathfrak{D}_X, o)$ must be established through physical identity and observable type. In an extensive table, this operation uses identity and provenance keys; numerical proximity does not serve as a key. The record preserves the particle or antiparticle label, the group possessing the observable and the identity of the measured property.

The independence of the selection procedure is examined by hiding or permuting the metrological column and checking that $\mathfrak{D}_X$ remains unchanged. This test is meaningful for a generator whose effective composition can be executed; merely preserving an output file does not by itself constitute such a test. The cryptographic hash fixes the artefact being compared and makes modifications detectable, but causal independence belongs to the operation and its dependencies.

Statuses are assigned to the domain and operation supporting each assertion. A row may simultaneously document an internal operator, an evaluation given a signature and an external comparison, without identifying these three levels. Every subsequent realisation is linked to the definition, proposition or table establishing its scope. In particular, the catalogue's documentary classifications apply to the snapshot specified by each record and do not change the mathematical status of a subsequent construction.

19.4 Metrological correspondence of the twenty-three classes

The following table preserves the source's 23 classes and specifies the form of reading required by each upon reaching the metrological interface. Its function differs from the operator classification in Proposition 16.10 and the documentary inventory in Tables 7 and 8: here the comparable quantity, realisation convention and accompanying data are distinguished. Flavour and sector names guide the correspondence; the transmitted mathematical object remains in the corresponding fibres, representations and readers.

For charged leptons, the reading leads to the mass of the central section or to evaluation of the selected fibre. For neutrinos, it also preserves the transformation between mass and flavour and the mass-squared differences; it therefore does not reduce to an isolated scalar. For quarks, every figure is accompanied by the renormalisation scheme and scale. The electromagnetic and chromatic classes respectively express an experimental bound and the null gauge section; the W , Z and scalar classes require the declared pole or resonance convention. Composites preserve mass, width and constituent identity. The three topological classes are compared through their fusion, braiding and gap laws once the dynamical realisation is fixed; their dimensions and characters belong to their own codomains. The virtual class preserves the pole, obstruction and prolongation horizon. This reading determines which object reaches metrology without conflating the twenty-three classes with twenty-three universal scalar masses.

Table 3: Metrological reading of the twenty-three classes and type of transmitted object.

Class	Representatives	Object documented for the reading
Charged lepton 1	Electron and positron	Central route, conjugation and beta coordinate.
Charged lepton 2	Muon and antimuon	Fibre, readers and evaluation of the declared signature.
Charged lepton 3	Tau and antitau	Fibre, readers and evaluation of the declared signature.
Neutrino 1	Electron flavour and conjugate	Neutral fibre and mass–flavour realisation.
Neutrino 2	Muon flavour and conjugate	Neutral fibre and mass–flavour realisation.
Neutrino 3	Tau flavour and conjugate	Neutral fibre and mass–flavour realisation.
Up-type quark 1	$u, \bar{u}$	Flavour and colour fibre; scheme and scale chart.
Down-type quark 1	$d, \bar{d}$	Flavour and colour fibre; scheme and scale chart.
Up-type quark 2	$c, \bar{c}$	Flavour and colour fibre; scheme and scale chart.
Down-type quark 2	$s, \bar{s}$	Flavour and colour fibre; scheme and scale chart.
Up-type quark 3	$t, \bar{t}$	Flavour and colour fibre; scheme and scale chart.
Down-type quark 3	$b, \bar{b}$	Flavour and colour fibre; scheme and scale chart.
Electromagnetic gauge	Photon	Null chart and helicity representation.
Chromatic gauge	Gluons	Null chart and adjoint colour representation.
Charged gauge	W^+, W^-	Route, signature and pole convention of the realisation.
Neutral gauge	Z^0	Route, neutral mixing and pole convention.
Scalar	Higgs	Scalar representation, signature and pole convention.
Mesonic composite	Mesons and conjugates	Constituents, binding record and composite reading.
Baryonic composite	Baryons and conjugates	Trilinear composition, binding and composite reading.
Fibonacci topological	$1, \tau$	Fusion ring and Frobenius–Perron dimension.
Ising topological	$1, \psi, \sigma$	Fusion rules and sectoral representation.
Cyclic topological	$\mathbb{Z}/6\mathbb{Z}$	Winding characters and braid representation.
Virtual	Intermediate sections	Finite horizon, prolongation and obstruction.

19.5 Metrological snapshot, units and transcription accuracy

The snapshot identified by the corpus as Particle Data Group 2026 is used, with a cutoff date of 15 January 2026 and a documented consultation on 26 July 2026. This section reproduces that documentary state: it does not incorporate a subsequent catalogue update. The references in the evaluation supplement and those in the historical grid retain their individual provenance; they are not automatically assigned the edition of the general inventory.

Each value is preserved as its original decimal string, together with its presentation, unit, uncertainty and provenance key. This decision prevents an intermediate floating-point conversion from changing the last digits or removing significant zeros from the transcription. The original string and the typographical representation are distinct data: the former reproduces the file’s arithmetic exactly; the latter communicates the result with precision appropriate for reading. The digital representation digits of a reference do not increase its experimental precision.

Conversion between MeV and GeV uses the exact factor 10^3 , applied to both the value and its uncertainty. To present a mass, the unit is written MeV/c^2 or GeV/c^2 ; to present rest energy, MeV or GeV. The relation $E = mc^2$ is the chart allowing passage from one reading to the other. Comparing coordinates requires fixing that chart in both columns.

The inventory snapshot expressly records the scheme and scale of its external records as not supplied. A property name may contain an indication such as “Breit–Wigner”; it is preserved literally, but that partial indication does not complete the metadata required by the comparison. Empty cells, unavailable-data markers and not-applicable markers remain distinct. A documentary unknown is not replaced by zero.

19.6 Individualised evaluations and reading conditions

Table 4 brings together 20 numerical presentations: the electronic beta coordinate, two null outputs and 17 evaluations conditional on declared descriptors. Among the latter, 11 correspond to elementary sectors and 6 to composites. This count expresses the content of that documentary presentation; the fibres, collections and records of the domain retain their own extent.

The table preserves all twenty entries. To facilitate reading, four modalities are identified: B, electronic beta coordinate; N, null chart; F, evaluation conditional on a complete signature; A, conditional angular evaluation. The subscript c denotes material composition. The condition of each entry belongs to its result type and is also preserved in the electronic catalogue accompanying the text.

Table 4: Twenty individualised evaluations, with their chart and comparison modality.

State	HMT coordinate, MeV/c^2	Subsequent documentary reference	Type
$e^\pm$	0.510998950690000809	0.51099895069(16) MeV/c^2	B
$\mu^\pm$	105.658360294435829303	$105.6583755 \pm 0.0000023 \text{ MeV}/c^2$	F
$\tau^\pm$	1776.860917631432295595	$1776.93 \pm 0.09 \text{ MeV}/c^2$	F
$u, \bar{u}$	2.165924536173907663	$2.16 \pm 0.07 \text{ MeV}/c^2$	A
$d, \bar{d}$	4.679550162948874172	$4.70 \pm 0.07 \text{ MeV}/c^2$	A
$c, \bar{c}$	1270.500838641911135286	$1.2729 \pm 0.0045 \text{ GeV}/c^2$	A
$s, \bar{s}$	92.940093907845640641	$92.9 \pm 0.7 \text{ MeV}/c^2$	A
$t, \bar{t}$	172597.479544019136261367	$172.60 \pm 0.27 \text{ GeV}/c^2$	A
$b, \bar{b}$	4190.119763299790218075	$4.186 \pm 0.006 \text{ GeV}/c^2$	A
γ	0	$< 10^{-18} \text{ eV}/c^2$, preserved bound	N
g	0	Null gauge chart; no experimental comparison row	N
$W^\pm$	80378.964909704547024865	$80.3625 \pm 0.0077 \text{ GeV}/c^2$	F
Z^0	91187.533444313407844855	$91.1879 \pm 0.0020 \text{ GeV}/c^2$	F
H	125292.466042536133000433	$125.13 \pm 0.11 \text{ GeV}/c^2$	A
π^0	134.976724327872125739	$134.976827767685 \text{ MeV}/c^2$, supplement	F_c
$\pi^\pm$	139.570485171572191375	$139.570390983681 \text{ MeV}/c^2$, supplement	F_c
$K^\pm$	493.677085649530612476	$493.676599458041 \text{ MeV}/c^2$, supplement	F_c
K^0	497.611186497750457359	$497.610767531258 \text{ MeV}/c^2$, supplement	F_c
$p, \bar{p}$	938.271751546984898299	$938.27208943 \text{ MeV}/c^2$, supplement	F_c
$n, \bar{n}$	939.565810472507023313	$939.56542194 \text{ MeV}/c^2$, supplement	F_c

In this table, reference values are presented with the decimal notation and units declared for each observable. The calculation record also preserves the decimal string used in the evaluation, including the difference between that string and its rounded presentation for the tau, Z and Higgs. The quark comparison remains informative until the scheme and scale chart is assembled. The W, Z, H sectors also require the resonance convention to be preserved. The six composite references come from the supplement identified in the evaluation records; their metadata are not completed by analogy with the elementary references.

The complete signatures documented in modalities F and F_c , in the order $\sigma = (n_A, s_C, k_{\Delta_4}, b_{120}, b_\zeta)$, are:

μ	42, -3, 8, 0, 2
τ	64, 0, 25, -1, 6
W	94, -1, 0, -2, 4
Z	95, -1, -11, 0, -4
π^0	44, -4, -25, 1, -2
$\pi^\pm$	44, 1, -2, 2, -1
$K^\pm$	54, -1, 17, -2, 2
K^0	54, 0, 32, 3, -2
p	59, 0, 9, 1, 0
n	59, 1, -34, 0, 1

The angular modalities preserve, respectively for u, d, c, s, t, b, H , the pairs

$$(n_A, s_C) = (11, 7), (17, 8), (61, 8), (41, -3), (100, -1), (71, -5), (97, 9).$$

The two declared components are not extended by three implicit zeros. The evaluation equation and the mechanism producing a descriptor are different assertions; the proofs of both are found in their respective definitions and propositions.

19.7 Relative differences and uncertainty propagation

For a fixed common chart and two positive coordinates of the same observable, the relative difference and logarithmic difference are

$$\delta_X = \frac{m_X^{\text{HMT}} - m_X^{\text{ext}}}{m_X^{\text{ext}}}, \quad r_X = \log \frac{m_X^{\text{HMT}}}{m_X^{\text{ext}}}, \quad \delta_X^{\text{ppm}} = 10^6 \delta_X.$$

A common unit conversion $m \mapsto am$, with $a > 0$, leaves both differences invariant: the factor a cancels in the quotients. The absolute difference is multiplied by a , as is its uncertainty. For a zero external value, a one-sided bound or an interval without a central estimate, the preceding quotient is replaced by the comparison appropriate to that observation type.

When a joint uncertainty model is available, let $d_X = m_X^{\text{HMT}} - m_X^{\text{ext}}$. Its variance is

$$u^2(d_X) = u^2(m_X^{\text{HMT}}) + u^2(m_X^{\text{ext}}) - 2 \text{Cov}(m_X^{\text{HMT}}, m_X^{\text{ext}}).$$

The formula follows by expanding $\text{Var}(U - V)$. If $u(d_X) > 0$, the normalised difference is $z_X = d_X/u(d_X)$. Its statistical interpretation requires the corresponding probabilistic model and covariance; an unrecorded field is not equivalent to zero variance. Asymmetric uncertainties are preserved as such and are not automatically converted into a symmetric deviation.

For several species, the joint covariance matrix is retained. With the discrete signatures fixed, if $f_X(\vartheta) = \log(m_X/m_{\text{ref}})$ depends on continuous coordinates already constructed and J is its Jacobian matrix, first-order propagation gives

$$\Sigma_f = J \Sigma_\vartheta J^\top.$$

This formula preserves correlations shared by the angular pair, Δ_4 , the action ratio and the towers. Its application presupposes the regime in which linearisation controls the remaining terms. An integer change of signature is recorded as a discrete variation of the construction, not as an infinitesimal Gaussian fluctuation.

The precision of an evaluation therefore has several sources: the experimental reference; the unit, scheme and scale chart; the route distribution; the tower approximation; and the selected coupling. If Σ_{ij} denotes the covariance between contributions i, j , the total covariance of their sum is $\sum_{i,j} \Sigma_{ij}$. Retaining the cross blocks avoids ascribing independence to several agreements that use the same antecedents.

19.8 Absolute scale and control of the harmonic approximation

The absolute scale and relative correction enter at different points in the construction. Fix an internally constructed basal reference section $m_\star^{(0)}$, and let $\chi_\star = \chi_{\text{full}}(\sigma_\star, \eta_\star)$. In the normalisation relative to that section, the expression is

$$m_X^\beta = m_\star^{(0)} \frac{\chi_{\text{full}}(\sigma_X, \eta_X)}{\chi_\star} R_{\text{act}}^{\kappa_X}.$$

The common basal factor remains explicit. If the reference is electronic, $m_\star^{(0)} = m_e^{(0)}$ is the dimensional realisation of $\mathbf{e}_0$, with all factors in its basal equation, and $\chi_\star = \chi_{\text{full}}(\sigma_e, \eta_e)$. This recovers $m_e^\beta = m_e^{(0)} R_{\text{act}}^{\kappa_e}$. If the common clock is used to write $m_\star^{(0)} = m_{\text{clk}} b_\star$, the factor $b_\star$ preserves that basal normalisation; it is not replaced by 1 or incorporated twice into the character. For two states in the same chart and with the same reference, the quotient is

$$\frac{m_Y^\beta}{m_X^\beta} = \frac{\chi_{\text{full}}(\sigma_Y, \eta_Y)}{\chi_{\text{full}}(\sigma_X, \eta_X)} R_{\text{act}}^{\kappa_Y - \kappa_X}.$$

Cancellation of $m_{\star}^{(0)}/\chi_{\star}$ is exact. It comprises the common basal prefactor and the common factors in the absolute branch of action, clock and velocity chart. Consequently, the action decade 10^{-34} retains its role in the dimensional realisation and is not inserted again into the relative corrector. Differences between routes remain in their signatures, readers and tower sections.

To control a truncation, it is useful to fix the normal form of the character. Let

$$\zeta_{270} = 135\Delta_4^2, \quad N_h = \eta_h + b_{120}\mathbf{1}_{\{h=120\}}, \quad \mathcal{S} = \langle 90, 120 \rangle \setminus \{0\}.$$

Then

$$\log \chi_{\text{full}} = n_A A_{\text{rad}} + \frac{s_C}{6} C_{\text{rad}}^* + k_{\Delta_4} \Delta_4 + b_{\zeta} \zeta_{270} + \sum_{h \in \mathcal{S}} N_h s_h.$$

In this convention, the height-120 term appears only once. The term $b_{\zeta} \zeta_{270}$ retains its quadratic origin in the record; $N_{270} s_{270}$ belongs to the harmonic reading. Equality of the numerical index does not identify the two objects.

Truncation bound and multiplicative error. Suppose that, for $h > H$, the already determined coefficients of a route satisfy $|N_h s_h| \leq C q^h$, with $C \geq 0$ and $0 < q < 1$. The logarithmic tail R_H satisfies

$$|R_H| \leq \sum_{\substack{h \in \mathcal{S} \\ h > H}} C q^h \leq \sum_{h=H+1}^{\infty} C q^h = \frac{C q^{H+1}}{1-q} =: \varepsilon_H.$$

If m_H is the coordinate obtained by truncating that tower while all other factors remain fixed, $m/m_H = e^{R_H}$. Monotonicity of the exponential yields

$$e^{-\varepsilon_H} \leq \frac{m}{m_H} \leq e^{\varepsilon_H}, \quad \left| \frac{m - m_H}{m_H} \right| \leq e^{\varepsilon_H} - 1.$$

The bound assigns precision to the evaluation from the route and its operators. The values of C, q, H are justified by that construction; the experimental residual selects neither the tower length nor the bound to be proved.

19.9 Complete preservation and reproducibility of the comparison

The catalogue preserves complete records in tabular and structured formats. Each row retains its source file, ordinal, original key and occurrence number of that key. This identification preserves even two literally identical rows: equality of content does not remove the second occurrence. Format conversion preserves the original strings, empty cells and statuses, and includes the hash of the input bytes.

The inventory of 855 observables is also linked to the metrological rows of the integral corpus. The correspondence is verified by the pair consisting of the observable type and its identifier; the check compares multiplicities, not merely sets of keys. The internal tables of cells, collections, multisecutions and routes remain separate from that inventory. The 23 classes, 20 numerical presentations and historical grid also retain their own files and statuses.

Transcription checks verify recovery of every field, row order, duplicates, decimal strings and correspondence with the reference tables. Their scope is documentary. The mathematical proofs of the atlas, readers and material realisation establish the internal outputs; metrological comparison completes that exposition through well-defined observables and charts. Documentary integrity and the sufficiency of a proof are checked separately.

Catalogue extension therefore follows an accumulative rule: a new evaluation incorporates the state, its record and signature, the tower section, reader, realisation, comparison chart and provenance. Earlier cases retain their content and cutoff date. Thus, inventory growth makes a greater part of the domain visible without retrospectively altering what each proof and evaluation established.

20 Conclusions

The principal result is the construction of a positive mass operator on the fibre of a persistent particle, with a determined scalar specialisation for the central electron. Positivity follows from the congruence transformation of the positive basal operator, with a self-adjoint reader and positive action ratio. On the electronic trajectory, the two readers produce (80, 54, 6) and determine the same return correction. In the adopted dimensional chart, the complete evaluation gives

$$m_e^\beta = 0.5109989506900008086 \dots \text{ MeV}/c^2.$$

The value combines the electronic prefactor and the return correction; it is compared with the reference value and uncertainty stated in Table 4. The scalar result specific to the centre does not replace the operators or individualisation conditions of the remaining collections.

The trajectory supporting that evaluation is not stationary: its cellular projection visits exactly the twenty-seven positions defined by $c-r \equiv 0, \pm 1 \pmod{9}$. The reversible register distinguishes histories sharing the same absolute-value histogram; the spectral decomposition of $(+++---)^2$ concentrates its power at $k = 2, 6, 10$ and exactly recovers $(\Omega, P_6) = (80, 6)$. The operator criterion on the fibre further establishes that a TPK transport preserves mass if and only if it preserves the complete coordinate of the character and the return exponent, equivalently $[M^\beta, V_F] = 0$ on the stable domain.

The regional ambient space of 729 positions is likewise not left as an isolated cardinality. Pairwise trit grouping gives the bijection $\beta : \mathbb{F}_3^6 \rightarrow E_9^3$, identifying it with the body S_0 , while the completion $\hat{E}^3$ decomposes disjointly into $729 + 243 + 27 + 1 = 1000$ positions. The complete shell contains 271 positions, the apex contributes exactly $1/1000$ of the uniform measure, and projective compatibility preserves normalisation in every dimension. The proof and the two figures in the main text separately exhibit the set partition and its cubic realisation.

The provenance of the coefficients is determined by the ordered evolution of 108 operators $\mathcal{U}_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t$. The chart $\mathcal{R}_{12}$ extracts the complete incidence ledger from the terminal state; $\mathcal{E}_{108}^{90,120}$ produces the channel coordinate; Π_H extracts the signature; and $H_{12}/4$ obtains K . The integral inverse restores the channels and sum exactly. The periodic record, normalisation of α , action scale, and mass readers act on those outputs. The same structural origin is therefore preserved from the kernel's operations through to evaluation and metrological comparison.

20.1 Persistent identity, atlas, and conjugation

Persistence is formulated over compatible extensions; the route preserves its observable history; the representation determines which sheet, orientation, spin, flavour, and colour data distinguish states. This identity precedes the mass operator and relates its readings without reducing the particle to a numerical coordinate.

The atlas provides a finite classification that can be traversed in full. Its 81 cells, 56 collections, 13 multisections and 324 oriented conditions are four censuses of different objects. Their organisation identifies the central exception of the electron and the non-central fibres. In the latter, preserving a multisection or an operator is a positive description of the result: it specifies what remains equivariant and which additional operation produces a spectral reading. Physical individualisation takes place through its representation and channel, preserving this distinction.

Conjugation likewise does not reduce to a single sign inversion. Cellular inversion, reversal of a route, the involution on words, charge conjugation and spinorial holonomy preserve their own domains and actions. The Möbius construction specifies the fibre and loop that generate orientation inversion. Return to the same phase may leave different memory, while restoration of a representation may require an additional circuit. These results describe how structure persists when its partial observations coincide. Antiunitary conjugation additionally requires a dynamical distinction: commutation with the Hamiltonian reverses the direction of the evolution group, whereas preservation of its real subspace at all times requires the corresponding anticommutation relation. Compatibility with the mass operator retains its own formulation.

20.2 Mass evaluation, composition, and occupation

Mass evaluation combines two levels. The refined character uses the integer signature and channel moments; the dimensional section composes action, clock and unit chart. The electronic prefactor remains present when expressing the electron relative to the chronological scale. The ratio of action sections acts in the return readers and allows fibres to be compared through exponents derived from their records. Equality of the two central readers has received a concrete proof on the electronic trajectory, with its scope distinguished from that of the other atlas profiles. Clock velocity and constitutive velocity coincide through the identity between the product of the two channels and the determinant of the response operator. Changing the unit chart transforms their coordinates while leaving the dimensional quantity invariant.

Particle composition has been placed before signature extraction. Binding records preserve boundary conditions, sheets and predecessors on which the functionals act. The binding term therefore has a defined place in the composite operator. Mesons and baryons are organised through their constituent spaces and chromatic representations; resonances additionally require open-channel dynamics and the definition of the pole. Their width and survival time are coordinated readings of that dynamics.

In the neutral antipodal sector, the extractor's two scalar contractions vanish exactly: $b_{120}'' = b_{\zeta}'' = 0$. This cancellation does not remove the oriented register, because the anti-invariant subspace retains the complex operator $J^2 = -I$; nor does it extend to general routes or modify masses or PMNS transport. In the longitudinal prolongation, comparison of ten 108-step words with their 1080-step concatenation includes the explicit seam term. The new boundaries are not crowns, so that $\Delta C_z = 0$ and $\Delta k_{\Delta_4} = \Delta T_z$, with -2 an attainable local-control value. Representation periodicity and the memory of the complete TPK state therefore remain distinct.

The graded completion develops occupation over the complete modes already declared. Antisymmetrisation constructs the CAR, from which Pauli exclusion follows. Symmetrisation constructs the CCR on the finite-particle domain and makes visible the boundary term introduced by a bosonic truncation. The distinction between equality of a mass and coincidence of a mode is expressed by the Gram determinant of double occupation. Spin, sheet or colour may preserve independent modes even when a partial observable has the same value.

Orbital realisations connect action to closure conditions and Kepler–Sommerfeld geometry. Their metric, central-dynamics and representation assumptions have been retained alongside the equations that use them. In the fusion and braiding sectors, the incidence ring, its positive dimensions, association-change matrices and exchange characters are made explicit, together with the local identities and realisation conditions used by each. This organisation makes it possible to recognise what belongs to an algebraic construction, a dynamical realisation and a physical identification.

20.3 From the spectrum to metrological comparison

Metrological comparison completes the progression through typed observables. The sector-by-sector comparisons are accompanied by the complete inventories and their sources. A running mass, a pole mass, a limit and a width retain their definitions, units and uncertainties. Each model evaluation is linked to its record, reader and chart; each experimental observation preserves its edition and provenance. The 855 documentary mass and width rows constitute the domain of the comparison catalogue, distinct from the number of individualised predictions.

The contribution brings together three determinations: the object that persists, the operation producing its spectrum, and the chart in which that spectrum is compared. Mass results from their composition; orientation, memory, and fibre structure remain available to distinguish states sharing a partial reading. The electronic case realises that composition in an explicit scalar section, while the atlas preserves the diversity of operator-valued sectors and their physical realisations. This relationship between structure and magnitude unifies the result.

The additional composition closes the longitudinal and gravitational dependence without altering the spectrum. Incidence produces $L_\alpha = \alpha_{\text{HMT}}^{16}(5759/23040)L_\star$ before the clock is constructed; hence $R_{108} = L_\alpha$. The energy $Q_L = \hbar c_{\text{int}}/L_\alpha$ jointly fixes m_{clk} and the cotransformed factor $b_e = E_e^{(0)}/Q_L$, so that the scale cancels in every already constructed mass. On massive fibres, the two conserved equalities $H\Lambda = \hbar c_{\text{int}}I$ and $\mathcal{R}_g\Lambda = L_\alpha^2 I$ uniquely force $G = c_{\text{int}}^3 L_\alpha^2 / \hbar$. This determination follows the mass operator; the expression in terms of t_0 is its circular corollary, not its generator. The null sector retains its non-invertible treatment, and no species-dependent decimal correction is introduced.

A Complete census of regional signatures and orbits

The following tables complete the finite proof in Section 4. Each signature is obtained from (4.2); its multiplicity counts the initial conditions of one cursor, not pairs of cursors. Pairs are formed subsequently through (4.3).

A.1 Signatures of the two sheets

Sheet	Six-window signature	Preimages
Additive	122564	18
Additive	122898	18
Additive	212645	18
Additive	212988	18
Additive	221456	18
Additive	221889	18
Additive	456221	18
Additive	465898	18
Additive	546988	18
Additive	564122	18
Additive	645212	18
Additive	654889	18
Additive	889221	18
Additive	889654	18
Additive	898122	18
Additive	898465	18
Additive	988212	18
Additive	988546	18
Multiplicative	000000	108
Multiplicative	177393	8
Multiplicative	177555	8
Multiplicative	177771	8
Multiplicative	339555	8
Multiplicative	339771	8
Multiplicative	339933	8
Multiplicative	393177	8
Multiplicative	393393	8
Multiplicative	393555	8
Multiplicative	555177	8
Multiplicative	555339	8
Multiplicative	555393	8
Multiplicative	555555	24
Multiplicative	555717	8
Multiplicative	555771	8
Multiplicative	555933	8
Multiplicative	717555	8
Multiplicative	717717	8
Multiplicative	717933	8
Multiplicative	771177	8
Multiplicative	771339	8
Multiplicative	771555	8
Multiplicative	933339	8
Multiplicative	933555	8
Multiplicative	933717	8

A.2 Orbits of the ternary image

The representative is the lexicographic minimum of the orbit. “Fibre” counts decimal emissions over a word, and “Weight” counts their joint initial conditions. Diagonals are collected over the entire orbit. The census $\nu = (n_0, n_1, n_2)$ preserves symbol multiplicity and is invariant under permutations.

Representative	Orbit	Fibre	Weight	ν	Diagonals
001002	6	2	288	(4, 1, 1)	{0, 1, 3, 4, 6, 7}
001011	6	2	288	(3, 3, 0)	{0, 1, 3, 4, 6, 7}
001012	6	2	288	(3, 2, 1)	{2, 5, 8}
001021	6	2	2088	(3, 2, 1)	{2, 5, 8}
001022	6	1	144	(3, 1, 2)	{2, 5, 8}
001100	3	6	864	(4, 2, 0)	{2, 5, 8}
001101	6	2	288	(3, 3, 0)	{0, 1, 3, 4, 6, 7}
001110	6	1	1944	(3, 3, 0)	{0, 3, 6}
001112	6	5	1008	(2, 3, 1)	{0, 1, 2, 3, 4, 5, 6, 7, 8}
001202	6	1	144	(3, 1, 2)	{2, 5, 8}
001210	6	1	144	(3, 2, 1)	{2, 5, 8}
001211	6	2	576	(2, 3, 1)	{2, 5, 8}
001220	6	6	864	(3, 1, 2)	{0, 1, 3, 4, 6, 7}
001222	6	1	144	(2, 1, 3)	{2, 5, 8}
002012	6	1	144	(3, 1, 2)	{1, 4, 7}
002112	6	1	144	(2, 2, 2)	{0, 3, 6}
002211	6	1	144	(2, 2, 2)	{1, 4, 7}
002220	6	1	144	(3, 0, 3)	{0, 3, 6}
011021	6	2	288	(2, 3, 1)	{0, 1, 3, 4, 6, 7}
011112	6	2	288	(1, 4, 1)	{0, 1, 3, 4, 6, 7}
011120	6	1	144	(2, 3, 1)	{0, 3, 6}
011201	6	1	1944	(2, 3, 1)	{1, 4, 7}
011211	6	1	144	(1, 4, 1)	{0, 3, 6}
011220	6	1	144	(2, 2, 2)	{1, 4, 7}
011222	6	1	144	(1, 2, 3)	{1, 4, 7}
012112	6	3	432	(1, 3, 2)	{1, 2, 4, 5, 7, 8}
012121	6	1	144	(1, 3, 2)	{0, 3, 6}
012202	6	1	144	(2, 1, 3)	{0, 3, 6}
012210	6	1	144	(2, 2, 2)	{2, 5, 8}
012220	6	1	144	(2, 1, 3)	{1, 4, 7}
012222	6	1	144	(1, 1, 4)	{2, 5, 8}
021022	6	1	144	(2, 1, 3)	{2, 5, 8}
021121	6	3	432	(1, 3, 2)	{0, 1, 2, 3, 4, 5, 6, 7, 8}
021202	6	3	432	(2, 1, 3)	{0, 1, 2, 3, 4, 5, 6, 7, 8}
022022	3	2	288	(2, 0, 4)	{1, 4, 7}
022112	6	1	144	(1, 2, 3)	{1, 4, 7}
022121	6	6	864	(1, 2, 3)	{0, 1, 2, 3, 4, 5, 6, 7, 8}
022202	3	2	288	(2, 0, 4)	{0, 3, 6}
022211	6	2	288	(1, 2, 3)	{1, 2, 4, 5, 7, 8}
022222	6	1	144	(1, 0, 5)	{1, 4, 7}
112112	3	6	1440	(0, 4, 2)	{0, 1, 2, 3, 4, 5, 6, 7, 8}
112211	3	4	576	(0, 4, 2)	{1, 2, 4, 5, 7, 8}
112222	6	2	288	(0, 2, 4)	{1, 2, 4, 5, 7, 8}

B Catalogues of multisections, route types and sector classes

This appendix organizes three distinct levels of the internal inventory. The 13 multisections group atlas cells; the 56 collections classify internal routes; the 23 classes document their articulation with physical sectors and representations. These cardinalities do not count the same object and cannot be added to obtain a number of particles. The tables accompany the definitions and proofs of the article: their purpose is to locate each record, its incidences and its readers, not to replace a proof with a census.

The documentary order of the rows, their identifiers and all repetitions appearing in the sources are preserved. Literal identifiers allow the tables to be checked against the JSON files. A dash denotes an empty source cell; it does not represent zero. The files also preserve every field and digit, the original statuses and verifiable provenance. The *Structural and Metrological Catalogue of Particles*, the data appendix to this delivery, lists the 324 oriented routes in its Section 2 and the 855 metrological records in its Section 3. The definitions and proofs of the present article are included in its main text and appendices.

B.1 The 13 multisections of the atlas

For a multisection M , let $\mathcal{C}(M)$ be its set of cells and let $\gamma_{B_1}(c)$ be the primary route preserved by the atlas in cell c . In this structural appendix, the fiber is defined by

$$B_1(M) := \text{span}\{e_{\gamma_{B_1}(c)} : c \in \mathcal{C}(M)\} \subseteq \ell^2(\mathcal{R}_{\text{adm}}).$$

The primary routes of distinct cells determine distinct basis vectors; consequently, $\text{rank } B_1(M) = \dim B_1(M) = \#\mathcal{C}(M)$. This definition fixes the object to which the table's rank refers; the collections column counts the route configurations present, not physical species. The integer spectra of the readers K_D and K_Ω are preserved separately. In the literal notation, x indicates multiplicity. The 13 rows have the documentary status of an exact diagonal operator on the fiber B_1 . Their representation as a physical scalar requires the declared transport, except for the rank-one selection provided for in the record itself. This statement reproduces the recorded scope and does not turn every spectrum into a unique mass.

Table 5: The 13 multisections, without row selection.

Multisection	Rank and incidence	Classification and depth	Integer readers
charged_lepton_ G0	<i>Rank:</i> 1 <i>Cells (identifiers):</i> 41 <i>Collections:</i> 1	<i>Collection:</i> charged_lepton <i>Depth:</i> G0_center <i>Color:</i> none	<i>K D:</i> (80,54,6)x1 <i>K Ω:</i> (80,54,6)x1
charged_lepton_ G2	<i>Rank:</i> 8 <i>Cells (identifiers):</i> 21,23,25,39,43,57, 59,61 <i>Collections:</i> 8	<i>Collection:</i> charged_lepton <i>Depth:</i> G2 <i>Color:</i> none	<i>K D:</i> (0,24,0)x1; (40,78,27) x2; (80,54,7)x1; (80,66,2)x1; (80,66,3)x1; (100,66,0)x1; (100,66,2)x1 <i>K Ω:</i> (80,54,6)x8

Continued on the next page

Continuation

Multisection	Rank and incidence	Classification and depth	Integer readers
charged_lepton_G4	Rank: 16 Cells (identifiers): 1,3,5,7,9,19,27,37,45,55,63,73,75,77,79,81 Collections: 16	Collection: charged_lepton Depth: G4 Color: none	$K D$: (0,24,0)x1; (40,24,2)x2; (40,54,6)x2; (40,84,30)x2; (60,78,25)x3; (80,66,2)x2; (80,66,3)x3; (80,66,4)x1 $K \Omega$: (24,54,2)x1; (56,54,5)x4; (80,54,6)x11
neutrino_G1	Rank: 2 Cells (identifiers): 40,42 Collections: 2	Collection: neutrino Depth: G1 Color: none	$K D$: (40,24,2)x1; (40,78,27)x1 $K \Omega$: (40,54,4)x1; (80,54,6)x1
neutrino_G2	Rank: 4 Cells (identifiers): 22,24,58,60 Collections: 4	Collection: neutrino Depth: G2 Color: none	$K D$: (0,24,0)x1; (40,84,30)x1; (60,78,25)x1; (80,66,3)x1 $K \Omega$: (4,54,2)x1; (80,54,6)x3
neutrino_G3	Rank: 6 Cells (identifiers): 20,26,38,44,56,62 Collections: 6	Collection: neutrino Depth: G3 Color: none	$K D$: (40,24,2)x2; (40,78,27)x1; (60,84,28)x1; (80,54,6)x1; (80,66,3)x1 $K \Omega$: (36,54,4)x1; (56,54,5)x1; (80,54,6)x4
neutrino_G4	Rank: 8 Cells (identifiers): 2,4,6,8,74,76,78,80 Collections: 8	Collection: neutrino Depth: G4 Color: none	$K D$: (0,24,0)x1; (40,24,2)x1; (40,78,27)x2; (40,84,30)x1; (80,54,7)x1; (80,66,2)x1; (100,66,0)x1 $K \Omega$: (36,54,4)x1; (56,54,5)x1; (80,54,6)x6
quark_down_G1	Rank: 2 Cells (identifiers): 32,50 Collections: 2	Collection: quark_down Depth: G1 Color: B G	$K D$: (40,24,2)x1; (40,78,27)x1 $K \Omega$: (40,54,4)x1; (80,54,6)x1
quark_down_G2	Rank: 4 Cells (identifiers): 30,34,48,52 Collections: 4	Collection: quark_down Depth: G2 Color: B G R	$K D$: (0,24,0)x1; (40,84,30)x1; (60,78,25)x1; (80,66,2)x1 $K \Omega$: (4,54,2)x1; (80,54,6)x3
quark_down_G3	Rank: 6 Cells (identifiers): 12,14,16,66,68,70 Collections: 6	Collection: quark_down Depth: G3 Color: B G R	$K D$: (40,24,2)x2; (40,78,27)x1; (60,84,28)x1; (80,54,6)x1; (80,66,4)x1 $K \Omega$: (36,54,4)x1; (80,54,6)x5

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Continuation

Multisection	Rank and incidence	Classification and depth	Integer readers
quark_down_G4	<i>Rank:</i> 8 <i>Cells (identifiers):</i> 10,18,28,36,46,54,64,72 <i>Collections:</i> 8	<i>Collection:</i> quark_down <i>Depth:</i> G4 <i>Color:</i> B G R	<i>K D:</i> (0,24,0)x1; (40,24,2)x1; (40,78,27)x2; (40,84,30)x1; (80,54,7)x1; (80,66,3)x1; (100,66,2)x1 <i>K Ω:</i> (36,54,4)x1; (80,54,6)x7
quark_up_G1	<i>Rank:</i> 4 <i>Cells (identifiers):</i> 31,33,49,51 <i>Collections:</i> 3	<i>Collection:</i> quark_up <i>Depth:</i> G1 <i>Color:</i> B G R	<i>K D:</i> (40,54,6)x1; (60,78,25)x1; (80,66,2)x1; (80,66,3)x1 <i>K Ω:</i> (80,54,6)x4
quark_up_G3	<i>Rank:</i> 12 <i>Cells (identifiers):</i> 11,13,15,17,29,35,47,53,65,67,69,71 <i>Collections:</i> 12	<i>Collection:</i> quark_up <i>Depth:</i> G3 <i>Color:</i> B G R	<i>K D:</i> (40,24,2)x2; (40,78,27)x2; (60,84,28)x1; (80,54,6)x3; (80,66,3)x1; (80,66,4)x1; (100,66,0)x1; (100,66,2)x1 <i>K Ω:</i> (56,54,5)x3; (80,54,6)x9

B.2 The 56 route configurations

The signature identifying a collection is preserved literally, with its signs and separators. The cells and multisections make explicit that a route collection is not, by definition, equivalent to a physical species. The joint profiles count the combinations recorded within the collection. The decimal spectra of the character and the complete signature spectra remain intact in `catalogo_familias.json`; the incidence and integer readers expressed here provide a means of navigating that information.

Table 6: The 56 internal route configurations.

Collection and signature	Incidence	K_D	K_Ω
1 24 0 12 0 -12 +++---++---	<i>Number of cells:</i> 5 <i>Cells (identifiers):</i> 21,31,41,51,81 <i>Multisections:</i> charged_lepton_ G0 charged_lepton_G2 charged_lepton_G4 quark_up_ G1 <i>Collections:</i> charged_lepton quark_up <i>Color:</i> R none <i>Joint profiles:</i> 5	(0,24,0)x2; (40,54,6)x1; (60,78,25)x1; (80,54,6)x1	(24,54,2)x1; (80,54,6)x4

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Continuation

Collection and signature	Incidence	K_D	K_Ω
2 24 0 4 0 -12 +++-----	<i>Number of cells:</i> 2 <i>Cells (identifiers):</i> 11,61 <i>Multisections:</i> charged_lepton_ G2 quark_up_G3 <i>Collections:</i> charged_lepton quark_up <i>Color:</i> R none <i>Joint profiles:</i> 2	(60,84,28)x1; (80,54, 7)x1	(80,54,6)x2
3 24 0 8 0 -12 +++-----	<i>Number of cells:</i> 2 <i>Cells (identifiers):</i> 1,71 <i>Multisections:</i> charged_lepton_ G4 quark_up_G3 <i>Collections:</i> charged_lepton quark_up <i>Color:</i> R none <i>Joint profiles:</i> 2	(60,78,25)x1; (80,54, 6)x1	(80,54,6)x2
4 28 -6 -2 -2 -8 0++---0++---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 53 <i>Multisections:</i> quark_up_G3 <i>Collections:</i> quark_up <i>Color:</i> G <i>Joint profiles:</i> 1	(80,66,3)x1	(56,54,5)x1
5 28 -6 -4 -2 -8 0++---0++---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 20 <i>Multisections:</i> neutrino_G3 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 1	(80,66,3)x1	(56,54,5)x1
6 28 -6 0 0 -12 +++-----	<i>Number of cells:</i> 2 <i>Cells (identifiers):</i> 30,70 <i>Multisections:</i> quark_down_G2 quark_down_G3 <i>Collections:</i> quark_down <i>Color:</i> G <i>Joint profiles:</i> 2	(40,24,2)x1; (80,66, 2)x1	(80,54,6)x2
7 28 -6 10 0 -12 +++-----	<i>Number of cells:</i> 2 <i>Cells (identifiers):</i> 33,64 <i>Multisections:</i> quark_down_G4 quark_up_G1 <i>Collections:</i> quark_down quark_up <i>Color:</i> G <i>Joint profiles:</i> 2	(40,24,2)x1; (80,66, 2)x1	(36,54,4)x1; (80,54, 6)x1

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Continuation

Collection and signature	Incidence	K_D	K_Ω
8 28 -6 12 0 -12 +++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 10 <i>Multisections:</i> quark_down_G4 <i>Collections:</i> quark_down <i>Color:</i> G <i>Joint profiles:</i> 1	(100,66,2)x1	(80,54,6)x1
9 28 -6 2 0 -12 +++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 3 <i>Multisections:</i> charged_lepton_G4 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(40,84,30)x1	(80,54,6)x1
10 28 -6 4 -2 -8 ++0---++0---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 50 <i>Multisections:</i> quark_down_G1 <i>Collections:</i> quark_down <i>Color:</i> G <i>Joint profiles:</i> 1	(40,78,27)x1	(80,54,6)x1
11 28 -6 4 0 -12 +++-----	<i>Number of cells:</i> 2 <i>Cells (identifiers):</i> 9,40 <i>Multisections:</i> charged_lepton_G4 neutrino_G1 <i>Collections:</i> charged_lepton neutrino <i>Color:</i> none <i>Joint profiles:</i> 2	(40,24,2)x1; (80,66,2)x1	(40,54,4)x1; (80,54,6)x1
12 28 -6 4 0 -8 +0+---+0+---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 80 <i>Multisections:</i> neutrino_G4 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 1	(40,78,27)x1	(80,54,6)x1
13 28 -6 6 -2 -8 ++0---++0---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 74 <i>Multisections:</i> neutrino_G4 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 1	(40,78,27)x1	(80,54,6)x1

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Continuation

Collection and signature	Incidence	K_D	K_Ω
14 28 -6 6 0 -12 +++-----	<i>Number of cells:</i> 3 <i>Cells (identifiers):</i> 13,43,63 <i>Multisections:</i> charged_lepton_ G2 charged_lepton_G4 quark_up_G3 <i>Collections:</i> charged_lepton quark_up <i>Color:</i> G none <i>Joint profiles:</i> 3	(40,24,2)x1; (80,66, 2)x1; (100,66,2)x1	(80,54,6)x3
15 28 -6 6 0 -8 +0+---+0+---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 23 <i>Multisections:</i> charged_lepton_ G2 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(40,78,27)x1	(80,54,6)x1
16 28 -6 8 0 -12 +++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 60 <i>Multisections:</i> neutrino_G2 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 1	(40,84,30)x1	(80,54,6)x1
17 28 6 -10 0 -12 +++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 19 <i>Multisections:</i> charged_lepton_ G4 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(40,84,30)x1	(80,54,6)x1
18 28 6 -2 0 -12 +++-----	<i>Number of cells:</i> 5 <i>Cells (identifiers):</i> 18,29,49,59,79 <i>Multisections:</i> charged_lepton_ G2 charged_lepton_G4 quark_down_G4 quark_up_G1 quark_up_G3 <i>Collections:</i> charged_lepton quark_down quark_up <i>Color:</i> B none <i>Joint profiles:</i> 4	(40,24,2)x1; (40,78, 27)x1; (80,66,3)x2; (100,66,0)x1	(80,54,6)x5

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Continuation

Collection and signature	Incidence	K_D	K_Ω
19 28 6 -4 0 -12 +++-----	<i>Number of cells:</i> 2 <i>Cells (identifiers):</i> 2,42 <i>Multisections:</i> neutrino_G1 neutrino_G4 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 2	(40,78,27)x1; (100, 66,0)x1	(80,54,6)x2
20 28 6 -6 0 -12 +++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 69 <i>Multisections:</i> quark_up_G3 <i>Collections:</i> quark_up <i>Color:</i> B <i>Joint profiles:</i> 1	(80,66,4)x1	(80,54,6)x1
21 28 6 -8 0 -12 +++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 52 <i>Multisections:</i> quark_down_G2 <i>Collections:</i> quark_down <i>Color:</i> B <i>Joint profiles:</i> 1	(40,84,30)x1	(80,54,6)x1
22 28 6 0 0 -12 +++-----	<i>Number of cells:</i> 4 <i>Cells (identifiers):</i> 22,32,72,73 <i>Multisections:</i> charged_lepton_ G4 neutrino_G2 quark_ down_G1 quark_down_G4 <i>Collections:</i> charged_lepton neutrino quark_down <i>Color:</i> B none <i>Joint profiles:</i> 3	(40,24,2)x1; (40,78, 27)x1; (80,66,3)x2	(40,54,4)x1; (80,54, 6)x3
23 28 6 2 0 -12 +++-----	<i>Number of cells:</i> 2 <i>Cells (identifiers):</i> 8,39 <i>Multisections:</i> charged_lepton_ G2 neutrino_G4 <i>Collections:</i> charged_lepton neutrino <i>Color:</i> none <i>Joint profiles:</i> 2	(40,24,2)x1; (40,78, 27)x1	(36,54,4)x1; (80,54, 6)x1
24 28 6 4 0 -12 +++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 12 <i>Multisections:</i> quark_down_G3 <i>Collections:</i> quark_down <i>Color:</i> B <i>Joint profiles:</i> 1	(80,66,4)x1	(80,54,6)x1

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<i>Continuation</i>			
Collection and signature	Incidence	K_D	K_Ω
25 28 6 8 0 -12 +++---++----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 62 <i>Multisections:</i> neutrino_G3 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 1	(40,24,2)x1	(80,54,6)x1
26 30 -6 -4 -2 -8 0++---0++---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 4 <i>Multisections:</i> neutrino_G4 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 1	(80,54,7)x1	(56,54,5)x1
27 30 -6 -8 0 -8 +0+---+0+---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 55 <i>Multisections:</i> charged_lepton_G4 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(60,78,25)x1	(80,54,6)x1
28 30 -6 0 -2 -8 ++0---++0---	<i>Number of cells:</i> 2 <i>Cells (identifiers):</i> 24,34 <i>Multisections:</i> neutrino_G2 quark_down_G2 <i>Collections:</i> neutrino quark_down <i>Color:</i> R none <i>Joint profiles:</i> 2	(0,24,0)x1; (60,78,25)x1	(4,54,2)x1; (80,54,6)x1
29 30 -6 0 -2 -8 0++---0++---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 75 <i>Multisections:</i> charged_lepton_G4 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(40,54,6)x1	(56,54,5)x1
30 30 -6 0 0 -12 +++---++----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 14 <i>Multisections:</i> quark_down_G3 <i>Collections:</i> quark_down <i>Color:</i> R <i>Joint profiles:</i> 1	(80,54,6)x1	(80,54,6)x1

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<i>Continuation</i>			
Collection and signature	Incidence	K_D	K_Ω
31 30 -6 0 0 -8 +0+---+0+---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 54 <i>Multisections:</i> quark_down_G4 <i>Collections:</i> quark_down <i>Color:</i> R <i>Joint profiles:</i> 1	(0,24,0)x1	(80,54,6)x1
32 30 -6 4 0 -12 +++++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 65 <i>Multisections:</i> quark_up_G3 <i>Collections:</i> quark_up <i>Color:</i> R <i>Joint profiles:</i> 1	(80,54,6)x1	(80,54,6)x1
33 30 -6 8 0 -12 +++++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 44 <i>Multisections:</i> neutrino_G3 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 1	(60,84,28)x1	(80,54,6)x1
34 30 6 -4 0 -12 +++++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 68 <i>Multisections:</i> quark_down_G3 <i>Collections:</i> quark_down <i>Color:</i> R <i>Joint profiles:</i> 1	(60,84,28)x1	(80,54,6)x1
35 30 6 -8 0 -12 +++++-----	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 17 <i>Multisections:</i> quark_up_G3 <i>Collections:</i> quark_up <i>Color:</i> R <i>Joint profiles:</i> 1	(80,54,6)x1	(80,54,6)x1
36 30 6 0 0 -12 +++-----	<i>Number of cells:</i> 2 <i>Cells (identifiers):</i> 38,58 <i>Multisections:</i> neutrino_G2 neutrino_G3 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 2	(60,78,25)x1; (80,54, 6)x1	(80,54,6)x2
37 30 6 0 0 -8 +++0-+++0-	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 78 <i>Multisections:</i> neutrino_G4 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 1	(0,24,0)x1	(80,54,6)x1

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Continuation

Collection and signature	Incidence	K_D	K_Ω
38 30 6 0 2 -8 ++++-0++++-0	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 27 <i>Multisections:</i> charged_lepton_ G4 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(40,54,6)x1	(56,54,5)x1
39 30 6 0 2 -8 +++0-+++0--	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 48 <i>Multisections:</i> quark_down_G2 <i>Collections:</i> quark_down <i>Color:</i> R <i>Joint profiles:</i> 1	(0,24,0)x1	(4,54,2)x1
40 30 6 4 0 -12 ++++-++++-	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 7 <i>Multisections:</i> charged_lepton_ G4 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(60,78,25)x1	(80,54,6)x1
41 30 6 8 0 -12 ++++-++++-	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 28 <i>Multisections:</i> quark_down_G4 <i>Collections:</i> quark_down <i>Color:</i> R <i>Joint profiles:</i> 1	(80,54,7)x1	(80,54,6)x1
42 34 0 -12 -2 -8 0++++-0++++	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 45 <i>Multisections:</i> charged_lepton_ G4 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(80,66,4)x1	(56,54,5)x1
43 34 0 -12 0 -12 ++++-++++-	<i>Number of cells:</i> 2 <i>Cells (identifiers):</i> 57,76 <i>Multisections:</i> charged_lepton_ G2 neutrino_G4 <i>Collections:</i> charged_lepton neutrino <i>Color:</i> none <i>Joint profiles:</i> 2	(40,84,30)x1; (80,66, 2)x1	(80,54,6)x2

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<i>Continuation</i>			
Collection and signature	Incidence	K_D	K_Ω
44 34 0 -16 0 -8 +++0-+++0-	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 37 <i>Multisections:</i> charged_lepton_ G4 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(40,24,2)x1	(80,54,6)x1
45 34 0 -16 0 -8 +0+---+0+---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 15 <i>Multisections:</i> quark_up_G3 <i>Collections:</i> quark_up <i>Color:</i> B <i>Joint profiles:</i> 1	(40,78,27)x1	(80,54,6)x1
46 34 0 -20 -2 -8 0+---0+---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 35 <i>Multisections:</i> quark_up_G3 <i>Collections:</i> quark_up <i>Color:</i> B <i>Joint profiles:</i> 1	(100,66,0)x1	(56,54,5)x1
47 34 0 -4 -2 -8 ++0---++0---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 56 <i>Multisections:</i> neutrino_G3 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 1	(40,24,2)x1	(36,54,4)x1
48 34 0 -4 0 -12 +++---++++---	<i>Number of cells:</i> 3 <i>Cells (identifiers):</i> 6,25,46 <i>Multisections:</i> charged_lepton_ G2 neutrino_G4 quark_ down_G4 <i>Collections:</i> charged_lepton neutrino quark_down <i>Color:</i> B none <i>Joint profiles:</i> 2	(80,66,2)x1; (80,66, 3)x2	(80,54,6)x3
49 34 0 -4 2 -8 ++++-0++++-0	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 67 <i>Multisections:</i> quark_up_G3 <i>Collections:</i> quark_up <i>Color:</i> G <i>Joint profiles:</i> 1	(100,66,2)x1	(56,54,5)x1

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Continuation

Collection and signature	Incidence	K_D	K_Ω
50 34 0 -8 -2 -8 ++0---++0---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 66 <i>Multisections:</i> quark_down_G3 <i>Collections:</i> quark_down <i>Color:</i> B <i>Joint profiles:</i> 1	(40,78,27)x1	(80,54,6)x1
51 34 0 -8 -2 -8 0++---0++---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 77 <i>Multisections:</i> charged_lepton_G4 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(80,66,3)x1	(56,54,5)x1
52 34 0 -8 0 -12 ++++---++---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 36 <i>Multisections:</i> quark_down_G4 <i>Collections:</i> quark_down <i>Color:</i> G <i>Joint profiles:</i> 1	(40,84,30)x1	(80,54,6)x1
53 34 0 -8 0 -8 +0+---+0+---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 5 <i>Multisections:</i> charged_lepton_G4 <i>Collections:</i> charged_lepton <i>Color:</i> none <i>Joint profiles:</i> 1	(40,24,2)x1	(80,54,6)x1
54 34 0 0 -2 -8 ++0---++0---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 26 <i>Multisections:</i> neutrino_G3 <i>Collections:</i> neutrino <i>Color:</i> none <i>Joint profiles:</i> 1	(40,78,27)x1	(80,54,6)x1
55 34 0 0 0 -8 +0+---+0+---	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 47 <i>Multisections:</i> quark_up_G3 <i>Collections:</i> quark_up <i>Color:</i> G <i>Joint profiles:</i> 1	(40,78,27)x1	(80,54,6)x1

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Continuation

Collection and signature	Incidence	K_D	K_Ω
56 34 0 0 2 -8 +++0--++0--	<i>Number of cells:</i> 1 <i>Cells (identifiers):</i> 16 <i>Multisections:</i> quark_down_G3 <i>Collections:</i> quark_down <i>Color:</i> G <i>Joint profiles:</i> 1	(40,24,2)x1	(36,54,4)x1

B.3 The 23 documented classes

These rows bring together public sectors, residences and representation. The status key refers to the adjacent legend and preserves the status of the source. In particular, an exact internal operator, a conditional evaluation and a closed scalar realization remain distinct assertions. The table neither updates their status nor turns a conditional record into a prediction.

Table 7: Original class statuses.

Key	Literal status	Documentary reading
C01	CLOSED_ELECTRON_TWO_WAY	Electron: documented bidirectional closure.
C02	CLOSED_NULL_CHART	Null chart closed at this stage.
C03	NOT_A_UNIVERSAL_MASS_SCALAR	The class is not expressed as a universal mass scalar.
C04	OPEN_FLAVOR_COLOR_SCALARIZATION	Flavor and color scalarization open at this stage.
C05	OPEN_MIXING_AND_SCALE	Mixing and scale open at this stage.
C06	OPEN_NAMED_STATE_SCALARIZATION	Scalarization of the individually identified state open at this stage.
C07	OPEN_PHYSICAL_SCALARIZATION	Physical scalarization open at this stage.
C08	OPEN_ROUTE_AND_SCALARIZATION	Route and scalarization open at this stage.

Table 8: The 23 catalogue classes, in their documentary order.

Class	Representatives and residence	Incidence and representation	Status
SM-LC-1 lepton_cargado	<i>Representatives:</i> e^- / e^+ <i>Residence:</i> center (5,5) and antisection	<i>Selected cells:</i> 1 <i>Multisections:</i> charged_lepton_G0 <i>Charge:</i> $\pm 1 e$ <i>Representation:</i> colorless	C01
SM-LC-2 lepton_cargado	<i>Representatives:</i> μ^- / μ^+ <i>Residence:</i> charged-lepton multisection	<i>Selected cells:</i> 8 <i>Multisections:</i> charged_lepton_G2 <i>Charge:</i> $\pm 1 e$ <i>Representation:</i> colorless	C07
SM-LC-3 lepton_cargado	<i>Representatives:</i> τ^- / τ^+ <i>Residence:</i> charged-lepton multisection	<i>Selected cells:</i> 16 <i>Multisections:</i> charged_lepton_G4 <i>Charge:</i> $\pm 1 e$ <i>Representation:</i> colorless	C07
SM-NU-1 neutrino	<i>Representatives:</i> $\nu_e / \text{anti-}\nu_e$ <i>Residence:</i> neutrino multisections	<i>Selected cells:</i> 20 <i>Multisections:</i> neutrino_G1 neutrino_G2 neutrino_G3 neutrino_G4 <i>Charge:</i> 0 <i>Representation:</i> colorless	C05
SM-NU-2 neutrino	<i>Representatives:</i> $\nu_\mu / \text{anti-}\nu_\mu$ <i>Residence:</i> neutrino multisections	<i>Selected cells:</i> 20 <i>Multisections:</i> neutrino_G1 neutrino_G2 neutrino_G3 neutrino_G4 <i>Charge:</i> 0 <i>Representation:</i> colorless	C05
SM-NU-3 neutrino	<i>Representatives:</i> $\nu_\tau / \text{anti-}\nu_\tau$ <i>Residence:</i> neutrino multisections	<i>Selected cells:</i> 20 <i>Multisections:</i> neutrino_G1 neutrino_G2 neutrino_G3 neutrino_G4 <i>Charge:</i> 0 <i>Representation:</i> colorless	C05
SM-Q-U1 quark	<i>Representatives:</i> $u / \text{anti-}u$ <i>Residence:</i> up-type multisection	<i>Selected cells:</i> 16 <i>Multisections:</i> quark_up_G1 quark_up_G3 <i>Charge:</i> $\pm 2/3 e$ <i>Representation:</i> R,G,B = Z/3 orbit	C04

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Continuation

Class	Representatives and residence	Incidence and representation	Status
SM-Q-D1 quark	<i>Representatives:</i> d / anti-d <i>Residence:</i> down-type multisection	<i>Selected cells:</i> 20 <i>Multisections:</i> quark_down_G1 quark_down_G2 quark_down_G3 quark_down_G4 <i>Charge:</i> $\pm 1/3$ e <i>Representation:</i> R,G,B = Z/3 orbit	C04
SM-Q-U2 quark	<i>Representatives:</i> c / anti-c <i>Residence:</i> up-type multisection	<i>Selected cells:</i> 16 <i>Multisections:</i> quark_up_G1 quark_up_G3 <i>Charge:</i> $\pm 2/3$ e <i>Representation:</i> R,G,B = Z/3 orbit	C04
SM-Q-D2 quark	<i>Representatives:</i> s / anti-s <i>Residence:</i> down-type multisection	<i>Selected cells:</i> 20 <i>Multisections:</i> quark_down_G1 quark_down_G2 quark_down_G3 quark_down_G4 <i>Charge:</i> $\pm 1/3$ e <i>Representation:</i> R,G,B = Z/3 orbit	C04
SM-Q-U3 quark	<i>Representatives:</i> t / anti-t <i>Residence:</i> up-type multisection	<i>Selected cells:</i> 16 <i>Multisections:</i> quark_up_G1 quark_up_G3 <i>Charge:</i> $\pm 2/3$ e <i>Representation:</i> R,G,B = Z/3 orbit	C04
SM-Q-D3 quark	<i>Representatives:</i> b / anti-b <i>Residence:</i> down-type multisection	<i>Selected cells:</i> 20 <i>Multisections:</i> quark_down_G1 quark_down_G2 quark_down_G3 quark_down_G4 <i>Charge:</i> $\pm 1/3$ e <i>Representation:</i> R,G,B = Z/3 orbit	C04
SM-GA-EM boson_gauge	<i>Representatives:</i> γ <i>Residence:</i> null gauge sector	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> 0 <i>Representation:</i> colorless	C02
SM-GA-GL boson_gauge	<i>Representatives:</i> g <i>Residence:</i> color gauge sector	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> 0 <i>Representation:</i> gauge octet	C02

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Continuation

Class	Representatives and residence	Incidence and representation	Status
SM-GA-W boson_gauge	<i>Representatives:</i> W^+ / W^- <i>Residence:</i> massive gauge sector	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> $\pm 1 e$ <i>Representation:</i> colorless	C08
SM-GA-Z boson_gauge	<i>Representatives:</i> Z^0 <i>Residence:</i> massive gauge sector	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> 0 <i>Representation:</i> colorless	C08
SM-H-1 escalar	<i>Representatives:</i> H <i>Residence:</i> scalar sector	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> 0 <i>Representation:</i> colorless	C08
COMP-MESON meson	<i>Representatives:</i> $\pi^0, \pi^\pm, K^0, K^\pm, \eta, \rho, \varphi, J/\psi, Y...$ <i>Residence:</i> product of section and antisection	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> dependent on the composite state <i>Representation:</i> singlet or other representations	C06
COMP-BARYON barion	<i>Representatives:</i> p, n and baryonic configurations <i>Residence:</i> product of three sections	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> dependent on the composite state <i>Representation:</i> physical singlet	C06
TOPO-FIB anyon_no_abeliano	<i>Representatives:</i> $1, \tau$ <i>Residence:</i> topological sector	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> defined by the topological codomain <i>Representation:</i> not applicable	C03
TOPO-ISING sector_ majorana	<i>Representatives:</i> $1, \psi, \sigma$ <i>Residence:</i> topological sector / zero	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> dependent on the realization <i>Representation:</i> colorless	C03
TOPO-Z6 anyones_ abelianos	<i>Representatives:</i> boson, fermion and four braided sectors <i>Residence:</i> topological sector $Z/6$	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> $Z/6$ definida por el trenzado <i>Representation:</i> colorless	C03
VIRT-1 virtual	<i>Representatives:</i> virtual sections <i>Residence:</i> TPK inverse system	<i>Selected cells:</i> 0 <i>Multisections:</i> — <i>Charge:</i> dependent on the prolongation sector <i>Representation:</i> according to sector	C03

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